

Regina Sipos

Critical Technical Practice in Grassroots Innovation

Proposals of Change through
Indonesian Collective Intrinsic Design

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Indonesian Collective Intrinsic
Design

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Abstract

This study aims to extend the term critical making—previously used in engineering studies and design and as a term for DIY citizenship—to grassroots innovation communities in the Global South, specifically in Indonesia. My research was motivated by the need to investigate non-Western approaches to bottom-up, democratic design experiments that resemble making, and to uncover techno-solutionist promises and ideological colonialism as previously highlighted by other critical researchers. My hypothesis is that such a shift of perspective allows researchers to better understand critical-material practices that have been previously neglected by academia.

My research has also been inspired by recent socio-technical shifts caused by an *era of participation*, including the rapid growth in the number of maker spaces and the revival of open-source movements. These are claimed to have improved the innovative capabilities of everyday citizens and caused a growing interest in the research of participatory practices. At the same time, imaginaries and promises of technological development have been critically investigated. It is still rare, however, that academia researches such practices without intervening (e.g., through participatory research), or attempts to understand autonomous, intrinsic grassroots design processes. Those, especially in the Global South, so far remained under-researched.

This is the main research gap the study aims to address. By mapping grassroots critical makers on the island of Java, conducting criterion-based analysis, and combining analytical frameworks that were created to investigate grassroots innovation movements and volunteer-based intrinsic design projects from a techno-social perspective, this study answers three research questions. The first

research question aims to understand the *grassroots* dimension by building on the hypothesis that such communities do not constitute a homogenous movement, but are small, situated, and diverse. The second research question is concerned with understanding the *criticality* of grassroots innovators, including how critical knowledge is produced and shared. The third research question is about *making*; how is the term received in the Global South, and how does the practice manifest in intrinsic design projects?

To map potential projects, I conducted field research in Indonesia. The aim of this criterion-based research was to exemplify what grassroots critical making might look like in the Global South. Contributing mainly to the fields of grassroots innovation studies and post-colonial computing, this study found that there exists a pluriverse of grassroots communities. These grassroots communities are highly political and critical in their framings, especially of blueprint developmentalism and techno-solutionism that does not account for local context. This criticality informs the critical-material, self-directed projects such grassroots communities conducted. The societal relevance of their practices is that they contribute to reflexivity in wider society, inspire future generations of grassroots innovators, and generate opportunities for the creation of “hopeful interventions” that might, in the long term, contribute to good, sustainable, self-directed development in Indonesian communities.

This study’s novel approach contributes to multiple fields by extending the term “critical making” to the grassroots by revealing the pluriverse of such under-researched practices, and by investigating intrinsic design processes without interfering in them.

Contents

1	Introduction, Research Questions and Prospectus	1
1.1	Introduction	1
1.2	Situating Innovative Practices	5
1.3	Research Design	27
1.4	Conclusion and Prospectus	38
2	The State of Critical Making	41
2.1	Critical Making as Innovative Scholarly Practice	43
2.2	Extension Towards Design: Making as Critique	47
2.3	Exploring Citizen-Driven Projects: DIY Citizenship	49
2.4	Grounding Critical Making in Grassroots Practices	51
3	Methodological Tools and Theoretical Arguments	55
3.1	Critical Making as a Research Framework	55
3.2	Extension Towards Grassroots in the Global South	61
3.3	Conclusion	87
4	Research Methodology and Analytical Framework	91
4.1	Participatory Research Design and Qualitative Data Collection	92
4.2	Case Study Selection and Analysis	97
4.3	Ethical Considerations and Reflexivity	108
4.4	Scopes and Limitations	110

5	In the Field: Grassroots Critical Making	113
5.1	Preliminary Research: Grounding Critical Making in Grassroots	114
5.2	Mapping Grassroots Critical Making	128
5.3	Grassroots Critical Making	130
5.4	Case Study I: Combining Ancestral Knowledge and Rural Community Infrastructures	154
5.5	Case Study II: #hackfromhome Hackathon	182
6	Cross-Case Analysis and Discussion	221
6.1	Pluriverse of Grassroots Communities	223
6.2	Co-Production of Critical Knowledge	231
6.3	Making As Intrinsic Design	244
7	Conclusion	255
7.1	Towards a Grassroots Critical Making	255
7.2	Contributions	256
7.3	Future Research	257
7.4	Final Thoughts	259
	Bibliography	261

Abbreviations

AT	Appropriate Technology
ANT	Actor-Network Theory
CM	Critical Making
CoP	Community of Practice
CTP	Critical Technical Practice
D4D	Digital for Development
DIWO	Do It With Others
DIY	Do it Yourself
GIM	Grassroots Innovation Movement
HC4D	Human-Computer Interaction for Development
ICT	Information and Communication Technology
ICT4D	Information and Communication Technology for Development
ISP	Internet Service Provider
ITU	International Telecommunication Union
PD	Participatory Design
R&D	Research and Development
SCoT	The Social Construction of Technology
STI	Science, Technology and Innovation
STS	Science and Technology Studies
Tech4Good	Technology for Social Good
UN	United Nations
UNCTAD	United Nations Conference on Trade and Development
UNDP	United Nations Development Programme
UNRISD	United Nations Research Institute for Social Development

List of Figures

Fig. 1.1	Dimensions of Making. Based on Doyle (2013)	18
Fig. 1.2	Made: Technology on Affluent Leisure Time (Hertz 2012b)	25
Fig. 1.3	Relevant Aspects of Grassroots Critical Making	33
Fig. 1.4	Research Questions and Attributes of Communities Researched	40
Fig. 2.1	A Timeline of the Term Critical Making	43
Fig. 2.2	Three Stages of Critical Making	45
Fig. 3.1	Methodological Considerations of Researching Critical Making	56
Fig. 3.2	Theories and Methodologies Extended Towards Grassroots Critical Makers	62
Fig. 3.3	Industrial and Grassroots Innovation	71
Fig. 3.4	Theoretical Framework: Theories, Concepts and Methodologies	89
Fig. 4.1	Infrastructuring in Intrinsic Project Design	107
Fig. 5.1	Four-step Approach of the Scientific Inquiry	115
Fig. 5.2	Hertz's Slides at the STARTS Talks on Critical Making	116
Fig. 5.3	Presenting the definition of Critical Making	117
Fig. 5.4	"What is Critical Making for you?" at C&T'19	121
Fig. 5.5	Hotel Purgatorio by Fajar and Murti	126
Fig. 5.6	How Interviewees Recommend Local Initiatives	129

Fig. 5.7	Interpretations of Reusable Cloth Pads: as Enabler of Informal Economy in Surabaya (04.01.2020) and as Critique and Human Rights in Jogjakarta (14.01.2020)	134
Fig. 5.8	“Search, Sweeping, Confiscation?” What to do if a Police Raid Comes	137
Fig. 5.9	Triptych of a Colonial Past: Tinted Glass Window Advertising of a Dutch Insurance Firm	140
Fig. 5.10	Rural Bamboo and Woodworking Workshop	143
Fig. 5.11	Traditional Handmade Clay Brick Factory	144
Fig. 5.12	Traditional Arts and Sockets, Pins and Jacks	146
Fig. 5.13	The Collective’s Van	148
Fig. 5.14	Water Filtration System of a DIYbio Collective	151
Fig. 5.15	Fields and Traditional Rice Storage Huts of Kasepuhan Ciptagelar	161
Fig. 5.16	The Chief Leader of the Indigenous Community in his Workshop	165
Fig. 5.17	Voucher for the Community Network’s Internet Hotspot	168
Fig. 5.18	Infrastructural Adjustments to the Community Network	175
Fig. 5.19	Generative Prerequisites of the Indigenous Community Network Project	176
Fig. 5.20	Indigenous Community Network Artefact Ecology	178
Fig. 5.21	Hackathon Applicants’ Professional Background	208
Fig. 5.22	Screenshots from the Mentoring Sessions: Prototyping and Mock-ups	210
Fig. 5.23	Timeline of the #hackfromhome Hackathon	211
Fig. 5.24	#hackfromhome Hackathon Artefact Ecology	216
Fig. 6.1	Input from the Field in Cross-case Analysis and Discussion	222
Fig. 6.2	The Fate of Ideas and Development of Critical Artefacts Over Time	236
Fig. 6.3	Sustainability Dimensions of an Intrinsic Design Project	238

List of Tables

Table 4.1	Data Collection Methods	96
Table 4.2	Overview of Thematic Analysis Process	99
Table 5.1	Field Research Timeline	125
Table 5.2	Timeline of Mapping Grassroots Critical Making in Indonesia	129
Table 5.3	Skills, Problem Framings and Interventions Shaping the Artefact Ecology	198
Table 5.4	Overview of Rejected and Accepted Platforms	203

Introduction, Research Questions and Prospectus

1

1.1 Introduction

Grassroots ingenuity combines critical thinking and technological practice to investigate and address social injustices. For example, WoeLab developed 3D printers out of Western e-waste from Togolese landfills¹, data_labe involved the local communities of the favelas in digitally mapping their own neighborhoods to create transparency about basic sanitation in Brazil², and the Singapore-based Engineering Good refurbished over 10,000 laptops for children who would otherwise be left out of the digital classroom during the pandemic³. However, these grassroots efforts have been historically mostly overlooked by academia.

In terms of researching innovative capabilities and processes, historically academia has mainly been interested in institutionalized research organizations. In his Critical Theory of Technology, Feenberg strongly criticized the exclusion of the majority from technology development (Feenberg 1991). In the current era of participation (Smith et al. 2017), academic interest in exploring more inclusive methods such as participatory design has increased. Albeit still niche, some researchers look beyond human- and user-centered design led by designers, and instead turn to grassroots conducting “democratic design experiments” (Ehn et al. 2014). Activist academia has started engaging in reflexive processes of decolonizing itself (de Castro Leal et al. 2021) by critiquing mainstream practices of technology development and the top-down innovation processes of

¹ <https://www.facebook.com/woelab/>, see also (Remond, Abba, and Useille 2019).

² <https://cocozap.datalabe.org>, see also (de Lima Santos and Mesquita 2021).

³ <https://engineeringgood.org>

international development projects and agencies (Talhoun et al. 2019). In a similar vein, researchers have explored novel models of researching innovation and technology development for plurality through actors in the Global South, and “appropriate technology” or “computing within limits” (Escobar 2018; de Valk 2021). Indeed, Escobar and the Indigenous leader Krenak (cited on page 203 in Szaniecki et al. 2020) and others provide alternative ontological views of (technology) design, opposing the idea of development based solely on industrial and economic growth. They oppose this idea because it excludes social groups and alternative approaches from the development process. The shifting of innovation practices from actors with large budgets to everyday people, enabled by more accessible, affordable, and “hackable” technologies, has brought a return to exploring grassroots innovation movements (Smith et al. 2017). The activities that constitute grassroots innovation are defined as follows:

“In North and South, in cities and rural settings, networks of activists, development workers, community groups and neighbours have been working with people to generate bottom-up solutions for sustainable developments; solutions that respond to the local situation and the interests and values of the communities involved; and where those communities have control over the processes involved and the outcomes. [...] Whether born of material and economic necessity, or motivated by social issues marginalized by the conventional innovation systems of states and markets, networks of people promote and coordinate alternative activity attentive to these needs and issues. They develop discourse and mobilize supportive resources among wider publics” (pp 3-4, Smith et al. 2017).

Simultaneously, the idea of bringing critical thinking and critical theory closer to technology development is gaining academic attention. Research and development (R&D) of high-tech solutions like Artificial Intelligence (AI) demand critical technical practice (Agre 1997). Critical making (see e.g. Ratto and Hockema (2009) or Hertz (2012), builds on the idea of CTP, and is defined as a method of combining critical thinking and material practice in engineering studies and design.

Outside of academia and traditional innovation, everyday people are equipped to open the “black box” of technology, and are *not* finding it empty (Winner 1993). In response, advocates and practitioners of open-source software and hardware are attempting to make the box as transparent as possible. A combination of two simultaneous developments has broadened and intensified laypeople’s access to science, innovation and technology (STI) and technical-material practice. On the one hand, people have gained access to knowledge through high levels of

internet penetration (ITU 2021). The internet has been supporting the democratization of technology and innovation through access to knowledge, including the knowledge and blueprints openly shared by innovator communities worldwide in open-source repositories⁴, social media groups, and messenger applications (Ratto and Boler 2014). On the other hand, people now have access to tools and offline communities. the spreading of makerspaces—community spaces where rapid prototyping and experimentation is supported by the cumulative knowledge of the community and enabled through analogue and digital tools. Such shared tools include 3D printers, laser cutters, CNC machines, and affordable open-source electronics like Arduino and Raspberry Pi computers. The term used for this material practice is “making”.

Artefacts for socially innovative maker projects include 3D-printed prosthetics and cheap sensor-based solutions for agriculture that make technological solutions affordable for low-income people. In terms of their framings, open innovation, knowledge sharing, and peer-to-peer learning in these spaces are central to the ideology of making (Capdevila 2013). At the same time, the open-source hardware (OSHW) movement (e.g., the Gathering for Open Science Hardware⁵) is inspiring many to upload blueprints for societally relevant prototypes into open repositories. These repositories include the Global Ecovillage Network Solution Library⁶, the “Whole Earth Catalog”,⁷ and the 3D Printing of Open Source Appropriate Technologies for Self-Directed Sustainable Development (OSAT) project⁸ (Pearce et al. 2010).

This rapid democratization of the access to technology is a landmark in the modern history of humanity. Through this process, citizen-driven R&D, citizen science and other participatory practices contribute to innovation. Over the past decade, technical-material practices like making have become visible, especially in the West. Maker communities have been thoroughly researched for its potential to assist distributed manufacturing and sustainability (Kohtala 2017; Berglund and Kohtala 2018; Klemichen et al. 2018) (Fleischmann, Hielscher, and Merritt 2016; Millard et al. 2018), post-capitalist practices (Baier, Hansing, and Werner 2016), the diversification of the economy (T. S. J. Smith 2019), education (Unterfrauner et al. 2019; Monfredini and Frosch 2019; Blikstein 2013; Blikstein 2017; Halverson and Sheridan 2014), openness (Ahmadi et al. 2019;

⁴ See e.g. GitHub, Thingiverse or Wikifactory

⁵ <http://openhardware.science>

⁶ <https://ecovillage.org/solutions>

⁷ <http://www.wholeearth.com>

⁸ <https://www.thingiverse.com/jpearce/collections/open-sourceappropriate-technology>

Saari et al. 2021), and citizen and open science (Fisher and Gould 2012; Chan 2019). Social scientific analysis of makerspaces and grassroots digital fabrication/innovation movements has only started to emerge. However, a focus on criticality, reflexivity, or responsible innovation—the main focus of this study—has been rare.

Critics and those concerned about digital rights say improvements are necessary before the “black box” of technology is truly opened. Examples of overreach include biased algorithms causing discrimination (Hajian et al. 2016) and technologies designed to make the users dependent on them (Oliver et al. 2011). Nevertheless, grassroots innovators around the world are engaging in the combination of critical thinking and material practice. However, research has neglected discourse on the critical, bottom-up approaches that can be found in the Global South, where societal issues are seemingly more imminent and sustainable solutions are not expected to come “from above” from development agencies, the government, or companies. As I will demonstrate, such grassroots innovators ask questions about socio-technical inclusion and justice, and take political stances on access and the morality of existing technologies when negotiating which fit ones into their artefact ecology.

From an academic perspective, it is thus timely to explore the critical technical practices of grassroots innovation communities. Such processes are participatory and interdisciplinary, and shed light on local issues, because the movements

“produce knowledge from below, information about society which is inconvenient to and resisted by those above: the wealthy, the mighty and the learned (or, as we might say, states, corporations and disciplines)” (page 1, Cox and Fominaya 2009).

The main goal of this study is to engage with grassroots innovators and describe their critical-material practices in academic terms. It thus extends the term critical making to the grassroots, and reveals their practices through a novel lens. The novelty of this approach is driven by learning from what structural dependency theorists call *the periphery* (Prebish 1950; Furtado 2020). This study researches the politico-economic Global South, specifically Indonesia, as supplemented with other examples from around the world. Here, I would like to point out that the novelty is *not* that grassroots innovators in the Global South are newly designing or innovating solutions; these are not emergent practices (Chan 2014). Moreover, as Szaniecki et al. put it,

“presenting participatory practices from the Global South leads to the scientific understanding of this ‘South’ as powerfully plural, inhabited by radical differences of all

kinds—cultural, social, ethnic and, ultimately, ontological” (page 204, Szaniecki et al. 2020).

The novelty of this study is that, inspired by the work of critical postcolonial thinkers like (Escobar 2018), locally specific and critical grassroots innovation processes and practices that lead to social and environmental sustainability are thought to contribute to the scientific knowledge base. Subsequently, comparing lessons learned from historical attempts like the development of appropriate technologies in international development derive insights for more sustainable context-relevant practices.

In the next sections of this chapter, I will briefly touch upon mainstream narratives of STI, including the historical dichotomy between innovation as practiced in the *center*—or top-down, mainstream, Western, innovation—and between the *peripheries*—grassroots innovation practiced in rural areas and by people in the Global South. After this, I will outline insights about the era of participation, which represents an interesting and novel shift in the democratization of innovation. I will then unite two topics under the pluriverse of grassroots innovation. Finally, I will outline in detail how citizen-driven innovation practices started through the digitally extended material practice of do-it-yourself (DIY) movements, referred to as making. Here, mainstream maker practices are contrasted with critical making practices. These considerations paint the background for the second part of this chapter, my research design. I will here briefly explain the theoretical background and methods, and present my original research questions, which are based on academic literature and research findings. In the closing part of this chapter, I offer a conclusion and brief prospectus for this study.

1.2 Situating Innovative Practices

1.2.1 Innovation in the *Centre* and the *Peripheries*

Traditionally mainstream innovation activities are understood as exclusive to industry, military, and academia, and conducted by formally organized research institutions (Smith et al. 2017). The term “innovation” has been seen by economists as a catalyst of growth (Schumpeter 1911), with entrepreneurs and startups promising jobs and economic prosperity in exchange for venture capital. The same narrative applies to every continent, driven by the global spread of so-called unicorn, decacorn, and hectocorn companies (valued over USD 1, 10 and 100 billion, respectively).

As opposed to popular narratives, mainstream innovation—greatly influenced by Schumpeter (see e.g. Schumpeter 1911 and Schumpeter 1934 in Alvedalen and Boschma (2017) or in Moulaert et al. (2014)—is not about invention. Instead, mainstream innovation is about entrepreneurship that implements new technical, organizational or market solutions, thus generating economic competitiveness. Innovators are known as the creators of ideas, and this narrative seemingly neglects that the most important aspect for any innovation to succeed is its adoption (Denning and Dew 2015). Many people have ideas, but few know how to realize them. Those with the capacity to move beyond the idea stage rarely account for the negative, unintended, or indirect impacts of their work. Academia is a big driver of innovation, and conventional science has also been criticized for the lack of reflective processes. Lack of reflexivity and responsibility has led to various issues (e.g., environmental destruction) while reproducing and fueling existing prejudices and discrimination (Wajcman 2010). There have been attempts to address the lack of reflexivity and responsibility, for example through the European Responsible Research and Innovation (RRI) framework, as well as courses on “design refusal” in engineering education⁹. Both would have been useful for example to avoid the “diesel dupe” scandal of 2015, in which engineers designed software that would change cars’ performance to cheat emission tests. However, it is predictable that engineers would rather keep their jobs than refuse to design the software, and there are no measures currently in place to hold companies accountable for their impacts on society and environment.

Whether done by academia, the public sector, development agencies, or, techno-fetishism (Hertz 2015a) in innovation and technosolutionism in international development (ICT4D and D4D, see e.g. (Ames 2019) are both widespread. The fascination with the utopian promises of technology have a long history. The best example of fiction warning about the consequences for others is *Metropolis*, the German futuristic dystopian science-fiction film from 1927. Today, the societal divide predicted by the movie is reality in many places, for example between the Global North (or Western countries) and the Global South. It is important to note that even though this study uses the term “Global South”, it is a term that contributes to the arbitrary division between “developed” and “underdeveloped”, “first world” and “third world”, or “here” and “there”. It is a polythetic category that is used as antonym, “with *‘the global north,’* an opposition that carries a

⁹ See for example the course titled *The Art of Saying „No“: Design Refusal* at the Olin College of Engineering <https://www.olin.edu/news-events/2021/the-art-saying-no-design-refusal>

great deal of imaginative baggage congealed around the contrast between centrality and marginality, capitalist modernity and its absence” (page 45, Comaroff 2016).

Building on the idea of the “capitalist modernity” and its absence, many see technology as a solution to societal problems caused by unjust technosocial arrangements. Despite repeated calls for more critical engagement with technology development (Agre 1997), simultaneously pursuing technical and sociopolitical agendas remains a methodological blind spot for academia. Either the possibilities of technical innovations or careful sociopolitical critique is prioritized (Lindtner, Bardzell, and Bardzell 2016).

Following the Second World War, the UN announced the “development decade” to counter the vast unemployment and underemployment in developing countries. The UN based their development on a Keynesian approach to economic growth and interventions transferring capital from developed to developing countries. This approach set the stage for all subsequent development decades (Koehler 2015). However, as an article published on the website of the UN shows, the decades leading up to the Millennium Development Goals were “four decades of struggle.” The gap between developed and developing countries continued to widen, and by the end of the 1990s, the UN realized that a focus on economic growth by itself was insufficient. New institutional preconditions for development, including decentralization and participation, were established (Jackson 2007). However, by then, the neoliberal policy of globalization, transfer of capital, and subsequently technology and science, created dependence on the West (Kelly Moore et al. 2011).

Focusing on the “geek diplomacy” of open-source hardware model for open science, Kera summarizes:

“[...] we can finally question the [...] whole idea of technology transfer rooted in the unreflected colonial views of the Global South as recipient of science knowledge leading to development (Byerlee and Fischer 2002) (Forero-Pineda 2006). The discussions about science in the Global South perpetuate a form of ‘epistemic violence’ (Spivak 1988) that defines technologies and science as things that are always transferred and applied in the developing countries with the help of various donors, corporate responsibility programmes or other innovators from the ‘west.’”(Kera 2015)

Decades have reinforced the center-periphery dichotomy through globalization and centralized, top-down development. It’s only recently that research agendas have recognized that activities are worthy of research in the periphery, too. Eder (2019), for example, in his critical survey of the geography of innovation,

found that until recently research has focused on successful core regions, “implicitly assuming that there is no innovation in peripheral areas” (page 1). Social scientists in rural studies such as Eder are challenging neoliberal globalization, because much like the Global South, rural areas are prone to being neglected by globalized development. Scholars discuss the challenging of global (food) chains through relocalization (Hendrickson and Heffernan 2002), highlight attitudes towards local embeddedness (Winter 2003), and localize movements and their apposition to global economies (Hess 2009b). Such efforts in the West are thus receiving renewed attention, and not only rurally. As Campbell summarizes in his book on grassroots innovations in urban societies “Making Massive Small Change”, *massive small* includes constructive activism for bottom-up change. As an alternative to ‘bigness’, it coalesces the collective power of many small ideas and actions. Massive small is a way to deliver radical incrementalism—including protocols, conditions, and behaviors—to help form urban neighborhoods (Campbell 2018).

In terms of the Global South, critiques of international development aren’t new. There have been centralized attempts to develop locally relevant technology-based solutions over the past decades. However, these attempts did not have the effects expected, because they were short-lived and overshadowed by other developmental approaches. Such attempts have recently started to be revisited, and there are now visible efforts towards engaging in post-colonial and localized practices of innovation. At an UNCTAD panel, Gillian M. Marcelle, the Executive Director of the University of Virgin Islands Research and Technology Park, building on her decades long work as a research scholar and advocate of technology and human development, summarized critical imperatives for redefining innovation in and for the Global South. She set the challenge to disrupt the status quo, in terms of science, technology and innovation (STI) supported by UN agencies, to better “draw on the resources, creativity and insights of all groups, not only those occupying positions of privilege, to tackle global problems” (Marcelle 2016). She suggested that we reconceptualize STI by diversifying and democratizing policy making and develop more effective measures to better align STI with the challenges of development.

There are at least a dozen adjectives describing different local bottom-up practices of technology development, hacking, and participatory innovation, as also briefly hinted at by Marcelle. These terms include frugal, user-centered (or human-centered), open, social, and grassroots innovation. The widespread adoption of these terms shows that the agents, arenas, agendas, and processes around innovation are opening up in society (Elliott and Turner 2012). There is a world of innovation “out there” separate from the historical institutions of

R&D. However, grassroots innovations remain under the radar of academia and industry, or traditional innovators. Opposed to mainstream innovation (and the policies supporting it), which is supposed to “keep-up with an apparently universal techno-economic frontier” (page 1, Smith et al. 2014), grassroots innovations are created in response to social justice and environmental issues that often arise in conventional innovation models (ibid.). These critiques of mainstream innovation, including ICT4D, are prevalent. The United Nations Research Institute for Social Development (UNRISD) published an online article on the “darker side of innovation for development”, which starts with the potential of innovation to contribute to positive social change. The article also warns that innovation disrupts existing systems and creates winners and losers, or people resist change because previous policy changes made things go “from bad to worse” (Twerenbold 2017). This is a general warning for innovators to be more careful about technology’s negative impacts and their local context. An example of the negative effects of the top-down innovation practices that grassroots activities attempt to counter are described by (de Valk 2021) and her critique of *green capitalism*. Here, she advocates for more research on and visibility of small-scale practices that are “challenging ‘business as usual’ (but) aren’t part of public discourse because they are small scale, less visible, often hard to access, easily appropriated by and seemingly unable to compete with the powerful lobby of large corporations” (page 1, ibid.). Thus, the people who are left behind by for-profit and business-model-driven innovation must try to solve the own issues arising from mainstream practices. Their practices continue to remain invisible, because the small scale is deemed irrelevant when observed ‘from above’.

Escobar, reflecting on how the world is being transformed through climate change, calls for a confrontation with “the design disaster that development is” (page 7, Escobar 2018). He calls this disaster *elimination design*, which erases vernacular design and local knowledge and replaces it with technologies developed elsewhere¹⁰. Such solutions are supposed to be universal, developed for everyone regardless of their social, political, economic, geographic, or religious context. Because of their supposed universality, these solutions often fail to serve communities. Recognizing the lack of impact in particular cases, there have been attempts to move away from this *blueprint developmentalism* (Smith et al. 2014), including development programs built on the work of the economist Schumacher, who advocated for intermediate technology in his book “Small is Beautiful” (Schumacher 1973). Intermediate, or “appropriate” technology was aligned with

¹⁰ It is important to note here that UNDP Accelerator Labs are attempting to address this by creating databases of grassroots innovations.

the UN's development decades and was supposed to free people from poverty and provide them with meaningful work. Schumacher's ideas were criticized for putting the creation of jobs before output and lacking output (profits) that placed commodities out of reach of the poor. Critique were also expressed about the approach constituting a version of neo-colonialism (Weidel 2015). In reality, the approach turned out to be the opposite of this critique; intermediate technology fostered a culture of well-being through meaningful work, and radically placed technology choice in the hands of the people (ibid).

Appropriate technology was also supposed to replace the development practices of simply importing large-scale, expensive, and inadequate technologies developed elsewhere. This idea appealed to the frustrations development workers in the field had with post-War industrialization blueprints. Schumacher's Intermediate Technology Development Group advised multiple UN agencies on how to ensure more appropriate technologies are being developed and deployed, see e.g. Rist (2011) and Willoughby (1990) in Smith et al. (2014). After a large uptake, however, the programs were abandoned and different approaches to top-down development took over.

Grassroots practitioners however continued to foster innovation movements. As described in detailed cases by (Smith et al. 2017), the appropriate technology movement in South America continue and became an instrument to foster social empowerment. Other notable examples include the People's Science Movements in India in the 1980s, the Honey Bee Network in India, started around 1988-1989, and the Social Technology Network in Brazil, established in 2004 (ibid.). These movements represent large-scale communities, however, and there are also smaller communities and collectives that do not necessarily belong to a movement.

In their comparison, Smith et al. (2017) compare such historical movements and today's grassroots movements for social inclusion through technology (or *grassroots innovation movements*). Through this comparison, they reveal three enduring challenges for grassroots innovation, which include: 1) attending to local specificities while simultaneously seeking widescale diffusion; 2) being appropriate to existing situations that the projects/communities ultimately seek to transform; and 3) working with project-based solutions to achieve social justice goals whose root causes rest in structures of economic and political power (Smith et al. 2014). These challenges lead to producing knowledge such as grassroots ingenuity, empowering inclusion, and structural critique, which contribute to plurality and reflexivity (ibid.). In terms of exploring plurality, I argue in this study that it isn't essential for researchers to look to major grassroots innovation

movements to uncover interesting findings for science; the diversity of projects that make up the “global assemblage” is an interesting approach in itself.

Small is not only beautiful, it also contributes to science. Hyperlocal solutions excel in local knowledge production and self-directed development, as they don’t take their frameworks and directives from larger movements. Ideas and blueprints might be shared globally, as it is the case in maker projects. However, the lack of financial incentives and scaling keeps them participatory and small. These projects rarely engage with international development agencies or profit-driven industries and when they do, they try to engage on their own terms. The people behind the project are solving the issues caused by those actors and develop solutions for communities where such solutions aren’t profitable because the scale is deemed too small to generate return on the investment. These are also the reasons why it is difficult for research to engage with such grassroots innovation communities; their diversity, plurality, and scattered distribution makes it more complicated to research such small communities, see e.g. Sipos et al. (2019) or de Valk (2021). However, in their processes, individual grassroots communities also generate knowledge that would not have otherwise been generated.

Marcelle (2016) in the aforementioned speech mentioned jugaad innovation, reverse innovation, and frugal innovation as sources of considerable optimism. Conceptually relevant, grounded, and “out of the box” thinking can develop conceptually relevant solutions to wicked problems, especially in settings rife with uncertainty and scarcity (ibid.). Indeed, jugaad and frugal innovation inherently appear in contexts that are resource-constrained, but there is more to grassroots innovation in the Global South than constraints. One of the main goals of this study is to find, analyze, and derive lessons from the practices of grassroots social innovation, including reflexivity and criticality, as opposed to top-down innovation practices and blueprint developmentalism.

1.2.2 Recent Shifts in Innovative Practices: Era of Participation

Academic attention on Grassroots Innovation Movements (GIM) has been increasing in recent years, fueled by the “era of participation” (Smith et al. 2017). This era of participation includes activities and approaches that involve citizens in design, innovation, and decision-making processes.

Innovation, an economic buzzword promising prosperity, has also been adapted to describe participatory processes. There is growing interest amongst researchers to explore participatory innovation such as grassroots innovation,

user-centered innovation, open innovation, social innovation, frugal innovation, and community innovation. This growing interest, however, mainly focuses on grassroots innovation as movements, potentially ignoring small groups with community roots that focus on hyper-local solutions. Of course, there is a long history of citizen participation and collaborative approaches such as democratic consultation processes. Some aspects of current directions in grassroots innovation projects might be considered novel, but it is important to research them while acknowledging their historical origins: e.g. citizen science as a revival of historical movements like the People's Science Movement in India (Smith et al. 2017) through digital technologies.

It is difficult to draw a line when this new era began, but these different approaches to innovation reflect the fact that agents, arenas, agendas, and processes of innovation have again been opening up in society for the past decade (Elliott and Turner 2012). This time, most of them are based on digital technologies. Interest in Fablab workshops as spaces of bottom-up innovation has been growing, together with imaginaries around the seemingly fantastic possibilities of the maker movement. Narratives have included bringing about a new industrial revolution (Anderson 2012), and democratizing innovation and technology design by empowering the consumer (Tanenbaum et al. 2013) to make their own products through *prosumerism* (Paltrinieri and Esposti 2013). Maker communities have been celebrated for their potential for far-reaching social (Unterfrauner et al. 2020; Bosse et al. 2019), political (Maxigas 2012), and environmental (Lange 2017; Kohtala 2016) impacts.

Whether promises truly deliver positive impacts is debated, critiques of makers' technosolutionism and ideological colonialism (Lindtner, Bardzell, and Bardzell 2016), involvement with the US military (Finley 2012), and concerns about "forgetting open hardware" (Benchhoff 2016) have been expressed. The movement's rhetoric of self-empowerment through entrepreneurship in the US has also been criticized. The "maker industry" has become an example of capitalism's ability to instrumentalize any movement that might undermine it (Morozov 2014). The communities that follow the underlying principles of making but are not explicitly part of a "maker movement" are significantly less visible in the mainstream narrative, but still interesting in terms of scientific research on grassroots innovation. One of the ways to reconcile these tensions was the term critical making, which stands in opposition to the corporatized, consumerist, mainstream narrative of making, in which "Everybody Just Buys the Kit!" (vinyl sticker, Hertz 2012).

During the COVID-19 pandemic, the promises of digital technologies suddenly became reality. It brought opportunities for remote work and online

education but has also revealed the limitations of the diverse socio-technical situations of people around the world. These people were often left behind by a lack of access and deepening of the digital divide, especially for people who lack technical knowledge or physical infrastructures to obtain data. Grassroots innovators responded rapidly to challenges while keeping diverse socio-political and technological realities—such as privacy, access, and national legislation—in mind. Researchers witnessed rapid responses to issues, some of which were not explicitly critical, but still highlighted major societal issues, such as social injustices, inequalities, and the breakdown of global supply chains. Thus, grassroots projects stepped up and refurbished pre-used computers for schoolchildren¹¹, developed healthcare solutions such as protective gear and parts for ventilators produced in local makerspaces¹², and invited people living in favelas to share their data for transparency, improving the tracking of outbreaks¹³. These responsible innovation capacities included rapidly prototyping, testing, documenting and reproducing helpful new products (Kieslinger et al. 2021).

While the potential of social and reflexive innovation in the maker movement has always been interesting for research, this pandemic has made it especially important to research and produce a better scientific understanding of how grassroots innovators develop solutions. Innovation has been the provenance of entities with large budgets, and the word itself has a strong neoliberal connotation (Schumpeter 1934). With the digital divide, the cracks of technology-based approaches to development are also showing, highlighting how centralized, large scale, top-down solutions have little sustainable effect in local contexts (Schumacher 1973).

The pandemic has shown the risks and limits of relying solely on large, global, and centralized corporations and governments. COVID-19 has triggered unprecedented interest in local, distributed, and openly accessible design and manufacturing to solve the massive unmet needs that have surfaced during the crisis.

1.2.3 The Pluriverse of Grassroots Innovations

The dispersed, heterogenous nature of grassroots projects makes them difficult to research, and acquiring funding for researching small-scale projects is often hard

¹¹ <https://engineeringgood.org/digital-inclusion/cac/>

¹² <https://www.careables.org/>

¹³ <https://datalabe.org/>

to obtain. Nevertheless, international and academic interest in grassroots innovation has been increasing. In terms of international development, the United Nations Research Institute for Social Development (UNRISD) has sporadically published studies on grassroots movements and innovations. For example, they have published on grassroots movements, political activism, and social development in Chile and Brazil (Foweraker 2001), and grassroots voices in social policy in Rwanda (Ezeanya 2017). For this study, perhaps the most interesting development is the United Nations Development Program's (UNDP) growing interest in grassroots.

In 2019, the UN launched the UNDP Accelerator Labs, which they call a large-scale experiment “designed to close the gap between the current practices of international development and the accelerated pace of change”¹⁴ by reimagining development through frugal/grassroots innovations (terms they use interchangeably). These, according to the UNDP, can accelerate sustainable development because grassroots innovations are sustainable by design, the people who suffer from a problem have crucial knowledge, and grassroots innovation signals unmet development needs. They argue that while such innovations might seem small, a series of small innovations might lead to larger breakthroughs (Gustilo Ong and Gustale 2020). In addition to this, the London School of Economics Capstone team recently released a policy recommendation report written for the UNDP Accelerator Labs titled, “Grassroots Innovation: An Inclusive Path to Development” (Bernstein et al. 2021). Today, the UNDP has 91 such accelerator labs in 115 countries, financed by the Government of Germany and Qatar, through the Qatar Fund for Development¹⁵. In the Grassroots Innovation Database co-curated by UNDP India and the Honeybee Network, they gathered to date 1554 practices in India only¹⁶.

One of the most prominent academic directions is the analysis of Grassroots Innovation Movements (GIM), including the analysis of historical movements such as the appropriate technology movement in South America (pp. 56-79, Smith et al. 2017), as well as novel grassroots innovation practices, like the phenomenon of “Hackerspaces, fablabs and makerspaces” (pp. 100-122, *ibid.*). They define GIM as able to occur virtually anywhere, in any sector, with the participation of any ordinary person to generate bottom-up solutions to solve local issues

¹⁴ <https://sdgs.un.org/partnerships/accelerator-labs-network>

¹⁵ The initial investment was over 81 million USD, then Germany invested another 20 million EUR in 2021: <https://www.undp.org/news/germany-invests-additional-eu20-million-undp-accelerator-labs-going-joint-venture-reaches-new>

¹⁶ <https://grid.undp.org.in/#4.1/22.82/82>

neglected by conventional innovators or out of necessity, involving the broader public in its discourse and resource mobilization (pp. 3-4, *ibid.*).

It is important to note that not all grassroots innovators see themselves as part of a broader movement, and not all phenomena which researchers call grassroots innovation movements are homogeneous or worldwide. A recent paper on the Global Open Science Hardware (GOSH) movement shows that the term “open science hardware” includes different framings and actors with different backgrounds, from community science projects to social innovation, and from critical pedagogy to trans-hacktivism (Arancio 2021).

Individual communities might work to alleviate specific topics or social issues, including sustainable agriculture (Norton et al. 2019), farming in urban settings (Heitlinger et al. 2013), beekeeping and citizen science (Phillips et al. 2014), and community networks (Bidwell 2020). Innovation is context-relevant, situated, diverse, and heterogenous, and many such projects do not fall under the umbrella of a movement.

For this study, the most relevant types of critical grassroots innovations are those within the realm of alternative technologies. These include participatory design (PD) practices and Information and Communications Technologies (ICTs) (e.g. Ghoshal et al. 2019; Ghoshal et al. 2020), the development of appropriate technologies (Smith et al. 2013), or the appropriation of technologies (Egash et al. 2004). The sustainable appropriation of technologies by grassroots communities particularly emphasize, “how socio-technical aspects of technological design, development, and implementation, contribute to ongoing transformations of power relations, living conditions, and our collective future” (Teli et al. 2021).

This study focuses on critical technical practice in grassroots communities, especially on deriving scientific observations from small collectives that produce locally relevant solutions and knowledge that are often neglected. This is an opportunity to illuminate practices independent from large funders. As I will outline later, without necessarily knowing, grassroots innovators manifest relevant ideas of “Small is Beautiful” (Schumacher 1973) and some even address its shortcomings. Their ideological freedom can be preserved by refusing to take funding that would influence their practices. Instead, they support the values of their communities¹⁷. From such genuine actions, I extrapolate knowledge on what

¹⁷ These case studies will reveal how the values of the community are placed before financial interest; in Case Study I, the Indigenous community decided against producing essential oils because they felt it harmed their environment and their health. In Case Study II, the collective decided against taking funding from a major sportswear company for an “innovation lab,” as they felt that the company’s demands conflicted with their values.

sustainable and participatory innovation might look like, based on the practices of local, bottom-up innovation communities.

This is especially relevant to keep in mind when analyzing grassroots innovations in the Global South. Still today, populist notions about center-periphery dynamics are constantly reinforced. There exists a strong distinction between innovation and knowledge produced through Western approaches, and research on the projects and communities “on the periphery,” which remains scarce (e.g. Avle and Lindtner 2016). Narratives of such dynamics need to be viewed through a critical and post-colonial lens. The capacity of grassroots innovators to ask critical socio-technical questions while enabling other local and international actors’ creativity and agency is particularly ripe for scientific analysis. During this process, they cultivate plural ideas about what constitutes sustainable, inclusive and social development (Smith et al. 2017). This study contributes to a scientific understanding of how designs for the pluriverse, or “a world where many worlds fit” (page xvi, Escobar 2018), and the subsequent pluriverse of local worlds or small-scale practices, challenge *business as usual* (de Valk 2021).

The starting point for this study is inspired by Smith and his colleagues’ aims “to make grassroots innovation movements more visible and learn from their experiences, in order that we better understand, appreciate and engage with them in the pursuit of sustainable developments” (page 3, Smith et al. 2017). Engaging with the Global South, it is necessary to use decolonial thinking while investigating the practices grassroots innovators use. Decolonial thinking highlights how they address their communities’ wicked problems (de Castro Leal et al. 2021), which stem from being entangled in the colonial past and existing power structures.

1.2.4 Democratization of Technology: Making and its Narratives

The previously mentioned *jugaad* and other types of Indigenous material practice are often equated with a local type of *making*, which is a term I will be referring to throughout the study and would like to explain further. The uptake of the so-called “maker culture” in the last decade represents a shift in innovative capabilities. It is important to note that uncritiqued technosolutionism—the belief that technology can unilaterally solve all of society’s problems—also applies to making (Lindtner et al. 2016). Mainstream making builds on the promises of digital fabrication and rapid prototyping conducted in FabLabs. Mainstream making has been inspired by Make: magazine articles, presented at Maker Faires,

and promises self-actualization and income through entrepreneurship. Making promises to change the whole world by offering universal livelihood, sustainability, and education, similar to the promises of international development. But making focuses on the West, rather than accounting for non-Western makers who are artists or make things out of necessity. Other makers are political, “DIY citizens” (Ratto and Boler 2014) or “critical makers” (Hertz 2012). Such inherently critical approaches have recently been approached by researchers and grassroots innovators as answers to the optimistic promises and overstated manifestos, see e. g. Fonseca (2015) or Morozov (2014). Ideological colonialism neglects *jugaad*, *gambiarra* and *ngoprek*, the Indian, Brazilian, Indonesian versions of grassroots hacking and repurposing hardware out of necessity (respectively). It also dodges critical questions about who gets to innovate and who is left out due to their gender, age, and class.

Similar to mainstream innovation, critiques of mainstream maker culture are also becoming prevalent, drawing attention to critical making (the de-sanitization of mainstream making) and post-colonial computing (observing human-computer interaction through the post-colonial lens). Researchers recognize that technology is only a means to an end. The artefacts of different material practices represent the people, cultures, and issues that shaped them. Whether by leading open academic discussions about the utopian vision of making and technosolutionism (Lindtner et al. 2016), uncovering promises of making—such as the democratization of technology or sweeping social and political changes (Lindtner and Lin 2017)—or attempting to bring together criticality and material practice in critical making (Ratto and Hockema 2009) and looking beyond the idealized picture of the maker, reintroducing a sense of criticality from counterculture-DIY back into post-2010 maker culture “to un-sanitize, un-smooth and re-politicize it” (Hertz 2015a), there is much to learn from such critical observations of socio-technical developments.

1.2.4.1 Situating Making

The definition of making is still contested. Thus, what *counts* as or *constitutes* making is also relatively up for discussion. The most concise definition is attributed to Doyle (2013), who described it as a contemporary culture or sub-culture that’s a technology-based extension of DIY culture. In addition to a mix of analogue and digital tools, maker culture is known for its intersection with hacker culture and commitment to openness. In their shared spaces for communities, not only are the tools communal, but the skills and knowledge are also generously shared (Fig. 1.1).

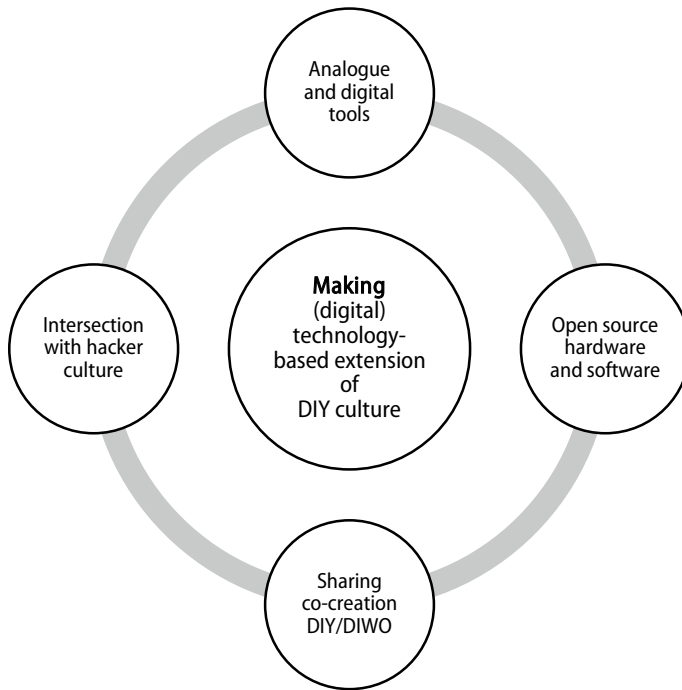


Fig. 1.1 Dimensions of Making. Based on Doyle (2013)

Making is not a homogenous activity that looks the same worldwide. In fact, a clear distinction is possible between a very visible, commercialized, mainstream practice—mainly the provenance of white male hobbyists in the West—and other types of making, such as grassroots, community-based making and making out of necessity. While the study will focus on the latter, I would like to briefly touch upon the mainstream version of making.

Buechley describes Maker Media’s commercialized version of making as attempting to expand markets and become profitable by targeting a specific audience. She found that 85% of the magazine’s covers featured white boys and men, alongside robotics, electronics, and vehicles (Buechley 2013). Other studies identified further persistent gender gaps and found that women and gender-non-conforming (GNC) people are undercounted because they do making differently

and are spatially excluded from makerspaces that support the prototypical maker found in Make: magazines (Melo 2020)¹⁸.

In terms of its narratives, Bean and Rosner asked themselves whether making is a movement or a brand, and concluded that it is a brand. By their reading, Dougherty's *maker movement* falls is commercial rather than a social movement (Bean and Rosner 2014). Similarly, *mainstream maker culture* has been defined by Barton and Tan as a "culture that re-inscribes white, male, middle class norms" (page 2, Barton and Tan 2019). They contrast this practice with community-engaged and community-centered making that is locally grounded and expansive.

These mainstream and commercialized approaches exclude crucial elements of making that embrace diversity. One doesn't necessarily have to buy the Make: magazine, join a FabLab, or 3D print to be a "maker". Ratto and Boler, for example, in their edited volume on "DIY Citizenship" (Ratto and Boler 2014), include worldwide groups of knitters who convene online. Following this logic, one might say that the timeline of making starts with a group of ladies (Gowanda Ladies Social Society) gathering to quilt, knit, sew, socialize, and talk about books in late-nineteenth-century New York. According to Yu (2016) this marks the beginning of the "embryonic period" of the history of making. Of course, if making is defined as any hands-on creative process, and if makerspaces are community spaces (or third places) where people can make things, the Gowanda Ladies Social Society might indeed be a starting point. Clearly, a digital element that many consider a crucial part of making is missing from this example, but the craft and community aspect are strongly represented.

Stoyanova, in her paper locating criticality in DIY and making, goes even further back. She starts her discussion about humanity's relationship to making with the Greek philosophers Plato and Aristotle. They distinguished between *technê* (craft) and *épistêmê* (knowledge), and Stoyanova explains that Marx's *homofaber* laid the foundation of the instrumentalist-utilitarian definition of making with its focus on using tools to dominate Nature (Marx (1967) in Stoyanova 2017).

It is not the goal of this study to trace thousands of years of material practice, and the very recent history of making, maker movement and maker culture is very well documented. The examples above show that making is rooted in the do-it-yourself (DIY) movement. Certain aspects of the maker movement have existed for a long time. What makes making especially interesting for this study is that it is composed of DIY material practices extended with digital tools. This means

¹⁸ As an answer, specific inclusive, feminist and queer-friendly hacker and makerspaces were founded. Dr. Wuschitz introduced me to Mz* Baltazar's Lab in Vienna, Austria, and Dr. Schrock informed me about Double Union in San Francisco, USA.

that it has the potential for people to share their ideas worldwide and to open the “black box” of technology. Mainstream narratives, however, tend to forget the diversity and plurality of what maker culture can be. Below I will summarize key developments from the past decade in order to underscore the most relevant narratives surrounding making, and highlight some discrepancies.

The current discourse about the so-called “digital fabrication movement” or “maker movement” highlights three developments originating from the United States that have contributed to its worldwide success: awareness, access, and engagement. These prerequisites seem to be discussed mostly in institutionalized and branded forms, such as Maker Media and Fab Labs.

The word *maker* took on a specific meaning in 2005, when Dale Dougherty founded Maker Media Inc.¹⁹, which published *Make: magazine*. He used the term to rebrand and sanitize the term *hacker* to be more acceptable to the public, schools, and potential sponsors. The concept of maker started out by highlighting ordinary people making things in their garages and backyards (i.e., DIY culture). Making eventually also included work in the experimental media arts, open-source hardware, and hackerspace culture (Bogers and Chiappini 2019). Maker Media Inc.’s magazine and events have been crucial in spreading the so-called maker movement, here used as a marketing term. These have reached millions of people worldwide. *Make: magazine* published articles and instructions for DIY projects that people could make themselves. Additionally, under the same brand, Maker Faire events allow makers to showcase their creations and learn about new tools and gadgets. Maker Faires have been organized in around 100 countries and have attracted millions of visitors (Dougherty 2012b). The revival of DIY through digital tools was also helped by a course at the Massachusetts Institute of Technology, teaching students “How to Make Almost Anything” (Gershenfeld 2012). The course inspired what eventually evolved into the international Fab Lab Network, which provided a blueprint of low-cost fabrication laboratories (today approximately numbering 1750 in 100 countries). This blueprint was accompanied by a set of values that situated the FabLab as a laboratory that allowed individuals to share analogue space as well as digital tools, such as 3D printers and laser cutters. In Fab Labs, all-ages education on how to use tools was also provided, and a version of the original course under the name Fab Academy was offered to provide this education (Martin 2015; Peppler et al. 2016; Blikstein 2017).

¹⁹ Although Maker Media Inc., the venture capital backed for-profit company halted operations and laid off its staff 2019, it managed to restart some of its operations and considered continuing as co-op or non-profit, relaunched as Make Community (Constine 2019)

Many other types of workshops worldwide also offer similar set of tools, as well as education, workshops, and training tailored to the needs of their communities. Their proliferation worldwide shows that there is no branded monopoly on such spaces, their tools, communities, or programs; many non-Fab Lab branded workshops can be found worldwide, such as makerspaces, hackerspaces or repair cafés offer access to similar tools, as well as education and training tailored to the needs of their communities. What counts as making is situated and diverse (Braybrooke and Jordan 2017; Lindtner et al. 2016). It has different definitions, most of which emphasize the digital, technology-based extension of DIY. Yet, “DIY” includes metalworking, woodworking, traditional arts and crafts, DIY biology, and naturalism. “Digital” includes electronics, robotics, 3D printing, and CNC tools. Digital DIY communities value openness (e.g., the sharing of blueprints of open-source hardware online), and DIWO (“Do It With Others”). DIY practice often intersects with hacker culture and there are many crossover activities; many hackerspaces are also makerspaces²⁰.

Defining making as a collective DIWO or DIT (“Do It Together”) practice, instead of a more isolated “DIY,” has become popular around the world in recent years (Lou and Peek 2016). The diversity of makerspaces reflects the popularity of “making” as well as the diverse interpretations of the term. It refers to a wide range of places, from the safe space of an activist community that utilizes practices of hacking and making to a shared workshop where one pays for the use of machines to build an enterprise.

1.2.4.2 An Answer to Mainstream Practices: Critical Making

For this study, it is necessary to ask some critical questions: how can the utopian vision of making that is resting on technosolutionism and ideological colonialism be unmasked (Lindtner et al. 2016), and how can research look beyond the maker hype and its promises (Lindtner and Lin 2017)? Furthermore, what does a critical material-technical practice that utilizes maker practices beyond those utopias look like, and who is conducting it?

Likewise, the assumption is that critical thinking and a critical theory of technology has been reserved for academia, and innovative types of technical practice for traditional innovators. There have been various attempts to bridge the two, such as the Responsible Research and Innovation (RRI) framework of the European Union, which has so far mostly been applied in institutionalized research

²⁰ See www.noisebridge.net or <https://x-hain.de/en/> “hack+ makespace”. For deliberations on the differentiations and similarities between these two terms, see (Richterich and Wenz 2017).

organizations. However, the concepts of critical technical practice, critical making²¹, and RRI have rarely managed to address the particularities of community- and citizen-driven innovations taking place outside research organizations (Sipos and Akerman 2022).

What the discourse rarely mentions is that people have been engaging in DIY activities for a long time and in very diverse ways. Historical developments have also contributed to the range of open workshops we are seeing today. Indeed, Stewart Brand's "Whole Earth Catalog", published 1968–1998, is associated with the foundations of the maker movement. In Germany, the number of non-profit open workshops is estimated at around 1000, many of which have existed for decades and can be found in smaller cities and rural areas. Aside from Fab Labs, there is a compelling diversity in the global and German open workshop landscape. It includes not only hackerspaces and makerspaces, but also DIY and DIWO spaces, biologist and feminist spaces, repair cafés, mobile makerspaces, and Fab Labs (Sipos and Franzl 2020). The *global assemblage* of makers, as I will explain later, is even more diverse (Lindtner, Bardzell, and Bardzell 2016).

What terms and are used by the practitioners to describe the activities, processes, and aims of grassroots innovation communities focusing on critical and collaborative technology development in the Global South? As I will outline in more detail later, during a participatory workshop I organized in preparation for the field research, many participants confirmed that they did think there was an inherent criticality in DIY and making. However, they also expressed that they saw making as a privileged activity—a hobby for those who could afford it. They understood critical making to be directly related to frugal innovation, non-Western viewpoints, and local contexts, and brought a culture of community-based design approaches, discourse, reflection, and responsibility (Sipos and Wenzelmann 2021).

It became clear from my experiences that a combination of criticality and material practice was present in grassroots, it had just barely been documented. The critical potential inherent in maker communities comes from their cumulative and shared knowledge, and the technical possibilities that the democratization of innovation and access to technology through rapid prototyping offers, especially when combined with critical thinking and reflexivity (Ratto and Boler 2014; Sipos and Wenzelmann 2021). My goal in this study isn't to impose Western practices

²¹ Throughout the study, I will use the term critical making as well as critical technical practice to designate different modes of engagement. While the original idea was to explore critical making within grassroots, due to the pandemic, many things that normally would have been "made" by hand in a makerspace had to turn into a remote, digital practice.

onto grassroots, but to understand how and why they infuse their technical practices with criticality. My hypothesis is that by exploring their processes rather than their outcomes, we might learn how local development and the peculiarities of self-directed development can be made more sustainable.

1.2.4.3 Narratives, Mythologies and Promises: “Why I am not a Maker”

With its reach and growth in the past decade, making has become popular and widely available. Thousands of makerspaces worldwide—in schools, libraries, or as standalone spaces—attract large groups of people. Expectations have been high, one example is Chris Anderson, who equated the spread of making with the new industrial revolution (Anderson 2012). But even with tens of thousands of workshops around the world, making remained a fringe phenomenon and the new industrial revolution was nowhere to be seen (Troxler and Maxigas 2014), confirming that *making almost anything* was indeed possible, but begged the question: “we now have the means of production, but where is my revolution?”

In 2020, long-awaited revolutionary acts were enabled by makers in makerspaces. When supply chains started breaking down during the COVID-19 outbreak, open-source global design of healthcare equipment and local production were combined. This fusion led to the fabrication of personal protective equipment (PPE), parts for ventilators, and many more necessary objects (Kieslinger et al. 2021). But before this major accomplishment could be achieved, providing that makerspaces and the communities can alleviate such crises, the development path of making was rocky.

The first sign that making was becoming mainstream was when Barack Obama endorsed making and hosted a Maker Faire at the White House in 2014. His endorsement also meant that high-level politics were suddenly playing a role in a seemingly grassroots practice. There had been other concerns voiced before, for example, when the Defense Advanced Research Projects Agency (DARPA) gave funding to *Make: magazine* to fund 1000 makerspaces in US-high schools (Finley 2012). Well knowing that a funders’ agenda will affect practices, Finley suggests that a line was crossed when the military “infiltrated” the maker community (ibid.). Similarly, when the MakerBot, “the darling of the Open Source Hardware movement”, became closed-sourced after it was acquired (Benchhoff 2016), all the people who contributed as volunteers to the previously open-source 3D printer saw it as a breach of trust.

Smith has also pointed out the makers’ dilemma of asymmetric power relations. While there are yet few working examples of sustainable business models for makerspaces, they often resort to sponsorship. Sponsor-sourced funding for

makerspaces and hackerspaces often clashes with their community values. One of my interview partners in Indonesia told me that they refused to take money from a major sporting goods manufacturer because the “innovation space” they were supposed to design was far afield from the values of their community:

“Hacker visions and values for autonomous social innovation and critical involvement in open technology is co-opted by an agenda to educate, train and entertain people, and where the fear is that it reinforces compliance with conventional innovation agendas. Responses to this criticism point to the mainstreaming of makerspaces but gloss over the asymmetric power relations between partners. In terms of innovation democracy, the critical question becomes the conditions under which makerspace participants can really challenge, and even reshape, the agendas of sponsors and partners” (Smith 2017).

The social impact of makers in Western countries may be limited by the mainstream rhetoric of the entrepreneur maker and the standard engineering practices of their educational backgrounds. Reproducing industry practices that have created the problems in the first place is a wicked problem. If innovating for society is the goal, practitioners should rethink how they innovate and engage in multidisciplinary processes that include critical thinking, reflexivity, and co-creation with others. In this sense, critical making has been developed to unite critical thinking and the material practice of making (Ratto and Hockema 2009).

Amongst others, Garnet Hertz is an opponent of the depoliticized direction the maker culture has taken, which has turned it into a market-driven segment of grassroots technology development. The original, countercultural aspects of the DIY movement (such as the DIY activities of punk culture²²) seem to have faded from the discourse (Sipos and Franzl 2020). Hertz designed a front cover for a fictional *Made: magazine*²³, which explicitly and satirically highlights problems with mainstream making (Fig. 1.2):

²² Bennett and Guerra (2018) describe the emergence of punk as an organized DIY, countercultural phenomenon. It included setting up bands, creating media such as zines and alternative radio stations, and developing unique fashion, sales, and marketing structures. Cheap and accessible reproduction methods such as photocopying, copying music cassettes, and screen printing became tools of a new alternative movement that built on DIY to express political views (Furian and Becker 2000).

²³ *Made: Technology on Affluent Leisure Time* (2012)—Garnet Hertz. Format: Full Color Rectangle Vinyl Stickers, UV coated (single-sided) Dimensions: 3.5 inches (W) x 5 inches (H). The sticker was produced as a parody and critique of *Make: magazine* by O'Reilly Media. Distributed for free. Included as insert in the first edition of the “Critical Making” zine.



Fig. 1.2 Made: Technology on Affluent Leisure Time (Hertz 2012b). (Created by Garnet Hertz, public domain)

Hertz lampoons out the importance of staying away from any political activities (Join the Arduino Revolution but Avoid Civil Disobedience), the exclusive nature of maker culture (101 DIY Gadgets for White Males), and how such innovative technologies like the 3D printer are used for unsustainable activities (How to Use a MakerBot to Make a Three Cent Piece o' Plastic) (Hertz 2012). In Hertz's definition (Hertz et al. 2014), this version of maker culture is:

“makers = hackers – controversy”

His most prevalent critique is that when practitioners take part in predefined activities and only build artefacts out of kits, there is little space left for creativity, speculative processes, or reflection. Nevertheless, makerspaces can play important societal roles. They create third places for communities²⁴, support wellbeing and civic engagement through activism and critical thinking, and unlock grassroots capabilities (Bar-El and Zuckerman 2016). Makers also create innovative artefacts by using inclusive design methods. They support the creation of socially relevant prototypes, some of which can be innovative solutions (e.g., frugal innovation and appropriate technologies) to societal problems (Beltagui et al. 2019, Smith 2017). It is thus undeniable that makerspaces have improved education and citizen science, and have taught people to open the “black box” of technology. As later explained in this study, makerspaces that built their business models on selling gadgets were short-lived in Bandung. For example, when participants asked the workshop leaders in the first Fab Lab of Indonesia in Jogjakarta how to use a 3D-printer for real-world needs, they weren’t sure how to reply.

When looking beyond the commercial façade, countercultural aspects like “DIY citizenship” (Ratto and Boler 2014) are present in making today. Particularly in the Global South, some makerspaces already have realized their role as societal and political actors, as can be seen in the citizen-driven projects of Lab-COCO at the Umbigada Cultural Center in Brazil, the Hack-a-Toy workshops run by Engineering Good in Singapore, the ASKOtec mobile makerspace in Uganda, and Lifepatch, a grassroots initiative exploring science, technology, and DIY biology in Indonesia. However, while academia might call these makerspaces, they themselves do not necessarily identify as such.

Since the term critical making was first published by Ratto and Hockema (2009), the academic critique of maker culture has strengthened. Subsequent to its embeddedness in academic, scholarly practice and discourse, it had reached select grassroots practitioners. The practitioners’ critiques included complaints that mainstream making only accelerates overconsumption and creates more unnecessary objects without the user necessarily even engaging in the technical design practices. They celebrate the term critical making, because they don’t identify with this wasteful type of mainstream maker culture (Fonseca 2015). When organizing a session for a citizen science summer school, I was asked by

²⁴ Third places are welcoming spaces outside of our work and home that have been created to promote social experiences among like-minded people that are vital for civil society and democracy (Oldenburg 1989).

a Brazilian participant not to refer to her as a maker, as she perceived it as a very restrictive term that describes a Western practice embedded in start-up culture. Thus, certain practitioners refuse the maker label to describe their activities (Chachra 2015) as they feel like it only captures a Western narrative of innovation (Fonseca 2015). Fonseca and others choose to use the Brazilian word *gambiarra*, while Indonesian practitioners prefer to focus on the co-creative processes of knowledge production and tinkering of *ngoprek* rather than the ownership of a particular outcome (Siagian 2016). *Jugaad*, the Indian term used can be translated as “hack” or “resourcefulness” (Butoliya 2015), and *shanzhai* “copyleft” production practices in China are also getting more attention in an attempt to decolonize Western information technologies (Braybrooke and Jordan 2017; Lindtner and Lin 2017).

Thus far, research on such critical grassroots practices has focused on the technological and sociopolitical analysis of the “global assemblage” (Ong and Collier 2004 in Lindtner et al. 2016) of hackerspaces and makerspaces for collaborative, collective or community-based design (Bonvoisin et al. 2018; Buitenhuis, Zelenika, and Pearce 2010) and grassroots innovation (Smith et al. 2017) and attempted to highlight the diversity and situatedness of what counts as making (Braybrooke and Jordan 2017; Lindtner et al. 2016).

To build on this discourse, this study reflects on making as a restrictive, Western concept and explores whether critical making could describe the activities, processes and aims of grassroots innovation communities focusing on collaborative technology development in the Global South. It will thus challenge the mythologies of mainstream narratives of ICT4D (Ames et al. 2018) and making (Lindtner and Lin 2017), thereby contributing to the scientific knowledge base on non-normative making and the diverse practices of critical grassroots innovators.

1.3 Research Design

The original research plan of this study was to build on maker communities gaining access to rapid prototyping and digital fabrication methods. My hypothesis focused on exploring whether increased access to tools could shift innovation practices through the emancipation of citizens. If so, could there be proof that the potential application of critical thinking and multi-disciplinary collaboration for purpose-led innovation leads to critical making in grassroots innovation communities?

When taking a closer look, I recognized that the *global assemblage* of the maker community represented different practices and goals. As summarized earlier, aside from not-for-profit makerspaces (e.g., educational spaces in libraries), makerspaces with hybrid business models and commercial makerspaces exist that act like startup incubators and lead to the creation of numerous startups. In the same spaces, there are also examples of social innovations created using rapid prototyping tools that improve accessibility or sustainability. Some of these practices became even more visible during the pandemic, as supply chains broke down and people needed to locally produce life-saving equipment.

During my field research, I understood that grassroots innovators could contribute more to societal development than merely generating startups, social innovations, or proving that distributed production can be more efficient than globalized chains in emergency situations. The critical-material, socio-technical, and multidisciplinary practice²⁵ that drives these outcomes is rarely visible, and often more typical outside of mainstream makerspaces. In this study I will explore the collaborative practices, experimentation, iteration, critical identification, and representation of the needs and values of local communities. I argue that approaches that lead either to tactical interventions or to sustainable development are the next step in a scientific understanding of grassroots critical makers. The practitioners are contributing to more than just transitions, development, and the preemptive decolonization of possible futures. Inquiry into understanding reflexive practices that lead to those transitions is the basis of my research design.

1.3.1 Theoretical Background

In terms of its theoretical background, this study builds on the existing approach of critical making, which has so far been mainly applied to engineering studies, design, and a few citizen-driven projects. This study explores the existence and possibilities of critical making in a new field: grassroots innovation in the Global South. It also furthers the scientific understanding of grassroots innovation communities and contrasts their work with top-down, traditional innovation approaches, contributing to the scientific knowledge of grassroots innovation. It contributes findings about sustainable innovations on the ground, and the processes of those actors that make these innovations more sustainable. To better

²⁵ As I will demonstrate, these processes do not involve only hackers and makers, but the initiators also include people from other areas, like scientists and affected citizens, creating opportunities for truly participatory and self-directed development.

understand and scientifically analyze the participatory processes that support critical making, this study applies existing theories of STS and HCI to new areas of grassroots innovation. Most of these have been rarely applied to participatory technology development projects on the ground. This study also builds upon and contributes to the field of post-colonial computing.

1.3.2 Research Methods and Analysis

In terms of methods and analysis, this study further develops the Grassroots Innovation Movements (GIM) framework (Smith et al. 2017) through the additional exploration of plurality and criticality. By conducting participatory research and outlining findings across case studies, it contributes to experimental methods of real-world research (McNiff and Whitehead 2011) by demonstrating the processes of grassroots innovators in the Global South. This is particularly relevant for research, as intrinsic participatory processes of grassroots are seldom experienced in their authentic environments without the intervention of academia, which designs processes of inquiry that change the trajectory of grassroots innovation projects (Sipos et al. 2019). Such intervention is avoided in this study to better demonstrate the participatory nature and processes of such grassroots projects, and to highlight the critical thinking aspect behind them. The outcomes are only there to demonstrate relevance to societal issues of interest, and not to measure impact or other types of success.

1.3.3 Research Motivation

The original research proposition and research questions I submitted to DAAD in my funding proposal for field research aimed to develop a comparative study. I described analyzing grassroots critical making practices through a lens of social innovation, based on testing the aforementioned imaginaries. I was interested in exploring the democratizing power of maker practices, how they might empower everyday people to innovate, and whether open-source practices of participation and documentation could generate wider societal influence. However, I found this initial set of questions to restrict a better understanding of what practices are relevant on the ground. The original research questions were aimed at understanding value creation and were impact-driven.

Through critical analysis, I realized that my early questions represented a different motivation. They reflected a weak tradition of social innovation that sees

social innovation as any increase in aggregate individual utility arising from an innovation, rather than the strong tradition of social innovation, which focuses on the process of collaboration between different groups and the subsequent the restructuring of power relations (Ayob et al. 2016; Moulaert et al. 2014). This latter approach is especially relevant for the actual research motivation, that is, to understand the processes of grassroots innovation. This motivation is supported by two frameworks, both designed to understand processes; participatory and intrinsic design (PD) focus on collaborative processes, and grassroots innovation inquires how movements came to be (Smith et al. 2017). Thus, from the viewpoint of scientific inquiry, the process of how communities develop sustainable solutions and knowledge is significantly more interesting.

As I was engaging with grassroots communities on the ground, I realized:

1. First, that understanding their histories and generative contexts was crucial to be able to scientifically understand the origin of injustices.
2. Second, that criticality was often not explicitly built into the projects, or was discussed behind closed doors in personal exchanges, as being openly critical can lead to the loss of funding.
3. Third, that their activities are diverse. Those that seem to have achieved sustainability might deploy various design and knowledge production strategies, that other innovators can learn from. All these activities are embedded in the possibilities, needs, values, and interests of the local communities, and are most often considered above financial goals. These activities include:
 - Tinkering with different technologies until the appropriate solution is found, and openly sharing their practices in online documentation and presentations.
 - Inviting multidisciplinary groups to ideate, design, and deploy solutions in participatory manners, especially with those affected by the issue at hand.
 - Account for strategies that make the projects more sustainable on financial, social, and environmental levels.

There was thus a need to rethink my initial research questions, in order to better represent the societal relevance of these projects, and what the aforementioned traditional innovators can learn from them. The original questions were still relevant but needed to be extended, and the framework needed to be reconsidered to better reflect a new scientific inquiry.

My project shifted from “social innovation” to “grassroots innovation”, as the latter was less burdened by definitional confusion (see weak and strong traditions of social innovation in Ayob et al. (2016)). There also existed research on such

practices in the Global South that I could use as a starting point (Smith et al. 2017). I shifted from outcomes to processes, which is not only an important aspect of grassroots innovation research, but is also a central point of inquiry in participatory design (PD). Postcolonial research (Freire 1982) advocates for participation and to increase the relevance of scientific inquiry about the era of participation (Smith et al. 2017). At the same time, as a subcategory of PD, intrinsic design gives researchers a better idea of how self-directed grassroots design practices are *infrastructured* (Bødker et al. 2016). Finally, one of the main contributions of grassroots to scientific inquiry is that the knowledge produced in the process would not have been produced otherwise (Smith et al. 2014).

1.3.4 Research Gaps and Extended Research Questions

The research motivation of this study is to investigate the utopian visions of industrial innovation and mainstream making from a different perspective. As both of these vision rely on techno-fetishism, technosolutionism, and ideological colonialism (Hertz 2015a; Lindtner and Lin 2017; Lindtner et al. 2016), it is timely to direct our attention to non-Western approaches of democratic design experiments like *jugaad*, *gambiarra*, and *ngoprek*, as well as collectives and communities instead of makerspaces and movements. As Els et al. put it:

“Users are much too often not only taken hostage by neo-liberal capitalism but also patronized by advocates of human-centered design. (...) The neglected objects strike back” (page 8, Ehn et al. 2014).

This shift of perspective allows researchers to unearth and learn from critical making practices that have been “in the making” for decades, but out of sight for academia. In this study I explore the possibility of further evolving the term critical making to highlight locally relevant critiques of socio-political realities through grassroots technology innovation practices. My interest in critical making is motivated by wishing to identify, analyze, and understand the processes that contribute to plurality and reflexivity (Smith et al. 2017) through self-directed community-based knowledge production processes.

The first research gap is based on the observation that grassroots communities often aren't part of a movement. They are diverse, situated, and heterogeneous. Due to the particularities of academia, it can be difficult to justify the research of small-scale and locally relevant innovations (Sipos and Wenzelmann 2021). This study aims to address this lack of research of grassroots on a smaller and more

appropriate level, as opposed to focusing on movements and large-scale solutions (e.g., Smith et al. 2017). The so-called maker movement is also heterogenous and situated. Conducting research as if every maker was reproducing the same practice is counterproductive and contributes to the exclusion of those who don't fit under the normative umbrella, especially makers in the Global South (Kaiying and Lindtner 2016). It is important to note that we are not faced with a uniform activity that follows one central blueprint and should be reproduced anywhere in the world exactly as prescribed. Following Ong and Collier's definition (2004), we rather face a "global assemblage" (Lindtner et al. 2016) of hackerspaces and makerspaces. These spaces for collaborative design and grassroots innovation are brought to life by offline and online communities that make use of the tools they house. Examining their diversity and situatedness (Lindtner and Lin 2017) in local socio-political realities can extend and redefine what constitutes as making and shed light on societal challenges. But the lack of uniformity also makes research complicated. Some of these practitioners refuse the maker label (Chachra 2015) or use the term "critical making" to describe their practices (Fonseca 2015). There's no standardized term for these practitioners, which makes researching heterogeneous groups difficult. Innovative activities of groups of people are often carried out "beneath the radar" of academia and industry (Smith et al. 2017).

The second research gap is based on the learning that while innovative capabilities are shifting towards the "periphery", intrinsic design processes of grassroots innovators mostly remain invisible. These are, in many cases, ad hoc and happen as an immediate reaction to outside threats or opportunities. It is rare to observe such processes without the planned intervention of academia. This study contributes to the scientific understanding of such processes by describing intrinsic design processes, as opposed to participatory design methods where researchers intervene. Traditionally, collaborative innovation practices of the grassroots are seldom accessible to academia, unless it intervenes and designs processes of inquiry (Sipos et al. 2019). There is also a lack of accompanying research on this topic. However, the added value of grassroots approaches is strongly supported by researchers (Defila and Di Giulio 2018). To better understand the processes of design in critical grassroots communities—especially in the Global South—it is important to approach these without imposing externalities, and to study their intrinsic design processes instead.

The third research gap is based on the distinction between what is seen as knowledge in academia, and "other" types of knowledge production that are rarely part of the academic canon. Critical knowledge—knowledge that questions the status quo—is especially neglected, as recognizing critical knowledge might reveal uncomfortable truths about societal issues, injustices, and development.

Uncovering, recording, and adding this knowledge to the academic knowledge base produced by the “peripheries” in the Global South is rare. Thus, this study aims to collect important examples of knowledge that would not have been produced without the practices of grassroots participants.

This study is thus concerned with three equally relevant aspects of grassroots innovation: understanding the plurality of approaches, projects, solutions possible to any given local issue; critical societal and socio-technical knowledge these approaches produce; and the intrinsic-reflexive and participatory processes of grassroots making (Fig. 1.3).

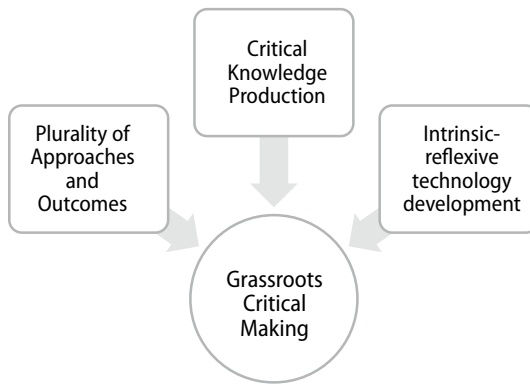


Fig. 1.3 Relevant Aspects of Grassroots Critical Making

The research gaps (and therefore, the research questions) are all intertwined. This study addresses the above questions by building on existing theories, methodologies, and critiques. Specifically, this study will focus on three key research questions that build on each other. These are defined and outlined in the next subsections.

1.3.4.1 First Research Question: Pluriverse of Grassroots

While neoliberal, capitalistic, and startup-oriented makerspaces are common throughout the Western world (Hertz 2012; Ahmadi et al. 2019; Barton and Tan 2019), many communities engage in a different kind of making. Their approach is community-oriented and aimed at raising awareness about or solving local issues and problems deemed too small for mainstream innovation. Some are part of a global assemblage and are local entities that deploy similar tools for making,

while others experiment with different strategies. Another important dichotomy is the aforementioned separation of center from the periphery—those marginalized by hegemonic infrastructures (Ehn et al. 2014). This study attempts to investigate beyond the notion of grand social movements and large-scale innovations, and instead approach communities in the spirit of “Small is Beautiful” (Schumacher 1973). By approaching them not as a movement (GIM) but as communities, this study can demonstrate their plurality and diversity, reflective of local context and needs, different communities’ values and interests, as well as their individual approaches to knowledge production and sharing. The latter is especially what distinguishes them from mainstream innovations, which often ensure that their intellectual property is protected. Critical maker communities openly share their knowledge on their websites, in online fora and repositories, and amongst each other (Saari et al. 2021).

As the goal of the study is to extend critical making towards the grassroots, it is important to provide an overview grounded in empirical research of what critical making might look like in grassroots communities. In a comparative manner, using the retrospective method of analyzing Grassroots Innovation Movements (Smith et al. 2017), I trace the history of the two oldest Indonesian collectives. These cases demonstrate relevant sustainability strategies and struggles. I compare the reflexive, long-term *strategies* of the Network Collective in Case Study I with the *tactical interventions* of the Science-Art Collective in Case Study II, illustrating workarounds for wicked problems. These represent two possible frameworks for sustainable, technology-based, self-directed development projects.

The first set of research questions focuses on understanding the “grassroots” dimension of grassroots critical making. It builds on the hypothesis that the grassroots are not a homogenous movement, but small, situated, and plural communities. Therefore, individual communities will have been shaped by their contexts, developed particular ways to produce and share relevant knowledge, and address societal issues beyond shared grievances, and try to make alternative futures happen:

- What are the similarities and differences between the communities? What are the specific characteristics of their members and practices, and what external circumstances or contexts have shaped them?
- What can be learned from Indonesian grassroots about the societal issues they uncover, the critical framings they generate, and the critical knowledge they produce? What does this knowledge production practice look like, and how is it done?

- How do these practices of critical making contribute to values being realized and alternative futures envisioned?

Furthermore, this research question is built on the hypothesis that by conducting empirical research on the island of Java, more critical grassroots innovation projects become visible than what the current academic literature suggests. This knowledge helps me record and map an Indonesian grassroots pluriverse.

1.3.4.2 Second Research Question: Critical Knowledge Production

While knowledge that is not produced by academia or other institutions is often not taken seriously, grassroots innovators produce different types of knowledge that are very interesting for postcolonial inquiry. The grassroots are known for open approaches to knowledge sharing (Smith et al. 2013) and participative knowledge production, which allow communities find each other and create opportunities for peer learning, accelerating the development of appropriate solutions to local issues by adopting the existing solutions of others. From a market-driven innovation or academic point of view, such innovation might seem slow, not replicable, and irrelevant on a large scale. But such dismissal might be too hasty:

“Responses to each challenge by grassroots innovation movements create forms of knowledge of considerable social value [...]. The capturing and understanding of the knowledge being produced and strategies ensuring that knowledge feeds into such debates is necessary, but without a too prescriptive or rigid framework to avoid distorting or bracketing out inconvenient forms of grassroots knowledge” (Smith et al. 2014).

Smith et al. define three framings of knowledge production: grassroots ingenuity, empowering inclusion, and structural critique. Although the knowledge produced might be “inconvenient to and resisted by those above” (page 1, Cox and Fominaya 2009), movements’ critique of existing inequalities enhances the reflexivity of society (Buechler 2000 in Smith et al. 2017). They are thus political actors, in the sense that they show alternative pathways for socio-technical development that raise questions about technological needs in societies, appropriate directions of technological change, and who is enabled to access, design, and own technology. They challenge specific directions and forms of knowledge production,

technological change, and development (Smith et al. 2017). Knowledge production (Jamison 2001) and co-creative knowledge sharing are relevant activities of maker communities. The decolonial scholar Escobar has highlighted different kinds of knowledge production approaches employed by social movements (Escobar 2004), and Leach et al. (2010) described the politics of knowledge involved in social movements. Discourses about technological controversies can also be seen as informal forms of technology assessment, as they protest and debate the very framings of the technology. Similarly, grassroots innovation processes and practices generate knowledge about sustainable innovation, where sustainability means socially just and environmentally sustainable development (World Commission on Environment and Development 1987 in Smith et al. 2014).

The second set of research questions of this study thus focuses on the criticality of grassroots critical making. It builds on the hypothesis that critical grassroots communities are political actors and develop and share critical knowledge. This knowledge is underrepresented in academic literature either because it is seen as produced by non-experts, or because it is only shared amongst trusted grassroots communities:

- How is critical thinking exercised, and what are the particularities of critical knowledge development and sharing by the grassroots?
- Is this knowledge used to enhance reflexivity and criticality in wider society? If so, how?
- If critical knowledge about societal issues or socio-technical questions is developed, how does it translate to real-world projects?

1.3.4.3 Third Research Question: Making as Intrinsic Design Process

Hackerspaces and makerspaces attract inquisitive people, early adopters, and creators of innovative artefacts. However heterogeneous, these spaces for grassroots technology innovation practices provide researchers with opportunities to observe processes of collaborative, collective, and community-based design (Bonvoisin et al. 2018). Makers and maker communities often engage in grassroots innovation, thereby acting as litmus tests to measure the discontentedness of society, pinpoint emerging societal needs, and propose improvements (Sipos and Wenzelmann 2021). As grassroots innovators react to issues, injustices, or more positive windows of opportunity, they unite to develop innovative, technology-based solutions for their own communities. This might happen quickly, or by tinkering on

solutions for years. They may bring in multidisciplinary knowledge from locals, or from around the world. Such intrinsic design processes are interesting for science because they shed light on democratic design processes that reflect the values of the communities. However, these processes are almost impossible to access as a researcher (who is almost always an outsider) without interfering with the process. Working with the assumption that the solutions developed in an intrinsic, autonomous manner differ from the innovations that are developed and deployed in a top-down approach, what can science learn from the grassroots about these democratic and more sustainable processes?

Two intrinsic design cases are outlined in this study: a retrospective analysis of a Community Network Project, and a real-time investigation of a hackathon organized by three Indonesian grassroots organizations. Intrinsic design is particularly interesting, as it is known for rejecting outside input (Bødker et al. 2016), meaning that no external values influenced the design of the project. The reflexive and participatory nature of this research was especially relevant because it presents an important alternative: decolonial research practice in development as suggested by Paulo Freire (1982). While Participatory Action Research (PAR) aims to intervene to improve practices, this was not my goal. Instead, here I investigate critical projects built on *tactics*, *strategies*, and often, *happenstance* (Bødker et al. 2016). This investigation provides insights into the self-directed practices of Indonesians and how technology is used to reflect upon and solve issues their own society.

This project is an interesting scientific inquiry for 3 reasons. First, through a comparison with mainstream hackathons, the study critiques issues with those practices, and sheds light on barely documented mutual aid, participatory design and decision-making processes (*gotong royong* and *nongkrong*) through which socio-technical issues and values of local relevance are articulated. Second, the design and infrastructuring work demonstrates how the community artifact ecology (Bødker et al. 2016) is shaped, and how different socio-technical injustices are addressed in the discursive process. Finally, how criticality is expressed throughout the discussions and outcomes shows how the participants reflect and respond to important societal inequalities.

The third research question explores making in grassroots critical making. Therefore, it builds on the first two research questions by asking questions about the actors who generate the plurality of local solutions and their critical thinking processes, and how these translate to material practice:

- In intrinsic, autonomous, self-directed practices, how does making compare to making in Western countries? What does making look like in Indonesian practice?
- How is making used as a practice? Are there different modes? If so, how do these compare to local needs and societal issues uncovered in the first research question?
- How do values and goals influence the intrinsic design projects and how is this reflected upon in the process of making?

1.4 Conclusion and Prospectus

The research questions outlined above are intertwined, and all lead up to the main goals of this study: extending the scientific knowledge base on grassroots innovation, further developing the concept of critical making by advancing our understanding of critical-material practices of grassroots innovators, building on the term with a post-colonial lens, and moving away from neoliberal approaches. These advancements however go beyond being mere solutions to a problem at hand. They represent further addressing societal issues such as centralized manufacturing, profit-driven healthcare providers, or the problematic state of healthcare. The criticality in grassroots responses to these issues is often latent, because people in contentious political climates need to be careful about voicing dissent. Any impact or value created during the projects is not important for this analysis. What matters is the knowledge we can learn from identifying the problem and the developing solutions, and how these are developed through participatory process.

At the beginning of my research, I asked myself: Is there evidence that makerspace members critically design and implement people-centered, reliable, and sustainable solutions to societal problems—as they often are best positioned to recognize and tackle such problems within their own communities? Can inspiration provided by practices of critical technical practice, design, and engineering change how socio-technical developments are addressed within citizen-driven projects—or is intervention necessary, when those skills, interests might already be available on-site? In response to these questions, I decided to present cases that are profoundly different, from the societal issues they try to solve to their approaches to developing solutions. The ingenuity of human creativity to hack

and make for society is investigated and leads to lessons about their critical, iterative, participatory, and multidisciplinary practices.

Critical and non-mainstream making is as diverse as the makers themselves. They all enable everyday citizens to better understand the “black box” of technology, its societal implications, and how it can make a difference in the development of more just futures. I chose these cases to show that critical making as a practice is very heterogenous. As academic research continues to explore bottom-up movements, it needs to account for these heterogenous and decentralized groups, which means that they cannot and should not be researched as uniform movements. A further difficulty is that the taxonomy is also often scattered, and grassroots don’t always self-identify with the nomenclature academics use to describe²⁶. Such grassroots groups continue to strengthen democracy, attempt to change the status quo, and raise awareness around societal issues invisible “from above” (Cox and Fominaya 2009). Scientists might be inclined to research movements once they’re more institutionalized, because they become easier to research and start following official frameworks. However, their growth often means that funding influences their practices and thus their political-activist roles diminish.

This study aims to learn from the grassroots innovators by deliberately steering clear of normative language, practices, and expectations and attempts to understand how processes of critical making respond to realities on the ground. The goal of this study is thus to further develop the term critical making while advancing HCI, grassroots innovation theories, and development studies through postcolonial computing. I achieve this by analyzing intrinsic-participatory design using reflexive-interventionist methods to investigate the cooperative design, innovation, and knowledge production processes of bottom-up, grassroots, interdisciplinary projects. Their practices are characterized by the adaptation of technological tools to their sociopolitical realities through intrinsic design and appropriation. Specifically, they use rapid prototyping methods and critical thinking to create social innovations and knowledge that raises awareness or addresses hyperlocal problems (Fig. 1.4).

²⁶ For examples on the terms I used when mapping critical making projects in the field, see chapter 5.2.

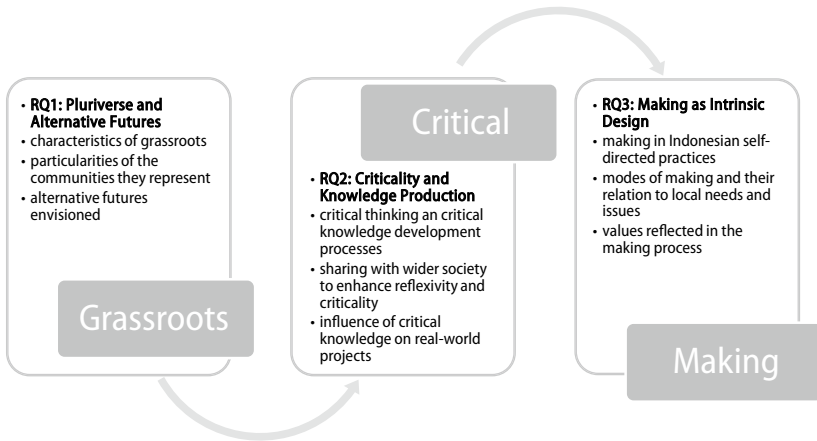


Fig. 1.4 Research Questions and Attributes of Communities Researched

This study investigates reflexive processes that reject Western academic approaches to innovation, technology-based development, and mainstream making. One of the most important hypotheses explored in the study is that while top-down innovation can be socially or environmentally unsustainable, critical grassroots innovation projects have non-market-driven, bottom-up approaches that constitute necessary prerequisites for long-term, sustainable technology adoption—even in rural areas.

The State of Critical Making

2

The aim of this chapter is to introduce the term critical making (CM), which originated from critical technical practice (CTP). One of the main goals of this study is to advance the term by applying it to grassroots innovation practices. CM is a relatively new and somewhat broadly used term. As I will demonstrate, different practitioners use it to define different activities. Since its conceptualization, CM is increasingly being adopted to describe design-oriented academic research practices and pedagogies. As Ratto and Hertz (2019) summarize, “engaging with definitions and articulations of critical making is important given its increasing circulation” (page 18, *ibid*). In this section, I trace the term critical making from its origins as an innovative scholarly practice (Ratto and Hockema 2009) and how it became a design practice to re-politicize Western maker culture (Hertz 2012). Ultimately, it was then extended to the social-activist practices of everyday people as “DIY Citizenship” (Ratto and Boler 2014)¹ and finally expanded to grassroots innovation practices (Sipos and Wenzelmann 2021).

The term critical making was coined more than a decade ago (Ratto and Hockema 2009), and it has been freely developed by researchers and practitioners because of its apparent flexibility. In concept, the term theoretically and practically combines two distinct, yet creative ways of engaging with the world: critical thinking (a cognitive practice) and making (a material practice). In practice, combining these two methods of engagement has proven to be rather difficult. As I will demonstrate, CM builds not only on CTP, it also relates to other similar concepts:

¹ However, most of this work has been focusing on the West, Global North, or minority world. Few linkages have been explored between making activities, grassroots innovation, and critical making in the Global South.

“The concept of critical making has many predecessors, all of which start with the assumption that built technological artifacts embody cultural values, and that technological development can be combined with cultural reflectivity to build provocative objects that encourage a re-evaluation of the role of technology in culture” (Ratto and Hertz 2019).

These predecessors include design practices such as values in design, reflective design, adversarial design, critical design, and speculative design. CM builds on these as a conceptual foundation that bridges critical practice and technological development (*ibid.*). In this sense, CM represents a type of conceptual-material work. This conceptual-material work and the terms CM and CTP are used in this study to describe self-directed grassroots design practices in Global South communities. Before this novel approach can be outlined, it is necessary to summarize its origins and development to demonstrate how it came to be that critical making became grounded in grassroots innovation projects.

It is important to repeat that current definitions of the phrase are not restricted to academic practices, so what qualifies as critical making is ambiguous. Ratto has said that critical making should be seen as a larger category which can link different (design, art, academic, engineering) practices—a hub for various material interventions (Hertz 2015a). In a recent interview I conducted with Hertz, he shared a similar notion:

“Critical making is a great term, but it’s not like it needs to be an exclusive club with a VIP list—people can come and go, bend things, and mess around.” (Hertz 2021)

This “bending” of the term thus can mean not only including different types of practices, but also different groups of practitioners. This is the goal of this study: exploring how CM, as a critical-material practice, drives grassroots innovation in the Global South.

There are 4 distinguishable steps in the complex and organic development process of the term, which represent how the term was extended to include new practitioners. From engineering students (Ratto and Hockema 2009) it was extended to designers, researchers, and makers who wanted to rethink their practice (Hertz 2012). Then it was further extended to describe citizen-driven critical activities (Ratto and Boler 2014), and finally, the critical technical practices of grassroots innovators in the Global South (Sipos and Wenzelmann 2021). This chapter will outline the evolution of Critical Making as briefly summarized in Fig. 2.1, to demonstrate how it can be extended to new audiences and practitioners:

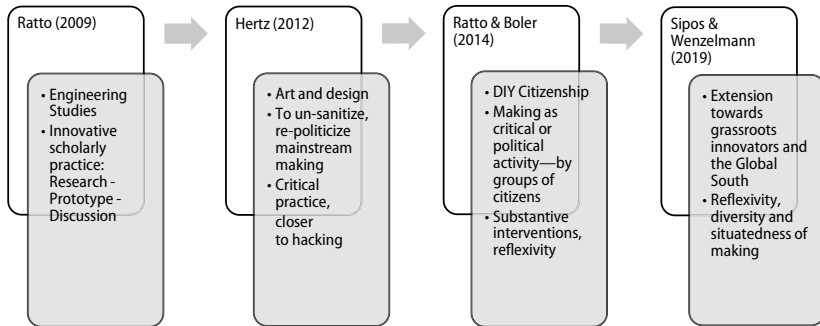


Fig. 2.1 A Timeline of the Term Critical Making

2.1 Critical Making as Innovative Scholarly Practice

Prof. Dr. Matt Ratto² coined the term critical making to describe a new type of constructionist scholarly practice for his engineering students:

“Critical Making is an elision of two typically disconnected modes of engagement in the world: critical thinking,’ often considered as abstract, explicit, linguistically based, internal and cognitively individualistic; and ‘making,’ typically understood as material, tacit, embodied, external and community-oriented” (Ratto and Hockema 2009)

As Ratto later explains, he sees the term “critical” originating in the critical scholarship of the Frankfurt school, from writers like Adorno and Benjamin. Their writing suggested that criticality did not only mean reflection, but also intervention in society (e.g., Hertz’s interview with Ratto in Hertz 2015a). They developed a critical method to innovative scholarly practice to address and discuss so-called wicked problems (Rittel and Webber (1973) in Ratto 2011, page 253) by using making to experiment with and develop a collective frame that highlighted

² Ratto is an Associate Professor in the Faculty of Information at the University of Toronto, and Bell University Labs Chair in Human-Computer Interaction.

and resolved disciplinary and epistemic differences (*ibid.*)³ Ratto aimed to critically explore the social issues inherent in technical systems, to acquaint students with the possibilities and problems of new physical and ubiquitous information technologies. He also developed the term to help them develop basic skills in designing, making, and evaluating information systems (Ratto 2009).

The paper that outlines the idea describes a workshop, called FLWR PWR, which used Arduinos (open-source, single-board microcontrollers) and craft materials to construct cellular automata that could communicate through infrared light. The communication between these cellular automata was programmed by the participants, enabling them to experiment with potential outcomes and exchange ideas. Debate about social problems can be very theoretical, but the physical objects acted as boundary objects (Star and Griesemer 1999) that represent processes and outcomes. Discourse among participants on potential social issues caused by the phenomenon of walled gardens could take place faster and more efficiently⁴. Elsewhere, in addition to STS, Ratto had also integrated the concerns of digital humanities scholars about digitality and information. He brought the emancipatory and constraining aspects of ‘technoscience’—as articulated by scholars Haraway, Star, Bowker and Latour—into academic critical maker projects (Ratto and Hertz 2019).

Ratto also provided an overview of how he developed the practice, which he sees as “a series of experiments in (...) critical making” (page 252, Ratto 2011) and explained its three stages. These are analytically but not functionally separable, as the project could start from any stage (Fig. 2.2):

³ Critical making can have different objectives. In the US and Canada, it mainly focuses on critically exploring the social issues inherent in technical systems, to acquaint students with some of the possibilities and problems of new physical and ubiquitous information technologies, and to help them develop basic skills in designing, making, and evaluating information systems that use these new technologies. The Critical Making course at the Technical University of Berlin is dedicated to gender issues and aimed at developing the critical thinking skills of students who will be leading future R&D projects.

⁴ Both the issue of walled gardens will be further elaborated upon in this study (see Sect. 5.5.2, artifact ecology discourses of the hackathon and the “free basics”), and I will refer to the usefulness of critical making to address so-called wicked problems later on (see Sect. 6.3.2.3, discussing grassroots innovators and their relation to wicked problems in society).

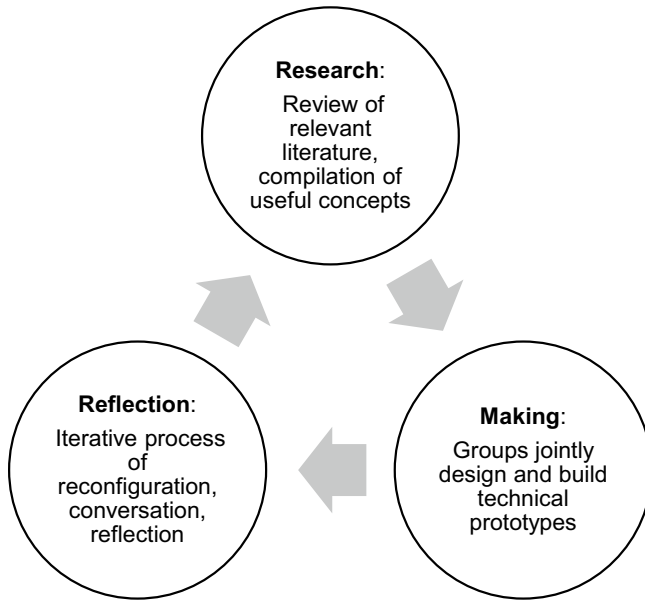


Fig. 2.2 Three Stages of Critical Making. Based on Ratto (2011)

The *research stage* reviews relevant literature and compiles useful concepts and theories on critical sociotechnical issues, especially in STS. This contributes to the next stage, as literature is mined for specific ideas that inform the construction of material prototypes, and fabrication (*making*) is used as a method of exploration. The *making stage* takes place when groups of scholars, students, and potentially other stakeholders design and build technical prototypes. It is important to note here that, rather than coming up with fully functional devices, Ratto encourages extending knowledge and skills through technical exploration; building working devices or solutions is not required. The *reflection stage* is an iterative process of reconfiguration, conversation, and reflection. In this stage, prototypes are used to explore various configurations and possibilities—the plurality of outcomes. These explorations express, critique, and extend relevant concepts, theories, and models to return the participants to the *research stage* to understand the concepts that the artefacts represent (page 253, *ibid.*).

Ratto's course at the University of Toronto (as of 2009) was titled "Critical Making: Critical Information Studies Meets Design Oriented Research". In

this university course, social issues inherent in technical systems were critically explored. Design was practiced through making in the Critical Making Lab, which was described as a shared space for opening up the practice of experimentation with embedded and material digital technology (Ratto 2009). Teaching critical technical practices and critical making can also be found in the US (Berkeley, Stanford, NYU) and Canada (University of Toronto and Emily Carr). The Critical Technical Practice course at the University of Colorado, Boulder equips students with theory and methods to think more critically, creatively, or radically about possible future human-technology relationships. It engages with design as a means of provoking discussion, ideas, and questions about technology and culture, as opposed to finding the ideal solution for a well-defined problem (Devendorf 2017). The “Let’s Make a Monster: Critical Making” course at Stanford builds on the notion of “technology-out-of-control”—a constant worry of modern societies. It defines critical making as a practice that utilizes technologies to think about humans’ constitutive entanglements with technology, while recognizing that insights often come from errors, glitches, and malfunctions (DeMarinis and Denson 2018).

At the Technical University of Berlin, I conceptualized their first critical making seminar, titled *Hacking Innovation Bias*. The course sought to teach interdisciplinary students how to think critically and responsibly about R&D and help them understand that technology development is political. It was built on Ratto’s three stages, and as expected, for students with a background in sociology, the technical-material design part was harder to grasp, and found an introductory session to the Arduino exhausting, even though many eventually enjoy its relatively accessible platform and the hands-on work. Engineering students tend to immediately jump into developing solutions and working prototypes and often found the conceptual explorations and discussions uncomfortable. Their discomfort is important to note because it shows that being confronted with an interdisciplinary mode of thinking can create tensions for people who have only experienced one discipline.

Ratto, in an interview by Hertz for the book “Conversations in Critical Making” expressed that there seem to be many different types of material engagements that previously already represented important processes for social intervention, such as critical design, critical technical practice, and participatory design (Hertz 2015a). However, in his conception (here he emphasizes that he doesn’t define the term) it differs from others because it focuses on the “lived experience of making” and how it can deepen participants’ ability to ask socio-technical questions. For him, it is a way to learn and explore the world, however, only in academia: “I was talking about this from a very academic perspective because when I was

first talking and thinking about this, my goal was to create innovative scholarly practice. I wasn't thinking about critical making as a more general form of social engagement" (page 37, *ibid.*).

2.2 Extension Towards Design: Making as Critique

In the same book, Hertz explains that his interest in the term came from the novel combination of hands-on technology development and studio practice, and its potential to encourage "makers—whether they are engineers, industrial designers, or technology-oriented artists—to step back and reevaluate the assumptions and values being embedded into their designs" (Hertz 2015a). Furthermore, Hertz says he felt that Ratto could have followed through with his process of making. In his opinion, objects are efficient as "things to think with" and can "travel through different contexts" (*ibid.*).

Hertz adapted the term to look beyond the idealized picture of the maker and to "reintroduce a sense of criticality back into post-2010 maker culture to un-sanitize, un-smooth and repoliticize it" (Hertz 2015b). He encouraged a type of making different from mainstream making, which dares to express political critique through art, activism, and social innovation. It has thus been the objective of Hertz to expand the term critical making as an appeal to hands-on makers to be more critically engaged with technology.

As he explained in an interview, he has also been interested as taking the term as a critique of maker culture, especially the brand of Maker Media. He saw this form of making as a sanitized version of DIY electronic practices in electronic art. For him, critical making is a partial embracing of maker culture, only with additional questioning and more nuanced consideration of why people make things (Sipos 2021).

He attempted to explore these ideas through two collections of partially citizen-driven projects that were published in zine-like booklets (Hertz 2012). As he mentioned in my interview with him on the present and future of critical making, he intended to inspire academics to engage in alternative publishing methods (Hertz 2021). His ideas reached the researcher-maker Fonseca, who was inspired to call himself a "critical maker" as opposed to a maker (Fonseca 2015).

Hertz collected critical responses to the maker movement, in which he found criticality in an "interplay between critical theory and hands-on practice, contemporary art, the process of developing new technologies, open-source hardware, tactical media and politics, interdisciplinarity and academic institutions, critical

and speculative design, mass-produced consumer culture, and hackers and hackerspaces” (page 5, Hertz 2015a). “Critical Making”, released in 2012, consists of 10 pocket-sized booklets and includes interviews with Phoebe Sengers—who develops new kinds of interactive technology that respond to and encourage critical reflection on the place of technology in culture—and Natalie Jeremijenko, who blends art, engineering, and environmentalism to create real-life experiments that enable social change. It also contains short descriptions of critical making projects, such as Soft Publics by Mike Manalo, facilitating temporary sites for legally protected speech, and the Hazmat Kite by Ken Gregory, which responds to the smog problem in Windsor, Ontario (Hertz 2012). These booklets were followed up by “Disobedient Electronics: Protest”, in 2016, a zine showing projects that point out how building electronic objects can be an effective form of social argument or political protest, and that the larger issues of what it means to be a human or a society today needs to be directly confronted. This zine format also consists of a number of project descriptions, such as the 79 percent Work Clock by Party, which calls attention to the gender wage gap, and Probots, a protesting robot by Csikszentmihalyi, Aguiar and Azevedo (Hertz 2017).

This lens of art and design in critical making is also reflected in many research projects. Conducted by a Dutch Critical Making consortium between 2017–2021, a project titled Bridging Art, Design and Technology through Critical Making asked questions such as: “How can 21st century creative practices art, design, making overcome the industry logic of techno-optimistic makeability and aestheticized products of creative industry?” (Cramer et al. 2018). In 2020, the workgroup Material Practices emerged from the project and organized the Making Matters Symposium⁵, which invited Garnet Hertz and the Jatiwangi Art factory from Indonesia⁶.

The Rhode Island School of Design and Creative Practice, instead of focusing on critical design, states that its students are immersed in a culture of critical making, and published a reader on the topic (Hermano and Somerson 2013). In Switzerland, the Critical Media Lab is an integral part of the Institute of Experimental Design and Media Cultures (IXDM) at the Academy of Art and Design FHNW in Basel. It’s a physical lab space and a conceptual vehicle, which, based on its website aims “to develop a contemporary notion of criticality towards

⁵ <https://research-development.hetnieuweinstituut.nl/en/activities/collective-material-practices-critical-times>.

⁶ https://research-development.hetnieuweinstituut.nl/sites/default/files/timetable-making-matters-symposium_0.pdf.

design, media practices and their cultures, but also aims for a playful and experimental proximity of practice and theory”⁷. Art, design and making as practices are thus somewhat interchangeable with criticality in these approaches.

2.3 Exploring Citizen-Driven Projects: DIY Citizenship

In the same interview with Hertz as I quoted above, Ratto reflects on how the term described activities that were considered “DIY citizenship” (Ratto and Boler 2014). This is to date the most relevant collection of academic essays on critical making in bottom-up, grassroots innovation. Based on a 2010 conference with the same title⁸, “DIY Citizenship: Critical Making and Social Media” was the first book to explore critical making in the everyday lives of people, particularly emergent communities of critical makers who engage in political discourse as “DIY citizens” (ibid. page 5).

In the interview with Hertz, Ratto recalls how interesting it was to see that in those years, what was considered making drastically changed. From his point of view, it became a more dominant cultural activity and activities included in the term broadened, encompassing more than just technical objects and including community gardening and “knit bombing” (Hertz 2015a). A move away from being merely a maker became visible through a desire to highlight political and conceptual attributes of the projects, in addition to their engagement with technology. Such grassroots approaches to critical citizenship are described through the terms “maktivism” and “DIY citizenship” (Ratto and Boler 2014). Other literature on grassroots critical making analyzes how the activist nature of such practices can also be found under the terms “DIY maker-activists” (Kohtala and Ede 2019), “maktivism” and “t inquiry” (Mann 2014).

Winner’s critique of a lack of public engagement and technocratic trends asking “How can and should democratic citizenry participate in decision making about technology?”, is turned around in the book. The editors instead ask: “How do we engage with society politically through technology?” (page 343, (Winner 1992) in (Ratto and Boler 2014) page 19). This question leads them and the contributors to projects of “DIY citizenship”, which they understand as productive

⁷ <https://criticalmedialab.ch>.

⁸ Organized by Prof. Boler and Prof. Ratto November 12–14, 2010. The dedicated website <http://diycitizenship.com> was not working during the writing of this study, but further information can be found at <http://sites.utoronto.ca/csus/Conferences.html>.

and socially interventionist practices that highlight a shared sense of responsibility. The groups involved include individuals and groups who are self-organizing and enabled by social media networks to scale up and cross the borders of the State. As Ratto and Boler highlight, they know that such communities might be dismissed as merely hobbyists and not taken seriously. However, their work creates online networks that provide opportunities for new forms of connective engagement to emerge and thus a more optimistic counterculture to form⁹.

As a next step, Ratto extended critical making as a practice “from academic critique to situated intervention” (Ratto 2016), working with a non-profit social enterprise, developing personalized 3D-printed prosthetics for people in Uganda and Cambodia¹⁰. In terms of research on purely citizen-driven (bottom-up) development, Research Group 2 on “Critical Maker Culture” at the Weizenbaum Institute for the Networked Society¹¹ in Berlin stands out. The research group originally focused on the production opportunities of maker culture, researching makerspaces and open labs in industrialized nations (USA, South Korea), emerging economies (Brazil, China), and developing countries. Today, they investigate maker and hacker communities as a decentralized network of producers, democratizing access to technology through diversity and inclusion. Their research is driven by questions of gender, sustainability, and international development¹². In addition, a Critical Making consortium was awarded funding by the European Union’s Horizon 2020 programme based on my research for this study. The focus of that project (running between 2021–2023) is slightly different from the focus of this study. It studies principles of Responsible Research and Innovation (RRI) in maker communities, with a special focus on questions of gender inclusion, youth, education, and open-source hardware¹³.

As the term started to become more widespread, it was adopted by maker-researchers (Fonseca 2015). Subsequently, people outside of academia started engaging with the concept. In 2015, a grassroots innovation collective in Indonesia was inspired to start a conference called “Transformaking”¹⁴. This was the first time the term was used in a non-academic setting.

⁹ I had a similar exchange with a maker after my presentation at the 35C3 hacker conference; she thanked me for including knitters in my presentation, as she felt that such groups were often excluded from what is considered making.

¹⁰ See e.g. <https://niatech.org>.

¹¹ <https://www.weizenbaum-institut.de/en/>

¹² <https://www.weizenbaum-institut.de/en/research/rg2/>

¹³ <https://criticalmaking.eu>.

¹⁴ <https://honf.org/transformaking-2015-open-call/>

2.4 Grounding Critical Making in Grassroots Practices

The International Summit on Critical and Transformative Making (or “Transforming”) was the response of a historical media art collective from the Global South to Western maker practices, and thus a starting point to ask other critical, post-colonial questions. The title shows the political nature of the event, and the description is similarly critical. The organizers of the 2017 conference asked how knowledge can be used to empower people in an exclusive system, and whether catastrophes caused by STI can be averted only through rationality¹⁵.

Learning about this conference meant that while grassroots innovation practitioners might rarely use the term critical making to describe themselves, the term does resonate with some of them. Hertz also pointed out that there were people responding to the term of critical making, especially those who were already doing this experimental hacking or electro-mechanical studio work under the guise of “electronic art”¹⁶ (Hertz 2015a).

It seems like research is moving towards a better understanding of grassroots practitioners through various forms of critical making. However, locating such practices remains difficult because terms used for self-identification in grassroots groups do not always overlap with mainstream or academic terms. It is important to note that some makers refuse the maker label to describe themselves and their activities (Chachra 2015). Some feel it is too exclusive and represents a Western narrative and practices of innovation. To describe his practice, instead of making, the free/open advocate and researcher Felipe Fonseca uses *critical making* (Fonseca 2015)—or local words, such as the Brazilian *gambiarra* (ibid.).

For the purpose of this study, it is important to understand what kinds of grassroots actors the term “critical making” resonated with, to ground the term in their practice, and finally, to describe cases to advance the scientific understanding of grassroots practices. Inspired by Ratto and Boler’s approach to DIY citizenship of, together with a co-researcher, I designed a participatory workshop (Sipos and Wenzelmann 2019) to see if the term resonated with grassroots practitioners, and ground critical making in bottom-up practices. The outcomes of this workshop were summarized in a detailed paper (Sipos and Wenzelmann 2021). For the purposes of this study, it is important to briefly summarize key points. The workshop titled Critical Making with and for Communities was held at the Ninth

¹⁵ <https://honf.org/transforming2017/>

¹⁶ As I will explain later, many of the Indonesian post-authoritarian collectives started out as media art collectives, engaging in electronic art, thus, Hertz’s observation also applies in the Indonesian context.

International Conference on Communities and Technologies (C&T),¹⁷ which had the goal of unmasking the utopian vision of making that relies on technosolutionism and ideological colonialism. It sought to look beyond the maker hype and its promises, as inspired by Lindtner et al. (2016). It also aimed to understand whether critical making could describe the activities, processes, and aims of grassroots innovation communities that focus on collaborative technology development in the Global South. We asked whether the term critical making could be extended to describe grassroots innovations, and how it resonated with the participants. Three questions were discussed: Does making democratize access to and practices of innovation? What is critical making for you? And finally, a two-part question: Aren't all makers critical makers? Why should they be critical at all?

While many of the participants confirmed that they thought there was an inherent criticality in DIY and making, in their opinion, mainstream making was a privileged hobby. Conversely, critical making for them was directly related to frugal innovation, non-Western viewpoints, and local contexts. It brought a culture of community-based design, discourse, reflection, and responsibility. Especially in the context of the Global South, making seems to inherently elicit criticality. The practice itself can serve as a tool to educate people, but often, making is done out of necessity (as in the case of *jugaad*). They also shared emerging topics of grassroots critical making, like as the importance of intersectional makerspaces and the need for guidelines for building technology for diversity and inclusion. They were also interested in how making could enhance educational curricula and create opportunities for skills development. In their eyes, making could bring technology literacy for all, democratize access to open-source innovation, and alleviate the difficulties commons-based organizations face in achieving financial sustainability.

With regards to whether critical making was a term that grassroots innovators used for themselves, we found:

“The term critical making still cannot serve as an overall description of the activities, processes and aims of grassroots innovation communities focusing on collaborative technology development in the Global South, however, it proved to be useful in discussions between these members of the global assemblage of makers. While being aware of the shortcomings of any one term to summarize all different practices, critical making provided a framework both open and narrow enough to identify, discuss, share and better understand the different hyper-local, scattered processes which

¹⁷ <https://2019.comtech.community>.

reflect the plurality and reflexivity of the global assemblage itself” (page 9, Sipos and Wenzelmann 2021).

This workshop improved our understanding of how to approach critical technical practitioners outside of academia and the West, which was a necessary step in further “bending” the term critical making.

This section summarized the development of the concept, and how it has expanded who can be seen as its practitioner—from Canadian engineering students to grassroots innovators in the Global South. In this next section I’ll highlight more relevant socio-technical theories and academic approaches. The next section builds first on what Ratto and his colleagues claimed critical making could look like as a research program (Ratto et al. 2014), and extends their ideas with theories about grassroots innovation and post-colonialism. Especially the latter would not have made sense in the analysis of Western practices. However, it is crucial for the analysis of the practices of grassroots innovators in the Global South.

Methodological Tools and Theoretical Arguments

3

Approaches to critical technical practice and critical making are not new, but remain open and shapeable. This literature review will build on the existing work of researchers like Ratto and Hertz and their interest in which methodological tools and theoretical arguments are relevant to critical making. In this interdisciplinary study, I build on relevant socio-technical theories by foremost considering their empirical, real-world applications and implications.

As a starting point, this chapter takes Ratto et al.'s (2014a) approach to critical making as a research program. However, as the novelty of this study lies in further developing and extending the term critical technical practice towards grassroots practitioners in the Global South, it is important to explore additional socio-technical theories and methodologies.

3.1 Critical Making as a Research Framework

The proposed approach builds on critical making as a research program as suggested by Ratto et al. (2014a). As I will demonstrate, additional research approaches can augment their analysis of critical making. After all, the authors' impetus was to better understand how the practice of making can be a core component of social critique and reflection (ibid.). "Critical making" is here understood as an umbrella term for different practices linking traditional scholarship in the humanities and social sciences to material engagement. A lot of interesting questions are posed in the other approaches as well that sharpen our critique of how technology is designed and developed. What unites the above approaches is critical making—material-conceptual processes that unite reflexivity and intervention. These approaches call for tactical interventions and a disruption of traditional models of technological development. They are mainly

aimed at engineers and designers, and only in some cases to the public (Sengers (2017) in Ratto and Hertz (2019)). If technology is to improve society, it must be critically reflective and designed to account for the complexities of what it means to be human (*ibid.*).

For this study however, the most relevant approaches are particular types of PD, which can improve our understanding of grassroots practices. These design and practice approaches have been highlighted by scholars who have shaped critical making, and have mostly addressed Western practices. This discussion will unfold in the second subchapter.

In this first subchapter, methodological tools for critical making as a research program as proposed by Ratto et al. (2014a) are explored. I detail the methodological recommendations in the paper that are especially relevant for this research: critical technical practice (CTP) and participatory design (PD). The latter is even further detailed, as PD has various sub-directions (including intrinsic design and values-in-design) that are of particular relevance to the study. Such approaches can help us better understand the innovation processes of participatory communities. In Fig. 3.1, I marked the approaches relevant to this study with dark grey:

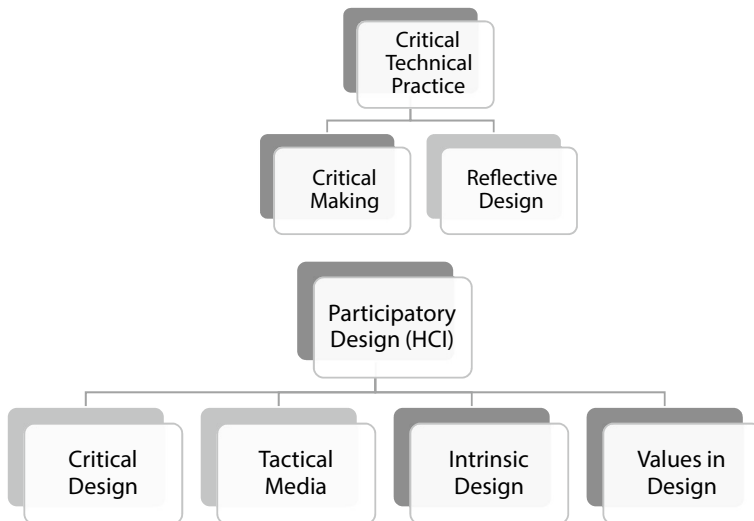


Fig. 3.1 Methodological Considerations of Researching Critical Making

In 2014, Ratto et al. edited a special issue of “The Information Society” (Volume 30, Issue 2), a Special Forum on Critical Making as Research Program. In the introductory paper, they specify that the goal of their program is to “articulate a research program that can help us better understand “making” as a core component of social critique and reflection” (Ratto et al. 2014). As this approach resonates most with the goals of this study, it is important to summarize the relevant theories through which knowledge and social organization can be conceptualized as a material practice of making.

The authors of the paper highlight that design studies and social research have been engaging in technoscience (e.g., Latour and Woolgar (1986); Haraway (2018); Ihde and Selinger (2003) on page 85 in Ratto et al. 2014). At the same time, dominant systems of information exchange and knowledge generation have focused on assembling rather than deconstructing, and the success of critical scholarship’s work on “scientific matters of fact” has impaired their ability to speak to “matters of concern” (Latour 2004b) in *ibid.*, pp 85 and 86)—except in the area of environmental justice (*ibid.*). The authors thus propose bridging theory and practice to revisit academic critique and call it critical making. In their non-exhaustive review of possible approaches, they highlight participatory design, critical technical practice, values in design/value-sensitive design, critical design, and tactical media. Ratto has also mentioned elsewhere (Hertz 2015a; Bogers and Chiappini 2019) that other material-conceptual practices bring practitioners’ attention to values associated with technology and its development, including critical design, critical technical practice and its extension into Human-Computer Interaction as reflective design.

There are thus many different academic and design approaches that overlap. What is relevant for this study is how critical making has inspired innovative scholarship (Ratto and Hockema 2009). Ratto’s inspiration for critical making was based on Agre’s bridging of disciplinary mindsets that were a precondition to hybrid multi-disciplinary practices. He saw the usage of critical literature from the humanities and social sciences “as a way to trouble and contextualize the instrumental logics associated with computer science and engineering” (page 21, Ratto and Hertz 2019). Ratto highlights this as the source of his term. Below I will focus on two main directions. First, on critical technical practice (CTP), the starting point for critical making, and a term that I will also use for the non-maker practices of grassroots innovations. Second, participatory design (PD) and its subcategories will be especially relevant for the case studies, as well as values in design (VID), which will be important to understand how grassroots innovators evaluate different technologies for inclusivity in their design processes.

3.1.1 Critical Technical Practice

Ratto was inspired to develop critical making by one of his teachers, Philip Agre. Agre's paper on Critical Technical Practice outlines a critical theory-based approach to technological design, designed to increase awareness and critical reflection on the hidden assumptions, ideologies, and values underlying technology design (Agre 1997). In this paper, he presents his own struggles to bridge his disciplinary mindset, not having had any critical tools to defamiliarize himself with dominant ideas or to imagine alternatives. He also discusses his attempt to stop thinking like an artificial intelligence (AI) developer and start thinking like a social scientist. He situates CTP as a critical theory-based approach to technological design, where critical and cultural theories improve the work of designers and engineers. Its ultimate goal is to increase awareness and critical reflection on the hidden assumptions, ideologies, and values underlying technology design, and to build reflection into the technology development process (ibid).

Critical Technical Practice, Science, and Technology Studies and Postcolonial Theory

The application of CTP, according to Foster et al. (2006), was always up to individuals and their own interpretation. Keeping this in mind, I would like to highlight the work of the Critical Technical Practice Lab at the Madeira Interactive Technologies Institute. This lab is especially interesting for this study, as their critical technical practice has been embedded in STS and postcolonial theory. They create long-term relationships with communities, reference African epistemologies, and develop technologies with and for the Global South (Csíkszentmihályi 2017). The Probots protest robot developed at the Institute was featured in Hertz's *Disobedient Electronics* zine (Hertz 2017), while RootIO, a project integrating ICTs into bucket-sized stations for grassroots communities in the Global South, was invited to the Future Innovators Summit of Ars Electronica in 2014.

3.1.2 Participatory Design

It is important to start a discussion of participatory design by making a distinction between two different design practices: one conducted by designers that creates an artefact that conveys a message (e.g., in critical and speculative design), and another that involves everyday people in the design of systems that concern them (participatory design).

Previously, when discussing the zines of Garnet Hertz, I gave some examples of critical and speculative design (Dunne and Raby 2013). This type of design

employs design fiction and speculative design to imagine possible futures through artefacts. It addresses the dominant focus on commercialization and for-profit interests within the world of design through the design and exhibition of objects. These practices reveal their interests and provide alternatives, while drawing consumers' attention to values, ideologies, and behavioral norms inscribed in the designs that are used in everyday lives (Bardzell and Bardzell (2013) in Ratto and Hertz (2019). Arguments about how design can critique the present or propose alternative futures go back decades. Bardzell (2018) summarizes key sources such as Schoen (1959), Simon (1969), Rittel and Webber (1973) and Ehn et al. (2014) for visions of design that aspire to improve society.

This study is concerned with the question DiSalvo asks in his adversarial design practice. He asks, how we can collectively make speculative representations and prototypes of possible futures as a creative process? What should come *after* those imaginations of possible futures (DiSalvo et al. 2010; DiSalvo 2012)? While grassroots innovators must stay open minded to imagine possible, just, inclusive futures, what next steps do they take after those imaginations to turn their ideas into reality?

The word “design” in this study refers to a practice known from Human-Computer Interaction (HCI) as participatory design (PD). To summarize briefly, PD enlists users in design practice (Hartson and Pyla 2019), and several interesting directions have been taken under the umbrella of PD in the last 50 years, which will inform this study. In recent years, hacker and maker culture and grassroots innovation has gained attention in the Human-Computer Interaction (HCI) community. But participatory design originated decades ago, with workers successfully calling for participation in the design of the systems that affected them (Ehn 1989). Based on the same fundamental values of democratization and empowerment, and with technologies becoming more ubiquitous and accessible, researchers within and beyond the field of HCI have developed frameworks and methodologies to support technology design for and with affected communities (Marquardt et al. 2017; Rohde et al. 2017; Smith, et al. 2017; Smyth et al. 2018; Taylor et al. 2017). Hackerspaces and the practices of DIY makers have been explored as “Emerging Sites of HCI Innovation” (Lindtner et al. 2014). Making has also been explored through post-colonial computing (Kaiying et al. 2019) and critical theories of colonial and postcolonial technoscience (Philip 2010). Utopianism has sparked speculative imaginings and explorations of alternative futures of making to reconstitute the utopian vision of making (Lindtner et al. 2016). These explore making in its societal setting and encourage critical reflection on the vast differences between narratives of mainstream and grassroots innovation processes, and to broaden the term to include non-Western making.

Participatory Design as Emancipatory Politics

Socio-political dimensions of technology design have been at the core of PD from its beginnings when it was developed by workers and unions to mitigate the profound transformations following the introduction of computers in workplaces (Ehn and Sandberg 1983; Kyng and Mathiassen 1980). PD emerged from a perceived of researchers and designers to better include users in the design process of new technological systems. Its origins can be traced back to a project that sought to understand how 1970s job distribution and wage systems were impacted by the development of production technologies (Asaro 2000). As Ratto et al. (2014) highlight, the logic of PD is distinctly Marxist, as it is concerned with how new technological systems deskilling and intensify the division of labor, shifting the control to managers. The 1980s UTOPIA project is an example of how PD addresses this issue by highlighting how workers need to participate in the design of such systems—albeit only on the interface level, not by accessing its deeper technological layers (Winner 1992). PD is thus an approach that democratizes technology design by giving system users a chance to participate in its design, and allows for learning between designers and those future users and/or stakeholders (Simonsen and Robertson 2013).

Community-Led Initiatives and Intrinsic Design

For a better understanding of grassroots innovation communities, PD's concern with community-led initiatives and intrinsic design are especially interesting. For example, Bødker, Korsgaard, Lyle, and Saad-Sulonen (Bødker et al. 2016) built on Kaptelinin and Bannon's (Kaptelinin and Bannon 2012) work in interaction design, distinguishing between intrinsic and extrinsic technology-enabled transformation. Bødker et al. (2016) focuses on how volunteer communities develop and infrastructure¹ their artefact ecologies (the constellation of technologies they own and use) through intrinsic design. In other words, "it is conducted by members of the community, with no input from the outside" (page 1, Bødker et al. 2016). This approach is especially interesting for both case studies, and is a major contribution to our scientific understanding of the innovation processes of grassroots practitioners—especially without the influence of designers, researchers, or development workers. In terms of community-led initiatives, Rossitto et al. (2021) investigates how socio-technical configurations can overshadow communities' key concerns, and how they balance efficiency and care.

¹ Infrastructuring refers to the process of actors gradually coming together, through which actor networks are stabilized and the negotiations are finalized (Star and Ruhleder 1996), or in terms of Latour, where "matters of concern" are addressed and resolved (Latour 2005).

Values in Design

I would like to briefly touch upon the term “values in design”, or “value-sensitive design”. This practice “takes into consideration the value assumptions of particular technical practices, the focus is broader and includes the negotiation of values by stakeholder groups, as well as the value trade-offs that are part of constructive activity” (page 87, Ratto et al. 2014). Research on values in design discusses how values (specifically biases) embedded in artefacts might end up playing out as disasters especially in information technologies used by the masses like Facebook or Google (Knobel and Bowker 2011). Advocates for values-led inquiries within PD, which support the emergence, development and grounding of values (Iversen and Leong 2012), draw attention to socio-political issues of accessibility, inclusivity, and identity embedded in community-based technology design (Fiesler et al. 2016).

Negotiations of values and unfortunate tradeoffs will be an important part of the analysis in this study. As I will explain later, grassroots innovators try to pay attention to the values built into the technologies they use in their artefact ecologies. While VID/VSD originally focused on the design processes of designers and engineers, it also touched upon grassroots innovators who repurpose or appropriate technologies. As the “intrinsic designers” of their own projects, it will be interesting to see how values shape the final outcome and what socio-technical injustices are uncovered in the process.

3.2 Extension Towards Grassroots in the Global South

In this second subchapter, relevant theoretical arguments and methodological tools extending critical making as a conceptual-material, socio-political practice for grassroots practitioners of technology in the Global South are presented. These theoretical arguments lay the foundation for a better understanding of critical making. Studies of critical making have been building on critical theory and the critical theory of technology, as well as Science and Technology Studies (STS). To scientifically ground and extend the term to the grassroots in the Global South, it is also necessary to explore grassroots innovation movement studies and post-colonial computing in Human-Computer Interaction (HCI), in order to be able to conduct an informed analysis. In the Fig. 3.2 I’m again marking the most relevant theories for this study with dark grey, and those that inform the study, but are not or less the scope of the study, with a light grey background:

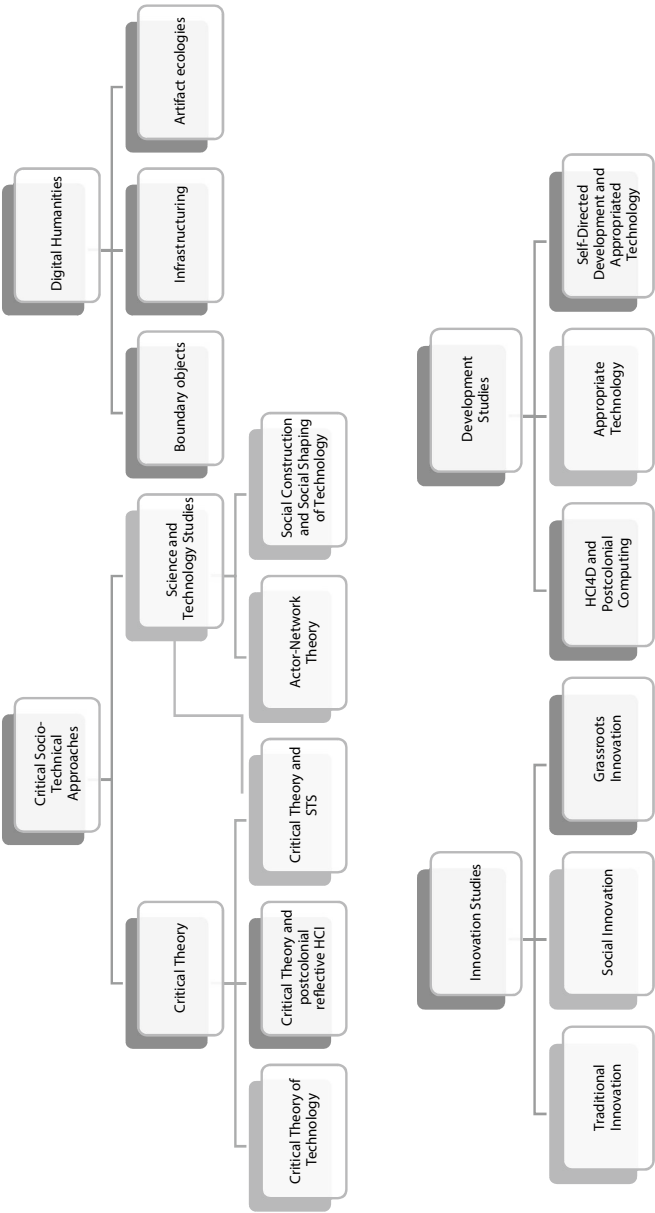


Fig. 3.2 Theories and Methodologies Extended Towards Grassroots Critical Makers

First, I'm outlining an extended literature review of critical socio-technical theoretical approaches. Criticality in critical technical practice is to be explored through a short outline of critical theory, followed by discussion of the critical theory of technology and other relevant theories of STS, for example of technology as a "black box" (as often highlighted both by Ratto and by Hertz, see e.g. Ratto and Boler (2014) or Hertz (2015a).

To extend the term to the grassroots, it is necessary to situate this research in innovation studies. Here, grassroots innovation is to be contrasted with conventional innovation theory, meaning traditional, mainstream theories and practices of industrial innovation. The term also needs to be compared with the term social innovation (Moulaert et al. 2014). Those few relevant enquiries where grassroots movements were explored as community-based versions of technology, science, and innovation (alternative science and technology) are also briefly outlined here, as well as grassroots computing, as theorized in HCI.

To extend the term critical making to the Global South, research also needs to be informed by development studies, especially concerning the usage of technologies for development and in development settings. Theories and pitfalls of HCI4D, the concept of postcolonial computing, appropriate and self-directed development, and appropriate technologies are outlined in this section.

Finally, literature on grassroots, postcolonial computing, and participatory design in Indonesia is reviewed. As I will discuss in the conclusion, this literature informs the study and the analysis.

3.2.1 Critical Theory

To briefly recap, the *critical* in the scholarly practice of critical making can be interpreted both as the cognitive practice of critical thinking and as an academic approach that learns from critical theory. The original definition of critical making is "an elision of 'critical thinking' (...) and 'making'" (page 62, Ratto and Hockema 2009). The initial goal of the technique was to conceptualize critical social issues through collaborative material construction (page 61, *ibid.*). Ratto and Hertz state:

"We believe that 'critical' is an essential attribute for such interdisciplinary practice, including reflective thinking about the value propositions and epistemic boundaries and practices within traditional disciplines. We also see criticality as including expansive thinking about the site and social import of the objects and forms of knowledge that emerge from one's work" (page 21, Ratto and Hertz 2019).

They also state that in their interdisciplinary learning practices, they build on the critical theory of the Frankfurt School. Its interventionist and transformative approach meant that criticality entails not just reflection but also intervention in society (page 37, Hertz 2015a). They highlighted the work of Marcuse, Adorno, Horkheimer (Ratto and Hertz 2019), and Benjamin (Hertz 2015a). Ratto and Hertz highlight the following key points from Horkheimer's (1972) summary. Critical theory is understood as distinct from the dominant academic understanding of theory. Instead of developing generalizable descriptions and normative categories such as better, useful, appropriate, productive, or valuable (page 207), it aimed to understand and establish a theory dominated by a concern for reasonable conditions of life (page 199) and consequently, social transformation and emancipation. The word critical is linked to the overt goal, "to liberate human beings from the circumstances that enslave them" (page 244, Horkheimer 1982).

The first generation of critical theorists in the early Frankfurt School was concerned with technology. However, according to Delanty and Harris, their engagement has been outmoded by historical developments. Moreover, contemporary critical theory has largely ignored questions of technology (Delanty and Harris 2021). Delanty and Harris summarize that Adorno and Horkheimer, similarly to Heidegger (Zimmerman 1990), were critical of the association of technological advancement with progress, and drew a connection between technology and capitalism. Their understanding of technology was shaped by Marx's theory of capitalism and Weber's theory of rationalization. Marx and Weber allowed for a view of technology not only as machinery, but also something connected with the question of nature itself that complements labor. Benjamin understood technology as a part of radical cultural transformation, which proves that their disparagement for modernity (or, capitalism) did not attempt to return to a pre-technological age, but rather was an expression of the disparagement for modernity (*ibid.*).

The Critical Theory of Technology

In terms of the Critical Theory of Technology, Stiegler argues for a technology based less on entropy (moving towards fragmentation, destabilization, and disorder) and more on *negantrophy*, a form of integration that enables human flourishing and care (Stiegler 2017; Stiegler 2018). The critique counters the neoliberal idea of "moving fast and breaking things" (Taplin 2017) by building inclusion, respect, and care into the processes of technological development.

Feenberg, building directly on the work of the Frankfurt School in his Critical Theory of Technology (Feenberg 1986), combines insights from philosophy of technology and constructivist technology studies (Feenberg 1991; 1999; 2005). He

argues that modern technologies are not neutral and embody the values of a civilization, thus their negative effects are not a mere unintended side-effect. Feenberg advocates for broader democratic participation, new approaches and technological choices to understand technology's democratic potential. He also touches upon tradeoffs between efficiency and participation and sustainability, which should to be considered in technology development (Feenberg 2005). How to broaden democratic participation and improve sustainability are relevant questions for grassroots innovation and critical technical practice, as I will demonstrate throughout this study.

Critical Theory and Post-Colonial Critical-Reflective Human-Computer Interaction

In the book "Critical Theory and Interaction Design" (Bardzell et al. 2018), new essays reflect on earlier work of critical theory to show how critical theory supports the work of interaction designers. Earlier authors include Benjamin (The Author as Producer, 1934), Marcuse (The New Forms of Control, 1964), Foucault's Panopticism (1975), and Judith Butler's Gender Trouble (1999) in *ibid.*). For this study, two ways that CT can influence HCI are especially relevant, as these touch upon critical reflexivity and post-colonialism.

Bruno Latour's famous essay titled "Why Has Critique Run Out of Steam? From Matters of Fact to Matters of Concern" (originally in Latour (2004b), reprinted in *ibid.*) is followed by an article by DiSalvo, who reads it as an unintentional expedition into design theory. His argument is that Latour starts a conversation about new modes of action and scholarship "with a twist"—critical impulse. While design has included critique for a long time, DiSalvo asks for more interdisciplinary work. Critical practices of design and critical impulses from the humanities and social sciences should inform each other. As significant contribution to critical-reflective HCI, he mentions articulating post-colonialism through interactive systems design (Irani et al. (2010) on page 476, DiSalvo (2018).

An essay from Said's influential book "Orientalism" (Said 1978), titled *Knowing the Oriental*, is followed by a paper outlining how HCI researchers are "representing others" (Grinter 2018). Said argues in his essay that texts were used to assert dominance over the colonized, and thus should be read in the context of their creation. Building on this, Grinter maps the power of texts threefold onto HCI practices, which correlate with the research questions of this study.

First, she examines how people who control the production of knowledge are empowered to represent through authorship; who is representing whom, and who controls the means of representation? Grinter points out that in HCI4D, research subjects are "marginalized people in the Global South" (page 724, *ibid.*), yet research is rarely produced by those affiliated with institutions in the Global South. A lack

of (in her study, African) input has cast locals as passive consumers of Western knowledge rather than active, innovative producers (Elovaara et al. 2006; Merritt and Stolterman 2012). Grinter also mentions challenges of access, such as bandwidth and high cost of publications (Gitau, Plantinga, and Diga 2010), in addition to the significant immigration barriers to attending conferences. Second, Grinter highlights how “othering” happens through representation—when the people who control knowledge production represent those who do not. She calls for reflection on how Euro-American HCI gives voice to others in their accounts, and how expressions like “throughout Africa” collapses the diversity of an entire continent². Grinter recommends that HCI4D embrace an agenda of emancipatory design that gives voice to the marginalized. For example, projects have enabled the sharing of information about exploitative employers (Irani and Silberman 2013) and street harassment experienced by women (Dimond et al. 2013). These projects help marginalized people exercise their voice, which is one of the main goals of postcolonialism (Said 1978). It’s important to note Spivak’s critique here. In “Can the Subaltern Speak?”, she ridicules the hypocrisy of postcolonial discourses that claim to raise the voices of those who were previously unheard, as the marginality of the oppressed is maintained, writing that, “and I will have recourse, perhaps surprisingly, to an argument that Western intellectual production is, in many ways, complicit with Western international economic interests” (page 271, Spivak 1988). Finally, Grinter critiques the methods of knowledge production in HCI4D research. She is leery about what it means to export methods (e.g., the movement of PD from Scandinavia to the US already proved problematic) and how such exporting exposes values people hold about participation. Reflecting on how to facilitate participation, she cites Puri et al.’s (2004) work comparing projects in different countries, concluding that while in South Africa, *ubuntu* (collective personhood and morality) drove participation, in India, respecting the hierarchy did so (Puri et al. 2004). Similarly, in my cases I will show that the hackathon was built on a *gotong royong* (mutual aid) approach, and there was a lengthy *nongkrong* (collective ideation) period. Still, in the rural Community Network Project, respecting the hierarchy and local beliefs while creating trust with the elders of the Indigenous community made the project sustainable.

A combination of critical theory and post-colonial critical-reflective observations derived from human-computer interaction drive this study. Investigating the

² For the sake of simplicity, I refer to Javanese grassroots innovators I write about as being Indonesian. In later analysis I will briefly reflect on how the cities and rural areas where the activities are conducted influence their practices, as well as on the important role the respect of the local customs and contexts played in making a community network project sustainable of in the analysis of Sect. 5.4., Case Study I. Describing these as “Indonesian” would be reductionist.

processes of intrinsic and self-directed participation is the main goal of Research Question 2. The analysis of critical knowledge production within grassroots communities is the goal of Research Question 3.

3.2.2 Science and Technology Studies

According to Delanty and Harris (2021), similarly to critical theory, certain aspects of Science and Technology Studies (STS) are anti-technocratic and an expression of radical thought built on postmodern theory. STS has long concerned itself with honing conceptual tools for locating the politics of decision-making in technology and science. It also has examined how computing is unevenly distributed, and the practical, local, situated elements of knowledge construction and technological development (Vertesi and Ribes 2019). Within the vast field of STS, numerous findings are relevant for this study, especially theories that reveal the interconnectedness of technology and society. For this study, it is relevant to note that the most STS studies still conceptualize the professional as the producer, and this approach is so common that it can obfuscate alternatives. For example, the knowledge of Indigenous people used in STI is often appropriated without compensation (Eglash et al. 2004).

Nevertheless, actor-network theory (ANT), the social shaping of technology, and the social construction of technology (SCOT) have been utilized to better understand grassroots innovation. For example, they can reveal the dynamics of maker culture, maker communities, and their innovation processes (Bertling and Leggewie 2016).

ANT connects design to complex social phenomena, and was developed to criticize social and technological determinism. ANT highlights the symmetry between human and nonhuman actants (Latour 2006), which allows for researchers to show the roles that developed objects play beyond the purely technological. Sometimes symbolically they reflect the societal purpose they play, which is especially important in critical making. ANT conceptualizes agency as derived from a network of actors and actants. Therefore, power is dynamic and changing, and authorities and capacities—including those of objects, regulations, and databases (DiSalvo 2018)—are constantly being delegated.

It is important to note that early on, ANT research was criticized for being apolitical and focusing on more powerful actors. For example, Star (Star 1991) argued that ANT forgets who benefits and who is marginalized as socio-technical relations are established, and asked researchers to investigate the margins where

actors struggle to change the status quo. However, by going beyond political parliaments and scientific laboratories (Latour 2005), ANT can help analyze grassroots innovation processes. For example, it can help connect infrastructuring and actor networking with participatory design (Björgvinsson 2014) to investigate how grassroots-based critical making technological development is done in co-creative processes between people and artifacts.

For example, Ekomadyo and Riyadi (2020) used ANT as a method for analyzing the design of a university community service project performed in cooperation with a local grassroots movement in Bandung, Indonesia. They investigate actor involvement by tracing the collective movement of human and non-human actors to demonstrate their social ties and constructions. The resulting panorama of all actors involved sheds light on the complex participatory design process that went into the project and the actants' influences on each other—with the project itself being the physical and metaphorical obligatory passage point, as defined by Callon (1986).

The social construction of technology (SCoT), as presented in *The Social Construction of Facts and Artefacts* (Bijker and Pinch 1987), outlines how technologies are socially constructed (Bijker and Law 1992). Through a SCoT approach, they challenged the presumptions of technological determinism. They argued that technology does not determine human action, but rather human action shapes technology (Bijker 2009). In addition to SCoT, theories of the social shaping of technology (Williams and Edge 1996) (Bammé 2009) have been developed to respond to the techno-economically rational and linear conceptions of technology and its negative consequences. Such social approaches give users a central role in developing technologies, casting them as users, agents of technological change, and “prosumers” (page 554, Oudshoorn and Pinch 2008).

Theories on socio-technical configuration processes emphasize the co-productive powers of the social and the material (Wajcman 2002; Oudshoorn and Pinch 2003; Bijker 2009). Translating these concepts to the practices in makerspaces, how people use digital technologies and what type of tools the community can access for prototyping is relevant. However, their technological practices will also be influenced and supplemented by the skills available to the community. Processes of maker communities are naturally co-productive, as makers engage in socially constructing and shaping technology through prototyping and coding. Building on these theories can lead to a better understanding of how the community responds to socio-political questions of technology, and how these questions shape their technological choices as they develop or appropriate technology-based solutions.

SCoT is also interesting because the approach often addresses how to “open the black box” of systems by understanding its internal workings. In “Science in Action”, Latour outlines a process for how a transparent box is translated into a “black box.” Science turns contested hypotheses into accepted results (Latour 1987) that are considered facts. When a matter of fact is settled and only input and output is deemed relevant, the internal complexity and workings are no longer questioned. When science and technology succeed, they become opaque and obscure. Winner famously wrote about opening this “black box” of technology (Winner 1993), and his approach can be used to interpret making and maker culture’s interest in open-source technology. Similar critiques have also been expressed by the Critical Engineering Working Group (Oliver et al. 2011) and Garnet Hertz, as he was extending the term critical making to mainstream maker culture (Hertz et al. 2014). Both advocate against the practice of “black boxing,” which they argue removes the agency of people and turns them into mere consumers. Some makers attempt to resolve this problem through technological literacy, repairability, and “prosumerism” (Fonseca 2015; Anzalone et al. 2015; Baier et al. 2016).

3.2.3 Conventional, Social and Grassroots Innovation

As this study builds on existing approaches of critical making and critical technical practice, it extends it to grassroots innovators. Therefore, it is important to situate grassroots innovation in relation to social innovation and alternative economic models, as opposed to market-driven industrial or mainstream innovation.

Social Innovation

Originally this study was supposed to investigate *social* innovation in critical technical practice. I will still refer to social innovation occasionally, as it shares similarities with grassroots innovation; both are non-institutionalized approaches to societal, technological, or environmental development. The strong tradition—a radical approach to social innovation—defines social innovation as “the political transformation of society through creating new social and power relations” (Moulaert et al. 2007; Moulaert et al. 2010; MacCallum et al. 2009). Non-market-driven innovations focus on the satisfaction of needs, reconfigured social relations, empowerment, or political mobilization, and innovative processes are at the forefront: “by focusing on

the innovation, rather than on just the person or the organization, we gain a clearer understanding of the mechanisms that result in positive social change” (page 36, Phills et al. 2008). However, due to recent uptake in its usage by various actors, especially in the private sector, the term social innovation carries unwanted connotations of entrepreneurship and utilitarianism (Lubberink et al. 2018; Moulaert et al. 2014). Currently, the term grassroots innovation isn’t associated with such notions, thus it better aligns with the goals of this study.

In addition to social innovation, grassroots innovators also develop innovative alternative post-capitalist economic models, such as community economy (Gibson-Graham 2006). Partly they adopt these models due to a scarcity of resources, but also often for political reasons. New economic models can be cooperative, time bank or barter-based, and build on commons and mutual aid. They also can advance the needs of the community over a desire for profit, thereby enabling the participation of marginalized people. As I will illustrate, they develop alternative futures (Kaiying et al. 2019) through sharing-based and socially centered economies. These futures amplify creativity, inclusion, and local actors’ agency, while cultivating plural ideas about what constitutes a sustainable, inclusive, and social development that improves democratic participation and inclusion.

Conventional Innovation

Conventional (often industrial) innovation is important to highlight because the term “innovation” is associated with markets, such as large-scale funding and the development of novel technologies. Thus, this type of innovation is a key driver of economies, alongside entrepreneurship. While von Hippel claimed that innovation had become more democratic after lead users gained access to information and advanced production tools (von Hippel 2005), it remains rare that lead users can compete in elite markets (see e.g. Björgvinsson’s (2014) arguments). As Smith et al. point out, making or grassroots digital fabrication “sits awkwardly with conventional innovation theory” (page 6, Smith et al. 2013). That is, social or grassroots innovation does not fit into traditional innovation frameworks because there are very strong contrasts between approaches to conventional and social or grassroots innovation. To analyze grassroots innovators’ goals and practices, economic theories that apply to traditional innovation simply do not work. These pertain to industrial innovation that results from processes within and interactions between major institutions like businesses, R&D centers, entrepreneurship and public policy of

governments (see e.g. Freeman and Soete 1997). Innovation is still seen as a practice driven by academia, government, or industry—major players with large budgets and seemingly infinite resources.

Grassroots Innovation

Opponents of large-scale innovations advocate for locally developed solutions for local problems instead of one-size-fits-all solutions deployed “from above” (Schumacher 1973) (Campbell 2018). Contrarily, the goals and practices of grassroots innovators are often political, and executed with a limited budget and a focus on values and participatory design (Fig. 3.3).

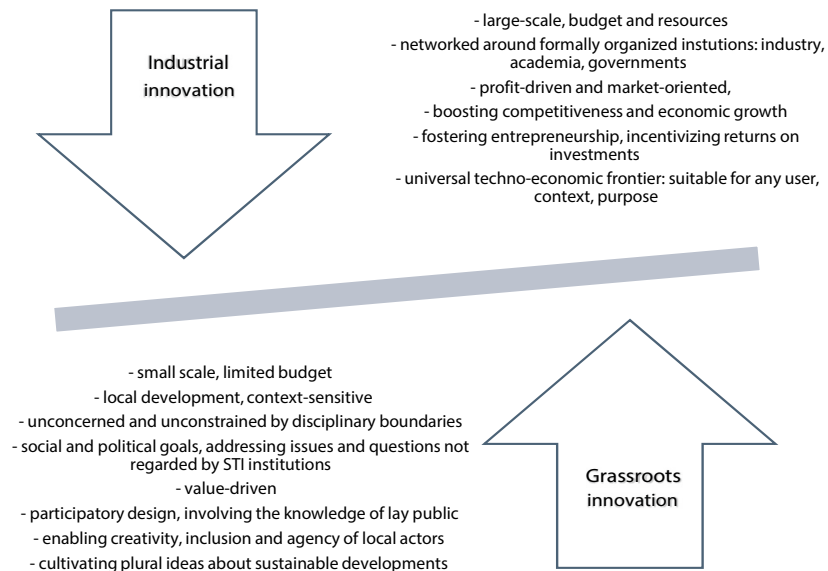


Fig. 3.3 Industrial and Grassroots Innovation. Based on Smith et al. (2017)

Recently, there has been growing interest among researchers to explore different versions of innovation: grassroots innovation, user-centered innovation, open innovation, social innovation, frugal innovation, and community innovation. The proliferation of adjectives shows that “alternative” processes of innovation remain relevant in society (Elliott and Turner 2012). However, to gain a more complete understanding of such alternative innovation processes, we need to further develop

theories centered around more participatory and democratic innovation practices e.g., of grassroots innovators.

Makers, for example, are finding interesting niches in community-based innovation and technology development. Many makers are addressing issues that fly under the radar of industry, government, and academia, because the solutions require personalization difficult to profitably achieve on a large scale. Consider projects that personalize equipment for people with disabilities³ or develop affordable assistive technologies together with the caretakers,⁴ combining R&D and social innovation. As the maker movement is a social movement heavily involved in community-based science, technology development, and innovation processes, more interdisciplinary research is needed to further develop our understanding of these interconnections. Smith and his colleagues place hackers and makers among grassroots innovation movements (see Chap. 6 “Hackerspaces, fablabs and makerspaces” in Smith et al. 2017). Thus, theories on GIM apply, and the practices hackers and makers are characterized by knowledge production (Eyerman and Jamison 1991 in Smith et al. 2017), particularly knowledge neglected by conventional innovation institutions and shared through non-proprietary forms of innovation. Social movements are social actors that learn by doing. Especially in the case of makers, they learn by doing together, meaning that they are “communities of practice” (Lave and Wenger 1991) that learn together through reflection and debate.

Although I outlined the full definition of grassroots innovation based on the work of Smith and his colleagues at the beginning of this study, it is important to repeat here some of the key factors mentioned there. Grassroots innovations are bottom-up solutions collaboratively generated and sustainably developed by groups of people. They often mobilize resources among wider publics and generate alternative activities that attempt to address social issues and needs neglected by major innovators (pp 3–4 Smith et al. 2017). These solutions consider the local situation, and also the interests and values of the communities that develop them. While many such innovations are born of material or economic necessity, some are motivated by injustices and social issues that are neglected by conventional innovation systems (see also Gupta et al. 2003 and Seyfang and Smith 2007 in *ibid.*). They build networks, develop discourses, openly share knowledge, and mobilize resources to reach their goals (*ibid.*).

However, relatively few studies have linked social movement literatures to studies of science, technology, and innovation (page 18, Smith et al. 2017). Examples

³ <https://be-able.info/de/be-able/>

⁴ <https://engineeringgood.org/assistive-tech/>

of successful attempts to make connections between social movements and mainstream R&D include studying the role of social movements and collective action in the construction of alternative science (Frickel 2004; Frickel et al. 2010; Moore 2006). In “Alternative Pathways in Science and Industry”, Hess wrote about issues neglected by conventional scientific institutions yet addressed by activists and the role of movements in developing alternative forms of technological change (Hess 2009a). In HCI theory, as previously briefly mentioned, the role of social computing technologies building grassroots movements has been explored by using the Freirean model of participatory communication theory of social movement organizing (Ghoshal and Bruckman 2019). Additional research has been conducted on how ICTs enable the outreach of grassroots social movements and shed light on inequitable socio-technical situations (Ghoshal, Mendhekar, and Bruckman 2020). These points will contribute to Research Question 2, which is partially concerned with the negotiations among members about which ICTs to use in the intrinsic design projects, and how.

In general, grassroots innovation projects aren’t globally unified, but scattered, forming heterogeneous projects in a bottom-up manner (Sipos et al. 2019). Their analysis requires analytical and methodological approaches different from mainstream innovation practices, and research is still somewhat scarce. Their similar goals make them part of a “global assemblage” (Lindtner et al. 2016). Not all grassroots innovators see themselves as part of a broader movement, and not all phenomena that researchers call “grassroots innovation movements” are homogeneous movements with uniform activities around the world. For example, the phrase “open science hardware” includes framings and actors with different backgrounds, from community science projects to social innovation, from critical pedagogy to trans-hacktivism (Arancio 2021). Individual communities might engage in specific topics or social issues they aim to solve, such as sustainable agriculture (Bidwell 2020; Norton et al. 2019), urban farming (Heitlinger et al. 2013), beekeeping and citizen science, (Phillips et al. 2014) and community networks (Bidwell and Jensen 2019).

The heterogeneity of grassroots innovation also feeds into Research Question 3. The strength of grassroots innovations is that many address issues of hyperlocal relevance. They produce knowledge in conversation with local laypeople, thus engaging traditional innovators and global networks in multidisciplinary ways. Their activities act as litmus tests to measure society’s discontentedness, pinpointing emerging societal needs and collaboratively proposing change (Sipos and Wenzelmann 2021). During such processes, they produce knowledge that otherwise would not have been produced (Smith, et al. 2014). They challenge hierarchies (Lindtner and Lin 2017;

Escobar 2018), as the knowledge from below produced by movements might be “inconvenient to and resisted by those above”(Cox and Fominaya 2009). The “era of participation” (Smith et al. 2017) requires researchers to engage with practitioners to design in participatory ways and to innovate with grassroots to understand better their methods.

3.2.4 Technological Approaches in Development Studies

As the study is concerned with analyzing critical grassroots innovations in the Global South, this subchapter briefly summarizes relevant approaches in development studies. For study, such approaches include postcolonial computing, HCI4D, and ICT4D, and relevant critiques of developmentalism.

Postcolonial studies is a field of research that examines the cultural legacies of postcolonialism (Ashcroft and Ahluwalia 2001). Edward Said, the founder of this academic field of theory and criticism (see e.g. Said 1978), was inspired by Fanon’s (1961) book “*The Wretched of the Earth*” and critical thinkers like Gramsci, Foucault, and Adorno (Buchanan 2010). Although postcolonial studies began with investigations of the historical conditions of nations and regions that were once colonies, it developed into reinterpreting and examining colonial ideologies concealed in literary text and investigating historical transformations that shape relationships today:

“Colonial relationships may have dissolved, and yet the history of global dynamics of power, wealth, economic strength, and political influence shape contemporary cultural encounters” (page 1311, Irani et al. 2010).

For this study on technology design in postcolonial settings, it is even more important to note the previous absence of research connecting postcolonialism and computing. Fernández quotes the artist Gómez-Peña, who said that “in discussions of electronic media, twenty years of post-colonial theory simply disappear” (page 59, Fernández 1999). Fernández asked for a reconciliation between postcolonial studies and electronic media, bringing a more global perspective to both and “temper imperialist initiatives presented in the guise of utopianism” (page 60, *ibid.*). The same year, Turk and Trees wrote a reflexive case study about the culturally appropriate and participatory design of an Australian Indigenous information system. By applying a postmodern and postcolonial perspective, they ensured that ethical issues were carefully addressed (Turk and Trees 1999).

These initial forays weren't however followed up on by HCI researchers, however. Merritt and Bardzell (2011) note that after around a decade of absence, postcolonial critique finally returned to the design discourse (also see Bardzell et al. 2009).

ICTD, ICT4D, ICT4B, and HCI4D

ICTD, ICT4D, ICT4B (billions) and other versions of the abbreviation of the term *Information and Communication Technology for Development* started appearing in international development organizations, academia and industry in the early- to mid-1990s. ICT4D is one of the few subfields within computer science that attaches a moral value system to its agenda, confronts inequalities through the design and deployment of ICTs, and includes interdisciplinary practitioners, government, and NGOs (Chen 2015). Technology for development, a precursor these variants, became widely used after the African Regional Symposium on Telematics for Development in 1995. After this symposium, international development agencies renewed their interest in the role of technologies in development programs. As Patra et al. summarize, the growth of public discourse of technology as an engine of growth paralleled the impacts of globalization in the Western technology industry in the West, and even in the developing world (Patra, Pal, and Nedeveschi 2009). ICT4D promised dramatic global change (Castells 1996; Negroponte 1996), renewing worldwide interest in the specificities of technology projects in development. For example, interest grew in the social implications of computers in Asian developing countries (Bhatnagar and Odedra 1992), linking of information-based global economy and socioeconomic development in Bangalore (Madon 1997), and development in the “information age” (Howkins and Valantin 1998). International agencies such as the UN, ITU, and World Bank increased interest in ICTD projects to create developmental outcomes for their recipients (see e.g. Nagy 1994; Mansell and Wehn 1998; Nulens and Van Audenhove 1999). While participatory approaches today are acknowledged for their potential, there were times when academia questioned whether local participation was even possible. Discussions warned designers to think critically about the “new orthodoxy” of participative approaches and how participatory development projects might still ignore context and reality (Heeks 1999) and perpetuate existing inequalities (Cooke and Kothari 2001). Huesca warned about the conceptual disarray of the field, which was impeding its goals to improve the human condition. Instead, he advocated for combining research findings from social movements with participatory communication for development (Huesca 2001).

A decade later, scholars expressed concerns about a lack of capability to transport design conventions and processes across cultures, because design aesthetics varied greatly depending on location (Marcus and Gould 2000) and designing for

resource-poor contexts shifted from user-centered design to “communitization” or community-centric design (Marsden, Maunder, and Parker 2008), rendering community engagement important to make projects sustainable (Brewer et al. 2005). Irani et al. highlight the different social, cultural, economic, and infrastructural challenges of HCI4D and called for HCI designers to learn about postcolonial theory so they could substantively and meaningfully adapt HCI methods and practices (Irani et al. 2010).

Postcolonial Computing in HCI

Postcolonial computing includes empowering postcolonial voices to represent themselves and countering hegemonic representations of the West (Irani et al. 2010; Taylor 2011). Although postcolonial studies began with investigations of the historical conditions of nations and regions that were once colonies, a shift in perspective later occurred. Researchers came to understand that the relevant topic was the historical transformation of conditions of cultural encounter. Postcolonial computing became an analytical orientation to better understand such challenges. This means that there has been an important and major move from a “development” discourse to postcolonial discourse in HCI4D (or Human-Computer Interaction for development). This shift centered on the questions of power, authority, legitimacy, participation, and intelligibility in the contexts of cultural encounter, particularly in contemporary globalization (ibid.).

The term postcolonial computing itself was introduced by Irani et al. (2010) to address the challenges of deploying Western technologies in new cultural contexts. They didn’t intend postcolonial computing to be a new domain, but rather an alternative sensibility to the traditional design and analysis process. They argue that building on STS and postcolonial studies is necessary to better understand complex issues of intercultural engagement around technology design. They name four key shifts that need to happen in HCI4D to achieve this goal (outlined on page 1314, ibid.):

- **Generative models of culture:** instead of classifying people on cultural dimensions, or using taxonomic views of “here and “there”—where “there” is *other, apart, disconnected* from the “here”—researcher should ask how the technological objects and knowledge practices of everyday life are dynamically generated through interaction.
- **Development as a historical program:** reflecting on the discourse of global technical solutions to problems, how development projects are aligned with the interests of commercial actors in industrialized countries, directionality of

product and monetary flows, and disempowerment through consumer-oriented development.

- **Uneven economic relations:** what counts as “legitimate design practice” is defined by economic conditions, such as how Indigenous technological capability of reverse engineering in Brazil has been reshaped by a powerful American economic actor.
- **Cultural epistemologies:** as HCI methods embody principles and models of knowledge sharing and representation, Indigenous knowledge should not be contrasted with scientific knowledge.

Building on postcolonial theory, critical theory, and cultural theory, Merritt and Bardzell proposed a new theoretical perspective for HCI4D design. They pointed out that with the growing body of technology design for “less technologically developed countries” came a rapidly increasing work of HCI for development. However, neither technology design nor HCI addressed inherent biases and Westernized thinking (Merritt and Bardzell 2011). They called for specific methodologies for design research in HCI4D, including appropriateness and a minimization of Western cultural interference.

In “Whose Participation? Whose Knowledge”, comparing PD projects in Sweden and Tanzania-Zanzibar, Elovaara et al. (2006) come to the conclusion that PD’s strength derives from its capacity to be translated to different locations, scenes, and technological skills. Participation and how to participate must be negotiated and adapted to the local setting. In the Swedish case, the relations between designers and users had to be renegotiated, yet in the Tanzania-Zanzibar case designers seemed to dominate, repeating two interrelated questions the paper builds upon: “How can the invisible knowledge, experience (...) be articulated and integrated into technology design? (...) How can the dominant notions of relevant expertise and views be challenged?” (page 16, Karasti 2003).

While digital activists and academics have been raising red flags about the negative impacts of technologies on our daily lives, humanitarian scholars have been proposing critical research agendas regarding how technologies are used. For example, in humanitarian settings, they point out the fairly limited scholarly engagement and warn about “tech-optimism”. Ultimately, they advocate for more reflectivity in practice, especially taking into account the fragile and vulnerable state of the communities that receive assistance (Sandvik et al. 2014), and embedding dignity in humanitarian PD (Talhok et al. 2019).

The crucial question is thus *who actually participates in PD*. This question has been urgently raised in HCI (Smith et al. 2017; Bødker and Kyng 2018), including feminist (Bardzell and Bardzell 2011; Ahmadi et al. 2018; Campreguer França

et al. 2021) and humanitarian PD (Almohamed et al. 2017; Talhouk, et al. 2019; Jensen et al. 2020), or HCI4D (Winschiers-Theophilus et al. 2012). For this study, HCI approaches like postcolonial computing (Irani et al. 2010; Taylor 2011; Avle and Lindtner 2016; Kaiying et al. 2019), pluriversal design (Escobar 2018; Hendry 2021), and reflexive PD (Strohmayr 2021) are especially important.

In their Introduction to the “Special Issue on Re-Imagining Participatory Design”, Bannon et al. (2018) state that they had received surprisingly few Non-Western (other than European, North American or Australian) submissions, and those they did were rejected in the review process. They list some possible explanations, including that Western biases or scientific biases (e.g., against political work) could be built into the peer review process. However, the scientific knowledge base about PD in the Global South seems to be growing, moving from the digital divide to digital justice (Heeks 2020), designing tools for “children on the margins” (Shahid et al. 2021), and empowering workers in the sharing economy (Keskinen and Winschiers-Theophilus 2020). Participation from the Global South is slowly becoming part of PD practice. There have been explicit calls inviting critical makers from the Global South to cooperate with researchers, participate in academic workshops, and share their experiences without being “exoticized” (Sipos and Wenzelmann 2019). Other recent developments in HCI include creating spaces to reflect on PD exercised in the Global South, and helping to map “participation otherwise” to design new worlds, as inspired by Escobar (Szaniecki et al. 2020).

Intermediate and Appropriate Technologies in International Development

Despite its significance in the twentieth century, research on technical assistance by the United Nations (UN) and other international development agencies remains scarce. It has played a relevant role in the global deployment of scientific knowledge, technologies, and practices by training of thousands of experts, impacting the redistribution and sharing of scientific and technological expertise (Mateos and Suárez-Díaz 2020). The UN has conducted education and training by specialists to help “under-developed countries” achieve the living standards of “developed countries” since 1949. Based on an international exchange of services in years prior, the Economic and Social Council recommended to increase the productivity “of material and human resources” to improve living standards. Building on this, in 1951 the United Nations announced its Technical Assistance Programme. This programme provisioned technical assistance as a “new and important service”, framing technical assistance as helping countries narrow the economic difference and improve their standards of living (Owen 1951). Survey missions found that administrative machineries that would help governments execute development programmes—as well as skills to apply for financing enabling such programmes—were lacking. Thus,

national centers to train government employees were established (page 329, *ibid.*) to build local public administration and develop vehicles for donor assistance. The primary actor of development was the State itself. The scale and topics of technical assistance varied significantly based on local and historical contexts⁵. However, science and technology were seen as essential to nation-building in Asia and Africa after the colonial times (Mateos and Suárez-Díaz 2020).

Schumacher and his colleagues founded the Intermediate Technology Development Group, which inspired the founding of a plethora of (short-lived) departments focusing on the development and deployment of appropriate technologies in international development organizations:

“International institutions like the Inter-American Bank of Development, the World Bank, the UN Environmental Programme, the International Labour Organization, the Food and Agriculture Organization, World Health Organization, the OECD, and UNIDO also established departments of appropriate technologies. Over this period, the plethora of programmes, projects and interests supporting The World of Appropriate Technology appeared substantial (Jéquier and Blanc 1983 for OECD). And yet, despite this build-up of interest and momentum, this was precisely the moment when the movement peaked” (pp. 5–6, Smith et al. 2014).

As Smith et al. (*ibid.*) summarize, appropriate technology peaked because the movement critiqued *blueprint developmentalism*, which meant that development practice was being reconsidered. Thus, the movement advocating for AT made itself obsolete. Nevertheless, in practice, there were persistent problems in how AT was deployed:

- **Generic AT solutions:** Local development contexts and goals were overshadowed by generic, turnkey solutions. These were easier to be included in the top-down produced handbooks and field guides, and practical for widespread diffusion. A one-size-fits-all focus on basic needs meant that further development or broader innovation dynamics were not included in the practice.
- **Lack of participatory practice and processes:** AT centers experienced challenges broadening participation in the design and innovation of the initiatives. Although the original principles included user-led participation and grassroots innovation, in practice it was built on engineers’ assumptions regarding what was needed on the ground. Innovations in participatory processes—including

⁵ In the Arab States, a seminar was held on community organization, including “education, village planning, rural development, health and hygiene, and the organization of co-operatives in rural areas”, while Libya, a “country (...) preparing for its independence” from its colonizers, was supported in understanding its economic potentials social welfare services, public finances, and industrial and administrative projects (*ibid.* page 330).

engagement with local ingenuity and the empowerment of local communities—would have been necessary. Skills, time, and humility was lacking to achieve these goals (Chambers 1997)

- **Lack of business models for sustainability:** Dependency on State and other funds could not be surpassed and autonomous spaces for the AT were not proactively developed (Abrol 2005), meaning that technologies could not be effectively marketed. Structural poverty and low local entrepreneurial capabilities inhibited the development of a sustainable business model.

Appropriate technologies that relied on aid and charity, and were motivated by non-market driven goals, were left behind as development policies moved toward neoliberal approaches. Structural adjustment agendas, integrating into the globalizing economy, and replicating the successful experience of East-Asian industrializing economies became the focus (Rist 2011; Kaplinsky 2010), instead of addressing the shortcomings of appropriate technology.

The decolonial thinker Escobar has criticized the UN's approach to international development and called it a "design disaster". He builds on Fry's work on approaching design as politics, and argued for a new type of design practice that recognizes how the world has been made unsustainable. He asks we include ethical, political, social, and ecological concerns in design, such as through the narrativization of design through stories that are immediate, engaged, and evocative (Fry 2011). Fry described the UN's approach as "elimination design", meaning that it scrapped vernacular design. Endogenous practices and local traditions were reduced by the UN as representations of "underdevelopment" and eliminated in the name of "development" (Fry 2009). Escobar instead calls for engagement that eliminates the structures of unsustainability that maintain the dominant ontology of destruction. This plurality of transformative visions to sustainability and social change drives Research Question 1.

Appropriate technology remains one of the potential solutions, as it's defined as the opposite of "elimination design":

"The technology of *mass production* is inherently violent, ecologically damaging, self-defeating in terms of non-renewable resources, and stultifying for the human person. The technology of *production by the masses*, making use of the best of modern knowledge and experience, is conducive to decentralisation, compatible with the laws of ecology, gentle in its use of scarce resources, and designed to serve the human person instead of making him the servant of machines" (page 128, Schumacher 1973).

As Chen summarizes, existing approaches of ICT4D emphasize contextually appropriate interventions—those with low cost, low infrastructure, high usability, and

development metrics. He mentions three reasons why contextually appropriate interventions are difficult to achieve. First, designing such interventions are complicated, as each requirement is complex and the technical challenges are often interrelated. Second, novel ICTs are expensive to manufacture, and infrastructural support and maintenance is complicated. While struggling with the other challenges (implementation, testing, deployment, evaluation), very little consideration is given to how the intervention can sustain itself.

Chen ultimately considers ICT solutions to be brittle, highly optimized, and difficult to adapt to new contexts. As a result, few projects survive the pilot phase. In response, Chen recommends using “computing within limits”, an approach that concerns itself with the context, users, and cost, with a focus on resilience (including simplicity, infrastructure independence and modularity) and sustainability (Chen 2015). Computing within limits has also been reviewed with the Escobarian lens of “designs for a pluriverse” as a potential approach to better understand small-scale practices that aren’t part of public discourse. Through an analysis of liberatory technology, appropriate technology, and convivial computing projects, de Valk (2021) comes to the conclusion that while their approaches might be different, small-scale practices remain fragile. She argues that their similarities create bridges and show common ground. They are similarly community-centered, small-scale, decentralist, and anonymously administered. Their modular and multi-purpose networking practices are made more resilient through diversity and simplicity, using minimal resources and producing open-source or commons-based outcomes. She also highlights how these approaches have fallen into disuse because they’ve been too politically divergent from neoliberal ideology or not political enough to approach local problems that are caused by global, wicked problems. Instead, they only approach technology as a solution (ibid) rather than as an opportunity to examine local societal contexts and communities.

As in past approaches to designing and deploying appropriate technologies, the designer remains an external person or project team, leaving participatory design in the hands of a researcher. While there are projects that try to support self-directed approaches to sustainable development, these remain the exception and are still designed by the “professionals”. However, similar to the recent shift in PD towards intrinsic design—in which projects are developed without external intervention, as I will demonstrate below—research in some cases recognizes that technologies do not always need to be designed appropriately by international development for communities. Grassroots innovators can appropriate technologies for themselves.

Self-Directed Technosocial Development: Appropriated Technologies

The book “Appropriating Technology: Vernacular Science and Social Power” collects essays that counter the narrative that projects need to be led by a *professional*, acting as the producer or designer of a technology development project. As Eglash et al. (2004) writes in his introduction, appropriation doesn’t always happen in a top-down manner⁶. Rather, the case studies show,

“how people outside the centers of social power—from white middle-class homemakers to rural Native Americans—have been able to use materials and knowledge from professional science for their own kinds of technological production” (page 12, Eglash et al. 2004).

Such cases of *appropriated* technologies—where artefacts of technoscience are adapted and appropriated to fit local contexts—are especially interesting for this study. In these projects, critical analysis of social issues is combined with goals of positive, sustainable application of STL. Eglash defines three ways technology can be appropriated: 1) *reinterpretation*, or a change in semantic association with no or barely and change in structure; 2) *adaptation*, which means the discovery of a latent function, or change in semantic association and use that requires flexibility and violation of intended purpose as technosocial features; and 3) *reinvention*, which means changes in semantics, use, and structure, or the creation of new functions through structural change (page 14, *ibid.*).

According to Eglash, the creativity necessary for people to adapt and reinvent technologies, to realize the potentials beyond the intended function receives insufficient theoretical attention. For this study, this gap is especially relevant. As demonstrated later, grassroots innovations and critical maker communities engage in sociotechnical experimentation to develop their communities. Going beyond tinkering with technologies, they reinterpret, adapt, and reinvent technologies to fit their goals and produce sustainable technological solutions.

⁶ The example he mentions is ethnopharmacology profiting off Indigenous communities’ knowledge without compensating them. A similar case was mentioned to me by a grassroots innovator in Indonesia, who said the different plans for healing were appropriated by foreign pharmaceutical companies. Such cases apparently led to research visa processes becoming more rigorous. However, I could not find further proof of this, thus, this note remains anecdotal.

Recognizing that grassroots communities sustainably appropriate different emerging/contemporary technologies, Teli et al. (2021) organized a workshop bringing together researchers and practitioners. Their goal was to understand how technology design researchers and activists can understand, promote, and design for sustainable appropriation. The researchers' interests included the role of existing technologies in infrastructuring initiatives and their limitations, and technical descriptions of grassroots-oriented technologies and their artifact ecologies. They were also interested in the conceptual contributions to values, and tactics central to the appropriation of technologies, like commoning and caring (ibid.). However, the workshop organized in Seattle was mainly focused on North American grassroots (e.g., the connection between facial recognition and the Black Lives Matter movement). How this translates to the Global South,—specifically to Indonesian and Javanese grassroots projects—will be explored in this study.

3.2.5 Sociotechnical Research and Participatory Approaches in Indonesia

Next, I will briefly survey existing research and relevant literature about sociotechnical developments, and especially reflexive and postcolonial approaches, in Indonesia. Indonesian researchers have adapted Western approaches, including STS (Barker 2017), to describe local phenomena. Within STS, Ekomadyo and Riyadi (2020) applied ANT to analyze community-based development design from a socio-technical perspective. A handful of Western researchers have also engaged in digital humanities to explore “nerd politics” (Postill 2018), “geek diplomacy” (Kera 2015), and postcolonial computing (Lindtner and Lin 2017; Kaiying et al. 2019). Both Indonesian and Western researchers have conducted research using participatory design. Most importantly Hasan Poerbo⁷ was the forefather of community-driven development in Indonesia since the 1970s (see e.g. Poerbo et al. 1995). These sociotechnical inquiries about Indonesian participatory technology development practices have greatly influenced the design of this study.

⁷ Not to be confused with Onno W. Purbo, who is Hasan Poerbo's son, and will also play an important role in this study.

Sociotechnical Developments in Indonesia

Reflecting on the late-colonial era in “Engineers of Happy Land”, Mrázek suggests an alternative way to study technology developments in Indonesia and their effects on people’s lives, space, culture, and identity (Mrázek 2002). The book’s vignettes interpret historical documents—including pictures, poetry, and statistics—to contextualize technology, reflect on colonialism, make sense of the world. The vignettes depict architecture and the sense of sheltering, and ICTs and the culture of listening.

In “STS, Governmentality, and Politics of Infrastructure in Indonesia”, Barker (2017) summarizes the work of a group of STS scholars with an interest in Indonesia and its infrastructures. He sheds light upon how sociotechnical change occurs in Indonesia, which stands in sharp contrast with how infrastructures were built and organized during the New Order period (1966–98) and how infrastructures are organized now. The book includes studies on the history of a State-led project to build an electrical grid in Bali during Indonesia’s New Order (Mohsin 2014), the deployment of a new DNA tracking system for Indonesian timber exports (Budiastuti 2017), and two community-based and participatory projects.

It is important to note the divide between pre- and post-1998. As Barker describes, how local infrastructures were built was influenced by decades of corrupt, authoritarian governance in the latter part of the twentieth century and the unpredictable years that followed (Barker 2017). Both main cases were also responses to the change in 1998, and the country has been struggling ever since. This is one possible explanation to the origins of the plurality of critical grassroots innovators in Indonesia. As Pushkin quotes from an interview with a local collective member:

“In a failed-state context, there’s no mainstream: so it gives us an opportunity to say this is how we believe stuff should be” (Pushkin, 2020).

Digital Indonesia is a collection of papers on recent sociotechnical developments. The most relevant ones for this study include Onno W. Purbo’s “Narrowing the digital divide”, which summarizes the current state of internet penetration in Indonesia. In it, he introduces failed top-down strategies by the government, such as the smart village program, which rolled out in 4700 villages. However, poor internet connection and limited availability of electricity caused major problems. The internet café program spent only Rp 800 billion of the Rp 2.4 billion total budget, as the rest was withheld by the governmental agency. He also describes how run-down or underused internet vans failed, leading to the program being shut down. In response, he advocates for bottom-up solutions with the need and funding coming from the villagers themselves. These solutions might include establishing neighborhood networks and narrowing the digital literacy divide (Purbo 2017).

Politics of Indonesian Hacking and Making

In terms of the politics of making and hacking, three main academic sources stand out. Kera has worked extensively with grassroots biohackers in Indonesia (Hertz 2012). She recognized Indonesian grassroots innovators' connections to Europe through "new media centers", as they called themselves in the 1990s⁸. In her analysis, she argues that hackerspaces embody what she calls "cosmopolitics"—new networks and alliances between human and non-human actors that extend political and social participation (Latour 2004a). Her later work on "geek diplomacy" shows how grassroots involvement in science bridging divides creates a more inclusive R&D. One of the cases she mentions is an open-source microscope and a simplified, cheaper microcontroller that a Central Javanese DIYbio collective has worked on in collaboration with a Swiss network for open biology since 2009 (Kera 2015). These cases show how open-source hardware (OSHW) designers challenge the status quo of global R&D by developing low-cost and reproducible hardware.

Tracing the particularities of the practices of post-authoritarian avant-garde collectives, Jurriëns highlights the Art-Science Collective's historically and socially specific principles. He summarizes how the collective breaks with the autonomy of art, engages with the praxis of life, provides utopian visions, and presents aesthetic innovation (Jurriëns 2013). He also looks at how members of the collective connect media (or digital art), hacktivism, and social engagement, shifting hacking from entrepreneurial individuals to alternative networks. Jurriëns also highlights how the Collective became sustainable and contributed to the sustainability of Indonesian hacktivism by inspiring a new generation through its spin-offs, such as an all-women collective and a "forking" of the DIYbio Collective (Jurriëns 2017).

Postcolonial Computing in Indonesian Grassroots

In terms of researching postcolonial computing, scholars such as Lindtner, Lin, Wuschitz, or Bardzell (e.g. Lindtner et al. 2016) have been critiquing the underlying technosolutionism in neoliberal utopias (also present in HCI) that endorses the innovation and democratization capacity of making. Kaiying and Lindtner (2016) apply STS and PD approaches and ethnographic research to describe how the legitimacy of DIYbio was established through boundary objects and experimentation. Indonesia could then position itself as a legitimate participant in international networks of knowledge production. Lindtner and Lin (2017) describe how DIY has been seen as a rejuvenating force in PD, capturing hopes for democratization. They

⁸ Both collectives described in Sects. 5.4., the Network Collective, and 5.5., the Art-Science Collective identified as such originally.

build on the example of a collaboration between the Indonesian DIYbio Collective and an international open-source biology network to advocate for a grounded acknowledgement of the situatedness of “hopeful interventions” attached to making, closely linked to the promises of the maker movement. Making, however, can only lead to political change if additional efforts are made. Kaiying et al. (2019) approach the DIYbio collective through the lens of designing alternative futures. In their paper, they challenge the Western historical assumptions of interaction design and computing and show that the care labor of Indonesian women had already created citizen initiatives for many decades. Their labor inspired members of the collective to envision tech futures through hackerspaces that run against the grain of the status quo (page 1579, *ibid.*).

Participatory and Community-led Approaches

The director of Arts & Creative Industries of the British Council Indonesia published a (non-peer-reviewed) article responding to the creative communities becoming a “post-pandemic” policy priority in England (Pushkin 2020). In this article, he writes about how key organizations in Indonesia call themselves “collectives”, removing the hierarchy associated with established organizations and embedding them in local communities. In terms of processes, he highlights that the cultural norm in Indonesia is *gotong royong*, or “mutual aid” (also discussed in detail by Bowen 1986). In these collectives, ideas are laid out and collective decisions on them are taken through lengthy discussions called *nongkrong*, or “hanging out”. Authorship is secondary, and solutions are developed without a focus on IP rights through *ngoprek*, or “tinkering” (*ibid.*).

These various expressions of an Indonesian collective society are important for this study because they demonstrate that the idea of collaborative design in Indonesia isn’t new. Historically, one of the most well-known forefathers of participatory design is Prof. Hasan Poerbo, who promoted sustainable development through new solutions to mass housing (Poerbo and Kartahardja 1979) and jobs in the informal sector (Poerbo 1979). Later, he became a driver of community-based development in rural areas (Poerbo et al. 1995), including the development of watershed management in the Ciamis uplands. A PAR-based “Human Action Model” (Jantsch 1972) was developed in the project, meaning that its objectives were defined by local norms and values, rather than goals and objectives prescribed from outside (page 7, Poerbo et al. 1995). “Participatory approaches to development” were seen as positive alternatives to top-down styles of development (page 12, *ibid.*). The case of the Jasinga-Leuwiliang Sustainable Development Project focused on cooperatives

for development, mobilizing local capital to retain control of its resources (page 35, *ibid.*).

3.3 Conclusion

This chapter started with Ratto et al. (2014), who recommended critical making develop into a coherent field of interdisciplinary research on material-conceptual critique. They suggested the fusion of design and making, and STS and PD. I have built on their ideas and outlined the basics of the two disciplines, tracing how they shifted from theorizing large-scale innovations and human-computer interaction to becoming frameworks for the analysis of bottom-up, grassroots sociotechnical practices. Agre's critique of a lack of reflection in technology development in industry and academia can be found in other forms of alternative innovation. While critical making has also been described as grassroots (Ratto and Boler 2014), it remains rooted in engineering studies, art, and design. Thus, this attempt at extending the term to describe the practices of critical grassroots innovation in Global South needs to be grounded in theory.

From this thorough and detailed literature review, I identified relevant theories and approaches to crystalize directions, which I summarize below. Important for this study are the following scientific starting points, which are strengthened by the literature review above and listed from the broadest to specific:

First, this study is concerned with socio-technical questions, which can be theorized through STS and HCI. I analyze their practices through infrastructuring processes, boundary objects, and artifact ecologies, which will improve our scientific understanding of how grassroots innovators respond to wicked problems (Rittel and Webber 1973). This approach supports RQ3: Processes of collaboration, participation, and design.

Second, this study focuses on a specific type of innovation called grassroots innovation, which was distinguished from industrial innovation and from social innovation, and better applies to making and hacking. As I will outline in the methodology chapter, the interdisciplinary Grassroots Innovation Movements (GIM) analytical framework of Smith et al. (2017) will be used to generate a retrospective outline of how the two cases analyzed in this study came to be. A GIM approach supports RQ3: Processes of collaboration, participation, and design, and RQ2: Production of critical knowledge.

Third, the projects uncovered and analyzed in this study are postcolonial, meaning that they are geographically and politically in postcolonial contexts, and their work *is* postcolonial; they work with marginalized people to support them in using their voice. Postcolonial computing as proposed by (Irani et al. 2010) is thus fitting for the analysis of these projects. I extend Irani with Escobar's (2018) invitation to investigate the plurality of solutions, approaches, and alternative grassroots innovators. This approach supports RQ1: Plurality of projects, approaches, and designed futures.

Fourth, the projects are discussed in this study because they represent grassroots critical making. Many such practices have been established in the Global South as participatory design, using happenstances, strategies, and tactics for self-directed, intrinsic design (Bødker et al. 2016) and to demonstrate a critical use of appropriated technology (Eglash et al. 2004; Teli et al. 2021). As opposed to traditional approaches of PD, including ICT4D and HCI4D, this study is interested in clear cases of grassroots ingenuity without any interference from researchers. Both intrinsic design and appropriated technology development provide more precise approaches. Moreover, the appropriated technologies are inherently critical, and their development is reflexive, as the hacks question the status quo instead of accepting it. This approach supports RQ3: Processes of collaboration, participation, and design and RQ2: Production of critical knowledge.

To outline how such theories and approaches were used in similar contexts to this study, it was important to explore how Indonesian and non-Indonesian researchers have approached Indonesian grassroots projects. What stands out is that most postcolonial computing research has focused on the work of a DIYbio collective also discussed in this study, and grassroots participatory and non-hierarchical development processes. There is also a lot to learn from historical Indonesian approaches to community-based development, which I apply to the rural Community Network project in Sect. 5.4.2.

As many concepts are interdisciplinary, it was difficult to simplify the relationship between various elements and concepts to illustrate them. Many of the overlaps and cross-pollinations—e.g., between self-directed intrinsic design, appropriated technology and critical making—are missing from this overview. However, Fig. 3.4 provides an overview of which practice falls under which larger theoretical framework:

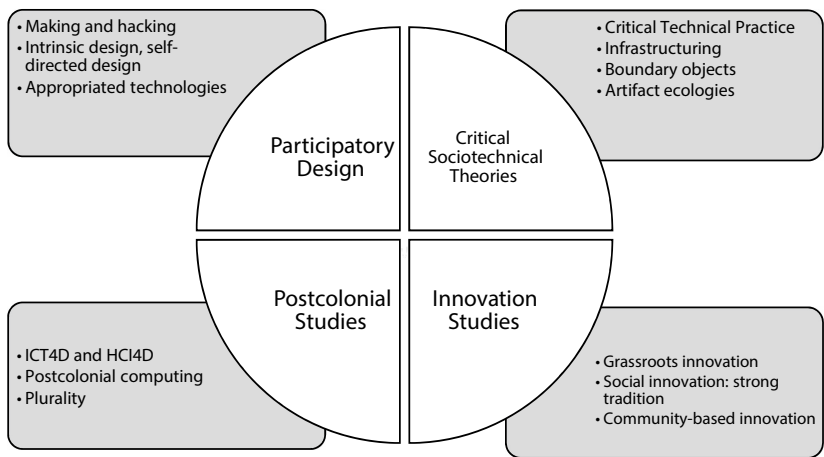


Fig. 3.4 Theoretical Framework: Theories, Concepts and Methodologies



Research Methodology and Analytical Framework

4

This chapter summarizes the research methodology used to collect and analyze data for this study and outlines the analytical frameworks and approaches I used. Before beginning the research for this study, I knew from over 10 years of working on ICT4D and grassroots social innovation projects that there was a rich diversity to critical grassroots innovation. The networks I had built and the hands-on understanding of the projects I gained during this time improved this study. However, my earlier observations needed to be justified scientifically. The research questions focus on understanding the plurality, processes, and knowledge production of grassroots critical maker communities in the Global South. Thus, I chose my research methods to support this scientific inquiry. Original empirical research was conducted, mainly using participatory methods and other qualitative methods. A quantitative method, an online survey, was only used when the number of research objects (in this case, the number of participants of a hackathon) was expected to be too high.

The novelty of this study lies in extending the term critical making to better understand critical grassroots innovation practices through a mixed-methods research design. Briefly summarized, I achieved my research goals through preliminary research, which established the legitimacy of and language for the field research, which then helped me create rapport for the two main cases: a retrospective example of intrinsic design and a fully digital example of intrinsic design.

The methods and analyses mainly consist of Western approaches, including analytical approaches of HCI and the GIM framework. However, as I will

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explain, these are supplemented with postcolonial theory, a reflexive approach (de Castro Leal et al. 2021; Lindtner et al. 2016), and systematic self-observation (Gorichanaz 2015).

4.1 Participatory Research Design and Qualitative Data Collection

I designed my research as a multi-step process. Qualitative research methods were mainly utilized, except for one quantitative method, a post-hackathon survey sent out to participants. As I will explain below, the methods I selected depended on the stage of the research. Qualitative methods primarily consisted of participatory engagement methods, which were inspired by the following considerations. First, in terms of the selection of methods, it is important to note that maker communities are Communities of Practice (CoP) (Lave and Wenger 1991), meaning that they learn, develop projects, and expand on their values as they go. Their activities are based in praxis, and they develop pathways for relevant technology-based projects through discourse and material engagement. They share a passion for activities they do, and improve as they regularly collaborate (Wenger-Trayner and Wenger-Trayner 2015), making their practices fluid and flexible. These characteristics of Communities of Practice make it possible that suggested improvements can become embedded in their everyday practice by conducting participatory research. Second, the practices of grassroots innovators in the Global South are investigated through a postcolonial lens. Based on the recommendations of the postcolonial scholar (Freire 1982), participatory methods are adequate. They engage the participants as experts of the underlying issues that are to be uncovered during the scientific inquiry. Third, the relevance of critical theory for emancipation through participatory action research had been explored. Consider how Kemmis, in critical future studies, built on Habermas's distinction of cognitive interests (Kemmis 2001). Participatory methods often start by establishing a connection (Howell 1991), an approach also used in decolonial projects built on long-term relationships with communities (de Castro Leal et al. 2021), which I strived for as a researcher.

I had originally sought to either 1) conduct participatory action research (PAR) a method in which researchers and practitioners work together to understand a situation and improve it in recurrent stages of planning, action, and reflection and evaluation (Kendon et al. 2009), or 2) use a reflexive-interventionist or multi-agent-based method (RIMA), which is advocated for in HCI in order to be able to create new theories around the maker movement. RIMA moves away from

neoliberalism and technosolutionism (Lindtner, et al. 2016), as it was developed by scholars with backgrounds in feminist utopianism and speculative sociology.

After careful consideration, I decided on the least “intrusive” method of moderate participation and allow the grassroots makers to develop their own intrinsic design project, inspired by Bødker et al. (2016). Moreover, intrinsic design is a fitting framework because the main goal of this study is investigating self-directed development (Pearce et al. 2010) and community-led or community-based design (Abrol 2005; Bidwell and Jensen 2019; Rossitto et al. 2021), both of which qualify as intrinsic design processes. Intrinsic design was previously used in HCI to study a volunteer community (Bødker et al. 2016) to better understand the design and infrastructuring work that shaped their community artifact ecology. The approach builds on the differentiation between intrinsic and extrinsic practice transformation, meaning whether changes are determined from within a community or external designers (Kaptelinin and Bannon 2012), as known from participatory design methods. The design work of the volunteer community described in this study is considered intrinsic design because it is conducted by members of the community, with no input from the outside. This study invites researchers to look further into communities where “in house” or local design activities are conducted, as volunteer-based communities shape technologies differently (Bødker et al. 2016).

The aforementioned participatory engagement and qualitative data collection methods contributed to three main research phases. First, preliminary research was conducted, exploring whether grounding the term “critical making” in the grassroots was possible. I also asked whether the term itself (or the critical-material approach behind it) resonated with grassroots practitioners. To achieve this goal, first, the term and the idea that it could cover grassroots critical maker practices was introduced at the Chaos Communication Congress (CCC), a major hacker conference in Germany (Sipos et al. 2019). This session showed very positive results in the Q&A and successive conversations. After this initial positive response, a one-day participatory workshop was organized at an academic conference for practitioners from the Global South. My co-authors (in the CCC presentation) and some participants (in the latter presentation) were recruited through a network of innovators from the Global South that I was also a member of, which had created trust and simplified my task of reaching the practitioners. This scientific grounding and mapping exercise helped make projects visible to academia and enable a better understanding of what terms are used by grassroots innovators to describe themselves (Sipos and Wenzelmann 2021). The goal for this study was to conduct a preliminary test of whether the term “critical making” resonated with the grassroots practitioners. It was also a covert exercise to

map different projects from the Global South that practitioners identified as CM projects to inform successive field research.

In the second stage I conducted field research on the first local projects based on feedback from the practitioners involved in the preliminary research. The field research started with a strategic mapping of further local initiatives to better understand and identify practices based on the three shared attributes I had identified for critical grassroots innovation projects. This stage was enhanced by my previous experiences, as I had met Indonesian grassroots practitioners during my professional engagements, so I had already gained credibility and rapport. The second goal of the field research was to highlight the plurality of grassroots innovators and to establish rapport for the next phase. The workshop provided enough evidence for designing the field research, which was to be conducted in Indonesia between November 2019 and June 2020. 22 initiatives were identified, and 13 of these initiatives were visited¹ and the founders and relevant core members present were interviewed, either *in situ* or as semi-structured interviews, depending on the time and the circumstances. In total, 16 interviews were conducted.

My focus of the visits and interviews was to establish whether the projects represented critical grassroots innovations based on the three key attributes. As is often the case with interviews, the information they shared with me was based on personal memories, thus wherever possible interviews are compared with other, more reliable written sources. The field visits were recorded as field notes ($n = 43$) and/or as voice recordings, with the consent of the interviewees.

The field research was designed to map, interview, and design a participatory design project to construct scientific knowledge about critical grassroots innovations. The scientific knowledge is here presented in two forms: ethnographic vignettes and case studies. Ethnographic vignettes are used to study how social change arises from conflict, using a description of a social situation (vignette), followed by analysis and theorization (Hohn et al. 2020). These vignettes illustrate the plurality of relevant grassroots approaches and the many social issues communities attempt to solve. They also demonstrate the lengthy process of finding projects that carry *all* attributes of critical grassroots innovations, which are later outlined in the two main case studies. The two main cases were selected because they carried all the predefined attributes. The case study method is used here, partly because “case study is nearly synonymous with STS” (page 1, Petrina 2019), but also because case studies allow for a comparative research approach. I was interested in comparing the two strongest cases of grassroots critical making, and thereby formulate heuristic generalizations based on principles such as

¹ I would like to note here that the other 9 initiatives were not excluded on any basis other than the outbreak of the COVID-19 pandemic.

community, place, perspective, and events (Knight 2001). It is important to note here that the main goal both of the vignettes and the two cases is not to define what *qualifies* as grassroots critical making, but rather to *exemplify* what critical making can look like in grassroots communities of the Global South.

Third, participatory research was conducted. For this study, two main cases were selected that clearly showed all three key attributes and could answer my research questions. One became a retrospective example in the applied setting due to the pandemic, the other main case was organized digitally in the second half of the data collection period. With regards to the participatory intrinsic project, participant observation ($n = 14$) was used as the main source of data gathering, and as a research strategy to better understand the phenomenon of grassroots critical making. This research was conducted with moderate participation because, according to (Spradley 1980), insider and outsider roles should be balanced. I contributed to the process only when explicitly asked to do so, because my technological guidance was necessary for the process to move ahead. With postcolonial reflexivity, I follow in the footsteps of excellent activist researchers (de Castro Leal et al. 2021) who stipulate that moderate participation does allow for being involved, yet not substantially interfering in the process.

While according to Postill, there is no agreed definition of the phrase “digital ethnography”, its use supports the study and analysis of groups using technology (Postill 2018). I had originally planned to conduct digital ethnography by mining online repositories of different collectives, as these contain years of knowledge production through open-source project descriptions and digital blueprints of artefacts in wikis and other online platforms. However, the pandemic forced me to move the second half of the research online starting at the end of March 2020. Online events included conference sessions ($n = 3$), which normally would have been held in person, but when moved online allowed for participation from the Global South and provided further insights into the framings of the grassroots innovators. The intrinsic design project, which was supposed to be conducted in person, became an online hackathon. Thus, the scientific inquiry turned from on-site research into remote research, and many of the discussions and interaction processes that would have happened between organizers, participants, and teams took place on instant messaging platforms, e-mail exchanges, and video sessions, making qualitative data easier to access retrospectively. All this data needed to be filtered and reflected upon. In addition to this, in the follow-up process, an online survey was sent to the participants of the hackathon². In this

² This was deliberately done later, in order to be able to understand mid-term effects of the hackathon on its participants, whether the excitement lasted longer than the award ceremony and whether individuals continued cooperating months after the event ($n = 13$).

research, new digital methodological approaches and analytical frameworks have assisted in better understanding the complex processes between critical grassroots innovators, which often happens informally and remains hidden from researchers.

Table 4.1 summarizes data collection methods used during the field research and participatory research.

Table 4.1 Data Collection Methods

Method	n	Description
<i>November 2019–March 2020: Field research period</i>		
Mapping	22	The mapping process resulted in an introduction to 22 Javanese makerspaces, collectives, and communities
Field visits	13	Out of these 22 projects, 13 were visited and interviewed ³
Field diary entries	29	These visits and other impressions gathered resulted in 29 field diary entries
Interviews	16	Individual interviews conducted and recorded on-site and online
<i>April 2020–August 2020: Participatory research period</i>		
Participant observation entries	14	Planning and hackathon sessions recorded between April and July 2020 ⁴
Post-hackathon survey	13	Out of 65 registered participants, 13 filled out the survey ⁵
Interviews with organizers	3	Key organizers interviewed in recorded online video calls to reflect on the hackathon
Online speaking contributions	6	In 2020 and 2021, 4 critical grassroots collective founders shared experiences in recorded international conference sessions I curated and moderated, two of them contributed twice

³ With the goal of identifying critical grassroots innovators, thus, focusing on the 3 key attributes.

⁴ Focusing on key decisions taken regarding the artifact ecology of the hackathon, its organization, negotiations, values, tensions, and infrastructuring.

⁵ A larger turnout was expected, but who that did not win were probably already disengaged at the time of the survey.

4.2 Case Study Selection and Analysis

This study presents multiple shorter vignettes and two main case studies in a cross-case, comparative manner. It is important to note that case study research has been criticized for a lack of rigor and basis for scientific generalization, according to Yin, mainly due to the “sloppiness” of the researcher (Yin 2009). To avoid this issue, this study aims to fulfill the four indispensable conditions for a rigorous case study according to Yin (*ibid.*) and Gibbert et al. (2008): 1) construct validity was achieved through multiple sources of evidence and triangulation, as well as a detailed outline of how data was collected and analyzed; 2) internal validity was ensured by establishing a clear connection between literature and the research framework; 3) external validity (or the analytic generalization of findings) was attained by outlining multiple cases, highlighting how they fit or don’t fit the research goal, and conducting a cross-case analysis; and 4) reliability was achieved through careful documentation of the evidence used in the study and the involvement of other researchers⁶ (Yin 2009; Gibbert et al. 2008).

The case studies presented below were selected based on my three research questions and their corresponding attributes. This targeted approach is referred to as “theoretical sampling” (Eisenhardt and Graebner 2007), “purposive”, or “criterion-based sampling”. In this sampling method, units (cases) are selected for their relevant features or characteristics, such as socio-demographic characteristics, relationships to experiences, or behaviors that enable detailed exploration of the central themes of the study (Ritchie and Lewis 2003).

In terms of the analysis of the data leading to the case studies, HCI researchers like McCarthy and Wright (2007) suggest that multiple forms of data analysis may be used, depending on the research questions being asked. Initially, all the gathered qualitative and quantitative data needed to be organized. As Wagenknecht et al. (2018) report from a workshop on STS research, data organization,

“starts with clearly defined concepts and research objects, this clarity never lasts for long. However, in the process of fieldwork and through the methods in use, things start to shake, concepts and objects become blurry, and another order of things emerges” (Wagenknecht et al. 2018).

⁶ In the case of this study, other researchers involved had been the supervisors. They include co-authors and a co-researcher in the exploratory phase, an Indonesian co-researcher during the field research, and two other Indonesian academics.

To find patterns and conclusions in the diversity of impressions gathered during the study, I followed Braun and Clarke's (2013) thematic analysis approach for identifying, analyzing, and reporting patterns across a dataset:

1. **The researcher familiarizes herself with the data.** The voice recordings were transcribed with an AI-based web application, then manually corrected as the application did not understand Indonesian words and had difficulties interpreting accents. Afterwards, I re-read the transcripts while listening to the recordings, as emotions are also conveyed through voice, and took note of these, as they carry secondary meanings. Digital field diary entries were organized and reviewed, and hand-written notes were typed up. Initial ideas, important insights, and interesting data were noted.
2. **Initial codes are generated.** Relevant parts of transcripts and notes were organized in a NVIVO 10 for Mac and attached preliminary codes to mark features that were relevant to the three research questions. The goal was to ensure the analysis and conclusion of this study wasn't merely anecdotal or built on insufficient evidence.
3. **Identify candidate themes.** In this step, I sorted the relevant data according to codes, subthemes, and larger themes, to highlight overlaps of the different cases according to the research framework of this study. What was helpful in this stage is that the interviews were already built around the research framework, however, the interviewees shared plenty of further insights that were not directly linked to the interview questions, thus, these needed to be organized thematically as well.
4. **Review themes.** This iterative stage involves a deeper review of identified themes, questioning whether to combine, refine, or discard initial themes to create more clarity and to meaningfully unite themes.
5. **Define and name themes.** Ongoing analysis further enhances the themes, provides theme names and clear working definitions, while also developing a unified story from the data. This was the step in which I moved beyond paraphrasing and created a narrative, allowing me to define the final structures of Chaps. 5 and 6.
6. **Produce the study.** In this final write-up phase, an interpretable piece of writing is produced, with examples that relate to the research framework. This resulted in Chaps. 5 and 6, as well as their coherent embedding in the complete study.

Based on this iterative approach to thematic analysis, the following analytical structure of findings crystalized. The plurality of critical grassroots innovations

is demonstrated through a collection of ethnographic vignettes. These vignettes are built on the three attributes, which were used as characteristics to find cases of critical grassroots technology projects on Java. While the cases might not all perfectly fit the three prerequisites, they illustrate ingenuity and critical knowledge and anticipate the two main cases. The two main cases are analyzed on two levels. First, the Grassroots Innovation Movement (GIM) analytical framework is reconsidered according to the research framework (presented in the next subchapter). Next, it is utilized to trace the generative conditions that highlight how these projects came to be, which creates a basis for the contextual understanding of the second part—the participatory intrinsic design projects.

This approach enables a thorough thematic analysis across the cases, which generates an understanding of critical making in grassroots innovation projects in the Global South (Table 4.2):

Table 4.2 Overview of Thematic Analysis Process

	Goal	Within-case Analysis: Chap. 5	Outcome: Chap. 5	Cross-case Analysis: Chap. 6
Sample cases	– Contributing to RQ1 – Identification of GCM projects	Analysis based on fulfillment of criteria: 1. Grassroots (RQ1) 2. Critical (RQ2) 3. Making (RQ3)	Vignettes	Cross-case analysis: – RQ1 – RQ2 – RQ3
Main case I	– Presentation and analysis of GCM project – RQs 1, 2 and 3	Retrospective analysis: GIM framework 1. Context 2. Pathways (RQ1) 3. Framings (RQ2) 4. Spaces and Strategies (RQ3)	Case study I	
		Retrospective intrinsic design project 1. Infrastructuring (RQ1) 2. Community strategies (RQ2) 3. Artefact ecology (RQ3)		

(continued)

Table 4.2 (continued)

	Goal	Within-case Analysis: Chap. 5	Outcome: Chap. 5	Cross-case Analysis: Chap. 6
Main case II	– Presentation and analysis of GCM project – RQs 1, 2 and 3	Retrospective analysis: GIM framework 1. Context 2. Pathways (RQ1) 3. Framings (RQ2) 4. Spaces and Strategies (RQ3)	Case study II	
		Digital intrinsic design project 1. Infrastructuring (RQ1) 2. Community strategies (RQ2) 3. Artefact ecology (RQ3)		

4.2.1 Within-case and Cross-case Frameworks

4.2.1.1 Within-case Analysis: Criteria

As mentioned above, the case studies presented in this study were selected based on the three research questions and their corresponding attributes, based on a targeted criterion-based approach, in which the cases were selected because they had relevant features or characteristics. For this study, the following attributes or criteria were relevant:

1. **Grassroots:** The cases had to represent bottom-up, grassroots communities or collectives, as opposed to academic, industrial, or governmental actors. This criterion contributes insights to RQ 1.
2. **Critical:** The cases had to demonstrate critical thinking in their processes, or express criticality through socially innovative projects, and thereby uncover social issues and gather and share critical knowledge about them. This criterion contributes to RQ2.
3. **Making:** The cases had to engage in technology development, such as making, hacking, or the appropriation of technologies for their own needs, as part of their design practice. This criterion contributes to RQ3.

4.2.1.2 Within-case Analysis: GIM Framework and Intrinsic Design Project

The two main cases will start with a retrospective review of their generative conditions, using the retrospective framework for the analysis of grassroots innovation movements, proposed by Smith et al. (2017). This multidisciplinary framework was developed to analyze movements that develop innovations in a bottom-up and participatory manner and produce knowledge in the process. It builds on earlier works of the authors, such as on the history of the appropriate technology movement and more recent fieldwork analyzing technologies for social inclusion (Smith et al. 2014). It also builds on findings from practical grassroots innovation, combined with values of social justice or environmentally sustainable developments (see for example Schumacher 1973; Hess 2009a; Smith 2005). Thus, it is highly relevant for this study. Specifically, the analytical approach proposed by Smith and his colleagues, based on four dimensions, is a suitable basis for the retrospective analytical part of this study. It was used to analyze historical and contemporary grassroots innovation movements. Chapter 6 on Hackerspaces, Fab Labs, and makerspaces shows the diverse landscape of making (pp. 100–122, Smith et al. 2017). It has also been used by Arancio (2021) to analyze the Global Open Science Hardware (GOSH) movement. Below, as I present the framework, I will use examples from these two sources, and later I will triangulate these with the cases.

Some of the assumptions of this analytical framework don't however, as Smith et al. also note, account for particularities. It combines various theories and concepts, including socio-technical transitions theory, STS, and innovation studies, with social movement literature. The authors highlight that in some Western social movement literature, resources were deemed a prerequisite for mobilization. A growing amount of literature on the distinct characteristics of social movements in the Global South, however, highlight resource limitations. Moreover, there have been extensive movements in resource-constrained communities in Latin America and Asia (Escobar and Alvarez 1992; Thompson and Tapscott 2010). Outlining these generative contexts in this study will have to be historically informed by economic, political, and cultural factors (page 21, Smith et al. 2017).

The other different aspect of this study is criticality. Critical thinking is not a key part of the GIM framework. Instead of focusing on criticality, the authors recognize that some grassroots actors are politically motivated, as they are

“interested in opening up (increasingly seamless) technologies to scrutiny, sometimes out of mere curiosity, sometimes for fixing and repurposing and in other cases as an overtly political act of technological citizenship” (page 100, *ibid.*)

The book dedicates a chapter to hackerspaces, Fab Labs, and makerspaces, which the authors conceive as a grassroots innovation movement because there is considerable activity outside formal institutions and their networks are committed to exploring the social possibilities of bringing tools to people (*ibid.*).

In this study, I will use the GIM framework to show two grassroots innovation approaches in depth. The goal is to highlight relevant aspects of the four dimensions of this retrospective framework and use it to underscore the contexts out of which the intrinsic design projects were born. The GIM framework suggests analyzing four interrelated concepts to understand grassroots innovation movements: contexts, framings, spaces and strategies, and pathways, which I will briefly explain below. These concepts are organized according to the Research Framework, and where applicable, supplemented with concrete insights from the aforementioned Chap. 6 on makerspaces and maker communities (pp. 100–122, Smith et al. 2017). It is important to note that this analysis, with a very few exceptions of short notes on development, describes Western practices. A different paper on the GOSH movement (Arancio 2021) is much closer to the cases analyzed in this study, as it also refers to practices in the Global South (mainly Africa and Latin-America). Both are briefly introduced below as both are indicative of examples that will be relevant for subsequent analysis.

4.2.1.2.1 Context

In the sphere of context, circumstances are mapped by utilizing a multi-level perspective (Geels and Schot 2007) in which “windows of opportunities” for grassroots innovation communities—such as sociotechnical regimes and socio-technical landscape developments—are identified. Most importantly, the context helps outline the conditions in which the movement is developing. Historical, political, economic, cultural, and religious contexts could be generative or constraining. Similarly, other circumstances, issues, and situations, including opportunities available within those contexts, could have had a generative effect on the movement.

4.2.1.2.2 Pathways

Pathways consist of opportunities that are constructed and assessed from multiple perspectives. How does the plurality of pathways contribute to alternative

developments over time? Ideas and aims are continuously developed and dismissed; objects and practices and their materiality, as well as the co-constitution of the social and the technical, also contribute to developments in different settings over time. These alternative pathways and their plurality show that there is not just one obvious “best” pathway. Rather, the political nature of grassroots movements might open new pathways with greater attention to issues of social inclusion, diversity, difference, and social justice, playing a key role in their RRI practices (page 28, *ibid.*)⁷.

4.2.1.2.3 Framings

In framings, future possibilities are negotiated collectively by establishing shared visions. Framing is the process of meaning production that helps communities connect to powerful narratives beyond shared grievances, which can then be expressed in critique towards mainstream practices. Framings are shaped by underlying assumptions, and can include problems, strategies, requirements, theories, knowledge, design criteria, exemplary artefacts, testing procedures, and user practices that emerge through social interaction. While framings can include technological frames, it also can include or exclude a broader set of framings (such as social, economic, or political questions), which can be important factors in how they design their practices.

In terms of existing literature, according to Smith et al. (2017), possible framings in makerspaces include free and open-source software and hardware, a new industrial revolution through personalized manufacturing and mass customization, the democratizing power of technological citizenship, and grassroots innovation and sustainable developments (pp 106–109, Smith et al. 2017). In the GOSH community, according to Arancio, framings include open science, environmental and cognitive justice, appropriate technology and critical pedagogies, hacker culture, and trans-hacktivism (pp. 7–12, Arancio 2021). For this study, the framings of GOSH are relevant and close to the framings of grassroots critical makers in Indonesia.

⁷ It is important to note here that as Arancio highlights, GOSH does not have a long history, as it was founded in 2015. Thus, pathways do not yet apply to it. Instead, she develops speculative future scenarios of pathways instead, based on the historical analysis of other similar GIM. For example, GOSH could get absorbed into another institution, or it could contribute to the opening of science so that in a future pandemic, makers don’t have to reverse-engineer but can adapt already tested and safe designs.

4.2.1.2.4 Spaces and strategies

Spaces and strategies crystallize novel strategies and cooperative forms. This concept captures actions communities take, and how those actions are influenced by the availability of resources. Spaces can be physical, but also social, discursive, and institutional. Locations and activities that enable communities to experiment and innovate differently can be analyzed by enrolling audiences, alliances, and users. They can improve their performance in a user-centered way by creating public engagement and making alternative spaces for said engagement.

In terms of makerspaces, according to Smith et al. (2017), spaces and strategies include maker movements and grassroots groups, educational outreach and skills provision, and business and state interests (e.g., public or private funding and donorship) (pp. 109–111, Smith et al. 2017). Arancio strategically divides spaces and strategies in GOSH. In terms of spaces opened up by members, she highlights community science projects, academic researchers, artists, educators, and social innovators (pp. 12–14, Arancio 2021). With regard to strategies, these are built on networks, intermediaries, and repertoires of action. Spaces and strategies include organizing global gatherings with both reflexive and material activities and the promotion of use cases as impactful developments. The mobilization of resources includes grants, in-kind infrastructures, and volunteer work (pp. 14–17, *ibid.*).

In terms of the research framework of this study, the context is important to help underscore the circumstances of the projects. Research Question 1—Plurality and Alternative Futures is addressed in Pathways, Research Question 2—Production of Critical Knowledge is addressed in Framings and Research Question 3—Processes of Participatory Intrinsic Design is addressed in Spaces and Strategies.

4.2.1.2.5 Pertinence of GIM Framework for Non-Western Critical Grassroots Communities

In terms of the GIM framework's suitability for this study, I would like to mention three observations: 1) its significance for the Global South, 2) the scattered, global-assembly nature of grassroots versus being official movements (or members thereof), and 3) the embeddedness of critique in the political nature of the grassroots.

In terms of the framework's usability to analyze the grassroots in the Global South, albeit many of the historical movements refer to originating from the Global South (e.g., India, Brazil), the case representative of the framework in the book was mainly reserved for Western maker culture. One could argue that being located in the Global South means the projects need to be discussed under the

dimension of context, as this is the geopolitical context they work in. However, only a few cases mentioned the Global South in minor remarks. Smith et al., referring to Escobar and Alvarez (1992) and Thomson and Tapscott (2010) highlight that “there has been increasing attention in the social movement literature to distinct characteristics of social movements in the Global South” (page 21, Smith et al. 2017). Despite apparent resource limitations, *deemed a prerequisite for mobilization in some Northern social movement literature*, there have been extensive movements across resource-constrained communities in Latin America and Asia. These movements suggest different forms of collective action and approaches to theories of social movements (Escobar and Alvarez 1992). In particular, Escobar and Alvarez argue that *economic, political and cultural factors are crucial* to Latin American social movements. So, in studying the generative contexts for grassroots innovation movements, as well as the opportunities that contexts provide, the lesson from the social movements literature is that this analysis should be historically informed (page 21, Smith et al. 2017). Thus, making here is seen from the lens of development, not as an activity that local communities engage in by themselves. Closer to the goals of this study is Arancio’s analysis of GOSH, which is a network that has been active in African and South American contexts. Hence, although Smith et al.’s framework is the original starting point for this analysis, most of the triangulation will be conducted with the findings in this article.

The second observation is that Smith et al. propose the analysis of the political nature of grassroots innovations under framings. Hence, I will also discuss the aspect of the production of critical knowledge (RQ2) under framings. This also sheds light on how criticality can manifest itself in different ways. In the case of this study, critique is of postcolonial settings and social injustices through critical-social innovation leading to technological developments or thinking about the technologies and their effects on their users have developed/appropriated in the process. It is important to note that these framings might not be loudly announced to the world in manifestos or on websites, but instead appear in quiet one-on-one discussions, where one would expect them the least. The reason for this being that critical goals must remain hidden in order to be achieved. They are hidden behind project goals communicated to funders and very subtly mentioned even within the trusted community, as systems are to be changed that are politically sensitive. To highlight the relevance of the critique, I considered adding this as a separate dimension. However, for the sake of simplicity, I decided to slightly adjust the GIM analytical framework to the research framework of this study. This adjustment is reflected only in the order of the dimensions, which is not a major change, as dimensions do not build on each other.

Finally, in terms of the plurality and diversity of grassroots innovation, it is important to note that many projects are not globally unified but scattered, forming heterogeneous projects in a bottom-up manner (Sipos and Wenzelmann 2021). The similar goals make them part of a “global assemblage” similar to makers (Lindtner et al. 2016). From an academic perspective, the dispersed nature of grassroots innovations and respective communities is a significant burden. It can be challenging to acquire funding for research without grouping separate communities into homogeneous global movements, as funding structures for academia often require that the research subjects deliver large-scale impacts. Smith et al. address these issues by grouping grassroots innovation projects into movements and by proposing the aforementioned comprehensive and multidisciplinary analytical framework for the analysis of such movements (page 3, Smith et al. 2017). This is convenient and fitting for the research framework of this study, but it is important to note that not all grassroots innovators see themselves as part of a broader movement, and not all phenomena that researchers call “grassroots innovation movements” have uniform activities around the world. Hence, while extracting generalized findings from the diverse group of grassroots innovators investigated in this study, I will highlight these individual projects through their respective vignettes and cases and only mention related movements where appropriate. The “Small is Beautiful” approach (Schumacher 1973)—i.e., to analyze the local embeddedness and sustainability approaches of projects in local communities—is more applicable here than theories of social movement building.

4.2.1.2.6 Intrinsic Design Project Analysis

As previously discussed, critical grassroots innovators constitute communities of practice (CoP). Hence, their practices are best investigated through participatory approaches. In the two case studies, a retrospective and a fully digital intrinsic design project (Bødker et al. 2016) is described (Fig. 4.1):

1. **Infrastructuring:** Dynamic aspects of grassroots communities unfold over time through the “form of irregular, opportunistic, adaptable, responsive and decentralized strategy of infrastructuring social innovations and reform that stand in contrast to existing power and economic configurations” (page 9, Vlachokyriakos et al. 2018). How the volunteer community infrastructures itself is a strategic endeavor, a matter of planning and deciding, and chance (Bødker et al. 2016).
2. **Sustainable strategies (Case I) and Tactical interventions (Case II):** While strategies and tactics are both present in the articulation work of a project, they are different types of approaches to making an artefact ecology work.

Strategic practices of design could be understood as extrinsic practice transformation. More importantly, these activities are momentarily detached from the project itself, while strategies are the design activities that a community collectively engages in, such as more formal community meetings (Bødker et al. 2016). Tactics are practices of use—situated activities that deal with happenstances and changes occurring in everyday community practices, reacting to changes in plans and designs (ibid.). These are inspired by Dourish’s notions of “strategies and tactics”, as adapted from De Certeau (Dourish 2006; de Certeau 1984).

Summarizing different strategies and tactics will contribute to what we know of how grassroots produce critical knowledge through their engagement with, appropriation of, and rejection of technologies, and thus contribute to RQ2.

- 3. **Artefact ecology:** The making of the tech stack reveals the physical infrastructure (community internet) or digital infrastructure (hackathon) of their projects, applied in similar ways as designers or researchers would (Bødker et al. 2016). Outlining the artefact ecology that the communities develop will help demonstrate grassroots’ engagement with software and hardware solutions, thereby contributing to RQ3.

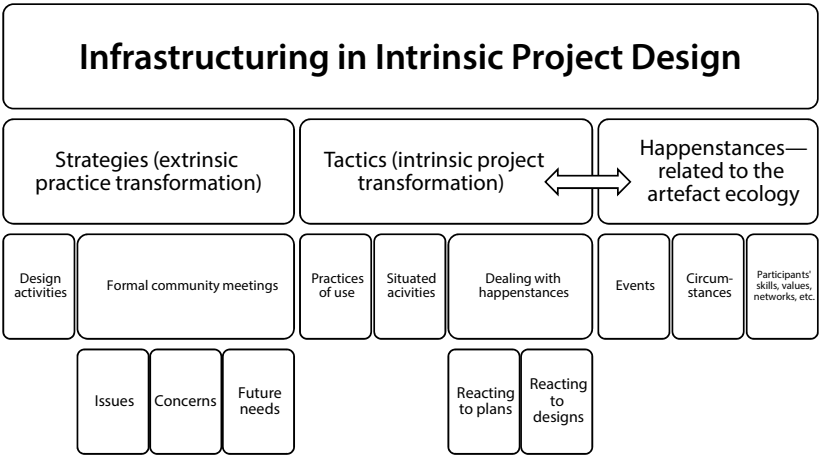


Fig. 4.1 Infrastructuring in Intrinsic Project Design. (Based on Pipek and Wulf (2009); Vlachokyriakos et al. (2018); Dourish (2006); de Certeau (1984) in Bødker et al. (2016))

4.2.1.3 Cross-case Analysis

For this study, the comparative perspective was selected because it has been a relevant approach in humanities, and it can be used as a heuristic tool. Stretton proposed using a comparative perspective to extend the researcher's experience, become aware of more possibilities, and help with question-prompting, cause-seeking, and effect-measuring—essentially as a system for questioning and not answering (Stretton 1969). However, there are methodological challenges to introducing a comparative framework, such as “many variables, small N” (page 159, Lijphart 1975), case selection, equivalence construction, cases or variable orientation, and causality (Mills et al. 2006). To address such concerns, I decided to conduct a cross-case analysis. The findings across the cases are summarized and triangulated with findings from other relevant studies, before I discuss them. Data from the field was collected and analyzed in the aforementioned thematic analysis process, at first separately for each case, also known as within-case analysis or writing up pure descriptions that generate insights (Eisenhardt 1989). Afterwards, these insights are analyzed in a cross-case manner, allowing for patterns and relationships to emerge, based on which hypotheses can be confirmed or formulated (*ibid*).

4.3 Ethical Considerations and Reflexivity

In terms of ethical considerations, the researcher carries a lot of responsibility when working with vulnerable groups. In this case, grassroots communities develop projects that represent societal issues and develop knowledge that decision makers might not want to acknowledge (Cox and Fominaya 2009). These issues of social justice—including homelessness, feminism, or the rights of Indigenous communities—are burning societal issues. Those who raise them might get into trouble or might cause trouble for those they attempt to help. Moreover, the situation of activists in Indonesia is unsafe, as human rights defenders are being criminalized (Janti 2022).

Careful consideration about whether to expose vulnerable communities is required. Many collectives benefit from international visibility and are proud of their work. However, given their immediate political circumstances, in this study, for their protection, I decided to use aliases and abbreviations to mask names. Thus, my approach is not anonymous, but pseudonymous (Scott 2005). Anonymizing is not a perfect process, but rather a continuum from fully anonymous to very nearly identifiable (Saunders et al. 2015). While attempting to keep

identifying details confidential, I recognize the challenges of this exercise. Someone who knows the people and collectives could still identify them, just as I immediately recognize people despite changed nicknames in other researchers' papers. I also am aware that full anonymity will not be possible because the collectives can be traced through the literature review and previous articles I have written.

In participant observation—the approach this study mainly utilizes—informed consent and voluntary participation are the most prominent ethical concerns (DeWalt et al. 1998). To address this, I prepared simple consent forms that were signed by the interviewees. In the case of the rural community, I met with the chief leader and explained my presence and goals, ultimately receiving the chief leader's approval. Finally, those video sessions that were recorded for the other case were webcast on the official social media page of the organizer. The participants were aware of this, as they knew of my presence as a researcher in addition to my role as a mentor. Throughout the research I was highly aware of my double role; I openly shared my research proposal, communicated my intentions, and invited interested parties to help shape my research. I helped with hackathon planning, went to the event itself in *gotong royong*-style, supported the teams as a mentor, and also gave a workshop afterwards. I did not strictly participate as an evaluator/jury member, as I did not want to influence any decision making. The research was especially relevant to exploring processes that often remain hidden to researchers. Other sources of digital ethnography (e.g., videos, websites and recorded conference sessions) are in the public domain and referenced as scientifically appropriate.

I also find researcher self-disclosure and further reflexivity, including disclosing intellectual and political beliefs, important (Herr and Anderson 2014). It is thus important to mention that my beliefs are aligned with those who have conducted similar research in the Global South and in Indonesia. Similar to them, I highlight grassroots innovators' work through a postcolonial lens, while avoiding technosolutionism and Western normativity (de Castro Leal et al. 2021; Lindtner et al. 2016).

I center my work around those who have been excluded by traditional power structures. While I don't think of myself as an activist, others might see my work as contributing to activist academia (de Castro Leal et al. 2021), as this study builds on critical theory and is political because it sheds light on social injustices uncovered by grassroots. My views are based on having worked with community radios in Hungary and Belgium before joining the UN, where I attempted to use the privilege that accompanied the job to empower grassroots innovators in the Global South through funding and networks. Afterwards, I joined Ashoka, the

world's largest network of social entrepreneurs, while setting up my NGO that supports the bottom-up development of systemic solutions to societal problems enhanced through technology. I also served as an executive board member of the Global Innovation Gathering, an international network of innovators mainly from the Global South. However, too much experience sometimes leads to me to assume that others already know what I know. To avoid this, I used free-format narrative recording to reduce the distance between the occurrence and the data collection, leading to more accurate and detailed data (Gorichanaz 2015).

I would also like to highlight that as a Western, white researcher, I must be reflexive on multiple levels. I know that many communities experience what I call *researcher fatigue*. In developing countries, local NGOs and communities suffer from donor fatigue due to unrealistic promises and expectations that have little to do with the situation on the ground (Anderson 1998; Awadari 2020; Kakupa and Shayo 2021). Similarly, grassroots communities enthusiastically share resources such as their time, space, and information, but do not get anything in return. At the same time, I know that my privilege created advantages for me, such as being a researcher at a Western university. I try to balance this power imbalance by building genuine, long-term relationships; I do not only reach out to ask for input, I also have personal discussions wherever appropriate, and I pay attention to how I might help others. Trust is difficult to build, and research participants might alter their natural behavior if they don't feel safe. Having been invited to events—some in participants' houses—allowed me to, in a sense, become a part of their communities. Understanding that objectivity and neutrality would be difficult due to personal involvement, I see my “insider” status as a major advantage that helps me conduct better, further-reaching research.

4.4 Scopes and Limitations

Before outlining insights from the field in the next chapter, I would like to briefly touch upon the scope and limitations of this study. This study focuses on exploring and investigating projects that could extend the term critical making to grassroots in the Global South. They represent grassroots ingenuity, diversity, and societal tensions, contributing important findings of this study.

Although most grassroots projects that become popular are still started by men (Chandra et al. 2021), I managed to achieve a relatively good gender balance. Albeit many of the vignettes represent projects started by men, all collectives in the second main case were co-founded by women, and the intrinsic design project (the hackathon) was started by three women. It is also important to note

that all the projects described in this study were implemented on the island of Java, the main island of Indonesia. Thus, they are neither representative of all Global South critical making projects, nor even all potential projects within the Indonesian archipelago, with its diverse cultures, peoples, and islands.

This study constitutes the start of a potential expansion of the term critical making towards the South. Building on existing research on grassroots and critical making, the study aims to mutually enrich both fields. The study also contributes to novel insights with its postcolonial approach, as both have been under-theorized in non-Western settings. An excellent example of a similar endeavor is Butoliya's (2015) work on Critical Jugaad (the Indian version of making, extended with criticality). However, she considers critical making from a design perspective rather than from a grassroots innovation or social innovation perspective.

In the Field: Grassroots Critical Making

5

This chapter presents the main findings from the field research. The goal of this chapter is to identify and deliver findings that validate or disprove the hypotheses presented in the research framework and the literature review. It also aims to support the main goal of the study—the extension of the term critical making towards grassroots through empirical evidence.

First, I present findings from the preliminary research period. The main goal of this exploration period was to ground the term **critical making** by exploring whether it resonated with grassroots practitioners (mainly hackers and makers). The research had been designed as participatory, so the findings in this subchapter feed into the following steps of this study.

Second, the design of the field research from Indonesia is presented, including a preliminary timeline and how that timeline changed. Partially due to the pandemic, but mainly due to the grassroots practitioners' own timeline within the participatory project, this provides interesting insights into the planning of intrinsic design projects. Then, a brief outline is presented of how the **pluriverse** of **grassroots critical making projects** in Indonesia were mapped and evaluated.

Then, key findings from the field research are presented. First, projects that are not grassroots critical making are briefly presented to show the contrast between such projects and those that fit the research framework. The latter are introduced in the form of ethnographic vignettes, providing insight into the aforementioned pluriverse of civic ingenuity. Each vignette is analyzed based on the three criteria to demonstrate how they represent *grassroots*, express *criticality*, and perform *making* within their communities. The ethnographic vignettes are followed by two case studies, which are organized as follows. First, the histories of the two collectives are analyzed retrospectively, based on the Grassroots Innovation Movements analytic framework, by underlining their origin stories and highlighting the contextual aspects of grassroots critical making in Indonesia. Then, by describing

and investigating concrete intrinsic design projects, the study demonstrates how each of the collectives practice grassroots critical making. The framework used for this exercise is based on an artefact ecology analysis composed of the happenstances, tactics, and strategies of the projects (Bødker et al. 2016). The aim is to uncover how values drive decision-making processes in these critical-material practices. The findings include examples regarding what critical making could mean in the Indonesian context. Specifically, grassroots critical making is an example of self-directed, **post-colonial computing** (Irani et al. 2010; Kaiying et al. 2019), so-called **good development** (Chambers 1997; Pieterse 1998; Smith, et al. 2014), and context-specific and (covertly) **politically motivated hopeful interventions** (Lindtner and Lin 2017). A cross-case analysis and discussion is offered in Chap. 6.

5.1 Preliminary Research: Grounding Critical Making in Grassroots

The original idea for this study was inspired by two recent developments: 1) the recent shifts in STS and HCI moving from a focus on industry towards researching grassroots, and 2) critiques coming from HCI and grassroots regarding maker culture and its promises. These motivated the main goal of this research and this study, to investigate what such critical-material practice looks like in the Global South.

In this chapter, I summarize the timeline of the preliminary research conducted for this dissertation, starting with learning about critical making from Garnet Hertz. This research on grassroots critical making is supplemented by my experience and existing networks, which have been crucial to extend the term critical making from the current academic usage. The current scientific framework of critical making is novel in engineering studies, critical viewpoints through art and design, and DIY citizenship in the West.



As researching **post-colonial grassroots innovation** through the lens of critical making is a novel approach, I decided to cautiously design my scientific inquiry. It was based on the hypothesis that are critical makers outside of academia and design, but might not call themselves by this name. First, I introduced the term to a broader audience of grassroots innovators—a worldwide network of innovators in the Global South and attendees of a major hacker conference. My goal was to investigate whether the term resonated with them. Second, it was necessary to ground the term in grassroots practices, thus I organized a mapping exercise in a one-day participatory workshop with diverse practitioners from the Global

South with a co-researcher. Having received positive feedback in these first two steps, I conducted field research, mapped critical grassroots projects in the field, then compared results against the three attributes of grassroots critical making. Finally, I identified two cases for a comparative case study of participatory-intrinsic design projects—one analyzed retrospectively, the other co-implemented with grassroots innovators—to generate scientific knowledge of grassroots critical technical practices (Fig. 5.1).

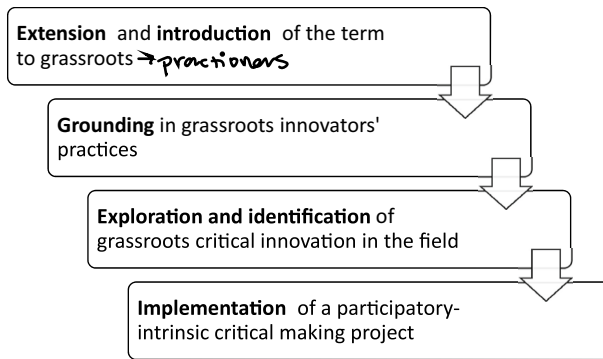


Fig. 5.1 Four-step Approach of the Scientific Inquiry

Below, I will explain these four steps in further detail, as originally planned and as events actually unfolded, affected by the pandemic and the intrinsic design practices of the grassroots innovators.

5.1.1 Learning About Critical Making

I had become aware of the term critical making in October 2017, when Garnet Hertz held a short presentation in Berlin at the STARTS Talks event¹. Hertz's talk critiqued the maker culture and was based on his academic work as the Canada Research Chair in Design and Media Arts, and Associate Professor at Emily Carr University in Vancouver. In this presentation, he explained the notion

¹ This event was organized by Ars Electronica and WAAG. These two entities, together with the Belgian Centre for Fine Arts (BOZAR) received EU Horizon 2020 funding to award innovative projects combining science, technology, and the arts, called the STARTS Prize. <https://waag.org/en/event/starts-talks-1-berlin>.

of combining critical thinking and making, as well as tracing various critical technical and design practices. One of his slides included a hand-written list of those practices, which was an excellent starting point for my research. He also highlighted his zine titled “Disobedient Electronics: Protest” (Hertz 2017) that I used in teaching the course “Hacking Innovation Bias”. It inspired my future plans on publishing this dissertation in a zine format in a collaboration with the Indonesian grassroots innovators that the ethnographic vignettes were based on. Hertz’s presentation showed me that there were alternatives to mainstream making and that criticality can be combined with material practice, allowing for people whose work is embedded in hands-on practices to express themselves. Engaging in alternative methods of academic publishing (e.g., zines) includes those who are typically left out of academic discourse (Fig. 5.2).

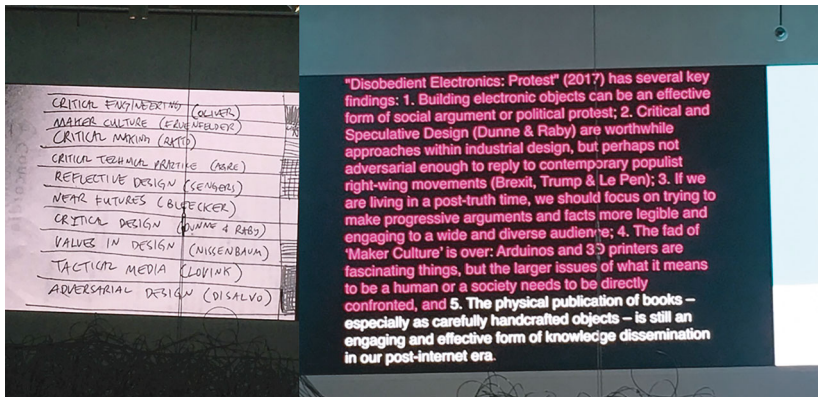


Fig. 5.2 Hertz's Slides at the STARTS Talks on Critical Making. (Source: The Author)

5.1.2 Extending Critical Making Towards Grassroots

Approximately 2 months later (December 2017) I started working at the Technical University of Berlin and submitted the topic of this thesis mid-May 2018. During this time, I started looking into the meanings associated with the term critical making. I realized that the way I understood it—as an approach to social innovation by grassroots actors—was not the leading academic approach. The term itself, however, was open to interpretation beyond engineering studies and

design research. Ratto and Boler had published a book of papers based on a conference titled “DIY Citizenship” (Ratto and Boler 2014).

To use the term critical making to identify and describe grassroots practices is a novel approach, which I first publicly presented in December 2018 at the world’s largest hacker conference, the 35th Chaos Communication Congress (35C3)² (Fig. 5.3).

Some scholars within critical making even point out that asking which maker practices arise out of survival, aren’t recognized for their criticality, etc., especially in the Global South



Fig. 5.3 Presenting the definition of Critical Making. (Source: The Author)

In a first attempt at exploring whether grassroots innovators identified with critical making, and how they understood the term, I sent an open invitation to the members of the Global Innovation Gathering (GIG)³, a network of innovators mainly from the Global South that I had been a member of since early 2016. The idea was to work with makers and hackers to co-create the session and propose

² <https://fahrplan.events.ccc.de/congress/2018/Fahrplan/events/9971.html>.

³ Global Innovation Gathering (GIG) is a global network of social and technological innovators, makers, and hackers. Most of their members come from non-Western countries. The author has been a regular Member since 2016, and between 2019 and 2021, she served as an elected Executive Board Member.

projects they wanted to present under the term critical making, and to explore whether my impression was correct that this term could resonate with some of them.

Titled “The Critical Making Movement: How Using Critical Thinking in Technological Practice can Make a Difference”, my presentation focused on the background and development of the term critical making, reflecting on the opportunity to extend it to citizen-driven and grassroots social innovation. The talk acknowledged that while groups of people who make critically might be seen as a broad and global phenomenon, research is needed to carefully examine its hyper-local relevance. I suggested following the notions Schumacher discussed in his book “Small is Beautiful” to understand the value of diverse, small-scale, locally relevant approaches (Schumacher 1973). The presentation briefly touched upon the difficulties researchers face, such as the decentral and heterogenous nature of grassroots innovator communities, the vocabulary that grassroots utilize being different from the academic vocabulary, and how institutionalized innovation becomes easier to research, but might also lose authenticity due to funding. In addition to my presentation on the background, two GIG members were supposed to share their practices. Ricardo Ruiz, a maker-researcher from Brazil, could not attend due to a lack of financing (which is a very common issue among practitioners in the Global South). In the paper based on the planned presentation (Sipos et al. 2019), he detailed the case of LabCOCO, a “critical making lab” that attempted to diminish prejudice and discrimination towards Afro-Brazilian religions through gaming. The other presenter was Saad Chinoy, an open-source hardware expert and maker from Singapore who was working with caretakers of people with disabilities, rethinking and hacking toys used in their work. He claimed these hardware hacking workshops “help to raise awareness about disability and how technology can enable and empower, if applied critically” (page 12, *ibid.*).

this research
focus is
however
reflected
in Sipos &
Kermap (2023)

The talk took place in the second-largest room of the conference with an estimated 500 people in the audience on site, meaning that the curators of the program expected a good turnout. As Ruiz was not able to attend, the Q&A session was extended by about 20 minutes, during which time the audience actively participated. After the session, another 30 minutes were spent off-camera discussing the topic with audience members, some of whom expressed appreciation for the fact that we presented cases from beyond mainstream maker and hacker culture, including knitting communities. In academic terms, this session is to be interpreted as a first attempt at extending the term to the grassroots, and a real-world attempt to understand how makers and hackers react to the term. The reaction was

overwhelmingly positive, thus, in the next step, a focus on a more participatory approach with practitioners from the Global South became necessary.

5.1.3 Grounding Critical Making in Grassroots

While the term undoubtedly resonated with German hackers and the two non-Western co-presenters, a more structured approach was necessary to understand whether there were grassroots innovators who self-identified as critical makers. It seemed evident from desktop research that such identification was sporadic. Even if certain practitioners used critical thinking while they developed solutions using methods that resembled mainstream making practices, it didn't necessarily mean that they called themselves critical makers. Therefore, having established that the term could resonate positively with grassroots innovators with diverse backgrounds and diverse projects, a proposal for a one-day workshop was submitted to the 9th International Conference on Communities and Technologies in June, 2019 in Vienna. The conference sought to foster an exchange with non-academic practitioners. Accordingly, this workshop titled "Critical Making with and for Communities: Community-Driven Critical Making Grounded in Practitioners' Perspectives on Definition and Praxis" aimed to learn more about the practice of those who might see themselves as critical makers, and promote a discussion around what the terms innovation and making meant to them. It also allowed for further grounding work in a more academic setting, while it allowing the term to be further explored with the grassroots innovators themselves.

The goal of this workshop was to further understand whether critical making was a term that could describe the activities, processes, and aims of grassroots innovation communities in the Global South. It was acknowledged from the start that many communities use their own words to describe their practices, which are embedded in their native languages, and many of those words have additional connotations. Indonesian practitioners prefer to focus on the co-creative processes of knowledge production and tinkering processes (*ngoprek*) rather than the ownership of a particular outcome (Siagian 2016). *Jugaad* is an Indian term that can be translated as "hack" or "resourcefulness", often stemming from a lack of resources, and critical jugaad practices have also been identified (Butoliya 2015). Shanzhai "copyleft" production in China is also getting more attention in an attempt to decolonize Western information technologies (Braybrooke and Jordan 2017; Lindtner and Lin 2017).

~ and this seems to be true for many in the maker movement in the Global North, too

local maker epistemologies

Although the following self-critical and reflexive questions were never asked directly, a goal of the workshop was to explore:

- What would it mean to further evolve the term critical make to highlight locally relevant critiques of socio-political realities through grassroots technology innovation practices?
- Would an umbrella term coming from academia allow scholars to identify, analyze, and understand the processes that are deemed too small and scattered, their relevance so hyper-local that they end up being overlooked by academia, governments, and corporations?
- Considering that local terms could be used for orientation and exploration in the field, is it even necessary to further evolve critical making as an umbrella term to incorporate those heterogenous and dispersed practices?

Another goal of the workshop was to further advance academic understanding of grassroots critical making, thus unmasking the utopian vision of making, which relies on **technosolutionism** and **ideological colonialism** (Lindtner, Bardzell, and Bardzell 2016). To achieve this, the organizers decided to look beyond the maker hype and its promises by inviting practitioners of non-Western approaches such as **jugaad**, **gambiarra**, and **ngoprek**. They also made sure to extend the invitation to collectives and communities instead of makerspaces. In doing so, the goal was to unearth critical making practices that might have simply been labeled with other local terms.

reconnecting
the utopian
vision of
making
HCI after
technosolutionism!

The invitation to participate⁴ inspired 16 practitioners to present seven cases, out of which ten made it to the conference and three projects were presented in detail. The participants presented diverse practices they use for critical, grassroots, collaborative processes. These practices included enhancing education in Nepal by introducing critical thinking and making, a hackerspace for women and trans* individuals to generate resilience in Austria, and a project using open-source technologies to build technology for diversity and inclusion in India. Projects from Hungary, Brazil, Spain, and Indonesia applied as well, but were not able to present as they encountered funding or visa issues. A problem of access remains pertinent even in conferences that explicitly involve non-academics in academic discourse (Fig. 5.4).

⁴ Published on <https://criticalmaking.globalinnovationgathering.org> and spread via social media, the mailing list, and messenger group of the community, as well as the conference's channels.

reintroducing criticality (as per (Hertz 2015a). The workshop also served as a rare opportunity for much-needed exchange between academics and practitioners, who together further explored the diversity and situatedness of what other practices could count as making (Sipos and Wenzelmann 2021).

This learning was very important to how I communicated with my local counterparts about which projects could be relevant for researching grassroots critical technical practice in the exploration phase of the field research. It gave me an overview of which terms were used by academia and which terms the practitioners themselves identified with. Finally, it confirmed that the concept of critical making, even if it might need explanation in different contexts, resonated with the practitioners.

5.1.4 Alternative Maker Conferences and Learning Formats

Before outlining the field research, I would like to briefly discuss two gatherings that shed light upon potential directions, practices, and language used within the practice of critical making. In the summer of 2019, I attended two different events with similar goals: participation, critical thinking, and material practice.

Nø School Nevers⁵ was organized for the first time in July, 2019 as a summer school for critical research around the social and environmental impacts of ICTs in rural France. Garnet Hertz told me about this event, where he was invited as a mentor. The summer school held an intensive 2-week curriculum based on a mix of theory and practice in hardware and software hacking, upcycling obsolete technology, and circuit bending. It featured theoretical lectures on digital colonialism, hacktivism, and oppressive algorithms. As a researcher who is mainly engaged in theoretical discussions and observing the projects of grassroots practitioners, this was an interesting opportunity to learn hands-on programming and hardware hacking skills⁶. As for the summer school's relevance for this study, there were only two non-Western presenters at the summer school. One of them, Sénamé Koffi Agbodjinou, was a Togolese architect and anthropologist, the designer of

information
communication
technologies

⁵ <https://noschoolnevers.com>.

⁶ The starting point for creating the critical artefact during the summer school was questioning the lack of media freedom and plurality in some countries, and speculating about the possibility of self-printing *samizdat* information material to counter state-driven propaganda. A video of the machine can be found at https://www.youtube.com/watch?v=IWv_VXeVmhE&t=16s. Coincidentally, such projects were already started by citizens <https://www.nyomtasssteis.hu/about-us-2/>.

a neo-vernacular approach to African cities, and the initiator of a critical technical project in which the Western e-waste polluting local landfills is repurposed into 3D printers. Although I already knew Sénamé, this gave me an opportunity to interview him about his future plans. Later on, I invited him to the kick-off of the Critical Making EU-project, to share his participatory and post-colonial visions. The other person relevant for this study was Fred Paulino from Brazil, who was one of the founders of the Gambiologia collective. The relevance of this collective is that they engaged in the Brazilian tradition of *gambiarra*, or finding creative solutions with limited resources, often by recycling or upcycling materials, thereby questioning supposed technological obsolescence.

André
miniz
highlights
this in
his work

Dinacon⁷ is an “alternative conference”⁸, meaning that it has a fluid timeframe and no keynote speakers—only node leaders and experts running workshops. There is a 4-week period for attendance, and the program is set by the node leaders and participants who are on site at any moment in time. The only set parameter is the topic of digital naturalism, meaning that projects must engage in field biology, wild hacking, or interaction design, and the main goals of the conference are experimentation and knowledge sharing. In 2019, it took place in Thailand for the first time, in 2020 it was organized in Panama, and in 2022 it was held in Sri Lanka. Local communities of digital naturalism are involved, and while in Panama the main local partner of activities was the Smithsonian Tropical Research Institute and most of the node leaders and participants were Western, the founder of a critical forest preservation project also presented his work. He provided me with important insights into how he works with Indigenous people, useful for the case of the Network Collective with the Indigenous community, presented in Sect. 5.4. Case Study I. Here I learned of a worldwide network of self-made, upcycled feminine hygiene products, which I also encountered later in Indonesia (see Sect. 5.3.2).

At first, I was skeptical about how these gatherings could contribute to my research because both were attended mainly by Western students, researchers, and designers. They could afford them without a stipend—including me, as I

⁷ <https://www.dinacon.org>.

⁸ Such alternative conferences or unconferences with project-based attendance and ad-hoc programs are a recent development. Similar conferences include the hacker gathering PIF camp in Slovenia, and the HackteriaLab in 2014 in Jogjakarta https://hackteria.org/wiki/HackteriaLab_2014_-_Yogyakarta.

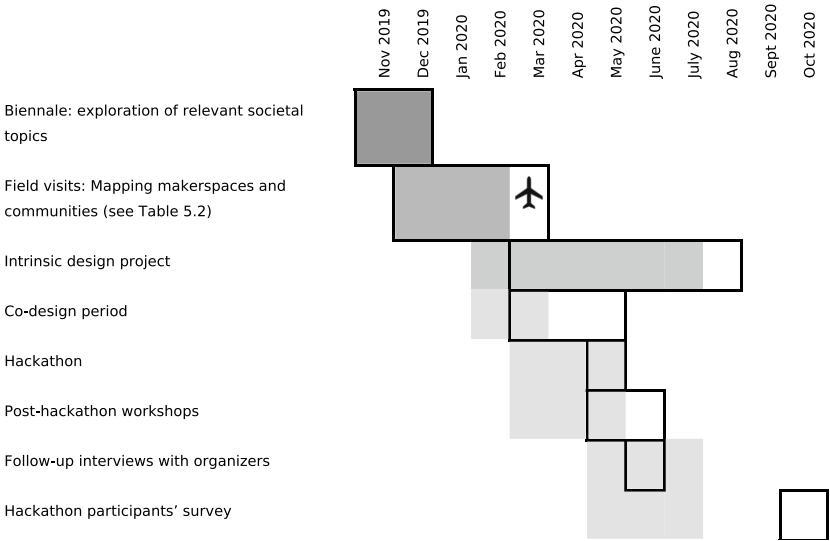
also had to cover the costs myself. However, I did learn new critical, reflexive approaches, questions to ask, and encountered critical technical projects. These experiences showed me again that patterns of similar societal and environmental issues can be recognized in the Global South, and that often similar solutions are being developed either with the help of open online repositories, or through local engagement and ingenuity. I also learned about the realities of “making” in non-Western climates. 3D printing might not function as well in heat and humidity, because they are calibrated and filament designed for continental climates. Setting up a makerspace, or a 3D printing room thus requires additional consideration.

5.1.5 Design of Field Research in Indonesia

In this subchapter, I briefly summarize the design of the field research. Data collection was conducted in Indonesia, on the island of Java. This starting point was selected based on my contact with the founder of the Art-Science Collective, whom I had known since 2015. She recommended other local initiatives to visit. Various media art experts, researchers, and makers who worked with Indonesian grassroots communities before also contributed to the list of potential initiatives. In November and December 2018, I received further input through an exchange with a potential co-researcher from Sanata Dharma University in Jogjakarta and an influential Indonesian curator also helped design the following research plan.

I received a scholarship from the DAAD to conduct the research that became the basis of Chap. 5 between November 2019 and June 2020. The goal of this field research was to map local critical grassroots initiatives and makerspaces and conduct a **participatory action research (PAR)** project with one of them. Here I would like to briefly note that my original timeline changed slightly, for two reasons. First, because of the pandemic, I had to leave Indonesia and redesign my research to continue conducting my research online, which caused delays. Second, although I originally planned a PAR project, I realized on site that following the **intrinsic design** processes of the grassroots innovators would lead to a much better understanding of their practice. Thus, the timeline of the project was not to be dictated by me, the researcher. Instead, I followed the grassroots innovators and their timeline when I scoped the project. This shift in the timeline is shown in Table 5.1.

Table 5.1 Field Research Timeline



Step 1: Understanding Local Societal Tensions: The starting point of my research was the “Biennale Jogja”, an international art and culture event. It asked critical questions about the future of urban planning, and whether cities were fair and democratic shared living spaces. I made a deliberate decision to start the field research from this event, as I was most interested in understanding the critical questions that local South-East Asian artists pose about their own societies and environments. The debate about such questions through material engagement seemed to be a sound starting point for understanding local societal issues and interests that could represent grassroots critical making. This event used techno-cultural hacking strategies to unite digital artists, activists, and makers from Southeast Asia to collaborate and prototype. Biennale Jogja is a sustainable network of practitioners and thus an excellent starting point for ethnographic research. While visiting the exhibition, I learned about so-called called “peripheral” issues, including locally relevant societal and environmental topics and concerns, and sociopolitical conflicts that people around the world live with. Many of these topics later came back in interviews and personal discussions.

For example, problems were inherited from economic reforms and infrastructure, which exploited nature and people during colonial times⁹ (Fig. 5.5).



Fig. 5.5 Hotel Purgatorio by Fajar and Murti. (Source: The Author)

An artwork addressed the labor rights of women working from home on tight timelines without any insurance¹⁰, which later was reflected in the discussions of the organizers of the #hackfromhome hackathon. Those who could not afford to stay home during a lockdown, like gig workers, were only paid when physically at work.

Step 2: Mapping Local Grassroots Innovators and Exploring Potential Projects: The second part of the field research was designed around two activities. First, I mapped the various potential grassroots critical maker projects on the island of Java that were recommended to me before the field research and during

⁹ Timoteus Anggawan Kusno: *How to exorcise the gravel in your shoe?*

¹⁰ Dian Suci Rahmawati: *Apakah Tubuh: Sebuah Ladang di Dalam Rumah.*

my visits locally. Members were interviewed to gain insight into the underlying context, history, motivations, practices, and processes of local grassroots innovators and to map and understand the communities, motivations, and actions behind the local maker ecosystem (Table 5.2). The second activity was the twentieth anniversary of the founding of the *Art-Science Collective*. “20 years: (...) We are Done” was the motto for this milestone, meaning the community was coming together to reinvent themselves once again, as they do every few years. Previously, such self-reinvention phases resulted in projects like their annual events. Their earlier international new media festival with participatory workshops inspired the world’s first and only week-long event and conference on post-colonial critical making, and the first official Fab Lab in Indonesia. The twentieth anniversary was thus an excellent opportunity to conduct research on the history, communities, activities, and the makerspace of the collective. Through this research I could understand the participatory practices, dynamics, and negotiations that lead to new projects.

Step 3: Project Period: During the third phase, I conducted project-based, participatory action research. The original plan was inspired by the non-traditional conference and learning formats I had encountered in France and Panama, including a co-design, co-creation, and co-evaluation period. Participants would have included interdisciplinary teams of scientists, makers, and citizens who would have worked on an existing project of the Collective or a new one they proposed. Instead, due to the pandemic, the founder of the Art-Science Collective reached out to me and two other women founders of grassroots organizations and proposed that we organize a fully digital and remote hackathon. As I had to leave Indonesia in March (marked in Table 5.1 with the airplane icon), conducting the participatory project remotely fit the research design. The timeline of the hackathon can be seen in Table 5.1, marked with thick cell borders as opposed to the original timeline marked with grey shades. The detailed investigation of this intrinsic design project is described in Sect. 5.5., Case Study II.

In the following subchapters, I fully explain the steps of the research that were briefly summarized here, with a focus on explaining the findings from the field in detail.

5.2 Mapping Grassroots Critical Making

It became clear during the workshop at the C&T conference that grassroots innovators do not identify with the term “critical making”. However, it did resonate with those who felt like they were not merely adopting mainstream practices (Sipos and Wenzelmann 2021). In preparation for the application for funding for the field research, during late 2018 and early 2019 I asked 6 people for recommendations: a community-based, DIY scientist who attended a Dinacon event in Thailand; an Austrian researcher who wrote her PhD dissertation on feminist hacking in Indonesia; a former media art event curator who is familiar with the work of Ratto and Hertz and has worked with media art collectives in Indonesia; and finally, a maker and hacker engaging in social good projects and communities in Singapore, who also co-authored the first paper on critical making in grassroots presented at the 35C3 Chaos Communication Congress (Sipos et al. 2019). I already knew an Indonesian critical art curator and a woman social innovator, and a critical making collective that grew out of a media art collective based in Indonesia. I also reached out to them for recommendations.

It was also confirmed during my field research that term “critical making” was not widely adapted or used to identify themselves, even by those who were recommended as critical makers. Thus, when starting to map potential projects, I had only a few starting points for who grasped the terms critical making, social innovation, and grassroots innovation. When asking others for recommendations, for example in the field, I had to describe the potential types of projects relevant for this research differently to find ones that suited the goal and the criteria. Depending on the background of the people I was talking with, I used the terms “grassroots innovation”, “innovation by everyday people”, “maker projects”, or described groups of people who use technology to solve a social issue and who think about technologies and society critically. Indonesian counterparts used vastly different vocabulary, such as NGOs, collectives, designers, media artists, and hackers. Even if the terms did not directly cover my understanding of the people, groups, and their activities, I did not exclude them from the research. Such words mean different things for people with different backgrounds, and the risk of missing a critical making project was significant (Fig. 5.6).

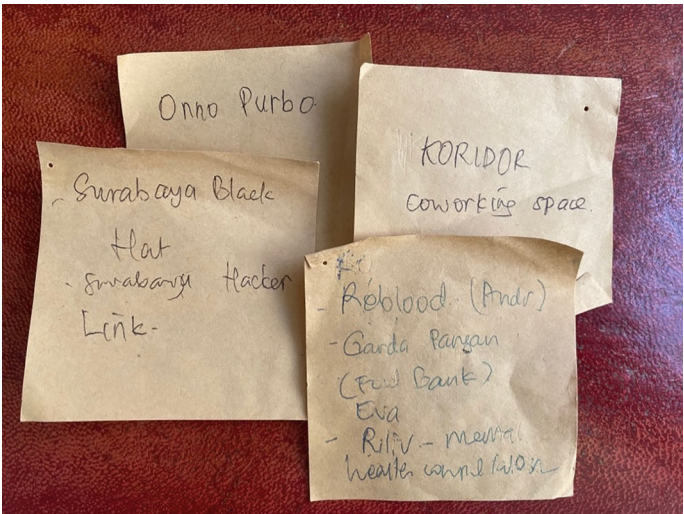


Fig. 5.6 How Interviewees Recommend Local Initiatives. (Source: The Author)

This helped me learn about various projects, some purely oriented around maker culture and entrepreneurial skills training, and others closer to media art. Most of them addressed a societal need or use, and many had a critical aspect. The relevant spaces, collectives, and individuals that I visited are summarized in the timeline table (Table 5.2).

Table 5.2 Timeline of Mapping Grassroots Critical Making in Indonesia

Community	Visit(s)	Location
Art-Science Collective	November 2019–March 2020	Central Java
DIYbio Collective	November 2019–March 2020	Central Java
Anarcha-Feminist Collective	December 2019–January 2020	Central Java
Visual Art Archive	February 2020	Central Java
Critical Artist Collective	November 2019	Central Java
Space and Place Makers	December 2019	East Java
Banned Books Library	January 2020	East Java

(continued)

Table 5.2 (continued)

Community	Visit(s)	Location
Traditional Art and Technology Collective	January 2020	East Java
Hacked Precious Plastic Machine	January 2020	East Java
Makerspace and Tool Library	January 2020	East Java
Critical-survivalist Collective	January 2020	East Java
Network Collective	March 2020	West Java
Social Innovation and Bee Project	March 2020	West Java

5.3 Grassroots Critical Making

In this subchapter, I will highlight cases of grassroots critical making. As I will demonstrate, the mapping process described above led me to projects that fit the research framework. Most of them address a societal issue or reflect a societal tension or injustice. As making is one of my criteria for analysis, I will briefly outline how the landscape of makerspaces looks different in Indonesia.

Then, I will describe projects that did not fully fit the framework of this study for some reason. Some had a low level of criticality or self-directed technology development, or were a social innovation project rather than a critical making project. This subchapter aims to support the study’s goal to extend the term critical making to the grassroots by showing what did not fit the framework through short examples.

Most of the projects visited and analyzed here contain some level of grassroots-ness, criticality, or making. Their activities are decidedly kept broad to allow this study to show the diversity of grassroots ingenuity. Afterwards, I will show critical grassroots innovation projects I identified on the ground. In ethnographic vignettes I will demonstrate how Indonesian collectives act as vessels for collaboratively developed ideas. Some of these collectives are more critical, while others have a stronger focus on technology, but all of them are rooted in grassroots communities. Finally, I will detail the two main cases of grassroots critical making and the respective intrinsic design projects that help answer Research Questions 2 and 3. These two cases—supplemented by findings from the vignettes—will then be compared.

5.3.1 Makerspaces in Indonesia

Before discussing critical grassroots innovation projects, I'll give a brief overview of the different types of spaces used by communities for collaborative technology development. When speaking of Fab Labs, hackerspaces, makerspaces, and innovation spaces, a fairly cohesive picture comes to mind. In the Western world, these spaces combine analogue and digital tools for activities such as woodworking and metalworking. Analogue instruments for DIY and repair share space with 3D printers, laser cutters, boxes full of microcontrollers and robot parts, and laptops for computer-aided design and hacking¹¹. In Indonesia, I was directed to very different places when I asked where I could find makers. Instead of some centralized blueprint, local contexts and the community gathering in the spaces generated the nature and type of makerspaces.

Bandung, for example, is a city of engineers and design. An interview participant mentioned that makerspaces started spreading here around 2017–2018. However, as he put it “these focused on selling gadgets for rich people, most Indonesians couldn't afford them” (G.H.I. 2020a). Their business models didn't work in the local context and many went bankrupt. Today, despite that trend, there are still 10–15 self-proclaimed makerspaces and local versions, like a “DIYfab co-lab¹²” in a city of around 2.5 million people. Since 2020, a new makerspace focused on education and incubation has been developed at the Bandung Institute of Technology by the US-American NextFab Foundation¹³.

Jogjakarta is seen as a cultural capital because it plays a central role in preserving Javanese fine arts. Many collectives here have a background in visual, performing, and media art from the Indonesia Institute of the Arts¹⁴. The city with around 374,000 inhabitants is relatively small and is known to be friendly and welcoming, but makerspaces are rare. One of the interview participants highlighted that the first local Fab Lab, supported by a Dutch organization and focused on education, folded because the host (the Art-Science Collective) could not identify or serve further local needs. After the basics of making were taught, participants started asking the organizers what they could do with the knowledge they gained. Unfortunately, the organizers did not have an answer to that

¹¹ A blueprint of FabLabs, including descriptions and zip files with layout designs, can be downloaded at <https://fabfoundation.org/getting-started/>. Even the target communities a FabLab might service are described.

¹² <https://diyfabcolab.com>.

¹³ <https://nextfabfoundation.org/training-academy-program-launch-duplicate-3/>

¹⁴ <https://english.isi.ac.id/profile/intro/>

question, and with funding running out, the Fab Lab was abandoned. **Simply importing maker practices, tools, and workshops without taking the local context into account did not suffice** (see Sect. 5.5.1.3.).

issues of sustainability when establishing makerspaces

Conversely, Surabaya used to be one of the biggest trading ports in Asia and remains a center of economic and business activities. The makerspace described in Sect. 5.3.2. utilized participatory sessions to define the needs of their users in an attempt to become sustainable, before they even set up the makerspace. With a woodworking company as their main sponsor, woodworking is the most prominent activity. However, they also cooperate with the neighboring co-working space to offer creative and micro-entrepreneurship skills training. They have set a good example on how to consolidate the global trend of makerspaces with local needs in the vast informal economy of Indonesia.

Throughout the island of Java, the most prominent spaces are small and specialized workshops that located in the houses shared by local collectives. The above-mentioned Fab Lab still has its remnants in a small room in the shared house of the collective, and young people from the neighborhood often curiously peek in and join them in learning how to make things. Also in Jogjakarta, in the house of the DIYbio Collective, a separate room contains the tools and artefacts (e.g., an open-source microscope developed to study E. Coli in the water) is accessible for researchers and makers spending their residency at the DIYbio Collective's house.

Other communities, like the *Belonging Collective* and the *Network Collective* (see following chapters) also have a house open for their members and for events, exhibitions, workshops, and most importantly, *nongkrong* (the practice of hanging out and ideating), but not for making. Their projects are developed in the *kampung* (village, or the neighborhood of a city) and in rural areas. Their version of making is based on traditional forms of handcraft and DIY, using local materials and skills, or (in Case Study I) appropriated technologies that are adjusted to fit local geographic circumstances.

5.3.2 Socially Innovative Practices

In this subchapter I offer insights about the projects that were recommended to me because they demonstrated social innovation, criticality, or making. Makerspaces had been failing in Indonesia for multiple reasons. In terms of the human factors, this Western concept did not resonate with locals. Even if they learned how to use 3D printers or CNC machines, they weren't sure how to utilize those skills beyond a hobby. Moreover, as one of my interviewees told me, many makerspaces

adapted the Western business model of “selling kits” that were too expensive and not seen as particularly useful by locals. Additionally, those who make critically most often become a member of a collective that reflects their skills and values and provides them with a space where tools and knowledge are shared in the community.

Before learning about failed attempts at establishing makerspaces in Indonesia, I asked my contacts to help me find Western makerspaces, meaning that I was looking for shared spaces with rapid prototyping tools in which people would develop socially innovative or critical making projects. While Western makerspaces exist, they are mainly in West Java, as that is where the oldest institute of technology is located and that’s where design (or graphic, interior, and product *desain* (Azali and Budiman 2016) is a common practice.

There is a relatively new makerspace in East Java too. Having learned from failed attempts of others setting up makerspaces elsewhere, the founders started off with a crowdsourcing-consultation process by inviting Javanese collectives and other experts to help them figure out how a makerspace could work in the local context. In this collaborative mapping process, they asked themselves what was needed or necessary, and how they could contribute to the Javanese society by finding a niche to fill. They realized that people didn’t understand the term “makerspace”. Thus, they decided to call themselves a “tool library” to make their offering easier to grasp. In addition to woodworking¹⁵, they offered workshops on 3D printing and sewing, and members can further develop their skills to become micro-entrepreneurs by selling the things they made. Much of Indonesia’s workforce generates income from these types of informal economies (Khoirunurrofik et al. 2022).

The founders of the makerspace continue looking for opportunities to further develop their membership and offerings. In the future, the makerspace could become a community of critical makers, because they continuously search for opportunities to engage with society in critical-material practices. This aspiration is noticeable in three examples. First, the founders reflexively co-designed their launch strategy with Indonesian collectives and civil society. Second, they cooperated with the anarcho-feminist collective to support homeless women in their region. Third, one of the founders joined the #hackfromhome hackathon and became a member of a post-pandemic circular ecology project in Indonesia.

¹⁵ Woodworking is the main activity because the makerspace received tools as a sponsorship from a company that produces them. Such practices were common in East-German workshops during the cold war (Sipos and Franzl 2020) and business and state interest still often influence what tools are available in makerspaces in the West too, by donating software and hardware (Smith et al. 2017).

However, currently, criticality is still low. This became visible when they collaborated with the aforementioned collective, which is known for their workshops teaching women how to sew reusable feminine hygiene products. A key insight here is that the makerspace's framing differs from that of the collective. In the case of the makerspace, the workshops are about teaching women living in a nearby homeless settlement how to sew and sell sanitary pads to garner income while helping save the environment. The makerspace aims to support environmental and social causes, but they are not particularly critical. The collective's framing is significantly more critical; its members run workshops to teach skills, which serve as an entry point for discussions with women and their education regarding taboo topics including such women's rights and reproductive rights, or the right to earn a salary and become self-sufficient. The difference in their interpretation is striking (Fig. 5.7).



Fig. 5.7 Interpretations of Reusable Cloth Pads: as Enabler of Informal Economy in Surabaya (04.01.2020) and as Critique and Human Rights in Jogjakarta (14.01.2020). (Source: The Author)

I also visited two projects, both socially innovative projects that aim to contribute environmental protection that use technology designed by students.

In Eastern Java, there is a group of young entrepreneurs who used the blueprint of the open-source Precious Plastic Machine[★] and hacked it to fit their purposes. In their attempt to alleviate Indonesia's massive plastic problem, they collaborate with the local "trash bank" or "waste bank"—a social entrepreneurial project, in which the waste banks act as intermediaries between households and waste buyers/recyclers (Ashoka Foundation 2010)—to acquire appropriate plastic. Trash is turned into 3D printing filaments produced by the modified Precious Plastic machine, a functionality they developed themselves. They also create upcycled plastic furniture in modern minimalist designs. These are difficult to market, as those who can afford them look down upon plastic, as the founder told me.

In Western Java, a "bee sanctuary" has been set up in an abandoned field in an urban area. IoT-supported beehives have been built, monitoring the bees via the internet and collecting vital data that is shared openly. On the same field, urban gardening and growing vegetables and fruits is also encouraged. This university-supported project inspires students to learn about agriculture, the environment, and technology development. They also participate in social entrepreneurship incubation, essentially educating the next generation of digital social entrepreneurs in a hands-on way. Topics that the capacity-building workshops focus on are in the entrepreneurial field as well, including methods like agile and scrum for purpose-led or social businesses.

These examples are how making is used to achieve societal and environmental goals. They use a type of technology that enhances their solution as their starting point, and their approach would be better categorized as "social entrepreneurial" (Moulaert et al. 2014) than "grassroots critical making". The key insights from these examples are how vastly differently two different communities can interpret the same artefact, whether as a tool for micro-entrepreneurship, or for discussions and awareness raising on taboo topics, essentially serving as grassroots activism.

5.3.3 Pluriverse of Grassroots Innovations

How we think and design, according to Escobar (and other decolonial thinkers) remains based on separation from the natural world, resulting in a disregard for local knowledge (Escobar 2016). Moreover, hegemonic practices of participatory design often silence forms of participation in the Global South. Such practices create ways to deal with historical, societal, and environmental topics through design (Szaniecki et al. 2020). Escobar thus recommends an ontological reframing of design, building on transition imaginations and positing the need for radical transformations. Such transformations are possible through the investigation of

autonomous design done in communities that design themselves, similar to the idea of intrinsic design in HCI (Bødker et al. 2016). He echoes the Zapatista and their invitation to construct a pluriverse, or “a world where many worlds fit” (page xvi, Escobar 2018). His call is in a similar vein as Schumacher’s ideas to redesign the economy by providing technological solutions on a small, context-fitting scale (Schumacher 1973). Ultimately, one of the most important problems with appropriate technologies was that they ended up being deployed in a top-down manner. It is thus timely to explore the “pluriverse” as designed from below. As previously outlined, grassroots innovation endeavors are plural, diverse, situated, and localized (Sipos and Wenzelmann 2021).

The second goal of this chapter is related to the research framework of the study. As previously explained, my research was inspired by the idea to extend the term critical making to grassroots innovators’ practices. By mapping practices and describing them as vignettes, I do not wish to exclude any project or normatively define what qualities as critical making. Instead, I exemplify what grassroots critical making might look like. Thus, in this chapter, I will portray some of the communities I visited, but that I did not end up analyzing further as a main case. Each provides researchers with interesting insights into the pluriverse of Indonesian grassroots projects.

These projects will be presented as **ethnographic vignettes**—short summaries of social situations often made sense of after the actual event, subsequently rewritten, and polished for use in a specific text (Hohn et al. 2020). Inspired by this approach, I will first briefly describe each project, add important contextual information, and then offer a thematic analysis based on the three criteria of the research framework. These vignettes help create a baseline understanding of what critical technical practice in grassroots might look like by highlighting how other projects deviate from the main cases and thus creating a contrast for the reader.

5.3.3.1 Access to Information: Banned Books, Digital Rights, Counter-Mapping

The freedom to access digital or printed information has been fluctuating throughout human history. In the Eastern Bloc after World War II, information that had not been approved centrally would be reproduced in self-published formats known as *samizdat* to evade censorship. Books that did not serve the ideological regime have been burned multiple times throughout history, such as in Nazi Germany

in the 1930s. Similarly, in Indonesia, books that “could potentially disrupt the public order” were banned for decades after a presidential decree in 1963, and even though the Constitutional Court ruled in 2010 that this was unconstitutional, the decree has lasting effects even today. In recent years, the Attorney General’s Office decided to conduct raids and confiscate books on the (now defunct) Communist Party of Indonesia. In East Java there is a small, citizen-run library that carries such books. They also have an extensive collection of zines and books on technology and classics. However, due to those banned books, they have been raided by the police in recent years. Thus, they have an information sheet hanging on a wall on what to do when that happens (Fig. 5.8).

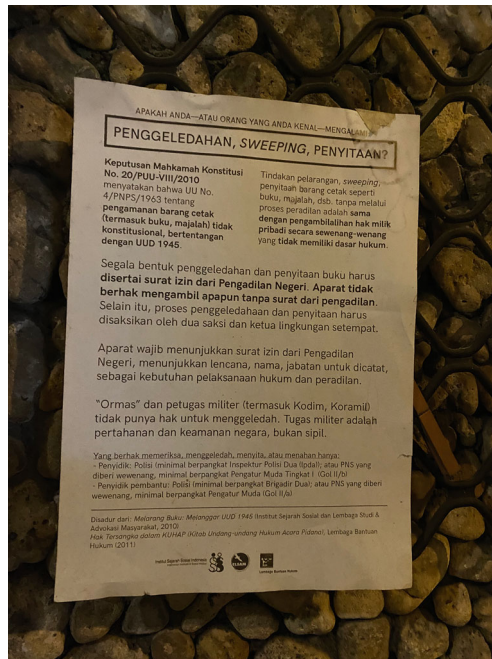


Fig. 5.8 “Search, Sweeping, Confiscation?” What to do if a Police Raid Comes. (Source: The Author)

It is not just this small, independent library that is being targeted. The demonization of communists started by the US government during the Cold War led to the mass killings of 1965–1966 under the pretense of tracking down communists (Bevins 2020). The anti-colonialist, anti-imperialist group Gerwani (Gerakan Wanita Indonesia, Indonesian Women's Movement) represented the largest feminist movement between 1920–1960 s. It had millions of members and ended with their imprisonment after they were blamed for the murders of six generals, which led to the justification of the aforementioned mass killings (Wieringa 1993; McGregor and Hearman 2007)¹⁶. The mass killings and how those who conducted them used ideologies to justify their actions is uncovered in the unsettling documentary interviewing former death-squad leaders, “The Act of Killing” (2012). The “Jakarta Method” of genocide—destroying the political left and movements attempting to reform the country developed by Washington—was so “successful” that it was replicated in other parts of the world too, including Latin-America (Bevins 2020). These were taboo topics for a long time, but the anti-communist sentiments remain in the Indonesian society and are being used to target communities today. As the founder of a local artist community in Central Java told me, they were raided a few years ago due to allegations of communist activities, and were so ostracized that they had to move¹⁷.

The library detailed in this vignette is not planning on moving. Instead, it has established itself in the local community and instead responds to raids as recommended by lawyers (Fig. 5.8). Libraries in general count as important social infrastructures around the world—places where people do not need to consume, they can just be or access information. The library is a place for communities, a “third place” (Oldenburg 2001) where likeminded people can gather and develop projects together. A part of the library also functions as a co-working space, an office for independent journalists (sharing the rent and contributing to the financial sustainability), and a space for workshops and exhibitions. They also do important educational work on digital issues and rights like privacy¹⁸, which is

¹⁶ See also Wuschitz's recent work <http://kunstraumproarte.com/stefanie-wuschitz-coded-feminisms/>

¹⁷ <https://mediaindonesia.com/nusantara/41339/menjual-isu-agama-dan-komunisme>

¹⁸ They hosted e.g. Mozilla's Glass Room, which is a travelling, interactive exhibition that aims to demystify technology <https://www.theglassroom.org>.

not a topic I encountered on many occasions during my field research. This work is crucial, as Indonesians' digital rights are barely protected. Laws are politicized and used to silence journalists that express critique (Tapsell 2019), mergers create monopolies of digital mega-corporations (Reuters 2021), and massages and online stores are able to collect data limitlessly about users. Streaming services and websites are banned under the auspices of protecting locals from harmful content. In reality, these efforts are corporate decisions enabled by the confusing legalities of the biggest telecommunication firm's owner being the State (Hermansyah 2016).

One of the projects developed by the library community is a decolonial mapping project. In terms of the framing of the library, they state that they are a community for an emancipatory, sustainable future, and the different resources and activities feed into this vision, including the mapping project. In terms of its practitioners, the mapping project is a grassroots project, as it is carried out by a group of citizens who crowdsource information from everyday people they encounter during their mapping sessions. They learn as they go, including about how to use the technology and which open-source tools could support their work. The contributors' main occupation is not working on the project, so the project itself is carried out without any substantial financial support. In fact, the library itself and the different projects around it are also all grassroots driven.

The project expresses critique in multiple ways. First, its main objective, as I was told by the initiator, is to decolonize maps by removing the colonial gaze from Indonesian mapping. What this means is that the mappers must be reflexive and self-critical, as they themselves learned about maps with a cultural layer inherited from foreign textbooks. Before colonialism, orientation was based on landmarks and wind directions, not on Western methods of orientation like cardinal directions, street names, and postal codes. Those are the tools of the colonizers and a completely different approach to representation. The initiator kindly took a day to walk us through workers' cafés. She also showed us the Europe quarter which still bears witness to the colonial times in the form of Dutch words and paintings on buildings (Fig. 5.9).

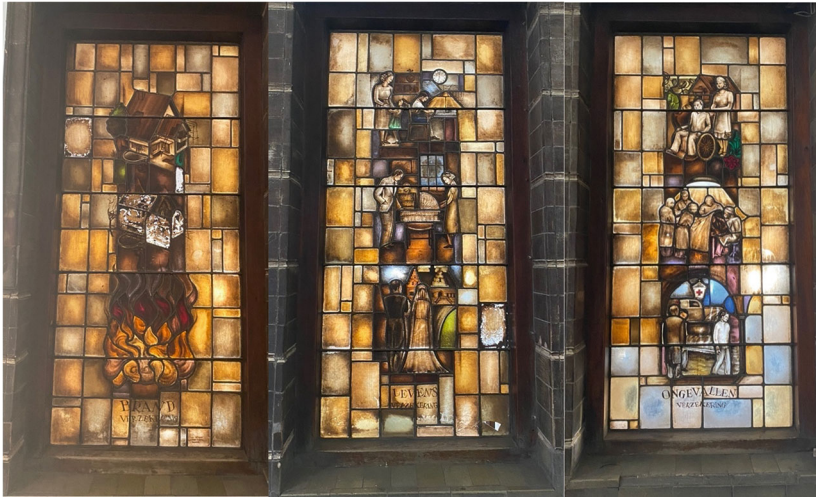


Fig. 5.9 Triptych of a Colonial Past: Tinted Glass Window Advertising of a Dutch Insurance Firm. (Source: The Author)

The walking map project inspired a follow-up project that builds on “counter mapping” (Peluso 1995) or “subversive mapping”, which is supposed to greatly increase the power of people living in a mapped area, who control representations of themselves (page 387, *ibid.*). The project maps spaces and places of local interest, like traditional coffee shops, markets, riverside villages, and railroad tracks. It aims to counter the limited access to contemporary knowledge about the city. At the same time, it asks questions about why women, children, and elderly have limited access to mobility, especially public transportation. The spirit of the project is to generate belonging through loitering for those who are not supposed to loiter in contemporary urban spaces, as inspired by Phadke et al. (2011).

In terms of technological design, it is a maker project in that sense that the maps are designed in-house with the decolonial lens in mind, forcing the design of the maps to be reinvented. An interesting example of mapping differently can be seen in the three walking maps of the old town (Europe, Arab and Chinese quarters). These maps don’t use metres to measure distance, and instead indicate by finger distance how long it takes one to get from one place to the next. Also, they contain only the names of the main streets, and the places to visit are represented by minimal illustrations. For the mapping practice, existing tools are used, such as the open-source OpenStreetMap (OSM) and StreetComplete. Open

source is important because today the IP of most maps is protected, thus, the knowledge is owned by a company. OSM contributors instead contribute to the open sourcing of critical knowledge about their environments.

5.3.3.2 Belonging in Urban and Rural Contexts

Grassroots mapping is also done elsewhere in Indonesia. In a different city in East Java, a local community arts collective has used Ushahidi, a popular open-source platform used for political mapping projects. It has been used for post-disaster relief in Nepal and Haiti, supporting fair elections in Kenya, and countering sexual violence in Egypt. It is a simple but efficient tool that helps people crowdsource geographical crisis information (Okolloh 2009). The founder of the Belonging Collective showed me large, hand-drawn maps in their collective's house depicting different neighborhoods (*kampungs*)¹⁹. This time they used Ushahidi to produce a “city map” of unmapped assets—such as small shops, facilities, areas of public interest—while highlighting urban challenges, such as areas prone to flooding and crime. Ultimately the idea was to inform political decision making through citizen reports, including how to use the available spaces to make the city more livable, and build urban resilience (Webber et al. 2021).

The Belonging Collective organizes events, collects zines, and cooperates with local artists to bring the *kampungs* to life. They did this for example by experimenting with augmented reality applications that project artists' works onto walls and spaces. They also developed a mobile “space and place making machine”—named after a popular local food cart—with which they can transport any equipment, and by opening the palettes hidden in it, create a stage for performances. As the founder told me, aside from such analogue making projects, they use existing technologies. Technology reuse had caused problems for them before. For example, the AR app they used was bought by an US corporation, they first renamed it and then discontinued it—showing one of the recurrent issues with closed-source software.

When I asked the co-founder of the collective what change he thought such citizen-driven projects would bring about, he said

“Nothing drastic, but they feel like they *belong* to the *kampung*. There is less crime and people do more creative projects.”

The feeling of belonging in alienated urban areas is just as relevant as in his other endeavor, which takes place in his rural home village. Here he works with

¹⁹ They also have an extensive collection of zines.

a local group of young people to reinforce their pride in their village, and thus reduce violence and the reducing number of inhabitants due to migration towards cities (also known as “brain drain”)—both major societal issues. They achieve this by creating third places and highlighting local heritage that can reinforce pride, especially in appropriate technologies, as I will explain in the research criteria below.

The project in this rural village is run by a local group of young people, kickstarted by the founder of the aforementioned collective, who comes from the village. It is truly grassroots, community based, and free of hierarchies—an ideology that is reflected in their motto, “no leader just together”. Their *grassroots* activities are ideated, designed, and carried out by the inhabitants of the village (mainly young men), in cooperation with the local authorities. They reinforce belonging by creating third places where people can gather. They have also reinvented traditional festivities, including a Ramadan market and the celebration of the rice harvest, when every member of the village cooks and invites others to their homes. To bring these traditional events closer to young people, they started intertwining them with art, music, and dance.

In terms of criticality, both projects build on the small scale of the *kampung* and the village. They reinvented the space and place to help people feel like they belong and just hang out (*nongkrong*), similar to the library described above. This supports wellbeing and democracy because it gives people a space to discuss and exchange. It also helped alleviate migration from the village; young people are now less likely to leave to work in a bigger city. Moreover, the project also showed that how the “public” spaces are used is exclusionary. On Java, people meet after the sunset because the working day is over and that’s when the heat becomes bearable. In this society, however, it is still inappropriate for women to leave the house after sunset, thus they are excluded from community life.

In terms of technology development or making, the project is an excellent case of how self-directed development can bring “new flair” to appropriate technologies. Ancient habits and spaces redesigned to be more accessible to youth, and local techniques and appropriate technologies like brick making, bamboo, and wood carving are presented in ways that highlight their advantages over modern technologies in the local context. For example, one place that celebrated vernacular technologies was a bamboo woodworking workshop, where traditional furniture, drinking cups, and straws were made, to be sold through fair trade

shops. Having this workshop located in the village creates jobs and improves environmental sustainability, which is especially important for the whole of Indonesia (Fig. 5.10).



Fig. 5.10 Rural Bamboo and Woodworking Workshop. (Source: The Author)

Another example of locally developed appropriate technologies is a small factory that produces hand-made clay bricks, sun-dried and finished with a traditional open kiln smoking technique. As they explained, this technology is going extinct, and the factory might go bankrupt soon because industrial bricks are cheaper and have been advocated for by the government as a prerequisite of a modern life. The local experience is that the clay bricks are not only more sustainable, but also “breathe” better and provide better protection against earthquakes as they’re more flexible than industrial bricks (Fig. 5.11).



Fig. 5.11 Traditional Handmade Clay Brick Factory. (Source: The Author)

On a much larger scale, the famous Jatiwangi Art Factory (JAF) in West Java managed to incubate contemporary art and cultural practices within local life in rural areas by hosting international residencies and turning the local ceramic factory into a place for exchange. As the JAF state on their website, the projects they host have critical missions, and are “specific projects in focuses of rural-urban issue. The character of an industrial society tends to be aggressive and hardworking, and we have discovered many traditional art forms that can also function as tools”²⁰. This rural community doesn’t want to become an international hub or festival. They would rather incrementally revive and reinvent the local traditions for themselves, so that the brain drain phenomenon is stopped, and locals are inspired to stay because they belong.

5.3.3.3 Traditional Crafts and Robots

Bringing vernacular knowledge and traditions to modern youth is also one of the key activities of a different collective in East Java. This multidisciplinary community is one of the longest standing communities that emerged from the New Order period, similar to the collectives in Case Studies I and II. This collective focuses on interdisciplinary art practices, and have built an oven in rural Indonesia for the local community that can smoke food and fire ceramics. More prominent in their work is their hardware hacking, as well as how they share ideas, skills, and knowledge in social processes through educational workshops and amongst themselves.

²⁰ See <https://jatiwangiartfactory.tumblr.com>.

They are a self-organized grassroots community, bringing together people with multidisciplinary backgrounds in art, music, and programming. However, on the surface they might not look like a grassroots community because they often take on professional projects. The main reason for their professional orientation is contextual. The city where their space is located is relatively expensive, and rent can be 8–10 times higher than in Central Java, according to an interviewee. Thus, they must generate income to upkeep their shared house, and this financial demand influences their practice, as they have to fulfill expectations. High and rising rents can make it difficult to maintain shared houses where people can safely experiment. As the aforementioned interviewee shared with me:

“[Central Javanese] collectives have more opportunities to play and experiment, [Eastern Javanese] collectives are financially more squeezed” (F.H. 2020a).

Their criticality lies partly with bringing DIY and hardware hacking to people through workshops. In these workshops, they remix digital and physical maker tools—such as robotics²¹, microcontrollers, sound and video—with traditional arts, crafts, and music. During festivals they also reach out to the younger generation and inspire them to learn about their heritage. Their focus is not on inventing new technologies, but showing people that instead of commoditizing digital and traditional art, creativity and curiosity can be fostered through improvisation.

In terms of *making*, aside from hardware hacking, the collective nature of the practice stands out. Everything from the ideation process to the outcomes are openly shared. As highlighted by the DIYbio collective, the terms making and hacking, in their understanding, erase hundreds of years of Indonesian practices built on mutual aid, exchanges, and community ideation. This collective highlighted their Indonesian roots as well. Their house consists of rooms that have individual workspaces full of artifacts of past and current projects. There is also a room reserved for the community to gather, share ideas, and hang out (Fig. 5.12).

²¹ They, for example, organized a “hebocon” in which the collective ran the library, as described above. A hebocon is a fun activity that also takes place in Western hacker conferences. It’s a fun robot competition for “poor quality” robots, meaning that high-tech robots are not allowed. Hence, many of the robots are made of recycled materials and throwaway parts, fighting to push each other off the table.



Fig. 5.12 Traditional Arts and Sockets, Pins and Jacks. (Source: The Author)

While the artefacts are the most visible outcomes of their activities, the process of “hanging out” (*nongkrong*) is what leads to those projects. When I asked one of the co-founders what this practice looks like in reality, he said they would meet frequently without a predefined agenda, “just sitting and not doing anything” (H.H. 2020). During *nongkrong* the community members talk collectively about their observations and ideas for projects. Some ideas are abandoned, others might be revisited later—“when the time is right”, as the co-founder told me (ibid.)—and some immediately developed collectively. Which route a project takes depends on whether the other community members find the idea interesting, and it builds on existing skills and available resources. Which ideas end up being realized and which ones don’t is also driven by whether there is an upcoming event or exhibition to showcase projects.

The collective also has a mobile makerspace. I briefly touched upon the Belonging Collective’s “space and place making machine”. The idea with this contraption was to fit a small stage and sound system into a box so that “pop up” events could happen anywhere. This collective share a van that they rebuilt for their own purposes, because they needed to carry expensive electronics and priceless art pieces to events and workshops and cannot travel there by train or flights. It can be seen as a type of mobile and modular makerspace. The maker tools are not permanently installed, as customary in Western mobile makerspaces, because the process of making and preparing hardware for workshops takes place mainly in the collective’s space. Only tools and things needed for the event are put into the van (Fig. 5.13).

making,
collectivity,
mobility



Fig. 5.13 The Collective's Van. (Source: The Author)

5.3.3.4 Biohacking for Safe Drinking Water

The first time I visited a DIYbio collective's house, my host (one of the co-founders) was nervously checking the level of water in the well. He told me that the level was very low and had been for a while, despite recent showers from the recent rainy season starting. He suspected two possible causes. The low water could have been a sign of climate change, or someone, most likely a hotel, dug deeper wells than residents, diverting water from the locals. If it was the latter, he thought out loud, which level of which bureaucratic office would he need to approach? If the first, what could be done at all?

This first encounter immediately highlighted three systemic, wicked problems:

First, many locals rely on water from their own wells, despite the town being urbanized. Some locals trust this source more than the water coming from the

pipes, though both are contaminated. Water from the pipe is supposed to be a “protected water source”, but 67% of household drinking water is contaminated with *E. Coli*, according to a report by the UNICEF (Cronin, Odagiri, and Aryad 2017). Wells are contaminated with coliform bacteria, cancer-causing nitrate (G. D. Smith et al. 1999; Muryanto et al. 2019) and lead (Musfirah and Rangkuti 2019). In reality, safe drinking water is only available from water dispensers, for which large (20-liter) bottles need to be bought at shops.

Second, climate change has caused numerous floods and landslides in some places, and heavy droughts in other. Both ends of the water disaster scale destroy livelihoods, erase biodiversity, and wreck natural ecosystems (Case et al. 2007). There is also proof that **least-developed countries (LDC)** are the most vulnerable to climate change and least able to recover from climate stresses (Vandeweerd 2011). Indonesia counts as a developing country, and the negative environmental and socio-economic problems caused by climate change are predicted to be major (Case et al. 2007), especially for poorer communities relying on agriculture.

Third, grappling complexity of Indonesian bureaucracy has been described as “pathological” (Andhika 2017). While contemplating where to start with his investigation, the founder of the collective explained to me that, in his experience it was unclear which bureaucratic authority is responsible. Was this a neighborhood or *kampung*-level problem due to the wells, or a city-level problem due to probably multiple neighborhoods affected? Could it be a regional problem, because the river runs through multiple municipalities? Which authority deals with it? Does this problem represent a question of water quality, health, or something completely different? Unsurprisingly, instead of dealing with complex bureaucracies, people use their energy to fuel grassroots ingenuity.

The collective describes themselves as “a citizen initiative in art, science and technology”. They are known for their open-science hardware projects of DIY biology and biohacking. Their members experimented with different organic technologies before they founded the collective. One of them told me he had engaged in biodiversity conservation in a local forest before becoming a member. In 2011 they organized a walk to explore the river, while organizing their first water quality testing with the microbiology lab of the local university. Their first official project took place in 2012, and was an experimentation with fermentation processes. This experiment was a reaction to the government increasing alcohol taxes, leading to people dying from consuming bootleg

methanol-containing liquor (*anggur pengangguran*, or “unemployment wine”). The grassroots answered with developing safe fermented wine from local fruits and sharing the fermentation process openly. Three founding members of the DIYbio collective “forked” from a historical collective I discuss in Case Study II during this time due to debates about recognition of authorship of an artwork. After that, they mainly worked to improve water quality, which resulted in a long-term project I will further elaborate below. It also led to the development of an open-source microscope hacked from a webcam and a low-cost microcontroller. In these community science projects, low-cost tools are developed that are easy to share openly.

The river project originally was aimed at helping people living in a homeless settlement get access to drinking water. As one of the founders explained to me, they managed to get media attention directed to the problem. However, media attention had unintended negative consequences; the illegal settlement was quickly removed by the authorities and the people had to find a new place to live. Afterwards, the collective became more careful about public communications. The local university also instructed them not to share or publish their findings regarding harmful materials in the drinking water *until they found a solution*. To the university, the results were so bad that publishing them would only have created “fear and chaos”. Since then, they have been working iteratively to develop the project into a solution that “really helps”, for example by influencing policy makers. This project has been put on hold because of the aforementioned siloed bureaucratic system. What is left in their power is to develop something useful. Over the years, the project has been built alongside locals, researchers, biohackers, and artists. It grew from measuring the levels of bacteria and chemicals to a potential project that could improve the health of residents. Technologically, UV lights have been experimented with to clean the water, as well increase cost efficiency.

One of the members sparked disagreement after she said the project grew organically. In response, the founder felt like this wording “erases everybody who collaboratively worked in there”. A collaborative approach thus does not mean that individual contributions are erased; without the input of a biohacker, they never would have thought about microplastics and hormones in the water. “Actually, every year, there is always someone who give us more knowledge”, she said (F.A. 2020) (Fig. 5.14).

hardware
design
“forking”

potential
risks &
consequences
of
visibility



Fig. 5.14 Water Filtration System of a DIYbio Collective. (Source: The Author)

The DIYbio collective is a grassroots critical maker community. They are a grassroots collective because they are an independent community-based organization. While many of them studied engineering or art, their collective is not associated with any universities or similar institutions of higher learning. They do collaborate with international networks that are interdisciplinary and include artists, engineers, and researchers. They rent a house, which the community shares, from a family member. Thus, the space is rather affordable for people who would like to experiment with open-source hardware or artistic residencies.

They are critical because they ensure that the technology they develop is oriented towards the community on multiple levels. First, it addresses issues that concern the local communities, and represent larger environmental problems such as the loss of biodiversity, loss of farmlands, bootleg alcohol poisoning, and lack of access to clean water. Second, by extending science to broader audiences through open science hardware, they ensure that there are low-cost alternatives

to education. The founder shared with me two crucial examples of how affordability can include or exclude people from learning. He said they organize their workshops for free because nobody would come if they charged even as little \$2 USD, as people could not afford it. He highlighted their “low-cost Arduino” project, a simple Arduino-compatible microcontroller with a USB interface that cost only \$3–4 USD, making it significantly more accessible (T.B. 2020). To put the pricing into context, another collective member mentioned elsewhere that as long as lunch from a local restaurant costs \$1 USD, the price of an original Arduino board still is more than a family’s warm meals for a month. Financial scarcity leads to novel solutions, which corresponds with Arancio’s observations about appropriate technology; while closed hardware for science attempts to be a universal solution, open-science hardware can democratize technology, reduce the divide, and ultimately improve development (Arancio 2021).

The third prerequisite is that the collective is a group of *makers*. However, this group rejects the maker label. Their impression is that hacking and making started in the United States, and that these biased words miss an essential Indonesian collectivity. Instead, they focus on collaborative technology development involving local communities and international networks. For this practice, they use the terms **DIY** and **DIWO**, and they experiment with open-source hardware. They have developed various hacker-maker projects around organic and ecological technology, while highlighting the importance of external input and knowledge sharing. They develop their own technologies, and appropriate solutions to better fit the local context.

Originally, developing the water quality project with the local community was one of the potential main cases for this study. Although the project operates on virtually no funding, I was supposed to receive a small payment for a presentation at the local university, which I intended to transfer to the collective for prototype development costs, to which the founder said, it was a lot of money for the project. This would have been an excellent opportunity to co-develop a citizen-science-based project that was already established, which could be elevated to the local governmental level. The water quality project is a rare and interesting example of a grassroots critical making project that could have been shaped with many different stakeholders. Building on other researchers’ long-term ethnographic observations and studies on the postcolonial computing aspects of the project, as well as its participatory and non-hierarchical development process, would have been very interesting. However, during the pandemic, there was simply no project to participate in. Nevertheless, following up could be interesting for future research on such “hopeful interventions” (Lindtner and Lin 2017) by grassroots critical makers with the potential for long-term change.

(intriguing example of funding considerations for community engaged, participatory, & grassroots embedded or collaborative research)

5.3.4 Discussion

This subchapter has offered more extensive insight into the plurality of projects that were mapped in my field research. The cases presented addressed wicked societal and environmental problems. These range from wicked problems negatively impacting the population globally to locally relevant, incremental issues that decision-makers might choose to overlook, or companies might find unprofitable. Some of these problems were inherited from the past as consequences of colonialism and the anti-communist sentiment that is still used to attack communities and raid libraries. Wicked problems like climate change and plastic pollution are just now starting to show the negative effects on everyday peoples' lives. Similarly, tensions are sparked by issues arising from the population's migration. Many are moving to high-density cities, alienating urban areas and vacating rural areas.

Collectives play important roles in addressing these problems. They reflect upon, attempt to solve, or direct attention to such issues. Even raising awareness is necessary, as many people do not yet know about the threats, as seen in examples of digital rights and water quality. Tools, from microcontrollers to maps, also need to be de-colonized and made accessible and relevant. Grassroots unite communities to exchange ideas, develop answers, and share knowledge by educating locals through critical-material practice, participatory workshops, countercultural zines, and open-source blueprints. Many collectives network and have frequent exchanges. They develop their answers in non-hierarchical, community-driven ways that mirrors their cultural context. Collective ideation and making is the direct and "natural" way of cooperation in a traditionally collective society.

Finally, it is necessary to reflect on the context of the projects. Making in the Global North and the Global South are fundamentally different, even in makerspaces. The Northern economic and socio-cultural dominance also emerge in hacking and making. Hacking is still seen as an illegal strategy to break out of corporate control in the North. In emerging Asian economies, hacking is a way to gain technical fluency by redesigning tools of a global technoculture. Making is framed as a fun hobby rather than a method of coping with resource scarcity (Kori and Novak 2020; Saari et al. 2021). In grassroots critical making in these Indonesian cases, resource scarcity might be a contributing factor, but there are stronger political messages of human rights, wellbeing, and impatience. As demonstrated in the vignette of the DIYbio collective, social issues might be recognized by locals, but the maze of siloed bureaucracies and general distrust caused by historically unreliable governments has led to "a failed-state

context, there's no mainstream" (Pushkin 2020). Thus, a pluriverse of grassroots innovations was spawned by the fall of the totalitarian New Order regime in 1998.

5.4 Case Study I: Combining Ancestral Knowledge and Rural Community Infrastructures



This case study and the next one are interconnected. This first case study investigates a West Javanese grassroots innovation collective that acts as a supporting network for Indonesian grassroots communities. It also engages in maintaining **community-based internet infrastructure** (better known as **community network**) projects and organizing rural ICT camps for skill sharing. The collective and their projects are highly intertwined, as I will demonstrate in this case study. Thus, because all cases are anonymized, I will be referring to them as the *Network Collective*. First, I offer a retrospective analysis of the collective based on the Grassroots Innovation Movements analytical framework, then I will describe their intrinsic design project.

5.4.1 Physical and Metaphorical Networks

5.4.1.1 Context: Flourishing Technology and Design Practices

The Network Collective was established only a few years after the fall of the totalitarian **New Order regime**, in 2001. During this time, electronic and digital consumer media became available to wider audiences, and the newfound freedom for people to express themselves through new media art provided innovative frameworks for socio-political change (Jurriëns 2013). Fresh graduates started various post-authoritarian collectives—*new media centers* and *media art collectives*—in these years of political change, and their practices were influenced by growing access to ICTs.

In terms of geopolitical context, it is important to note that the collective is headquartered in a relatively wealthy city due to its coffee and tea plantations and quinine production during colonial times. Historically, wealth also allowed for technical progress and rapid adaptation of new technologies from the 1920 s onwards. The Bandung Institute of Technology was established by the Dutch colonial government in 1920 (Institut Teknologi Bandung 2020), followed by the building of an observatory between 1923–28 by the Netherlands East Indies Astronomy Association (Bosscha Observatory 2020). The second radio station of the country was established in 1934, only a few months after the first one in

1947–1998, under leadership of President Suharto

Batavia (Jakarta) (Lindsay 1997). As an interviewee pointed out, Bandung rapidly became “a city of intellectuals” during these times (G.H.I. 2020a).

Today it remains city of engineers, but also artists and designers. The city had been historically conceptualized as *kreatif*, meaning anything new and different. Young people who enjoyed Western popular culture started a music magazine in the 1970 s, and independent fashion outlets appeared in the 1990 s (Yujin 2017). Creative industries in the form of bottom-up social movements were cultivated by the city’s government in the 2010 s (ibid.). These bottom-up, community-driven activities were organized by young people called *komunitas* (Lengauer 2021). They were heterogeneous in their size and activities. Even the people who made up the *komunitas* were diverse in terms of gender and age, with shared values such as peer solidarity, creativity, and friendship, also promoted in punk and indie collectives and peace movements (ibid.).

In terms of design, it is important to note that elsewhere in Indonesia, *desain* is still rarely considered in its expanded sense of an approach to problem-solving (Azali and Budiman 2016). The local university is known for their prominent faculty in arts and design practices, which inspired events on *design thinking* and *design action*, and birthed projects that apply participatory design methods to project development. As a Western Javanese interviewee currently living and working in Eastern Java explained, there are major differences between the practices because of their different local socio-cultural and educational contexts:

“...here [in Central Java], people more trying to reach the social innovation through art. But in [Western Java], they want to reach the social change [...] through design. So in different [approaches], you will meet different character of the city” (F.H. 2020a).

The Network Collective is known for having been one of the first “art and new media centers” that featured creativity as a fundamental element of their activities (Lengauer 2021).

5.4.1.2 Pathways: Reflexive-Iterative Development

In terms of its pathways, the Network Collective has a long history of development, which I would call a reflexive-iterative practice. The collective started as a “new media arts center”, a community of multidisciplinary students led by two artists, an architect, and a graphic designer who wanted to start something new and work together. During this time, their focus was to develop multidisciplinary arts and education in Indonesia, so they set up a space with a local bookstore to do activities that combine literacy and creativity. One of the founders received

a grant to visit European creative cities, which was one experiences that opened his eyes to creative industries and the idea that these could be a vehicle for developing Indonesian civil society. Over time, the platform grew into an organization that held workshops, festivals, concerts, events. Their goal during this period was already driven by what the internet can do for society, as they were trying to bring Web 2.0 to the community. The idea was to establish a blog-like platform and open it up to anyone and exercise little editorial control. As the founder told me, the resident artists and activists were very active during this time, openly sharing ideas and producing knowledge. During this time, they “incubated” the founder of the East Javanese collective, who was one of their residents, and now supports projects of belonging in urban and rural areas (as described in Sect. 5.3.3.2.). During these years, they also collaborated multiple times with the collective described in Case Study II and participated in a critical making event. Although they no longer organize trainings and residencies, they still regularly unite initiatives and collectives for exchange and cooperation.

A few years ago, an interviewee told me that the founders of the collective “disappeared for a while”. During this reflexive time away from the projects, they decided to take a strategic turn, with a stronger focus on concrete projects and facilitating smaller networks (I.N.A. 2020)²². In addition to the national-level network building, the collective also co-organized an event with Ethiopian journalists, data storytellers, open-source advocates, and feminist innovators to share their work and collaborate with Indonesian counterparts. While the event was co-organized with a German NGO, the key idea was to promote South-South exchange²³.

In terms of the aforementioned concrete projects, the collective decided to do more hands-on projects. Creative economy and media art was seen by the government as a good way to engage and get voters, so the collective started engaging with members of the government. They received funding to conduct research in

²² I witnessed such a one-day network building and knowledge-sharing event organized by the collective during my field visit. This event was attended by around 10 local NGOs, who presented their achievements and current challenges related to the rights of people with disabilities, women and similar societally relevant topics.

²³ The event, albeit significantly smaller, resembles the 1955 Asian-African Conference of Colonized Nations. Also held in Bandung, it was of major historical relevance as it is known as a turning point in the emergence of Global South’s solidarity movements and a collective quest for countries to liberate themselves (Tan and Acharya 2008; A. Phillips 2016). Its goal was to design and accelerate alternative futures by encouraging formerly colonized countries in the South to cooperate. The collective’s event was similar in the sense that it was focusing on South-South exchange, understanding that the local challenges—however different—might respond to similar solutions.

* example of south-south anti-colonial organizing

16 provinces in West Java to explore eco-tourism and similar possibilities. However, as the founder told me, when the politicians lost interest due to the election period, the projects stopped. However, the collective still had access to these rural and local communities and decided to continue to cooperate with some of the villages. For example, they mapped the cultural and agricultural landscape with the Indigenous community described in this chapter, which established a trusting relationship between the collective, the villagers and the local leadership, and together they decided to develop the Community Network Project described over the next chapters.

To summarize, over the last 20 years, the collective became one of the most influential grassroots organizations in Indonesia. It grew from a media art collective to supporting the development of other grassroots through exchange, and became an important contributor to sustainable rural ICT development in Indonesia. Interestingly, their core team remained the same throughout these two decades, meaning the pathways were enabled not through a change of team members, but through their framings.

5.4.1.3 Framings: Openness, Appropriate Solutions, Indigenous Justice

The framings contributed to an “organic” development because collective saw itself as an open platform for experimentation and creative collaboration. This means that they remained open and flexible to host and support new ideas, within the framework of core values, such as facilitation of public interest to develop creative knowledge and to help Indonesian communities collaborate.

The collective is a **strategic space** (Wylie et al. 2014) of **infrastructuring** for grassroots empowerment, because they develop civic empowerment using a toolkit of different methods, networks, and interventions. They focus on generating sustainable, long-term impacts by infrastructuring and helping develop the Indonesian NGO landscape. Similarly, in the Community Network Project, it will become obvious that long-term strategies have been developed and reflected upon in an iterative process aimed to lead to social, environmental, and financial sustainability. The project’s goals stretch way beyond the physical infrastructure, toward the empowerment of Indigenous communities to protect their own rights. Rather than developing short-lived projects, they formalized themselves into a foundation, and built on a blend of approaches:

- **Openness and documentation:** Knowledge and solutions are generated collectively through processes resembling peer production. The collective excels at documenting its workshops and projects, and making them freely accessible

as documentation and hands-on classes, so that their practices can be reproduced elsewhere. The community network infrastructure is also kept open, tweakable, and repairable.

- **Strengthening society by supporting grassroots innovation:** The collective provides networking, exchange, and infrastructures for grassroots innovators to meet, develop their projects, share ideas, and collaboratively overcome obstacles. By learning from others, the collective also lobbies to improve the situation of the grassroots.
- **Sustainable developments:** The collective's house is a space of sustainable experimentation. As I will demonstrate in the case of the community network, their goal is to intrude on nature as little as possible.
- **Appropriate technologies:** The Community Network Project is based on appropriate technologies, from the source of electricity to the ICT networks, which are affordable and maintainable by the community.
- **Decentralized, autonomous technology:** Resembling the ideals of hacker culture and many Western hacker projects, in the Community Network Project, the hegemony of one dominant perspective is challenged (Benkler 2006). Their technology infrastructure is designed as autonomous, decentralized, and thus, decolonized.
- **Criticality and cognitive justice:** The goal is to democratize technology, reduce the digital divide, remove asymmetries of urban-rural populations, show the viability of rural ICT solutions, and empower the Indigenous community by resolving injustices caused by limited access to ICTs. Boundaries between experts and non-experts are blurred as members of the community learn how to maintain the community network infrastructure.
- **New economic avenues:** Onno W. Purbo highlights that narrowing the digital divide could lead to economic growth (Purbo 2017). While this is not the most important aspect of this project, access to information has enabled experimentation with chili plants, to see which methods would increase yields and improve market prices.

5.4.1.4 Strategies: Civil Society, Open Documentation, and Self-Directed Development

The reflexive, long-term strategies of the collective are supported by the diverse networks they've built within Indonesia. Originally these networks were built for the development of civil society, including collaborating with other NGOs, but over time, academics, governments, and other supportive actors like critical makers developed visions and shared struggles.

Their repertoires of action include organizing events to share knowledge internationally, while learning from local intermediaries that support the development of projects on the ground. Events at the collective's house are still ongoing, however, their space has been extended to rural areas through the "rural ICT camps" and the Community Network Project. Their aim is to share sustainably produced knowledge is also reflected in their documentation practices. Material and reflexive activities are combined, similar to the idea of critical making, so that digital infrastructures are created and documentation is shared openly.

Other resources are mobilized. For example, they receive financial support in the form of grants, which are restricted to the projects themselves, so experimentation with new ideas is not funded. Currently, there are ongoing discussions with a UN agency about direct funding for ICT4D projects, instead of accessing these through the government as intermediary. Yet, the collective remains skeptical whether direct funding from the UN agency will be possible, as they have learned that most of the time, it is not, as the government prefers to keep control over funding.

In terms of spaces and strategies that the Network Collective engages in, the following intrinsic design project is the best real-world case. As I will explain, they carefully opened new avenues for collaboration by building trust, using their existing networks to mobilize context-appropriate resources, and building context-appropriate sustainability strategies that constitute what Smith et al. call *good development*. Good development is constituted by the inclusion of public participation and local knowledge in appropriate technology projects (Smith et al. 2014). Smith et al. refer to Pieterse, who reviewed alternative development, *post-development*, and *reflexive development* practices (Pieterse 1998). They also mention Chambers, who advocated for learning from locals and how they define their situation, as well as self-critical awareness in development (Chambers 1997). The Community Network project goes beyond these prerequisites of good development and seeks sustainable self-directed development.

L this skeptical stance makes a lot of sense given the role of northern powers in enabling mass massacres in the country in 1965-66

5.4.2 Intrinsic Design: Community Network Project

In the previous paragraphs, I highlighted some key aspects of the collective's work. Approaching the second layer of analysis, the most relevant aspects for the Community Network Project are its goals to remove asymmetries, reduce the digital divide, and support the Indigenous community's attempts to gain rights

while supporting sustainable development. I will analyze this case through the lens of **participatory-intrinsic design** and **self-directed development**.²⁴

It is first important to highlight the window of opportunity for this project. Internet access for all is envisioned by the International Telecommunication Union (ITU), however, as they put it, the **digital divide** between men and women, young and old populations, developed and developing nations, as well as urban and rural areas remains prominent (ITU 2021). Unfortunately, incentives for companies to expand networks into rural areas are low, due to geographical difficulties, low numbers of users, and lower income populations. People are turning to bottom-up solutions. DIY networks including mesh solutions can be found in the Global South (Bidwell and Jensen 2019) and in Western countries alike (Hardes, Dressler, and Sommer 2017). Academic interest in community networks has been growing in recent years (Braem et al. 2013) in urban and rural settings as an alternative solution. In 2020, the Internet Society Foundation (ISOC) suggested novel policy recommendations for the internet access of Indigenous communities. They highlighted crucial areas of need such as community consultation and engagement, engaging local capacity, digital literacy, infrastructure and ownership, as well as effective and accurate mapping (ISOC 2020).

mesh
networks

Some of these recommendations come to life in this case study. The setting is Kasepuhan Ciptagelar (*kasepuhan* is a Sundanese term that means *place of the elders*), a remote Sundanese cultural enclave located inside a nature reserve area, the Halimun-Salak National Park, 180 km south of Jakarta, Indonesia's capital city. The community group consists of around 30,000 individuals living in 568 villages scattered around Halimun mountain at elevations 700–1,200 meters above sea level (Dwiartama 2020). Being located hours away from the next city means these villages are hours away from the smog and noise pollution, in a harmonic, green, cool environment between mountains. It is also very challenging for electricity providers or telecommunications companies to build infrastructure there (Fig. 5.15).

²⁴ In terms of intrinsic design projects, my observation is that it is sometimes difficult to differentiate between outsider and insider. For example, in the case of the Col-lective: Is he external to the Indigenous community if he is from the same region and speaks the local language? According to a remark I received from a participant when I presented the case, he is an outsider, as he was not born in the village. However, he built a rapport so strong that the community considers him (and his family) a trusted insider rather than an outsider, even if he does not live in the village. The person responsible for ICTs (the “minister of ICTs”) was also born elsewhere, but is now an integral member of the local society with an important day to day role. I think as researchers it is not our place to decide, the answer lies in asking the individuals themselves.



Fig. 5.15 Fields and Traditional Rice Storage Huts of Kasepuhan Ciptagelar. (Source: The Author)

In March 2020, I visited this Indigenous community for almost a week. My trip to the village was made possible because the Network Collective had to attend to a technical and a social matter. The technical issue included maintenance work (adjusting the wireless tower's signal transmission) and adjustments to the community network's infrastructure. These adjustments needed to be done, as in the cloudy mountains, the solar panels were not providing enough electricity. Thus, micro-hydro-power cable had to be hooked up to the battery powering the network. In terms of the social matter, the wedding of a family member of the chief leader was taking place that weekend.

As opposed to some of the neighboring communities, this Indigenous community is open and has been the subject of research, mainly due to their sustainable and regenerative eco-religious agricultural practices and for implementing ancestral cultural patterns in modern times. In terms of agricultural practices, Indonesian Indigenous people consider rice as a symbol of life, and a gift from the God that they are not allowed to sell. Their farming practices follow ancestral principles and implement non-intensive agriculture methods. As I learned from an interviewee, in the local culture, 60% of the villagers' time is allocated to growing rice, leaving spare time for other activities. The restriction of

the area cultivated, strict rules in utilizing woods, and exact planting periods are, according to Dwiartama (2020) some foundations of social, economic and environmental sustainability, and most importantly, result in self-sufficiency. A long recorded history—approximately 650 years—of the village informs the ancestral cultural patterns that remain today. The leaders of the village, headed by the chief leader, are strictly followed by the community. They make decisions, and every few decades, the village moves. Where they move to, how the settlements are shaped, and where the houses are built, depend on the permission of the elders. As Susanti and Wahyuni (2020) speculate, customary values might guide the “modernization” of the community. They quote a paper written in Indonesian by Hermanto et al. (2012) that indicates that the change in the socio-cultural values of the Indigenous community is pragmatic in improving their quality of life.

These observations play a very important role in this study. The community has been self-sustaining for hundreds of years and it remains open to technological innovations. This includes access to technologies like community radio, community TV, and community internet—further opening the community to the outside world without having to travel. Opening up has kickstarted many changes in the community, such as experimenting with crops, maintaining infrastructures, and identifying technological self-directed developments (such as GIS for mapping) to better advocate for their Indigenous rights.

5.4.2.1 Infrastructuring Trust

To recap, the process of infrastructuring is driven by dynamic grassroots communities (Bødker et al. 2016). In terms of the infrastructuring process that is a prerequisite for the community network, I will outline three key infrastructures: one among the villagers, one that the collective generates, and most importantly, the trust that was infrastructured between the collective and villagers.

One starting point for the infrastructuring is the chief leader of the Indigenous community. Just like his father, he is very interested in technology and its potential benefits for his community. As echoed by the aforementioned ISOC report, for Indigenous internet access, accurate mapping is key. Recognizing the importance of counter-mapping (Peluso 1995), the community cooperated with an Indonesian Indigenous people’s organization and two other entities (Daro Mincharek 2019). However, the locals quickly expressed discontent and refused to participate in “participatory mapping” activities as they weren’t respecting the local culture. The chief leader decided to take matters into his own hands and decided to take more control over the process, by acquiring GPS units, GIS software, and learning how to use these from online video tutorials. Essentially, he started a grassroots counter-countermapping project. Due to their mapping processes completely disregarding local customs, leading to trust issues (Hendriatiningsih et al.

2010), as the community felt that outsiders could not be trusted anymore. The chief leader also expressed worries that several monoliths deep in the forest could be mapped and thus become targets for tourists and NGOs, possibly corrupting such spiritual places (Daro Mincharek 2019). By the time the Network Collective arrived to conduct their mapping process, they had to tactfully, politely, and patiently rebuild (or re-infrastructure) the community's trust.

A researcher of the Institute of Technology Bandung who joined the field visit highlighted the importance of mapping cultural information, thereby representing relationships between the community and the land, such as which directions particular landmarks could be approached from. While such information might be difficult to grasp with a different mindset, he said they decided to accept and map it because they believed it was important for the locals (R.A. 2020). The founder of the collective echoed the importance of this approach:

"In many cases, (the authority) sees map as an empty sheet of paper, only territorial map without any significance. But from our approach, learn and understand that inside any terrain, there's a story. And the story has significant relationship with culture or even civilizations, which in many cases been neglected. And that is how the development policy and agenda got disembedded (from) this reality" (G.H.I. 2020c).

The founder of the collective also highlighted the historical and colonial roots of this dis-embeddedness, as centralizing the tens of thousands of islands that make up Indonesia was done from a continental perspective. Such blueprint developmentalism, done in a top-down manner that erases the cultural, linguistic diversity through a shared language (Bahasa) leads to systemic problems. These post-authoritarian times, however, from a grassroots perspective, he said, were at least easier to navigate, as the government was not as powerful. To underscore the difference between the approaches of the government and the collective, he mentioned that when being allowed to respectfully access and record such cultural information, the members of the collective felt like it was an invitation to learn about the existence of the locals on a deeper level.

It is important to note here that it was a happenstance that the Network Collective was involved with the mapping project, and that it developed into the idea of a community network. This happenstance was then followed by the co-development of strategies to support the community in this endeavor (Bødker et al. 2016). According to the founder, it took approximately two years to get to know every community member, understand local power relations, and grasp the nuances of the local Sundanese dialect. Only afterwards did it become possible to collaboratively develop new projects with the Indigenous community.

At first, they did not start with a concrete project to be codeveloped, but instead focused on learning from the Indigenous community. Between 2013–2015, during the mapping project, they sought to gain trust and confidence, in stark contrast to how previous collaborators approached the community. It is important to note here that this infrastructuring process wasn't linear. One of the goals of the project was to develop sustainable income for the villagers from local products. One idea was to make essential oils from the forest, and an experimentation phase took place involving external experts. It did not take long for the local community to realize that the essential oil production would harm the local environment and exploit the sacred forest. When one of the local decision makers fell ill, the community took it as a sign. Despite the high market value and profit margin, the project discontinued; prospective financial gains mattered less than sustainability.

Only later, having decided with the locals to develop the community network, did they start infrastructuring a larger community of collaborators. Meetings between the locals, representatives of the national park, the ministry of ICT, and the National Agency for Research and Development were held. Plans were drafted, focusing on natural preservation and protection, cultural preservation, sustainable energies, economic development, and life improvement. The Network Collective facilitated the communication process, as they knew the specific protocols and processes that needed to be followed in the Indigenous culture. An example, the founder had shared with me how the daily activities are navigated around the rice; a backhaul tower could simply not be built any time, as locals could not be involved when the custom dictated that the sacred rice was to be taken care of.

Building trust would not have been enough for starting a community network with the locals, as the technological development also had to be supported by the chief leader. The role of the village head has been proven crucial to get the villagers on board with new technologies. Previously, the importance of leadership has been recorded when establishing renewable energy sources (Pratiwi and Juerjes 2022). As I will demonstrate in the next subsection, various technological developments had already been implemented in the village, which created the technological basis for the Community Network Project. These developments also show how the local community itself was infrastructured. The social structures of the villagers could be explained as a miniature society, in which everyone has their assigned role that they inherit or receive from the leadership. In this case, the person responsible for the telecommunications projects acts as the *minister of ICTs*, and there are young people who—similarly to the chief leader—teach themselves skills necessary to maintain the physical infrastructures.

Self-directed technological developments include the necessary prerequisites for the Community Network Project as well, which created the perfect window of opportunity for the project. The chief leader understood the urban-rural digital divide and that commercial companies would not be delivering internet infrastructure to the village due to a lack of financial incentives. At the same time, the necessary prerequisites were available on site, such as the willingness of the villagers to learn the infrastructure's maintenance. Due to the work of Mumpuni, a social entrepreneur, there was access to self-generated electricity. Purbo, another social entrepreneur, had created the legal basis for community networks.

The chief leader's interest in technologies and the separation between him as a private and public person became clear during my field visit. While we were waiting for our time to have the audience with the chief leader, we sat in an open-air workshop surrounded by tools. A young man showed up and started filing a high-performance coaxial cable while pleasantly chatting with the others (Fig. 5.16).



Fig. 5.16 The Chief Leader of the Indigenous Community in his Workshop. (Source: The Author)

Once he was ready, he went inside, sat down, and instructed us to join him in an official audience ceremony. We had to explain who we were, where we came from, and why we were there. He was the chief leader. As the founder of the collective explained to me, that is what makes the role so interesting; he can be very close with the villagers, but when it comes to formal protocols, the interaction becomes official before reverting to ordinary again (G.H.I. 2020c). He also mentioned that the chief leader is a connected man, not only in a physical sense, with all the people living in the village, but also virtually and digitally. The chief leader's father was already open to technology, and the leader himself, as an interviewee put it, is a "geek". He has been a member of the amateur radio community and was already in their group messenger before the community network existed, exchanging knowledge about technical issues. A couple of years ago, he learned that the community network had been attacked by hackers. His reaction was not to shut down the project. Instead, he asked the partners to consider building better cybersecurity and support the establishment of a rural ICT training center to spread awareness about potential threats. Other unintended impacts, such as online bullying or access to adult content started to happen, which immediately affected the wellbeing of the community. To address these issues, the community decided to focus on digital literacy by building a training center and organize rural ICT camps with the Network Foundation. Thus, instead of censoring content or limiting access, they chose education. Whether this approach solves the problem in the long term is yet to be seen.

The importance of religious protocols was also highlighted, such as testimonial processes that amplify the significance of the moment. This could be a blessing ritual of the community network infrastructure, or asking the gods for permission during the mapping process to take pictures. During my visit, once the maintenance work on the internet infrastructure was complete, a small fire was lit to thank the gods for their support. Respecting eco-religious customs is key to gaining and maintaining trust.

Finally, for the project to be successful, the collective had to infrastructure a community that could respectfully support the Indigenous community, and whom they could trust. During my visit, a co-researcher from the local university was present. They participated in the mapping and surveying, and had been following the development of the project. But there was also a need for other types of skills and knowledge. For example, an ICT company was needed to set up the internet connection between the village and the nearest access point, help maintain it, and share the necessary skills so that the locals would be able to maintain it in the future. To achieve these goals, the founder of the collective selected a small, local company that would deliver according to the expectations. A three-party agreement was signed between the village (the main beneficiary), an internet

service provider (ISP) company, and the Network Collective (facilitators on the technical and policy level).

To summarize, not all contributors are members of the collective or villagers. External people and companies with the right profile and mindset had to be infrastructured into the project. The collective was able to do this because they already had the experience and the necessary human networks.

5.4.2.2 Developing Sustainable Strategies

Most striking aspect of these approaches—both that of the village, and of the collective—was sustainability, both on the social and the environmental level. According to Bødker et al. (2016), inspired by Dourish's (2006) notions of strategies and tactics, intrinsic communities develop approaches to make the artefact ecology work. Such approaches include strategies—design activities that they engage in collectively in formal meetings. In this subsection, I will outline how the Community Network Project shows signs of sustainable strategies from the village and the collective.

Social sustainability: The artefact ecology of the Indigenous community is complex and developed. Its development needed various prerequisites, such as the interest and willingness of the chief leader, as well as his ministers and the villagers. They needed to work with external partners, including civil society, governments, and social entrepreneurs to realize their projects. As explained before, there had been cultural clashes and conflicts of interest. Thus, the community resorted to self-education and self-directed development to protect themselves from external threats. This self-education is key for the villagers to be able to maintain the artefact ecology themselves. As the collective's founder explained, this is a condition they also realized. Thus, they try to encourage the planning and work both to be done by the locals:

“[W]e only try to function ourselves as a facilitator, or maybe like a catalyst to make things more visible. But we would like to have everything in their hands. So they have to decide. They have to organize, and do the planning and everything” (G.H.I. 2020b).

Investigating the sustainability of the project, I asked him what he thought would happen if his collective stopped engaging. He confirmed that he thought the process would slow down, but because he ensured that the local community was directly connected with foreign funding agencies, government, and relevant institutions, he thought the project could continue on its own.

Even if the electricity does not need to be acquired from a provider and the villagers have the skills, there remain costs to maintain the devices, including purchasing new parts. The community itself works on the basis of *gotong royong*

(mutual aid), and their eco-religion does not allow them to sell the rice they grow. Thus, financial sustainability strategies needed to be developed, such as a business model. The source of income most directly linked to the community network itself is the voucher system. For 24 hours of access, the users pay less than the price of 1 pack of cigarettes. This price is a psychological threshold I had heard about during my previous field visits elsewhere, and it seems to work (Fig. 5.17).



Fig. 5.17 Voucher for the Community Network's Internet Hotspot. (Source: The Author)

Sustainable business models: The community is not allowed to sell rice, but they are allowed to sell other crops. The only other way to generate income would be to illegally mine. This is seen as disastrous both for the environment due to the use of mercury, as well as the mental health of the community (de Castro Leal et al. 2021). While the village itself could be mostly self-sustaining through agricultural production, there are still daily needs that need to be covered financially. So instead, new approaches are being tested and navigated to generate the necessary income. When selling produce previously, but not having had access to the market prices, middlemen took advantage of them. They thus needed to start generating local knowledge in terms of how they could optimize growing different crops and then sell them for a fair price.

The locals themselves had the necessary agricultural knowledge and skills, and with assistance and training from the South-East Asian regional hub of a Dutch development organisation, they were able to develop a system to distribute crops to the markets. Through iterative approaches, they selected agricultural activities that supplemented the local culture and thus, didn't intrude on the local community.

During my visit I experienced an abundance of different types of vegetables from different origins. In a civic science experiment, community members went online, researched different tips and tricks to grow chili²⁵, a popular spice that can be sold at a high price margin at the market. In two dozen rows, chili plants were growing, each using a different technique to see which worked best in the local ecological circumstances. The most successful techniques were to be reproduced during the next cycle. Crops are mapped in OpenStreetMap, and information about the development and coordinates is shared with other farmers on Facebook using the Timestamp app.

The key innovation in the second step was how to circumvent working with the dishonest middlemen, and instead directly sell the crops. Local youth participated in the training and are today coordinating trade, coffee production, and where to plant crops. They have a system to compare and keep track of market prices offline and online, and they sell the crops themselves. They have created transparency and openness about the market, and also coax other farmers who were also exploited by the middlemen to join them, starting a small local fair-trade movement. This effort is not about mass production, but bringing together the smaller quantities produced by local communities to reduce costs.

Technological sustainability: Finally, in terms of facilitating the infrastructure's long-term sustainability, the collective's founder had made the following strategic observations. His three rules for a technology to work in a rural area sustainably are, "low-tech, low-cost, [and] low-maintenance" (G.H.I. 2020c). When ensuring these attributes were represented in technology deployment, he said that the team can let the local community maintain it. This is possible by getting rid of technical barriers, and if some technical barriers persist due to their necessity, then knowledge and skills are required. For such cases, they established the Rural ICT Camp, which is a platform for exchanging knowledge on rural community internet infrastructures, not only their physical maintenance, but also threats such as cyber-attacks or harmful content. Technical skills and **digital literacy** are both part of the classes, especially since the local community had been exposed to such threats.

Good and Sustainable Development: The critique of the founder of the Network Collective towards **blueprint developmentalism** was described earlier. Smith et al. point out that the appropriate technology movement struggled with three persistent problems in its deployment. First, generic appropriate technology solutions were employed because they were easier to be included in top-down produced handbooks and field guides. The one-size-fits-all focus on basic needs

²⁵ They jokingly call this self-educational practice the "YouTube Campus".

and specific production was practical for widespread diffusion. Second, appropriate technology lacked participatory practices and processes. Although the original principles included user-led participation and grassroots innovation, in practice the solutions were built on engineers' assumptions of what was needed. Third, it lacked business models for sustainability, as dependence on State and other funds could not be surpassed. For this reason, the technologies could not be marketed because of structural poverty and local entrepreneurial capabilities were underdeveloped (Smith et al. 2014).

The Indonesian version of the appropriate technology movement for rural development was created in the 1990 s by Prof. Poerbo, father of Ono Purbo, who called it the Ciamis Model. It was designed and applied by PPLH-ITB²⁶ to initiate a process of sustainable development at the local level, by involving local communities in their own development—defined as change that is willingly introduced and internally driven. Poerbo defined a sustainable development strategy as:

- Exercises in social learning (groups of people learning from each other in the process)
- Information and communication to increase rationality in decision-making
- Starting small initially (hamlet or production group level) then replicating development to build a system
- Decision making is decentralized and based on democratic representation. The government serves as an enabler, and NGOs or universities act as intermediaries or catalysts
- Participatory action research can be used, “for limited purposes, at crucial times” (pp 37–38, Poerbo et al. 1995).

As demonstrated above, the Community Network Project addresses all these challenges. Here I would like to briefly summarize how they achieve good or sustainable development. Some of the strategies are veiled, such as the empowerment of local communities to defend their Indigenous rights and lands, while others are made clear. The overall goal of the learning grant is to create enabling ecosystems for community networks in developing countries through a multi-year and multi-donor strategy, while building human capacity for sustainability. The multi-year approach helps the Network Collective and the Indigenous community plan, experiment, and develop the project. At the same time, business models are developed by the community itself, in accordance with their own values. I also

²⁶ PPLH-ITB is the abbreviation of Pusat Penelitian Lingkungan Hidup Institut Teknologi Bandung, the Environmental Research Center of the Bandung Institute of Technology.

touched upon the prerequisites of good development before, which the Network Collective has championed. They became a facilitator, instead of a leader, and translated between the Indigenous community (whose value and belief systems they now understood) and the “outside world”, where the funding, knowledge, and physical infrastructure came from. The infrastructure itself was set up in an appropriate way, so that the local community could maintain it themselves after training. They respected local customs in the placement of the tower and responsibly utilized local resources such as solar and hydro power. The self-directed and appropriate nature of the project led to its overall sustainability and created inclusive innovation, which became a tool for Indigenous empowerment and social development (Smith et al. 2014). In these ways, the collective built a strong relationship with the local community to support democratic representation of needs and values.

5.4.2.3 Physical and Digital Artefact Ecology

The artefact ecology—the “tech stack” that constitutes the physical and virtual infrastructure of the project (Bødker et al. 2016)—is composed of multiple elements, some of which have been developed in the past decades and are crucial prerequisites for the Community Network Project, while others were newly added to bring the project to life. In this subchapter I will trace these historical developments and generative artefacts, as well as legislative struggles of grassroots to make internet accessible to everyone. Then I will outline the setup as it exists today and findings from the iterative approach applied by the locals and their supporters.

As previously briefly touched upon, the father of the current chief leader was interested in modern technologies himself. The micro-hydro project completed in 1997 provided electricity to the central village, followed by the community’s own radio station in 2004, and a TV station in 2008 (Daro Mincharek 2019). These are set up, run, and maintained by the local community and residents throughout the mountains. For the community network to come to life, there were two necessary far-reaching Indonesian social innovation efforts: the aforementioned hydropower plants, and Purbo’s internet liberation efforts. Both can be described retrospectively as starting from “hopeful interventions” (Lindtner and Lin 2017), which built movements around their bottom-up goals, and eventually became sustainable developments.

Sustainable and self-maintained hydropower plants: Since Indonesia ratified international agreements on reducing its CO₂ emissions, the government’s plan is to follow and create demand for renewable energy by developing industrial

zones, including mining activities and industrial smelting²⁷. While these projects are environmentally problematic, it is interesting for this study to trace how hydropower plants, including mini hydro and micro hydro, ended up producing around 9% of Indonesia's electricity.

Today, around 50 million Indonesians are without electricity. Even worse, half of them live in electrified areas, but need grid densification programs to get access. The other half live in remote, rural villages²⁸. Around 15 years ago, this number was around 105 million people, and improving electricity service through micro-hydropower generation has been attempted multiple times by actors including the Indonesian government. Government-constructed MHP-plants were criticized for their poor construction and maintenance, but the project was also scrutinized because it didn't involve the local communities: in 2006, 45 out of 56 government projects failed, and the remaining 11 were deemed only partially successful (Ashoka Foundation 2006). Local businesses attempting to run MPH-plants often closed as state-owned and subsidized companies entered the market to sell electricity at a cheaper rate. Thus, those were also rarely sustainable solutions and such a threat made it too risky for private investors to engage (*ibid.*).

Citizen-owned and citizen-driven renewable electricity projects remain rare, because gaining access to major power grids is difficult for Indonesian communities. It is particularly difficult for the Indigenous community described in this study, which is 2–3 hours away from the nearest town. Nevertheless, they are known and researched for their successful community-based renewable energy project. As there is no direct access to a grid but plenty of renewable solar and hydro energy, the villagers initiated the development of micro-hydro power plants themselves (Pratiwi and Jueres 2022) and continue to maintain them. For the Community Network Project, they even renovated micro-hydro plants that were out of commission.

Liberation of internet access through policy: According to the Indonesia Telecommunications Act No. 36 of 1999²⁹, telecommunications operations—including the operation of networks, services, and special telecommunications—can only be carried out after obtaining a license from the Minister responsible for the field of telecommunications. Radio frequency or any radio-frequency-emitting apparatus couldn't be used without the approval of the minister.

²⁷ <https://www.hydropower.org/blog/indonesia-promotes-hydropower-to-create-the-dem-and-for-industrial-development>.

²⁸ https://energypedia.info/wiki/indonesia_energy_situation

²⁹ <https://www.postel.go.id/content/EN/regulasi/telecommunication/uu/law36-1999.pdf>.

Onno W. Purbo describes the liberation process as “12 + years of struggle”. Its actors have been, on the one hand, corrupt environment/government officials, and on the other hand, a group willing to educate the people and institutions and demanding low cost internet access (Purbo 2006). Challenges to equitable internet access were identified as geographical, educational, and regulatory (Ashoka Foundation 2005). To address these issues, Purbo and his colleagues promoted Internet technology through their Computer Network Research Group in the 1990s (Barker 2017) at the Bandung Institute of Technology.

They outlined strategies to transform Indonesia into a knowledge-based society, including broadening access to the internet and publishing books and papers in Indonesian on the topic. Most importantly for this study, he advocated for adoption of public policies that enabled internet penetration. They also developed and experimented with low-cost technologies to extend internet beyond academia by refining affordable off-the-shelf wireless network (LAN) equipment and giving service coverage to entire neighborhoods with an external antenna. Most important for this study, he continued advocacy work with national government officials to ensure a more enabling policy environment for expanding low-cost internet access. He advocated for deregulation and ceasing the need for a broadcast license. Together with the Internet Service Provider Association and the Internet Café Association, they started a successful advocacy campaign to change the regulations. Since deregulation in 2005, the 2.4 GHz bandwidth is free for public use without any licensing requirement.

The policy change and experimentations with low-cost LAN equipment made it significantly more affordable than the dial-up systems. Back then, they were still a luxury for many low-income households, as at 15–45 USD/month, only middle and upper income families could afford it. An estimated 200,000 neighborhood users and many schools were connected to the internet by 2005 (Ashoka Foundation 2005). One of the founders of an Indonesian collective gave a presentation in Vienna in 2016 that underscores how people responded to the lack of access by developing “appropriate technologies” based on necessity. In this presentation, he compares the practice to *jugaad* and *gambiarra*, and introduces the Indonesian word *ngoprek*³⁰. In his presentation, he also speaks about the *wajan bolic*—a popular Indonesian hack that utilized a frying pan to create a wifi antenna in the early 2000s:

³⁰ *Ngoprek* is not an official word, but a slang word meaning modification, tinkering, hacking or cracking. It is derived from the word *oprek*, meaning searching, finding, disassembling, modifying, collaborating and creating something new <https://pujo.staff.ut.ac.id/ngoprek/>.

“In Indonesia also this is a common hack that (...) is called *wajan bolic* (...) basically a jar for cooking and you put some kind tube there. This is to enhance your internet. Until ‘98 I still remember that we have 64 kilobytes per second internet and this allows us to steal from the university internet connection and this *wajan bolic* is quite famous because they give tutorial to the people and how you can make it with your own plate—of course if you’re not your mother didn’t catch you on stealing the *wajan*—and for me it’s very interesting that there’s so many development of technology or hardware is developed by the citizens” (Siagian 2016).

Today, the hack is comprehensively documented online on a popular hardware repository³¹. It was confirmed to me by one of the participants of the research that the original tutorial Siagian mentions (the “they” in this quote) originated from Onno Purbo and his associates. There are also various sources on the Internet where Purbo is sharing information about how to build a neighborhood network hotspot using this method (Prayoga 2017). Purbo’s work has had a major impact on the hacker and maker communities in Indonesia. His efforts to liberalize the policy to allow for 2.4 GHz Internet in community networks has been one of the key prerequisites for the Community Network Project described in this study.

Moreover, his work had another effect that is crucial for the project. The ISP company working with the Indigenous community began as a grassroots initiative. It started as a community network built on self-taught skills and techniques from online sources, including those of Purbo. They started operating on the village level, and their first backhaul tower was built from bamboo and used the *wajan bolic*—a grassroots critical making project in itself. Their aim was to provide community internet access sustainably for their own village, using locally available materials. The project was only incorporated after having been invited to other rural villages to develop the same infrastructure. Today they partner with the Indonesian ministry of ICTs, developing internet access all over the country.

Further experimentation to adjust the infrastructure to the local circumstances has been necessary. The biggest problem of the artefact ecology enabled by these social innovations is how to make it environmentally sustainable while able to withstand the elements of nature. The main internet gateway is about 40 km from the tower that uses point to point access through radio waves. It uses MikroTik routers in addition to commercially available, off-the-shelf ZTE products not directly developed for particular climates or contexts. When I asked the representative if they had considered developing something like the African BRCK project (which is known for its sturdiness to withstand the elements of nature), he said this off-the-shelf solution worked perfectly for their needs. The advantage of MikroTik for this project is that the company focuses on ICT equipment

³¹ <https://www.instructables.com/How-to-Make-Indonesias-WajanBolic-or-PanciBolic/>

for emerging markets. These devices are affordable and readily available, with access to training. For this study, it was interesting to learn how the physical infrastructure needs to work with nature. The field visit took place during the rainy season, when there was not enough sunshine to power the main router of the internet infrastructure through the solar panels. Thus, the battery needed to be connected via cable to the micro-hydro plant, which is a more reliable source in the rainy season because it uses power generated from water. Access to that source of electricity was already installed by the local electric engineer from the village, and during the visit, it simply needed to be connected. In addition, the technicians of the ISP company had to direct the direction of the antenna towards the main gateway, as it had been moved by the wind (Fig. 5.18).



Fig. 5.18 Infrastructural Adjustments to the Community Network. (Source: The Author)

Currently the community network receives 200 Mbps from the main line, which translates to around 40–60 Mbps shared by up to 600 users. There are four towers in total, and each tower cost between 15–30 million IRP (950–1900 EUR), not including an additional monthly access fee. An international NGO funded the start of the project, and the income is used to maintain and expand the Community Network Project through small company set up by the villagers, with a business model that covers access, maintenance, and reinvestments. The locals received four-step training from the ISP company, which included instruction on how to set up a tower and the router, dashboard and monitoring, and the voucher system. Thus, new skills have been learned by the locals and jobs are created for them.

Many elements of the artefact ecology have been sustainably appropriated (Teli et al. 2021) and contribute to our understanding of how they can support the knowledge production, recording, and sharing of Indigenous communities. Knowledge is produced about the everyday life of the village, resulting in a continuous and thorough documentation of oral history and ethnographic knowledge with video cameras and drones. The location, yield of the crops, and their market prices are documented and openly shared online, using OpenStreetMap and geo-tagging, resulting in transparency and the empowerment of local farmers, who now can negotiate with large supermarket chains. Knowledge and skills are also shared as villagers regularly contribute to maintaining the artefact ecology of the project. These participants include the chief leader, the “ICT minister”, and a group of youth who have learned how to maintain the system. The local community has been infrastructured to sustainability. The Network Collective was merely a catalyst and facilitator for the projects, but the locals today cooperate with funders, capacity builders, and others relevant for the sustainability of the project.

To summarize, the generative circumstances and prerequisites for this Community Network Project to become sustainable included various necessary socio-technical developments. These are depicted in Fig. 5.19.

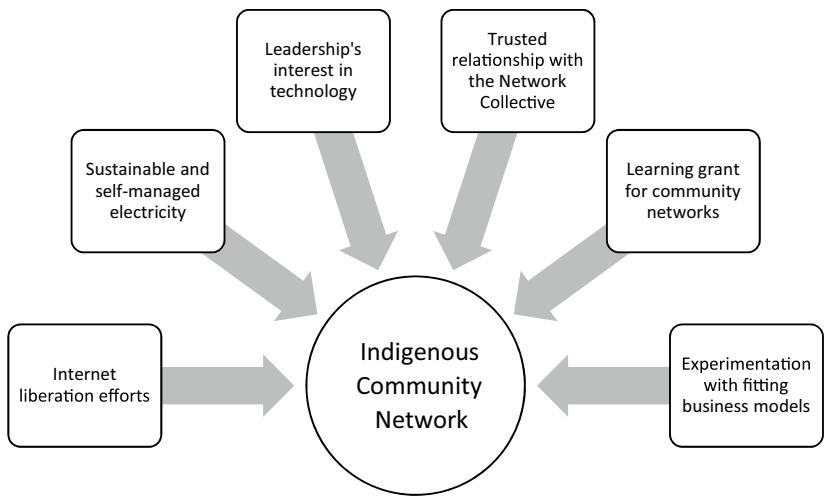


Fig. 5.19 Generative Prerequisites of the Indigenous Community Network Project

The Indigenous village's leadership's interest in technological development for the community, which lasted two generations so far, has been crucial to creating the sustainable artefact ecology within the village. Their disillusionment with external contributors, who in their top-down approaches (blueprint developmentalism) neglected the local customs, was generative too. In response, the Network Collective built trust with Indigenous communities by respecting local customs. Finally, a window of opportunity opened when a "learning grant" was made available for community networks around the world. It also supported the community's experimentation with business models that fit their culture, didn't cause any environmental harm, and empowered farmers.

5.4.3 Findings and Discussion

The Indigenous community has been developing appropriate and self-developed technologies to support their agricultural practice, for example through wind vanes, which combine local beliefs and appropriate technology:

"[W]ind vanes make the rice goddess happy, but they also help see the wind's direction for the pollination process of the rice" (G.H.I. 2020b).

The importance of the happiness of the rice goddess reminded me of the example shared by another collective regarding the demons that swallow houses on the mountain. I heard many similar examples where storytelling was used to convey knowledge. During the pandemic, for example, people dressed as ghosts were deployed to encourage locals to stay at home. Ghosts represent bad news in the local belief system, and informed them about the fatal effects of the disease (BBC 2020). Knowledge can be shared in different ways, such as through storytelling. The Network Collective's approach to these stories was to listen to them, record them, and try to understand their underlying ancient wisdom. It is also important to note that plans are being made for the future to use sensors to measure rain and seismic activities, as well as determine when to harvest. The ancient knowledge will remain central, not be erased or replaced, but merely enhanced with the use of novel technologies. Similar post-colonial computing strategies are illustrated in the artefact ecology map of the Community Network Project below (Fig. 5.20).

Brings to mind scholarship in cultural rhetorics

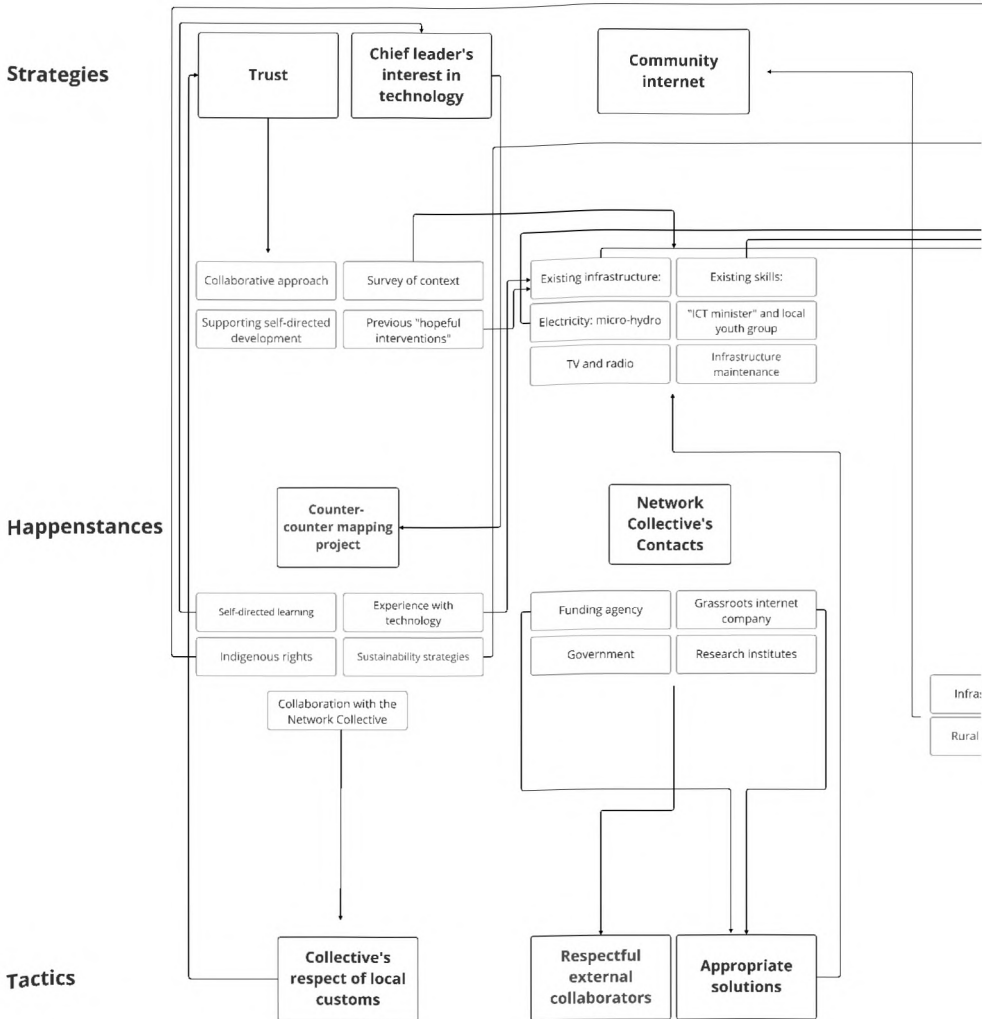


Fig. 5.20 Indigenous Community Network Artefact Ecology

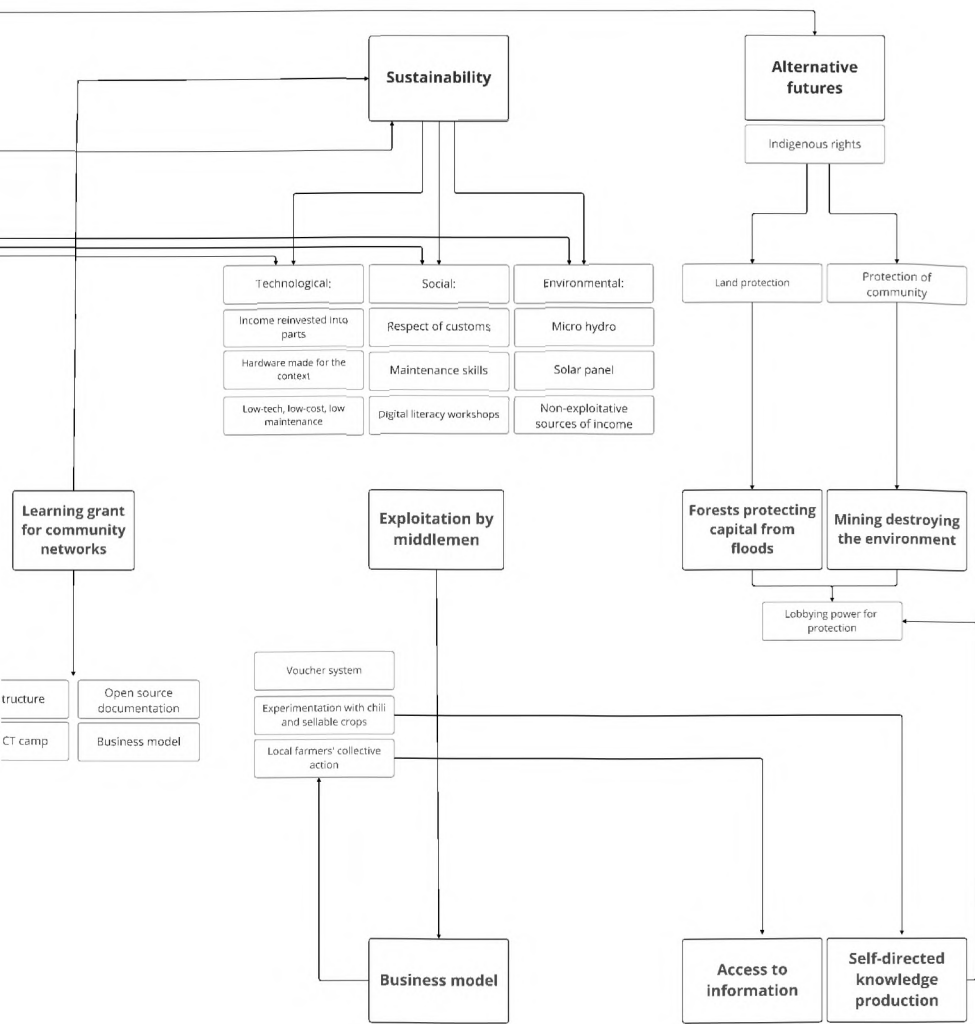


Fig. 5.20 (continued)

The artefact ecology of the Community Network Project is strategy-heavy. It is possible that this is due to the matureness of the project. Some interventions that are now strategies might have started out as tactics. While causality is difficult to determine retrospectively, a tactic that is currently maturing into a strategy is that of the business model. An element of it—the collective action set up by local farmers—had been developed as a tactic to respond to a happenstance. The middlemen had exploited a community that had no access to information about market prices. This sparked a search for strategies of sustainability, as the community learns to identify which opportunities are appropriate for their context, such as vouchers and experimentations with new crops like chili.

The Indigenous community's values mainly guide the Community Network Project, which became visible in the "counter-counter-mapping" project. The community's unwillingness to work with people who do not respect their traditions pushed them in the direction of self-development and experimentation with self-directed learning. The Network Collective's involvement in this process was a happenstance, without which the project would have taken a different pathway. Their respect of the local customs and willingness to learn and act as an intermediary and a translator between the Indigenous community and other actors was crucial in developing the project to be sustainable. However, without the Indigenous community's existing infrastructures and reflexive approach to their own values, as well as the chief leader's interest in technology, the project would have developed differently. The same insight applies to the Network Collective's values; their existing networks that allowed them to pick the right partners for the project; or to identify and obtain the community network learning grant.

These values also all enabled experimentation, leading to sustainability as a strategy. As shown in the table above and described in this case, sustainability is supported by different measures. The sustainability of the technology infrastructures is supported by context-specific hardware, which is low tech, low cost, and low maintenance. For those parts that are more difficult to maintain, the Indigenous community learned the necessary skills and gained access to the grassroots internet company. As the Indigenous community is eco-religious, environmental sustainability was one of the key values informing the artefact ecology. Finally, an alternative, non-exploitative business model was also developed that supports the maintenance of the system. Finally, access to information through ICTs also leads to the empowerment of local farmers through self-directed learning and experimentation.

To be able to access information is a prerequisite for survival in this case and constitutes the motivation for the community's work towards alternative futures, in which they have the power for self-directed development. The final strategy is

the covert goal of protecting Indigenous rights, that are under threat from multiple sides. When investigating what drives the Indigenous community to develop themselves and their ICT infrastructures, I was told that one key aspect is that their knowledge (e.g., regarding their environment) gets recorded. The second, more covert agenda was saving the land of the community. This was briefly explained in the counter-counter-mapping project, in which they taught themselves how to use tools to map their lands. Their eco-religion and culture dictates preserving the forest, hence they decided against selling essential oils, which they saw as destroying their forests. A similar problem exists with mining, which was seen as the only way to generate income before the community developed sustainable business models for themselves. Tourism, mining, and other activities could be detrimental for the Indigenous lands. Developing bottom-up strategies to fight the lobbying power and the deep pockets of the mining companies is crucial to protect the community and its lands. One key reason to “convince Jakarta” (i.e., the government) to create policies for Indigenous land protection has been that flooding leads to additional costs. If the ecosystem of forests is protected, it also protects the cities from flooding³².

Grassroots and social innovations that are *infrastructured* stand in contrast to existing power and economic configurations (Pipek and Wulf 2009; Vlachokyriakos et al. 2018). In this case, against external threats—such as the government still not acknowledging Indigenous rights and land—a fear of mapping exercises possibly leading to tourists destroying the land or making mining companies aware of mineable land, but also aforementioned middlemen banking on the lack of access to information regarding crop prices of the locals. However resilient, the official recognition of their land and their rights could help the Indigenous community protect it legally.

The artefact ecology also illustrates an important critique against top-down development expressed by the interviewees, which resulted in the aforementioned discontinuation of the first participatory mapping project. As highlighted by the founder of the collective, the “dis-embeddedness” of such development practices cause more harm than they help and are often a waste of funding. It is a general sentiment that the government’s offers to support entrepreneurship are rarely utilized. An example of this were creative hubs set up in Bandung and Jakarta. People were either not aware or they were skeptical of them, so they were barely used. In contrast, my interview partners listed what they called “non-ideological

³² The situation is so dire that the government has decided to move the sinking and polluted capital to East Kalimantan on the island of Borneo. This move ruins the basis for argumentation to gain Indigenous rights.

innovations”, such as frugal innovations of farmers. These, resembling *jugaad* and *gambiarra*, are represented by the example of a motorbike (a popular mobility tool in Indonesia) that’s been hacked to help separate the rice from its shell. As my interview partner put it, in Indonesia “people just need to make things work” (G.H.I. 2020a) because they can’t rely on the government. The huge gap in inequality drives people to innovate, and the artefacts of grassroots innovation is how people express critiques of the government.

This case contributed to the research framework and all three research questions. It shed light on how a grassroots collective was able to establish trust with a grassroots Indigenous community and involve another grassroots entity—an internet service provider—on the technology level. Criticality was expressed on multiple levels, including in the critique of past top-down and centralized approaches, the immediate empowerment by fixing issues of transparency and fairness in terms of the sales of crops and also the knowledge shared openly to inspire the reproduction of such projects elsewhere. Finally, it was expressed in long-term and covert goals to address human and Indigenous rights. The case reflected the importance of treating self-directed, intrinsic design processes with respect, as even “participatory design” projects can go awry if the local customs are not respected.

5.5 Case Study II: #hackfromhome Hackathon



This second case, like the previous one, also presents two separate, yet interconnected levels of grassroots critical making. Here I present the background of a Central Javanese collective that has been at the forefront of critical making for decades. This first part aims to shed light on how the same collective can experiment with terraforming a volcano, contribute to space research, and organize a civic **hackathon** in a pandemic. It is difficult to find one pseudonym for them, as their interests are exceptionally diverse, so I will be referring to them as the *Art-Science Collective*. This pseudonym is somewhat reductive as their art, science, technology, and maker practices have shifted a lot in the past two decades, but it still reflects most of their practices. This collective initiated the second layer in this analysis, the intrinsic design case study of the #hackfromhome hackathon. They and two other entities enabled the critical making process taking place within the hackathon through their own critical technical design practice: the organization of the hackathon itself.

The basis of the second layer of analysis is a hackathon organized as a response to the government's slow reaction to the pandemic. It was an intrinsic design project that supported the self-directed technology development of open-source maker projects reflecting on Indonesia-specific issues during the first days of the pandemic. As the co-initiators of the hackathon, the collective and the project were intertwined in this second case as well. The organizers did not only contribute design skills, technical knowledge, and professional networks; it also attempted to inspire the next generation of critical makers. The event included participants of the hackathon (or critical makers) and their intrinsic design projects, some of which they developed, while others were created through further design coalitions with each other, inspired by the collaborative nature of the hackathon.

This interconnectedness makes this case especially interesting for my scientific inquiry, as it helps us better understand the processes, skills, and strengths brought to the table by the organizers that shaped the vision of the project. The participatory-collaborative nature of the whole process, and the critical knowledge produced during the exchange are also relevant for my analysis. In this chapter, I offer a retrospective analysis of the collective's activities based on the Grassroots Innovation Movements analytical framework. Then I will describe the intrinsic design project that I joined in real time, albeit digitally due to the pandemic, as a researcher.

5.5.1 “We are Done”: An Art-Science Collective 20 Years in the Making

5.5.1.1 Context: Experimentation in the Cultural Capital of Java

To understand the experimental, creative, and artistic nature of the collective, one of the most important contextual influences is its location. Set in the Cultural Capital of Java, the Collective has many advantages that previously described collectives did not. The Special Region of Yogyakarta (or Jogjakarta) is a Javanese monarchy, namely, a sultanate. The royal families' patronage has been a major contributor to the development of arts in the region, funding song and dance, literature, and (most importantly for this case) European-influenced visual arts since the nineteenth century (Ricklefs 2008). It is also home to one of Indonesia's top-ranked public universities that the Art-Science Collective often collaborates with. The city and its inhabitants are known for being approachable, creating a welcoming environment, and the city itself is very affordable in terms of rent, food,

and other daily necessities. But in return, it also struggles with high levels of poverty (Harini et al. 2022).

These two generative contexts (a focus on arts and an open, affordable city) have fostered a flourishing creative scene in Jogjakarta. Having access to art and other higher-level education, the constant influx of students, and the low cost of living are important enablers of creativity. I already previously quoted one of my interviewees, who explains how the high cost of living stifles the creativity of some collectives in Eastern Java because they must focus on developing projects that help them make a living. By contrast, here, experimentation and playfulness (F.H. 2020a) is highly enabled by the local financial and cultural context.

The third important historical and political context that supported the Art-Science Collective is the end of the New Order, as it is one of the aforementioned post-authoritarian collectives (Jurriëns 2013). Like the Network Collective, it was founded in 1999 by young students almost immediately after the New Order ended. As fresh graduates of the art school, they started their collective as a small group of friends that engaged and supported each other in expressing themselves through visual art, graphic design, writing, and music. The emergence of the Art-Science Collective coincided with the time that ICTs became more widely available. With access to the internet, not only knowledge would become more accessible, but also bring new ways for people and communities to represent themselves. At first, they did not have a name for their interdisciplinary creative work. As one of the founders told me, they were only later inspired by their international networks from Europe who told them that their practice was “media art”. Thus, a “**new media** art collective” was unofficially but immediately founded when it became possible in post-authoritarian Indonesia.

5.5.1.2 Pathways: Media Art, Fab Lab, Self-Directed Projects

Today, the collective is registered as a foundation, and their focus is on art, science, and technology. In the more than two decades since its founding, the collective has taken different pathways and experimented with dozens of different projects³³. As one of the founders told me, the collective reinvents itself every few years. Their evolution is based on external inspiration, new technological possibilities, and the interests and wishes of the new members:

³³ A comprehensive summary of the collective’s first 15 years, based on interviews with the three founding members, can be found in Mr. Taikan Huang’s 2014 master’s thesis (Taikan Huang 2014). The thesis analyzed the collective’s activities from an open science perspective.

“We don’t have to be in a hurry to define what [the collective] is. We started it because we did not want to be put in any box” (I.W.A. 2020a).

The collective started as a community of friends who *gotong royong*, or engage in mutual aid to develop ideas based on their shared skills. During the first years, they used their capacities and creativity to experiment and play with different technologies, from VJ-ing (video jockeying) to performances with fashion. Developing ideas together, without hierarchies and in informal settings, plays an important part in their co-creative activities even today.

Their first serious engagement with societal issues was in the early 2000 s, when they organized creative workshops for children living with disabilities, using formats such as electronics, music, and origami to explore the topic of disability through interactivity and computers.

A few years later, a new wave of inspiration came from a Berlin community that experimented with electronic music, and told them about the concept of “new media”. During this time they expanded on their interest in music, and even today, the collective is well known among young people for the parties they organize and the avant-garde, experimental electronic music artists they invite³⁴.

As the collective matured and became a more formal group, its interests started turning to research and development. They started seeking more international and institutional collaboration with local and international civil society and academic organizations. These partnerships nudged the collective toward more scientific approaches. An educational program was also launched, which is basically their way of sharing the knowledge they collect, experiment with, and develop. During this time, as one of the founders told me, they tried calling themselves a hackerspace. However, that term had a negative connotation and was not well known by their prospective partners. Thus, they opted for “new media laboratory” instead. This term was indeed much more positively regarded in the early 2000 s, however, the naming doesn’t diminish the collective’s interest in hacking and making.

The next few years can be described as their exploration of science, open-source hardware and software development, and a growing focus on community development. Efforts of these years included the extension of their educational program, including science-based international media art festivals, and a range of open science projects that addressed locally specific societal issues. For example, prominent grassroots critical maker projects addressed the metal contamination of

³⁴ One example is that during my visit, a resident artist built an antenna made of a badminton racket to record electromagnetic activity, that she then turned into so-called *noise music*.

(pica! I've been to many a noise show in ne and was involved in that local subculture + communing for several years

the town's drinking water, re-nitrogenizing the soil of the local volcano so farmers could grow crops faster after an eruption, developing blueprints for open-source wine fermentation after Indonesians passed away from home-made alcohol due to the government raising taxes, and building DIY microscopes.

During these years, their international influence grew as well. As a member put it:

"At the first time, we grow as a group and then as a collective, and then after institution, we don't have any place here, even in Jogja, or in Indonesia in general. And that's why we always going abroad to do a network" (R.D. 2020).

In 2009, the group was given a grant and invited to visit nine European countries and organize workshops, performances, and exchanges. The collective experienced this as difficult times, as they had to move between countries every few days in the winter, when the Indonesian Rupiah was worth little when exchanged to Euro. It was also difficult to be pioneers and have to establish their networks from the ground up.

In the 2010 s, however, terms like hacking, making, and DIY started to become more generally understood and entered the mainstream. Having visited the Fab Lab in Amsterdam and built professional networks in the Netherlands, in 2011 they received funding to establish the first Western makerspace in Indonesia, a Fab Lab. The collective started experimenting with prostheses made from local materials like bamboo, wood, and pineapple fiber. However, they had difficulty translating the idea of the Fab Lab to the Indonesian context. A member mentioned their frustration with the technologies, starting with the pace:

"Okay. The revolution is still going... Because it's like, how can you make a prototype when you have to wait almost 24 hours just to have the mold? [...] 24 hours just to make the mold, people in China can make a thousand!" (R.D. 2020)

They also objected to the techno-solutionist approach:

"We [learn] that the technology is the solution. Yeah. And then it's not that. It's only a machine, that's the problem. That's why the FabLab, it doesn't make sense anymore" (ibid.)

She continues to explain how Western making did not make sense in the local context:

“[The Fab Lab] is not contextual (...) in our society because [people are] already crafting [...] because sometime here in Indonesia, crafting and making, it’s different things. It’s a different term. But why is different? [...] actually, it is embodied in our culture, I think [...], especially in the East, right?” (ibid.)

Thus, establishing a Fab Lab in the early 2010 s was an important experience that helped the collective understand that simply importing machines and teaching people how they worked wouldn’t be sufficient or succeed in the Indonesian context. Tinkering and DIY has been done out of necessity, as playful hacks using locally available tools (like the *wajan bolic*) to access the inaccessible. Arts and crafts have always been part of the culture. What making could add to the Indonesian society’s material practices remained unclear at this point.

Having learned from this experiment, the collective developed an international conference on “critical and transformative making”, inspired by Ratto’s and Hertz’s work. The conference was explicitly focused on critical making, and critical-material projects were completed and led to tangible outcomes—probably one of the firsts of its kind in the Global South. Unfortunately, the annual event came to an end when the founding member who organized it passed away.

These formative decades, in which the organization matured while the collective iteratively experimented with different pathways, continues even today. The collective announced that they were “done” in 2019, at the twentieth anniversary of their founding. The following period of complete renewal opened the collective’s self-definition for reinterpretation by the next generation of critical makers. In this current phase in 2022, the Art-Science Collective defines its work in four separate “micro labs” of DIY: self-space science/micro museum, self-print and publication lab, self-biology/micro wet lab and domestic hacking (conducted by the all-woman section of the collective), and self-digital fabrication/micro Fab Lab. During this time, the smallness or appropriateness of DIY came to the foreground in the collective’s activities in three ways. First, the collective continues to organize parties to attract new members. Second, it remains a vessel for people to share their ideas that would not be taken seriously elsewhere. And third, having made a name for themselves, they can attract funding for developing ideas.

5.5.1.3 Framings: Hacking, Openness and Education

The Art-Science Collective is just like many other collectives. It is a container for its members’ ideas, which are developed in a *gotong royong*-manner, as mutual aid or commons-based assistance. When the founder asked me to help develop the civic hackathon described in the second part of this chapter, she immediately

opened with this expression. In the ever-changing and iterated pathways, which also influence their framings, *gotong royong* is permanent.

Having started as a group of friends, which grew into a new media lab, then a foundation, the founders still see Art-Science Collective as merely a name that represents an open and welcoming place to share ideas. It does not want to become a finalized, static space or community. Through this freedom, and staying out of any box, the collective allowed itself to remain flexible. Throughout the past two decades, they have used the following framings:

- **Hacker ethos and experimentation:** the collective called themselves a hackerspace before hacking became palatable for the mainstream. They had to abandon this framing, but hacking remained a big part of their practice, which includes taking things apart, opening the “black box”, examining and altering its processes, and generating output from this knowledge through an educational program or a social innovation. Their hacker ethos became most visible during the critical and transformative making conference, which was explicitly framed around questioning hegemonies. It addressed how technological systems are designed to exclude people, dignity, and personhood in innovation, and whether catastrophe can be averted through rational means alone.
- **Open-source software and hardware:** the collective is known for its open-source approaches, meaning that they use open-source technologies in their projects and their research and developments are shared openly. However, there have been disputes around IP rights and the recognition of people’s contributions. The collective’s approach is somewhat loose when it comes to such questions, and their justification is that they simply say they “recycle” ideas. A founders’ interpretation is:

“There are also a lot of people [who] ask: Well, you want to keep your research open, why you want to give your knowledge, why you stick on the openness and why you fight for open culture, open science, open design or open-source technology? But I think the (concept) open means you give more rather than you take” (I.W.A. 2020c).

- **Open science and citizen science for survival:** In cooperation with diverse researchers and universities, the collective has worked on various open science projects, many connected to microbiology. These projects are turned into art pieces for exhibitions around the world and start conversations with people. The collective also cooperated with locals to further citizen science. For

example, their all-women collective turned liquid soy waste (due to the tofu and tempeh production, this is widely available) into edible cellulose, biofuel, and bio-leather, by working together with a community of poor women in a rural area. As one of the members put it, this project reflects an almost survivalist-like ethos of self-empowerment:

“We cannot (...) have, even I don’t have any hope [that] the government or (...) people will be taking care of me. But through this perspective, I think the hacking making perspective I can survive and that if you (...) cannot buy [clothes] under the global crisis I, can still wear clothes because I can grow it” (R.D. 2020).

- **Technological citizenship:** The collective aims to establish a stronger technological Indonesian citizenship by directly contributing to post-colonial development. During my stay, they held a “climate dystopia” workshop for citizens to learn about electronics. They used this as an opportunity to educate people on microcontrollers and the negative effects of the climate crisis, which is predicted to disproportionately affect Indonesia. They also used their education platform to spread the technological solutions developed in-house. But maybe most importantly—and strategically—they have an open-door policy. They hack and experiment in their space while leaving the doors open for the *kampung* to see, which has led to children often curiously coming by and learning by doing with the collective.
- **Appropriate technology, DIY, and self-directed activities:** A focus on the “micro-labs” is a recent framing, which was only announced on January 1, 2022. When I asked the founder how come they decided on this direction, she reasoned that they were attempting to be minimalist. This meant that the collective wanted to shed some weight to use minimal space, energy, and equipment.

The Art-Science Collective is thus a vessel for open-source software and hardware, open science, microbiology, feminist, educational and many other projects. These projects are used as framings for a broad self-identification and to foster discussions with local, national, and international partners, as well as to attract new members.

5.5.1.4 Spaces and Strategies: Inspiring the Next Generation of Makers

To better understand why the flexibility of a broad type of self-identification is necessary, it is important to investigate the Art-Science Collective as a dynamic community in which both spaces and strategies frequently change. In terms of spaces, there are the physical and metaphorical spaces of activities. The former currently consist mainly of the homes of the co-founders. The collective used to have its own house, where the Fab Lab was located. During my field research, the collective was renovating a house near the home of a founder, which had shared spaces on the ground floor for experimentation, exhibitions, and events, and small apartments for artist residencies. These rooms were turned into short-term accommodations for locals who had to quarantine during the early days of the pandemic but had nowhere else to go (essentially becoming homeless for the period). Then, the collective stopped renting the house as it was a big expense, and the makerspace moved into a back room of one of the founders. Another founder hosts a space exploration museum in his home. Thus, circumstances have had a major influence on the physical space the collective uses, and its size and location has changed often. Most importantly, these spaces are used for outreach and education, which inspires the next generation of makers. The collective runs workshops for different Central-Javanese universities that cannot provide their students with maker tools or skills, for example.

Their strategies have been developed over the past 20 years. The collective originally didn't have access to international networks, but as described in their pathways, they had been strategically and painstakingly developing them. Today, they are composed of diverse groups including academia, industry, and other collectives, but also "networks of networks" such as artists' networks or innovators' networks. The collective has been receiving a lot of visibility because they translate their R&D findings into artworks that raise critical societal and environmental questions, and have been exhibited around the world. Their strategy of openness allows them to "recycle ideas", as they put it, or be inspired by works they encounter. Low-cost rapid prototyping methods and open-source electronics also create a window of opportunity for artists working with low budgets. Open-source tools to combine art and science result in audiovisual experiences, organic textiles, and prostheses made of everyday materials, and DIYbio processes of fermentation happen live, in front of the audience's eyes. In addition to the interesting, novel topics represented in their artefacts, post-colonial and sociotechnical framings, and appealing aesthetics, the collective members' friendliness has probably also helped their success in building networks.

The resources for projects include the aforementioned open-source resources, which lower the costs, and the collective only rents a house when it has the means. It is still a major effort to mobilize financial resources, as this was one of the skills of the founder who passed away. The collective also mobilizes resources by bringing members on board who develop their projects on a voluntary, unpaid basis (or *gotong royong*). However, this has led to disputes over projects that were funded. Income is mainly generated on a project basis. As mentioned in the case of the DIY biology collective, it is impossible to charge for educational workshops for locals, but collaborations with local universities still generate income. The main independent means to generate income had been through paid residencies, while the collective rented a house with apartments. Independence is an important keyword; the collective tries to keep up their neutrality. For example, they have refused to accept sponsorship from a major sports company to start an “innovation lab”. Their impression was that working with the funder would negatively influence their work, which has been a legitimate worry expressed by other makers too (Dougherty 2012a).

(This citation is from around the time of the dispute between Mitch Altman and Tim O'Reilly about the implications of the maker faire receiving funding from the Defense Advanced Research Projects Agency (DARPA))

5.5.2 Intrinsic Design Project: #hackfromhome Hackathon

As demonstrated in this retrospective description of the Art-Science Collective, their organization has remained flexible. At first glance, this might make their practices seem somewhat chaotic and lacking structure, but it has been advantageous. Flexibility allows them to react quickly to windows of opportunity, engage in activities that interest the collective, and learn new skills. This was also the case in the intrinsic design project described in this case.

The window of opportunity had been rather grim. In the first months of the COVID-19 pandemic, governments around the world reacted differently. In hindsight, the WHO had been criticized for issuing contradictory statements on how to react, and scholars called it “The First Post-truth Pandemic” (Parmet and Paul 2020). After governments started closing borders in mid-March 2020 and I arrived back to Europe, having cut my Indonesian in-person field research short, I was contemplating how to move the case online. The founder of the Art-Science Collective reached out to me as her and two other founders were under the impression that while they saw friends and family members get sick and pass

away, the government was not acting according to the seriousness of the situation³⁵. As their impatience and worries grew, they talked amongst themselves and with their networks about what could be done. They shared the impression that people in their networks were interested in developing solutions themselves. However, because of the pandemic and the geography of Indonesia, an in-person event bringing them all together wasn't feasible. Thus, the idea they came up with was to organize an online hackathon. The main challenge was that although some had organized hardware hacking workshops before, none of them had experience organizing an online hackathon.

5.5.2.1 Issue-Oriented Hackathons

The word hackathon is a portmanteau of “hack” and “marathon”; basically, they encourage the development of technology solutions in a very short time, often without pauses or even much sleep. Challenges are traditionally set by companies who are either looking for a proof of concept for their apps, or engagement with their APIs. Ultimately, companies get developers to work for them for free, in exchange for a challenging and fun time with their peers.

In recent years, hackathons that do not focus on software development also started taking place. Humanitarian hackathons³⁶, scientific hackathons³⁷, and technology transfer hackathons³⁸ were organized. The latter, which I co-organized, was set out as a collaborative hackathon, rather than a purely competitive one. Although the winning team did receive a free trip to CERN, teams were encouraged to share their knowledge with each other and received extra points for collaboration. Traditional, competitive hackathons organized by companies for developers have received criticism and participants have expressed skepticism about them. In my experience, the term itself can discourage participation in people who have been disillusioned by the sleep deprivation and rush.

³⁵ To paint a picture of the situation: newspaper articles detailed the dire pandemic situation in Indonesia, including how the government was reluctant to reveal the breadth of the outbreak. The John Hopkins Centre for Health Security put the Indonesian mortality rate at 8%, highest in the world at that time (McBeth 2020). Indeed, a few weeks earlier, the Indonesian government still communicated that the country was free of COVID-19, but critics said this was impossible. A lack of testing and other protective measures, combined with a “fear of creating mass hysteria” (Hermawan 2020), led to a lack of transparency. Instead, they tried to shield the tourist industry and allocated around 72 billion IDR (5.2 million USD) to influencers to promote tourism, while due to the PPE shortage, healthcare workers were seen wearing raincoats to protect themselves from the virus (Tarigan and Milko 2020).

³⁶ <https://theport.ch>.

³⁷ <http://sciencehackday.org>.

³⁸ <https://www.f6s.com/attractyounghackathon>.

Conversely, participatory citizen innovation can motivate continued involvement. Research on issue-oriented hackathons has focused on shaping broader discourse and certain societal problems. Bringing together interdisciplinary teams points to the value of networking over technological outputs (Lodato and DiSalvo 2016). In the year 2020, there was a proliferation of online hackathons³⁹ addressing various societal issues caused by the COVID-19 pandemic, mobilizing thousands of people around the world. Some were open in terms of topic but country-specific (see for example India⁴⁰, Germany⁴¹, Switzerland⁴²) or regional (for example in the European Union⁴³ or the Asian Development Bank's hackathon for Asia and the Pacific⁴⁴). Some had a humanitarian focus⁴⁵, including social issues and mental health⁴⁶, while others focused on data science and novel technologies (e.g., artificial intelligence and blockchain⁴⁷). Comprehensive healthcare data sets were published in open-source repositories to support the participants' work⁴⁸.

It is not surprising that the year 2020 caused so many hackathons to happen. The pandemic accelerated internet penetration (ITU 2021), and due to the lockdowns, many people found themselves at home with unexpected free time⁴⁹. Thus, the impression of the organizers was that people were looking for activities and—as discussed later in this chapter—many have suddenly lost their sense of agency. By proactively developing solutions, they thought, they could feel like they regained their agency, while applying their skills for a good cause.

The main goals of the hackathon detailed in this case, as per a draft for a newsletter posted in the messaging group (on 4th June 2020), were to activate

³⁹ See e.g. <https://joinup.ec.europa.eu/collection/digital-response-COVID-19/hackathons-and-events>, <https://www.programmableweb.com/coronavirus-COVID-19-hackathonsor>.

⁴⁰ <https://coronathon.in>.

⁴¹ <https://joinup.ec.europa.eu/collection/innovative-public-services/news/COVID-19-we-vs-virus-hackathon-de>.

⁴² <https://codevscovid19.devpost.com>.

⁴³ https://ec.europa.eu/info/news/euvsvirus-hackathon-develop-innovative-solutions-and-overcome-coronavirus-related-challenges-2020-apr-03_en.

⁴⁴ <https://www.adb.org/news/events/digital-against-COVID-19-hackathon-winners-showcase-live-webinar>.

⁴⁵ <https://www.equalithon.io/equalithon-list/event-one-y2xjb>.

⁴⁶ <https://covid-global-hackathon-2.devpost.com>.

⁴⁷ <https://covidathon.devpost.com>.

⁴⁸ <https://github.com/aiformankind/COVID-19-hackathon>.

⁴⁹ Here it is important to mention that they were fortunate. All around the world many people suffered from difficult times: lost their homes, livelihoods, had family members or friends who got sick or passed away. Some of this will be addressed below.

people's creativity by forming interdisciplinary teams of artists, makers, hackers, designers and scientists to co-develop locally relevant open-source hardware solutions and culture hacks for Indonesian communities. Thus, this hackathon was categorized as issue-oriented, as it wanted to inspire critical making in a post-colonial manner—by giving voice to locals to share and address their concerns. The case in this study aims to investigate the hackathon as an intrinsic design project on the level of its organization, mainly the *nongkrong* or ideation process, negotiations, and conflicts that represent the values and interest of the organizers. It also aims to derive findings on issue-oriented hackathons based on the feedback of the participants.

5.5.2.2 *Nongkrong*: Strategies of Trust and Care

Assessing the pathways of the Art-Science Collective, organizing an online critical making hackathon was a logical next step during the pandemic. For two decades, their members had developed participatory workshops and conferences for people to learn how to develop technological solutions. They already had a community of potentially interested people infrastructured, including those that participated in such events previously, or people in their academic and civil society networks interested in sociotechnical questions. They felt comfortable with engaging local communities of hackers, makers, and artists. Similarly, N:ow also built a community, because people using their co-working space and other services were more interested in social entrepreneurship and social innovation. This meant that the technology- and science-oriented community was supplemented with people interested in the social side of innovation and development. At this time, STACK was still new, but the founder being a famous media personality, she also could meaningfully conduct outreach to citizens through social media and television.

The founders were certain that, based on preliminary discussions and using the existing channels and networks, a large number of interdisciplinary participants would come to the event. As a next step, the grassroots communities had to infrastructure themselves to unite as an organizing team. Not long after the first message from the founder of the Art-Science Collective invited me to become a co-organizer and share my knowledge on issue-oriented hackathons, a messaging group was established. This was the first step in the *nongkrong* part of the process. As learned from other collectives, *nongkrong* (hanging out) is the first step of a project. It means to sit, talk, and create trust through exchange, while ideating. This can be a lengthy process as different options and possibilities are weighed, but it is crucial for all parties to reach an agreement before a project is developed.

Between Urgency and Strategy: In the case of the hackathon, this was a difficult step. In Indonesia, the pandemic was killing people, and it also caused

immediate economic issues in the informal sector and in the gig economy. Loss of income was a major fear for many, and some gig workers even evaded quarantine (Fadli 2020). From the perspective of the Art-Science Collective's founder, there was no time to waste; she was already experiencing the negative effects first-hand. Within a few days, her and the other partner organizations decided to cooperate. As the founder put it, "We will do it together, we should fight this together" (I.W.A. 2020b).

They did not only come up with the idea of an online hackathon, but also with locally relevant topics to inspire participants. The hackathon was originally framed as an attempt "to help the government combat corona", and while this framing is very positive, concern and critique is also part of the message. The hardware solutions to be developed—DIY hazmat suits, thermal infrared cameras, and disinfectant sprays and liquids, as well as mass screenings—were supposed to counter the lack of action by the government. The co-ideation period also included brainstorming potential other partners.

With the urgency felt by everyone, one key question remained to be answered: how to make sure that the solutions developed make sense and can really sustainably spread, so that the hackathon is not simply a one-off and feel-good event for participants? The ensuing discussions caused some friction, as the Art-Science Collective felt like they were ready to launch the communications campaign, as they had designed brochures, presentations, and a captivating video. They wanted to launch the hackathon itself immediately, not to wait until an ideal solution is developed; any solution would be better than nothing. The other two co-organizers, bringing their social entrepreneurial thinking, focused on attempting to define systemic, long-term goals and impacts the project could have. Reacting to urgent needs on site at a surprising speed, the Collective turned their residency apartments into isolation rooms and just like many other makers, and started using their machines to make PPE for healthcare workers (Kieslinger et al. 2021). As they explained, taking action helped ease their anxiety.

Infrastructuring and Care: About a month after the first discussion, concrete roles were still unclear, but the infrastructuring between project partners started slowly started taking shape. Two organizers contemplated how to start building the artefact ecology of the hackathon and proposed involving a renowned Fab Foundation member living in Bali. The Art-Science Collective intervened and shared that they already did a lot of the necessary planning work in the background. They had already developed specific challenges pertaining to households, medical care, communities, culture hacks, and the circular economy, and developed a project plan. After the initial surprise, all somewhat relieved organizers agreed to go through the to-do list and define what else needed to be done, and

distribute roles. Major details were still to be clarified, including financing, mentorship, medical standards, and dates of the hackathon itself. As the hackathon was envisioned as a “collaborative hackathon”, ideas about how to make the collaboration happen online and support people who had never met each other was also still being ideated. If people were to collaborate without knowing each other, merely based on having submitted similar ideas, who was to lead the team? What profile would the mentors have to have, and how would they be assigned to each team? How were the guidelines for participation and desired outcomes to be formulated by the organizer team if the potential team leaders were going to be makers, and not businesspeople and project managers? The organizers agreed that complicated rules and big words would turn them off as most people would “be there and make things”. Taking a step back, the organizers debated if Indonesians would understand the word hackathon. Thus, two online live events were planned to explain the concept, and any questions would also feed into the FAQ being prepared. The founder of N:ow tried to ease people’s minds by saying:

“The most important thing now is to ensure the hackathon runs smoothly. There will be different expectations, and if we mess up, that is OK. We are in this together” (F.H. 2020a).

The organizers however also shared their worries about the increasing time pressure, as the annual Idul Fitri⁵⁰ was approaching, when people working in cities would return to their hometowns at the end of Ramadan. Scientists started calling for a lockdown and warned that a large number of travelers could heighten the risk of COVID-19 becoming a nationwide epidemic (Fachriansyah 2020).

5.5.2.3 *Gotong Royong* as Tactics: Hackathon Design and Development Process

In the first few days, a detailed and well-designed presentation that was supposed to be sent out to various entities and potential funders in Jakarta was assembled. To the outside world (funders, media, or potential partners), the look of a well-organized team with clear roles that organizations play listed was important when communicating the seriousness of the event. I was invited to include myself as board member, advisor, or member of the organizing committee, despite the internal agreement that the organization of the hackathon itself would not adopt a rigid structure. Instead, it would build on the interested parties, “just working together organically... in Indonesia we call it ‘gotong royong’” (I.W.A. 2020b).

⁵⁰ Or Eid-al Fitr in Arabic, is a religious holiday marking the end of Ramadan, celebrated in the community with food. In Indonesia, it is a national holiday when people return to their hometown (*mudrik*).

Gotong royong is a term that translates in organization theory roughly to “mutual cooperation” or “mutual aid” (Kropotkin 1902). Essentially, the community shares the responsibility and works together through reciprocity (Bowen 1986; Alatas 2010; Pushkin 2020). In my experience, this term is lived historically in traditional communities and also by most collectives I visited.

Stepping Up and Out of the Comfort Zone: A few weeks after the first call, many hands-on details (such as a project plan) were proposed by the Art-Science Collective. Yet, these were communicated late with the other organizers, who, as described in the previous section, had already raised issues about the hackathon’s sustainability, impact, and potential projects that it might be generated.

At this stage, the project was also burdened with a familiar constraint of volunteer-based community projects: having few resources, in this case, no budget and limited hours. In addition, although the organizers knew each other on a personal level, they had not worked together professionally on a project before. None of them had organized a fully remote, collaborative, issue-oriented hackathon before. They were being friendly and polite, thus the concept that the project would be organized non-hierarchically also meant that people were hesitant to step into leadership roles. Hence, working together “organically” when organizing a complex project slowed down the design period.

Tactical Interventions as Answers to Reflexive Questions: Developing the project design and translating it into concrete tactical interventions in line with the organizers’ values, skills, and experience was the first major task. Tactics are situated activities that deal with plans and designs (Bødker et al. 2016), and here refer to the answers given by the organizers to their own reflexive questions, which resulted in strategic actions. Once the strategizing and way-finding phase came to an end and roles were clearly defined, tactical interventions could finally begin.

While hackathons are traditionally time-based events most often restricted to 48 hours, this hackathon was organized very differently. Reacting to some of the questions asked in the previous section and others raised in the next phase, tactical interventions that deviate from how other hackathons were designed. The strategic questions were asked in terms of *how* to organize the hackathon to mirror the care and reflexive thought processes of the organizers. In the table below I offer an overview of how the skills, problem framings, and tactical interventions were combined, reflecting the comfort zones and different values of the organizers, as well as their negotiations. This table underscores how the happenstances were addressed through tactical practices (Table 5.3):

Table 5.3 Skills, Problem Framings and Interventions Shaping the Artefact Ecology

Organization	Skills (Happenstance)	Strategies: collaborative, reflexive problem framing	Tactical interventions
Art-Science Collective	Participatory tech events	How to organize the framework of the hackathon?	Sign-up process, selecting and onboarding mentors, jury selection, and process
	Tactical approach	How to deploy the hackathon fast, reacting to the urgency?	Quick and efficient project design
	Artist / maker / hacker networks	How to reach participants with a technical background?	Outreach: social media and newsletter
	Documentation	How to ensure the open-source blueprints are made accessible to a wider public?	Winning blueprints collected and published in a brochure
N:ow Space and Community	Systems thinking	How to build a hackathon that makes sense societally?	Strategic, detail-oriented approach reflecting societal particularities
	Social innovator networks	How to attract non-technical participants?	Outreach: social media and newsletter
	Systems thinking	How to support the newly formed winning teams sustainably?	Follow-up workshops for knowledge transfer; centralized fundraising campaign
STACK Foundation	Media experience	How to bring nationwide attention to the hackathon?	PR: traditional media and interviews

(continued)

Table 5.3 (continued)

Organization	Skills (Happenstance)	Strategies: collaborative, reflexive problem framing	Tactical interventions
	Access to potential funding	How to generate funding for prototypes and documentation?	Fundraising from the Friend Center and budgeting
Shared interests	Social / collaborative project development	How can a hackathon (which is a traditionally competitive format) become collaborative, when people don't know each other?	Strategic support to make collaboration happen, also through mentorship design
	Open-source software and hardware	What types of existing software solutions (free of charge and/or open source) could support the goals of the hackathon?	Proposing possible contributions to the artefact ecology

These questions and the tactical interventions given as answers suggest that the different organizations were able to *gotong royong* together in a way that might have taken somewhat longer than originally hoped for by the initiator. Still, the Art-Science Collective allowed organizers to build on each other's work and do their tasks in a way that reflected their values and skills. The organizers were also able to flexibly interpret and redesign the hackathon format to work for this particular context and the profile of the participants they envisioned would participate.

5.5.2.4 Developing the Appropriated Artefact Ecology

Flexibility and considerations for techno-social reality highly influenced the artefact ecology of the hackathon. In these considerations, a strong dichotomy of inclusion and exclusion was present from the first discussions. Due to the pandemic and the lockdown, organizing the hackathon online was the safest decision. From an optimist perspective, going online meant that more people, even those who are not living on Java, could participate. At the same time, two main sociotechnical burdens became immediately visible. As the deadline of the application process was approaching, the organizers received a worried note that participants on a small island near Sulawesi had not had electricity and internet for a few days and would need the deadline extended to be able to apply. With a weak connection, applicants predicted that they would also have trouble uploading their pitch to a video streaming service. Trying to find a solution, they said sending it compressed, via an online messenger could be a workaround, if the organizers allowed. Of course, they did.

Concerns about the lack of electricity and internet infrastructure in many parts of the country was coupled with the target audience mainly using mobile devices to get online. Using mobile data for multiple days means high costs, and inevitably would exclude many who would have been able to join the hackathon in person. Especially younger and more educated people use mobile internet because it is often faster than landline (Puspitasari and Ishii 2016; ITU 2021). Highly influential in the smartphone usage patterns of young people is the “free basics program” of an American social media company. In practice, this means that when buying a data package for \$8 USD, the user receives 23 gigabytes (GB) of data, divided into 11 GB of internet access and 12 GB for various Facebook apps and other streaming services⁵¹. During my field research it was almost impossible to buy data packages that did not include the “free basics”; I only found one package without it. If a user does not wish to download the apps of this social media company, they are paying extra for data they cannot use. Scholars have been warning about digital corporations creating “walled gardens” (Sen et al. 2017) and the same constraints also had a major influence on the artefact ecology.

⁵¹ Current prices in April 2022 from <https://www.telkomsel.com/en/paket/omg>.

Open source or free of charge? At the beginning of the discussions with the organizers, people agreed that those accepted into the hackathon should all commit to developing open-source technologies. The values of open source resonated with the values of the organizers and the hackathon, moreover the organizers aimed to attract prospective participants that would be driven by altruistic motives, which have been linked to open-source software development (Oreg and Nov 2008; Gerosa et al. 2021). Due to the aforementioned lack of funding and urgency, the organizers did not even consider developing their own platform for the hackathon. They were also not considering self-hosted open-source solutions, as there was a lack of funding and expertise.

Signing up meant that the participants would repeatedly have to give away their data to different entities. When shortly after the outbreak of the pandemic, work, leisure, and hackathons moved online, Sinderson (2020) produced a toolkit reflecting on threats and inconveniences including data safety and privacy for users in this online sphere. It asked people to recognize that they engaged in *digital duct taping* as they “glued” different platforms together for their needs. The artefact ecology of this hackathon was also digitally duct taped together from appropriated platforms because ultimately the socio-technical needs of participants came before any techno-political considerations. We needed to ensure that as many people signed up as possible.

Ngoprek-ing Appropriated Technologies: The word *ngoprek* means to hack. As demonstrated in the previous chapters, Indonesian grassroots innovators have hacked (Pushkin 2020) or found ways to open, alter, and appropriate different technologies for their own needs. In this case, the need was to organize a hackathon to save lives in a resource-constrained and urgent situation. Below I will discuss the artefact ecology, including the rejected tools, as rejecting the use of a particular tool is also an important part of the appropriation process (Bødker and Polli 2014). In this case, when the organizers started designing the hackathon, they looked at other hackathons and which platforms they were using. The German government for example was using the hackathon platform *DevUpload* and the team messaging and collaboration platform *Slop* popular among startups globally. *DevUpload* was immediately excluded because it didn’t support collaborative hardware hacking. The organizers set up a preliminary *Slop* channel and started using it, especially as the different workspaces would have facilitated

the group building and communication process. But then worries about lack of bandwidth caused the organizer team to abandon it:

F.H.: “[...] let’s anticipate for those who might be in more remote areas and might be hindered from joining in because of lack of connection to [*Slop*]”.

C.I.: “We can also create different groups on [*Howdy*] [...]”

I.W.A.: “[*Slop*] is unpopular here, but very popular abroad during the pandemic. We considered before because [*Slop*] create[s] a workflow that makes it easier for participants [...] First idea is just using [*Howdy*] and they make a document sharing in drive and group call via *Whoosh*. But then so many apps and platform they need to use, while [*Slop*] is accommodating everything they need”.

F.H.: “from my part, if we want to use [*Slop*], we need a live tutorial. [...] if some people find it too complicated we can add [*Howdy*] as comm[unications] platform [...] So there will be two comm/working channels depend[ing] on the group[’s] accessibility [...] Third, we can just decide to use [*Howdy*] group”.

I.W.A.: “Oke, we create [*Howdy*] group for them, then give them another form link to submit & upload documents and video”.

(F.H., C.I., and I.W.A. 2020)

In the process of weighing advantages and disadvantages, discussions regarding the hackathon’s artefact ecology first centered around how potential platforms were functional enough for the hackathon. The second concern was accessibility—whether the platforms were known and used by Indonesians, and if they were included in Meta’s “free basics”. The third concern was whether the platforms were open source and respected data privacy (GDPR). As I demonstrate in the table below, the latter was indeed taken as third-rate criterion and did not apply to any of the selected platforms. Ultimately, whether the functionalities could have been achieved with fewer platforms (and therefore, less duct-taping and giving away participants’ data to companies) was also less of a concern for the organizers. They opted for accessibility, as to exclude as few people as possible from the hackathon (Table 5.4).

Table 5.4 Overview of Rejected and Accepted Platforms

Platform Pseudo- nym	Accessibility		"Free basic"	Functionality			Privacy		Notes
	Popularity	Band-width		Registration	Repository	Documentation and presentation	Internal communications & collaboration	External communication	
Rejected									
<i>DevU- pload</i>	x	✓	x	✓	✓	✓	x	✓ (GDPR)	Accessible and used for hackathons, but mainly for software, not hardware
<i>Garage</i>	x	✓	x	✓	✓	lim.	x	✓ (GDPR)	Accessible, free of charge for NGOs, but no collaboration or update of ideas possible; re-upload needed after hackathon
<i>Forker</i>	tech niche	x	x	✓	✓	✓	x	x	Very technical, complicated to learn, repository mainly for open-source software
<i>Object- universe</i>	makers	x	x	✓	✓	✓	x	x	For hardware projects; makers know of it, but regular people don't
<i>VolP</i>	x	x	x	x	x	x	x	x	Voice over Internet Protocol videotelephony; registration is required, no breakout rooms, obsolete

(continued)

Table 5.4 (continued)

Platform Pseudonym	Accessibility		Functionality				Privacy	Notes	
	Popularity	Band-width	"Free, basic"	Registration	Repository	Documentation and presentation			Internal communications & collaboration
Open-source VolP	x	✓	x	x	x	x	✓	✓	Not popular or well known in Indonesia and integration into popular social media did not seem easily doable
Slop	x	x	x	x	x	x	✓	x	Limited use for this purpose but other tools could have been integrated easily. Disadvantage: duct taping, privacy. Tutorial would have been necessary.
Accepted and used									
Collective's website	✓	✓	x	x	x	x	x	✓	Low-cost, easy to access and to modify by the collective, thus, a convenient way to have a go-to site for information.
Whoosh	✓	x	x	x	x	x	✓	✓	Low-cost, videoconferencing, easy to use and to join group calls via links, breakout rooms possible for different teams.

(continued)

Table 5.4 (continued)

Platform Pseudonym	Accessibility		Functionality				Privacy	Notes	
	Popularity	Band-width	"Free basic"	Registration	Repository	Documentation and presentation			Internal communications & collaboration
<i>Social media</i>	✓	✓	✓	x	x	x	x	✓	Free of charge, existing follower base; easy integration of the videoconferencing app into popular social media sites.
<i>Howdy</i>	✓	✓	✓	x	x	x	✓	x	Free of charge, easy group building, invitation via links and phone numbers, accommodates the sharing of pictures, files, links. Very popular in Indonesia.
<i>Many Docs Editors</i>	✓	✓	x	✓	x	✓	✓	x	Easy integration and sharing through links, most Indonesians are familiar with the various tools the platform offers.

In the end, the organizers opted for a mix of platforms that they had experience with and assumed that the potential participants would also find these easy to join and work with. They thought a tutorial would need to be organized for some platforms, which would have been too much of a burden for the over-stretched team. This was not the case with the selected platforms. In addition to trusting participants to be able to use them, the organizers also considered how the different platforms could complement each other:

- **Through easy integration:** For example, the *Whoosh* video conferencing tool with two *social media* platforms allows “live events” allow organizations to immediately reach their existing follower base.
- **By easily generating and sharing links:** For example, links could be shared to documents in *Many Docs Editors*, which all team members can be invited to edit. Or links to *Whoosh* video calls could be generated whenever needed, which can easily be shared, recorded, and traced later in the *Howdy* online messenger.

Three of these apps are part of the “free basics” program and are extremely popular and widespread in Indonesia. Hence, their popularity enabled the organizers to duct tape a hackathon together which would be accessible to many people. As seen in Table 5.4, however, most of them raise major privacy issues, a concern that was also reflected in an online message exchange. One of the organizers asked me what it meant that the Estonian platform *Garage* explicitly stated in its terms that information might be shared with “other parties in aggregated form”. I answered that if the organizers decide to use the platforms, any sharing of the participants’ information with third parties was to be done in aggregate to comply with GDPR, and I (R.S.) added:

R.S.: “[In Europe,] we take our privacy very seriously”.

F.H.: “here, [digital privacy] is a myth” (F.H. 2020b).

The co-organizer’s answer showed why potential platforms that were more privacy aware but were less fitting to the socio-technical context of potential participants were rejected. The platforms that are popular, mainly due to the

“free basics” program, were preferred by the organizers, as they were assumed to be more accessible to a larger number of people without bandwidth, money, and specific training. This however meant that the hackathon—out of necessity and urgency—contributed to the “walled garden” phenomenon by encouraging people to stay on those platforms.

5.5.2.5 Hackathon Timeline

Having designed the artefact ecology, the organizers reflected on what tools would need tutorials. During this process they realized they could not simply just make an announcement on a website. If they wanted to engage people with a non-technological background or those who are disillusioned by hackathons, they needed to do more outreach. Thus, before the event, raising awareness was the main goal.

Aside from interviews in media, the organizers also held two online and interactive sessions on their social media, inviting people to “Ask Me Anything”⁵². In these, the organizers highlighted that this hackathon will be different from what one might know, as it reflects on issues of local relevance thus is not tech-solutionist, and that the format will be collaborative. The pre-designed challenges were also shared in these sessions in the hope that the topics would inspire people to reflect on how they can contribute, asking for solutions for the household, hospitals, and communities. Sub-topics within those challenges included affordable and safe UV light boxes, disinfectants, hazmat suits, masks, and ventilator parts, but also culture hacks and a “wild card” category for other ideas.

A total of 67 applicants applied with diverse skillsets, including 24 students, 7 artists, 7 designers, 6 engineers, 5 people who belonged to a “maker family” (including 2 parents and 3 children), as well social workers, environmental activists, teachers, a nurse, and a digital hygiene trainer⁵³ (Fig. 5.21):

⁵² AMA is an open online format, in which the presenter proposes a topic (this could range from their expertise to any personal situation), in this case, the idea of the hackathon, and the audience can pose questions.

⁵³ “Digital hygiene trainers” are certified by a Training of Trainers (TOT) educational program run by Google with the Indonesian Alliance of Journalists on fact checking digital information <https://aji.or.id/read/press-release/1081/training-of-trainers-tot-google-news-initiative-training-network-untuk-akademisi.html>.

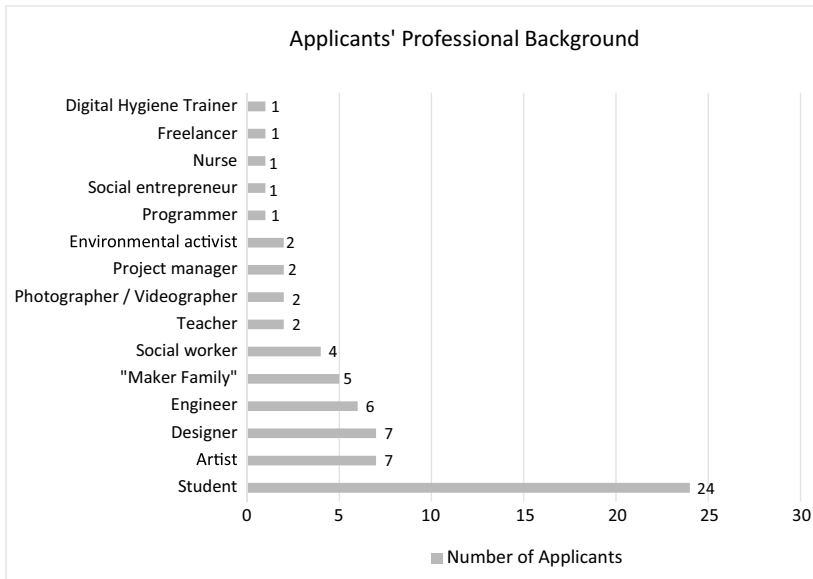


Fig. 5.21 Hackathon Applicants' Professional Background

The applicants were asked what types of tools they had access to. Computer ($n = 58$) and home printer ($n = 20$) were the most common, followed by maker tools like electronic components ($n = 11$), a sewing machine ($n = 8$), 3D printer ($n = 5$), laser cutter ($n = 1$), and photography and videography equipment ($n = 3$). The founder of the East-Javanese makerspace and tool library also joined, who indicated that he had access to wood and leatherworking tools ($n = 1$). Other creative prototyping tools indicated by the applicants included sculpture equipment such as a welder; grinder; drill, gas torch; screen printing, cooking utensils; knitting needles; soldering iron, electric iron; mini weaving loom, sewing utensils.

The applicants were divided into 12 groups based on their skills and the challenges they applied to. These included 2 “household perimeter”, 2 “medical perimeter”, 3 “community perimeter”, 3 “culture hack”, and 2 “wildcard” groups. During the hackathon, time and support was allocated to team formation, shared ideation, and the selection of a team leader, while mentors fostered a supportive environment and acted as intermediaries between teams who shared

similar issues, concerns, needs, or technological solutions. The individual groups then refined hardware projects, including (Fig. 5.22):

- **Open-source wearable technology in the form of an armband or watch with a body temperature sensor, a distance sensor, and a buzzer that alerts the wearer when others come too close.** This artefact reflects on the “issue” of many people not being able to afford commercial health tracker products. The other concern it addresses is that of collectivity and community in Indonesia, where people deliberately spend a lot of time close to others, a custom that helps spread the virus. The group also raised concern about the government’s plans to deploy a contact-tracing app that did not consider privacy rights⁵⁴.
- **A community alarm system to alert neighbors if someone has an accident or becomes sick.** The primary target audience for this solution is elderly people who stay home alone while their children work in agricultural settings or in large cities. The nearest neighbor can be alerted and call the police or an ambulance. This alarm system would also be useful in post-pandemic times.
- **A hat designed for kids that alerts its wearer when they are about to touch their face.** The most interesting aspect of this project is that it was designed and presented by young siblings, with the help of their father. In the process, a next generation of makers learned coding and experimentation with open-source hardware.
- **A virtual exhibition app to help art remain available to everyone as museums were closed during the pandemic.** It also helps local Indonesian artists, who the government seemingly forgot to support, create new revenue streams. Interestingly, this project was put together by participants from Jakarta, Pontianak, and Semarang, who wouldn’t otherwise have met each other.
- **A platform to share solutions for local production, circular economy, and ecology.** This massive project included collecting concepts of sustainable agriculture known to old Javanese farmers but not documented or available to the youth of today.
- Other ideas included a foot-pedaled handleless faucet, to help people avoid touching the same surfaces, and an affordable UV-C light box to disinfect doctors’ clothes in remote hospitals. The critical relevance of this project was that in Indonesia, most islands are located “remotely”, and critical infrastructures needed in a pandemic are scarce.

⁵⁴ Similar concerns had been raised by the CCC hacker community in Germany, e.g., <https://www.ccc.de/de/updates/2020/corona-tracing-app-offener-brief-an-bundeskanzleramt-und-gesundheitsminister>.

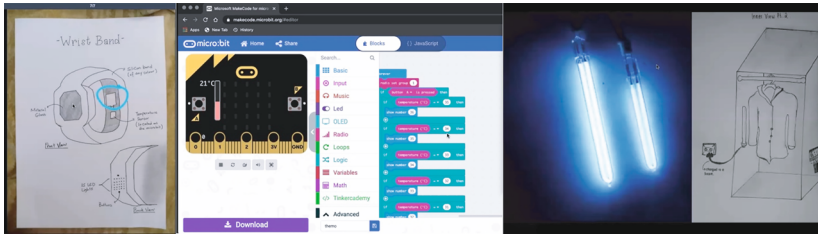


Fig. 5.22 Screenshots from the Mentoring Sessions: Prototyping and Mock-ups. (Screenshots: The Author)

At the end of the hackathon, the proposals were evaluated based on their pitch and documentation, across categories that reflected the values of the hackathon organizers: the appropriateness and simplicity of the technological solution proposed, affordability, reproducibility (especially with locally available materials), and finally, potential impact. Developing open-source solutions was a prerequisite when applying. The organizers' hope was that by defining these evaluation categories and sharing them with the team in advance, the projects would have a higher chance of becoming sustainable.

After the hackathon, the winning teams⁵⁵ received \$350 each for their prototype development and were invited to follow-up workshops. Such follow-up workshops aren't typical for hackathons. However, the organizers felt that they would extend the longevity of the projects, and by sharing knowledge, would benefit the Indonesian digital social entrepreneurial landscape as well. The workshop topics included fundraising, starting a social business, prototyping, and the legalities of creative commons. It was also discussed that, if necessary, a crowdfunding campaign could take place for those projects that get to the production phase, coordinated by the organizers, as another way to support the projects (Fig. 5.23).

⁵⁵ For the goals of this study, it is not directly relevant, but the 5 winning teams were the ones working on the alarm system, the virtual exhibition, the buzzer hat, the wearable temperature and distance tracker, and the circular ecology project.

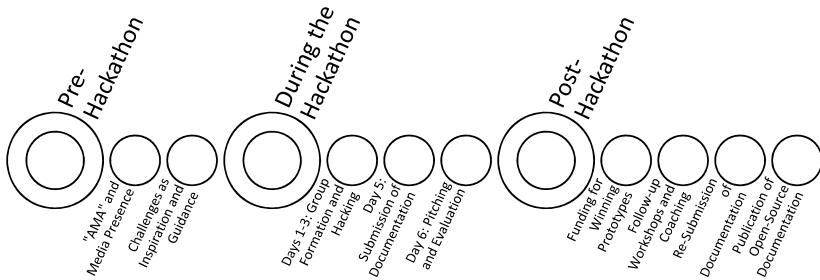


Fig. 5.23 Timeline of the #hackfromhome Hackathon

Unfortunately, the follow-up planned after the funding and post-hackathon workshops didn't take place. STACK repeatedly reminded the other organizers that they wouldn't have further capacity after July, while N:ow wanted to give teams time until the end of August to finish their prototypes to ready them for crowdfunding. Based on the reporting of the member of the Art-Science Collective who was in touch with the participants, the winners—some of them groups of students—were somewhat confused as to how they could spend the funding on their prototypes. The organizers tried to reassure them to use it however they thought was best, based on jury feedback. As a last step, the Art-Science Collective claims to have sent the documentation for a final review to STACK. During this period, STACK went through a shift in their mission, from building ecosystems to focusing on events. The open-source documentation of the projects, which was supposed to be the main output of the hackathon, and the tool to open the innovation process to the broader public for reproduction, was to date not finalized.

5.5.2.6 Participants' Perspective

A post-hackathon questionnaire was sent out to all participants in English and Indonesian. The link to the questionnaire was shared repeatedly between September and November via e-mail and on the instant messaging groups used by the participants (17 times in 5 different groups). The main goals of this questionnaire were to understand the motivations of the participants, how they identify

in terms of their expertise, whether they thought they developed an innovative solution during the hackathon, if the hackathon was truly collaborative, if they worked reflexively on their solution, and finally, if they will continue working on their solution or they would join another hackathon in the future. I deliberately sent out the questionnaire months after the hackathon because I was interested in seeing if people were still working on their projects six months later.

The age range of the participants was between 14–40, however the age limit was 17, so underage youth could only apply with the consent of their parents. Somewhat out of the average age range were the members of the “maker family”, which included two children aged 16 and 17 and a father who was 40. The average age of the other respondents was 31. Eight out of the 12 respondents identified as women, four as men and none as gender non-conforming.

The respondents came from Bogor, West Java ($n = 1$); an undisclosed city in West Java ($n = 1$); Jakarta, West Java ($n = 2$); Bekasi, West Java ($n = 2$); Jogjakarta, Central Java ($n = 1$); Surabaya, East Java ($n = 1$); Samarinda, East Kalimantan ($n = 1$); and Medan, North Sumatra ($n = 1$). The father of the “maker family” indicated that he was a foreigner. Non-respondent participants from Sulawesi also joined. While most respondents came from Java, an in-person hackathon could not have accommodated participants from Kalimantan, Sulawesi, or Sumatra, thus, the organizers’ assumption that an online hackathon could reach people an offline hackathon couldn’t have turned out to be true.

The backgrounds of the participants were similarly diverse, and included interaction design; computer science and linguistics; mechanical engineering; urban arts; interior design; and architecture. An English literature student also described himself as “zine maker, analog photographer, content writer and creative hub leader”, while a visual design communication expert described herself as “professional artist, freelance graphic designer and housewife”.

Regarding their motivations, the respondents said they were curious, loved creating something new, and enjoyed doing research to develop new insight and build networks. Four replied they wanted to help solve problems caused by the pandemic. One of them answered that she had been interested in participating in such an event for a long time but couldn’t because she worked full time. During the lockdown, as a teacher, she finally had free time at home.

In terms of the outcomes of the hackathon, the majority (7 respondents) said that they thought they had developed something innovative. A 28-year-old woman student, reflecting on how the project she worked on was innovative, included critical, political, technical, and societal aspects in her answer:

“I think our project is quite innovative because we started from the problems that existed when this pandemic started. We feel that many people are starting to get restless because they have to stay at home. If you are constantly at home, how do you earn money? We feel that money as an exchange rate is getting harder to come by. Then what about eating? How can we eat if our money is not enough for food needs. These questions kept popping up in our conversations. Had a thought, maybe we can barter our skills with food needs? Until finally we are at a point where we feel that food self-sufficiency is very important, especially in urban life. How can we all survive on food independently. Here we add insight through interaction with parties who have pioneered independent food initiatives. We also try to observe the social environment, especially regarding the need for food in terms of survival. We are here also trying to unravel the challenges of dealing with the limited space for physical interaction during the pandemic. And here, too, we try to overcome the effects of the vulnerability of casual and informal workers who have been active in the arts community.”

Her reply shows not only reflexivity, but it also describes the project development process between participants. It is a good example of what participatory grassroots innovation processes can look like. They can be based on exchange, iteratively build on each other's knowledge, experiences, and ideas, while reflecting on the societal context and vulnerabilities of particular communities.

The respondents also agreed that innovation should be democratized—made available as a tool to everyone, for example through education. A respondent thought that innovation is a mindset and could be done by anyone (“the roadside mechanic, construction guy, or gardener”), while two others replied that everyone deserves the same opportunities. That is, people with various skills should have easy and cheap access to a place where their innovations can be developed, as different ideas might become useful in the future. Moreover, a participant thought that innovation should happen regularly and become part of the Indonesian spirit. Thus, the respondents thought that their solutions were innovative on different societal, technical, political, educational levels, and agreed that innovative capabilities and tools should be made universally available.

In terms of the collaborative aspect of the hackathon, most of the respondents agreed that they worked well with their team members. They used various platforms, including ones used by the organizers, as well as online mind-mapping tools and 3D-modeling applications. The “maker family” held family meetings at home. People perceived each other as active, and team members not knowing

each other backfired only in the case of one respondent (“everyone was waiting for someone else to begin”). This of course does not mean that other teams didn’t have difficulties agreeing on an idea, developing a project plan, or distributing roles. Most respondents felt that the team members contributed equally. Interestingly, only five responded that the resulting project was very different from the original idea. The project’s evolution was attributed to prototyping, more research, new findings, and attempts to not reproduce existing solutions. As one of them summarized, showing a strong innovation mindset, it was “just the normal trial and error. Nothing dramatic”. Six respondents described the process as iterative, and five as linear.

Most respondents said they worked reflexively on their solution, however not all of them tapped a critical thinking process. Most of them answered that they took potential social and environmental impacts into account. One response harshly criticized capitalism and wondered how to reach people who have been “lulled by” it.

Nine respondents said that they wanted to continue developing their project, and one replied that he wanted to finish school first. Only two replied that they wouldn’t, but it is possible that questionnaire respondents remained engaged with their projects. Most participants were concerned with raising funding for their prototypes and one mentioned joining a larger movement with a similar goal. Most respondents said they felt more empowered to develop solutions after the hackathon, and that they would join a hackathon in the future. What they liked most about participating was networking, creativity, and learning.

To summarize, the hackathon fulfilled the original wishes of the organizers to inspire people to think about local issues, collaborate on ideas, and develop prototypes together. Participants included non-technical people and people from remote areas. Some participants who might not have been familiar with the hackathon format before were empowered to further develop their projects and are considering participating in other hackathons.

5.5.3 Findings and Discussion

This second case study described a Central-Javanese collective that has been actively developing, inquiring, critiquing, and making since it was founded in

1999. The Art-Science Collective might seem less strategic about their pathways and projects than the Network Collective, especially because they refuse to be categorized by one type of activity or skillset. Their approach however does not harm their sustainability because the annual or biannual renewal process allows new generations to shape their practices. **Tactics**, according to de Certeau, are the “weapons of the weak” (de Certeau 1984), employed by those without physical or social space to develop long-term strategies (Wylie et al. 2014). Based on this case, I argue against de Certeau; tactics might be a deliberate design choice for those who do not wish to be burdened by strategies, especially in a world where so many new impulses, learning, and knowledge-sharing opportunities and sources of inspiration are available for a collective to add to their toolkit. The Art-Science Collective might have formalized itself (as a “foundation”) into a strategic vessel for the skills and interest of their diverse community. However, this formalization took place mostly to align with expectations of other formalized entities, such as when applying for funding. Today, the foundation is a name they built a legacy around that allows new members better access to opportunities. The approach of having a broad framework of art, technology and society also allows them to remain flexible, react to changing circumstances quickly, and formulate critical and participatory responses in the form of tactics to those rapidly changing settings.

In the second layer, the case of the hackathon aimed to demonstrate an example of **grassroots critical infrastructuring**. This analysis builds on Bødker et al.’s (2016) framework for analyzing the technology appropriation practices of volunteer communities. This framework indicates that in intrinsic design or design done by volunteer communities developing their own artifact ecologies, the process is essentially a combination of happenstance, community strategies, and everyday tactics. Below I offer an overview and a brief recap of how the development of the hackathon artifact ecology took place (Fig. 5.24).

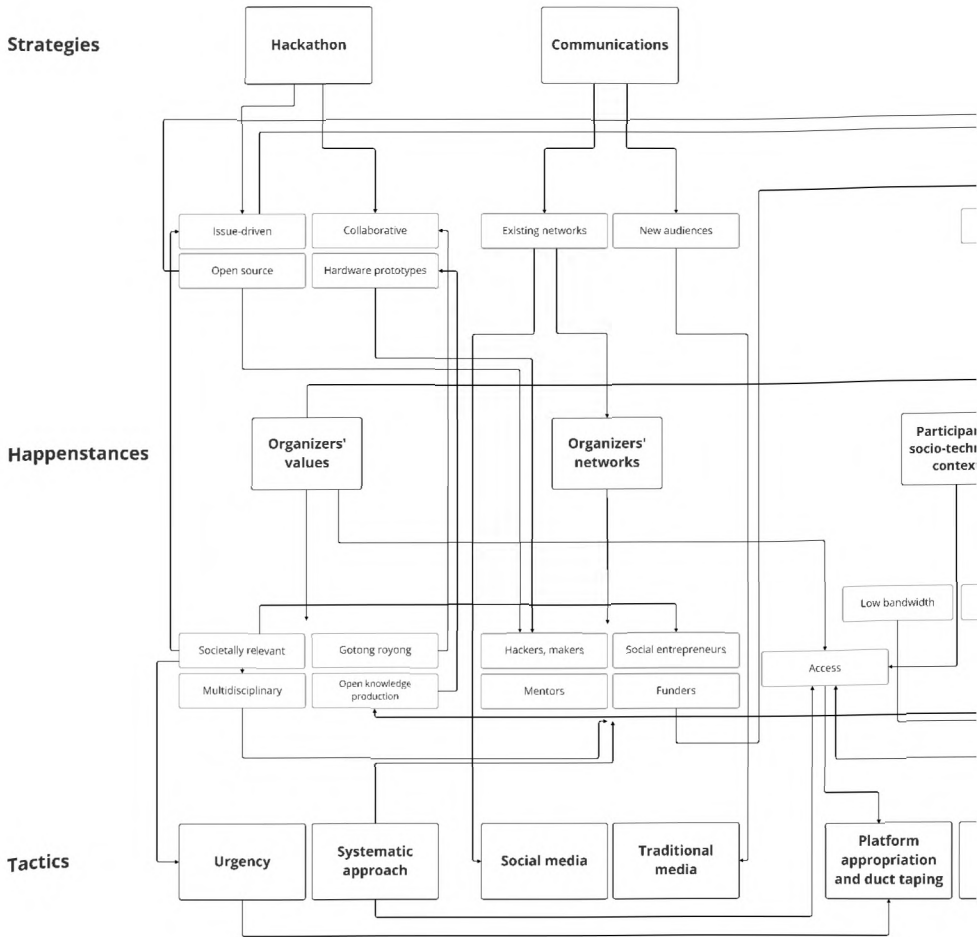


Fig. 5.24 #hackfromhome Hackathon Artefact Ecology

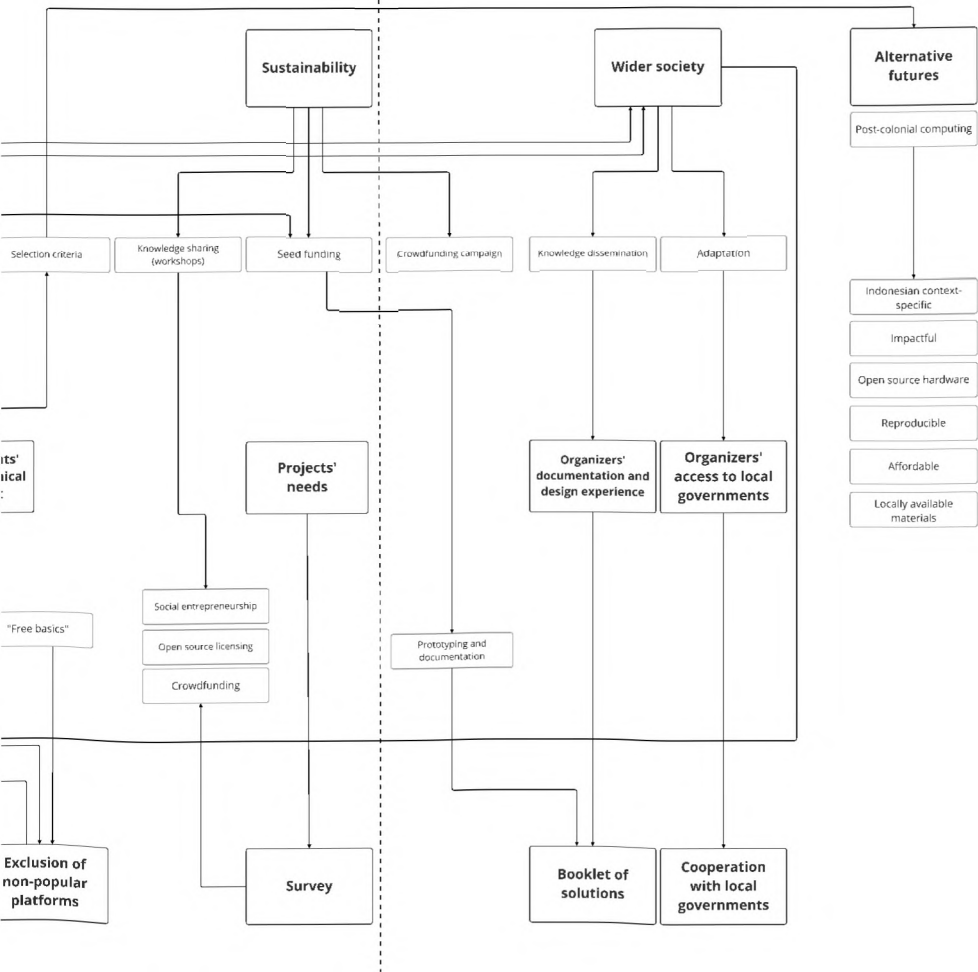


Fig. 5.24 (continued)

The artefact ecology of the hackathon sheds light on how the values of the different organizers (a happenstance) influenced the organization of the hackathon. Without the initiator (the Art-Science Collective), open-source hardware hacking would not have been in the foreground, while social issues, culture hacks, and general care for the sustainability of the projects developed in the hackathon would have been neglected. The diversity of the organizers' values made the artefact ecology complicated, but they also constituted its baseline and informed decisions throughout the whole process.

The strategies had been broadly agreed upon in the beginning, and developed collectively in calls. A lot of care went into designing the hackathon. This tradeoff between care and efficiency made the Art-Science Collective impatient. Although the founder of the Collective almost immediately felt ready to launch the hackathon, the other two organizers were more careful to ask strategic questions and shed light on broader societal matters. The clash here was twofold: 1) a feeling of urgency and immediacy, driven by the pandemic's negative impact on people's health and the direct economic impacts on the poor, and 2) a desire to do things right, support the development of financially, socially, and environmentally sustainable projects, and use the hackathon to build the Indonesian social innovation ecosystem, inclusive of rural areas.

Negotiations between urgency and careful design of strategies thus profoundly influenced the artefact ecology. With more time, funding, skills and volunteers, a self-hosted hackathon platform could potentially have been feasible. However, the hackathon was designed as a one-off event, and duct taping platforms that most participants were already signed up to allowed the organizers to deal with all happenstances. The organizers decided to appropriate popular platforms and fluidly integrate them. Rapid decisions on duct taping platforms instead of training potential participants to use others meant neglecting digital rights. Within the artefact ecology, tactics reacting to happenstances (in accordance with some of the strategies) informed the rest of the process. This case is thus more tactic-heavy than the previous one because most of the selected digital tools were selected as quick-fix tactics, and only representative of the overarching strategies. Unexpected happenstances were answered through tactics because the organizers did not have the opportunity to design a strategy of inclusion beforehand that could have respected data privacy as a strategy. Instead, the organizers invested in a more transparent digital future.

Ironically, organizers requested artefacts developed be open source, yet they did not organize the hackathon on open-source, privacy-respecting platforms. As shown under the strategy of alternative futures in the artefact ecology, they

built selection criteria on their shared values. By sharing the selection criteria beforehand, they aimed to inspire participants to develop socially reflexive and open-source solutions. These selection criteria—and therefore the winning artefacts—resemble the values of post-colonial computing (Irani et al. 2010): context-specific, impactful, reproducible, affordable, open-source hardware that uses locally available materials.

Combining open-source hardware⁵⁶ and socially innovative projects was also innovative about the hackathon (while expressing critique). This meant that for the sustainability of the projects, in addition to the aforementioned seed funding, further workshops needed to be organized. These workshops—based on a combination of what the organizers envisioned and the feedback of the participants—became tactics as well, though they were originally designed as strategies. Instead of designing an incubation-style program for the projects, one of the organizers pushed others to wrap up the project, as it took longer than anticipated and their resources were to be used elsewhere. In comparison with traditional hackathons, this care work of supporting the projects after the event is unusual and corresponds to the values of the organizers. Nevertheless, based on the participants' replies in the post-hackathon survey, their wish for sustainable hopeful interventions (Lindtner and Lin 2017) seems to have come true, as there were participants who aspired to continue working on their projects.

To summarize, just like the first case study, this second case also contributed to the research framework and all three research questions. The case showed an attempt at using open-source and privacy-aware solutions that was overridden by the local socio-technical context. It thus shed light on how a grassroots collective utilized its networks to engage two other grassroots entities to *gotong royong*—mutually support each other to organize a hackathon that would encourage people to develop context-relevant solutions. Issue-driven hackathons can be a strategy to develop the next generation of digital social entrepreneurs and engage with new people. Criticality is reflected directly in the framing of the problem that the hackathon aimed to address: the breakdown of supply chains and the government's hesitancy to act. On a more covert level, critique is also expressed in a post-colonial manner. The goal was not to import solutions, but to appropriate and develop technologies locally, and envision a different long-term strategy for rebuilding the economy. The organizers did not want to wait for a solution to come from “above” or “elsewhere”. Rather, they wanted to take matters into their own hands and inspire people to think about how to rebuild the economy in a more fair, sustainable, circular way—essentially, to envision alternative futures.

⁵⁶ The blueprints were to be released under a Creative Commons license.



Cross-Case Analysis and Discussion

6

In this chapter, the previously independently described ethnographic vignettes and the two main case studies are compared. To achieve this, cross-case analysis is used to mobilize knowledge from the individual case studies by comparing and contrasting them, thereby producing new knowledge (Khan and VanWynsberghe 2008). This comparison is done based on the research framework and is organized according to my three research questions. The cross-case analyses are followed by discussions of the findings, triangulated with relevant literature from the literature review and theoretical framework. Wherever appropriate, hypotheses are developed for further research.

This chapter is thus divided into three main sections, based on the research framework. To recap, the different types of input from the field have been structured as depicted in Fig. 6.1.

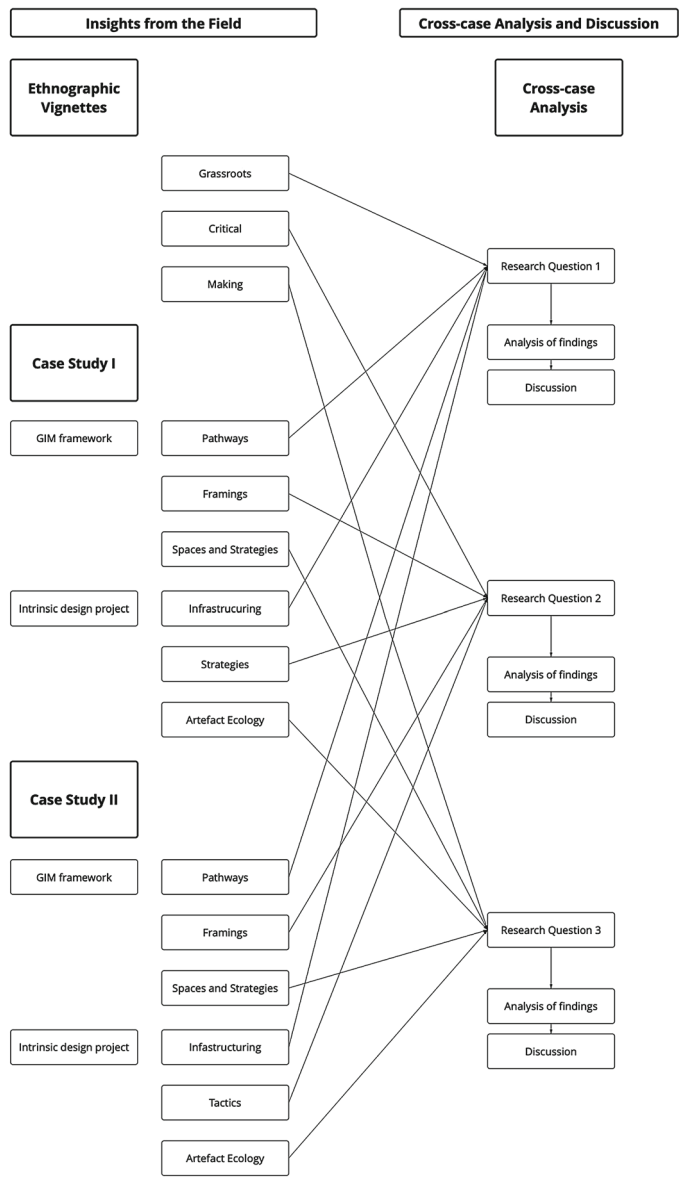


Fig. 6.1 Input from the Field in Cross-case Analysis and Discussion

6.1 Pluriverse of Grassroots Communities

The first research question concerns understanding Indonesian grassroots communities. To recap, the goal of this first research question has been to uncover the pluriverse of locally developed approaches and solutions, and the bottom-up communities that develop them. In this way, I show what the term *grassroots* means in research on *grassroots critical making*. It aims to illuminate the site-specific and critical knowledge developed according to the local context, and the societal issues that trigger the development of both. My hypothesis was that through empirical research, a wealth of diverse projects can become visible, informing the scientific knowledge base that had previously neglected bottom-up Communities of Practice and their innovations. Three interrelated questions were posed, inquiring 1) What are the characteristics of a *grassroots* critical making? 2) What are the particularities of the communities that make critically, and their shared capabilities? 3) What alternative futures do these communities envision?

6.1.1 Analysis

6.1.1.1 Context as Influencing Factor

Exploring the socio-political, economic, and geographic contexts of the collectives, it became clear that these settings shape the communities and their practices. The pathways of how the grassroots collectives develop highly depend on various external influences. After the fall of the New Order regime, a boom of post-authoritarian collectives emerged which were highly influenced by that history. For the first time in 30 years (since the Indonesian genocide of 1965–66), people could openly engage in collectives. Their emergence coincides with the massive growth of adoption and use of the internet in the late 1990s. The Network Collective, the Art-Science Collective, and the Traditional Crafts and Robots Collective were founded by students around this time. What makes their practices differ is their locations and the framings of the cities where they are situated.

The Art-Science Collective is based in the cultural capital where the art university is located that many of them attended. Thus, art and aesthetic presentations of their critical maker experiments at exhibitions is a major part of their practice. The Network Collective is based in a “creative city” with a major technical university that teaches people design. They approach social issues through design. Their location also highly influences the practice today. In the cultural capital, life is relatively cheap, so both the DIYbio and the Art-Science Collective can

experiment freely, as their monthly expenses are relatively low. Both collectives also recognize that resource scarcity can become a problem for citizens. As previously mentioned, the price of an Arduino can cover a month worth of warm meals, and most people can't afford attending workshops that cost money. The Art-Science Collective offers hardware hacking workshops, has an open-door policy for children in their neighborhood, and collaborates with local universities, making their knowledge and infrastructures available to students. The DIYbio Collective consciously develops low-cost open-source hardware, like a cheap Arduino alternative. In comparison, the Traditional Crafts and Robots Collective has to take on paid jobs to generate enough income in the relatively expensive city they are based in, which takes time away from playful exploration.

6.1.1.2 Diverse Membership and Citizen Participation

Grassroots communities are diverse and multidisciplinary. The collectives and their networks are composed of communities of people from different backgrounds (e.g., artists, designers, architects, engineers, programmers, and anthropologists). For example, in the DIYbio collective, water filtration and environmental monitoring is merely one activity. They are also building sound sculptures, audiovisual experiences, and artworks, similar to the Art-Science Collective and the making collective Traditional Arts and Robots. To recap, the collectives act as vessels for their multidisciplinary community members' ideas and projects. Although collectives are often founded by and attract university graduates, it is not a prerequisite for members to be educated. Rather, collectives are communities of like-minded people who share the same values. However, as being a member of such collectives is not remunerated, the practice will remain somewhat exclusive, although far not as exclusive as making in Western settings, as the collectives often engage in collaborations with citizens.

Collectives engage with everyday people through citizen participation. All of them tend to work with citizens in the immediate environments of the collectives' houses, which are located in bigger cities. However, many often work with smaller communities of citizens elsewhere, both in urban (*kampung*) and rural (village) settings. The Network Collective and Art-Science Collectives work both with *kampungs* in cities and villages. In the case of the Belonging Collective, they are based in the home village of the founder, where he established a separate collective community. The East Javanese makerspace and the DIYbio Collective worked with communities in homeless settlements in their respective cities. In some projects, the members of collectives temporarily take up roles of educators.

However, most often their goal is not to build a hierarchical situation of teacher and student, or expert and non-expert. Rather, they wish to work collaboratively

with everyday people, and learn from them. Examples of the collectives' conscious work to avoid hierarchy were shown in the Community Network Project and the water filtration project run by the DIYbio Collective. Even as initiators of critical making projects, the collectives try to remain facilitators and allow for self-directed development by those who know what they need.

During the hackathon, the critical maker "community" of participants extended their practice to anyone interested—students, a nurse or even a whole family working together. The participants (as demonstrated in an answer to the survey, see Sect. 5.5.2.6. on Participants' Perspective) showed attributes of grassroots critical making, including critical, reflexivity, shared knowledge, and skills that they used for co-learning during the hackathon. This learning was enabled by the Art-Science Collective and other organizers' work. Supporting citizen participation and self-directed critical making is a core component of the collectives' everyday activities.

6.1.1.3 Pluriverse within the Pluriverse

The collectives discussed in this study engage in particular societal issues (belonging, migration, digital rights) and critical maker activities (mapping, water filtration system, community internet). As briefly mentioned above, they also employ a multidisciplinary array of design proposals in based on interests, skills, and values. Two examples are the cases of the DIYbio or Art-Science Collectives. Another interesting finding is the decision of the Art-Science Collective to abstain from putting themselves in a particular box, or deciding on one key value, approach, or type of technology that they would commit to or be characterized by. This conscious decision to remain ambiguous allowed them to invite different people to be members and thereby accumulate different skills. These diverse skills, as described in the case of the hackathon, allowed them to respond quickly in an emergency. Their toolkit is vast, but it didn't account for some societal considerations. Therefore, what is even more interesting for this study was what happens when these main foci of grassroots actors are combined and consequently, novel types of activities are generated. An example for this is the hackathon, which combined skills of three different grassroots communities on open-source hardware hacking and social innovation, which inspired post-colonial computing projects to emerge. The novel combination of knowledge and values is a key component of grassroots innovation.

Another example is the Network Collective's work with the Indigenous community. Here, the collective's combined their interest in building and facilitating civil society networks with the villagers' and the chief leaders' interest in ICT

infrastructures. This fusion resulted in a sustainable community network that led to good development and Indigenous peoples' empowerment.

6.1.1.4 Nongkrong Process to Infrastructure Trust

It is interesting for this study to understand how the collectives are *infrastructured* together. Together is the keyword, as the framing of the hackathon was centered around the organizers, who would say “we should fight this together” (I.W.A.) and “we are in this together” (F.H.). This togetherness is achieved through long-term trust building, exchanges, and discussions, from establishing rapport to building friendships. Trust is often *infrastructured* through *nongkrong*, a lengthy ideation process comprised of multiple sessions of hanging out and exchanging ideas. In the process, people get to know each other, understand each other's values, explore the shared skills, and collectively evaluate ideas. They discuss which ones are timely and possible, and which others are to be revisited later. *Nongkrong* can be understood as the time when people develop and decide on the strategies. In the case of the hackathon, the urgency almost eliminated this essential step for grassroots, which led to frustrations when the project was to be launched, as the necessary strategies were not collectively agreed upon yet.

Trust within the collectives and with other communities is thus carefully created over long periods. This was found in the case of the Indigenous community. In the project, local customs were to be respected by the different partners and the appropriated technologies. However, to be invited to learn the local customs, the Network Collective first had to create trust. Blueprint developmentalism (Smith et al. 2014) and a lack of respect for Indigenous customs led to the community's mistrust of external people. The Indigenous community initiated its own counter-counter mapping project (a post-colonial critical making project in itself) and met the Network Collective. In the process, it took the collective two years to understand the local workings of the customs and the society and create the necessary trust for cooperation. This is an example of a grassroots community trading efficiency for care (Rossitto et al. 2021), or blueprint developmentalism for trust and an understanding of local customs. In hindsight, this tradeoff was necessary, as it enabled good development (Pieterse 1998). The project was intrinsically organized and self-directed, only facilitated by the collective, and thus today remains sustainable.

Care, patience, and *nongkrong* (collective ideation) is what allows for trust to be created, which holds the community together. In the next subchapter, I will also explain how *nongkrong* leads to a safe space for criticality in a country where activists are increasingly often jailed.

6.1.2 Discussion

6.1.2.1 Defining Grassroots

At the beginning of this study, a provisional definition of what grassroots innovation looks like was offered (pp 3–4, Smith et al. 2017). This first research question concerned itself with a deeper understanding of Indonesian grassroots, building on that detailed definition.

My analysis has included examples of the researched communities, and answered questions about who the grassroots critical makers are and how the collectives came to be in different contexts. My findings resonate with the aforementioned definition, which mentioned community groups and neighbors. The definition also mentioned networks of activists and development workers, but none of the members of the collectives would describe themselves as such¹. In my analysis I outlined how the collectives engage with non-community members in their practices, or with other collectives to form new pluriverses. Collaboration, network building, and citizen participation is thus another a key element of Indonesian grassroots innovation, resonating with theories on how the broader public is mobilized by grassroots (Smith et al. 2017). Finally, the infrastructuring of trust through *nongkrong* was described, which feeds into theories about how discourse is developed and the work of grassroots communities contributes to plurality and reflexivity in wider society through co-ideation with local communities (ibid.). Building on this, I would also like to add that they mobilize people by taking the time to create mutual trust.

I would like to add another important note here regarding why some collectives don't specialize in particular topics, become country-wide movements, or develop their pathways towards major international festivals. It's a conscious decision whether collectives remain incremental or irrelevant due to their hyper-locality. The Belonging Collective's rural project does not want to become another Jatiwangi Festival. Instead, it would rather create local third places and events for the villagers to enjoy. The Art-Science Collective explicitly decided not

¹ Here I would like to recap the difficulties encountered when using academic nomenclature: grassroots was not a term everyone understood or identified with. Thus, I experimented with using similar terms such as “everyday people”. My contacts directed me to “collectives”, which is not a term I would have used myself, but it was a term used by the groups I was looking for. Although some collectives formalized themselves into foundations, my contacts called these formations collectives as well. In Indonesia, collectives are fluid, open, flexible groups of people who meet regularly, mainly in the evenings (often after work, as the daytime heat subsides) to share insights, ideate, and develop projects together. For future research on grassroots critical making, this might be an interesting aspect to consider.

to specialize on a particular type of technology or practice, but allow themselves to remain open and free to experiment. Still others like the Network Collective, strategically, reflexively, and iteratively built their pathway to become an expert on networks (both NGO and internet). However, this is not a decision taken by all grassroots communities; some instead observe a culture of pluriverse within the pluriverse. Thus, building on Escobar's (2018) proposal and Szaniecki et al.'s (2020) mapping of plural participatory practices, which represent powerful approaches to change, this study uncovered that even within individual groups of people, pluriverses of critical makings and social innovations can exist, and these can strengthen the capacity of grassroots to answer urgent or emergency situations.

6.1.2.2 Grassroots Shifting Innovative Practices

After decades of authoritarian government, with their newfound freedom, people found that the failed State gave them an opportunity to redesign their world (Pushkin 2020). As other scholars have highlighted, hacking, and making are seen in Indonesia as cultural traits. They can be attributed to patterns grassroots innovators saw from their mothers, such as “the values of collectivism, open sharing and tinkering” (page 1574, Kaiying, et al. 2019). This notion is relevant to understand why so many grassroots communities see these attributes—described in the West as hacking or making—as a natural part of their practices. Thus, since the late 1990s, everyday people react by establishing collectives that combine the freedom to design with the skills and values they saw in their parents.

As demonstrated in the cases, various contexts can either support or hinder grassroots innovation. The critically made artefacts investigated in this study can be born of material or financial necessity, as a response to the local context. For example, a low-cost alternative microcontroller allows for experimentation and the involvement of local people in educational workshops of the DIYbio Collective. Or, collectives can be motivated by social injustices and neglect by conventional innovation systems (Gupta et al. 2003; Seyfang and Smith 2007). Others are motivated to express critique through experimentation and artistic installations, like the Art-Science Collective, which is also highly enabled by low-cost hardware and open-source documentation (Arancio 2021). Uncovering the enabling factors of grassroots innovations in a resource-limited context contributed to a better understanding of why an abundance of resources is *not* a prerequisite for mobilization (page 21, Smith et al. 2017).

It is also important to note that the grassroots and social innovations explored in this study contrast with existing power structures (Vlachokyriakos et al. 2018), especially those that neglect communities. The most prominent example of this

can be found in the Indigenous community in this study. The community had its local customs disregarded, and had to undergo blueprint development practices (Smith et al. 2014) on multiple occasions. These practices were disembedded from local reality, therefore not only were they unsustainable, they also created distrust. As explained earlier, the grassroots Network Collective decided to do the exact opposite of what the government and development workers did, facilitating not only the creation of a sustainable infrastructure and model for community networks in rural areas, but also creating a stark contrast for HCI4D and ICT4D to learn from.

As the collective facilitated the process, they managed to shift the practice of innovation into the hands of the Indigenous community. Similarly, many other collectives also organize free workshops and have open-door policies so curious people from their neighborhoods to join them. Open knowledge sharing and open-source documentation is a way that the grassroots shift innovative practices to everyday people. In the case of the hackathon, the grassroots managed to reach far and invite people from all over Indonesia and with the most diverse backgrounds to design context-relevant solutions for themselves, other communities, and post-pandemic Indonesian society. As (Jurriëns 2017) put it, the grassroots move innovation from solo entrepreneurs in the West to alternative networks in a post-colonial praxis of knowledge production and innovation. In the cases, they went beyond the networks of collectives and generated citizen participation and post-colonial citizen innovations.

Utilizing cost-efficient, hacked, self-developed (sometimes pirated) hardware and software, grassroots communities open up avenues for self-directed and bottom-up innovations. Through this, they blurred the artificial divide between experts and non-experts of technology and innovation, democratizing the practice. From the cases it seems that they learned from their mistakes. The Art-Science Collective “imported” the idea of a Fab Lab precisely as prescribed, and like many others, failed to contextualize the makerspace in the Indonesian society. Now new makerspaces are now more inclined to find expressions that help locals understand what they do, and to develop use cases for prototyping in the form of micro-entrepreneurial training, even for homeless communities. This might seem close to the idea of Western makerspaces enabling start-up companies, yet it is also very far from it. The goal is not designed to make anyone rich, but to make survival easier, and potentially, lead to empowerment.

6.1.2.3 Designing Alternative Futures

From the beginning, post-authoritarian and avant-garde collectives that started out as experiments and new media centers (Kera 2012; Jurriëns 2013) imagined a future different from the authoritarian times they were born into. Some, like the Art-Science Collective, used aesthetic innovation to communicate its utopian visions (ibid.). Even today, they extend reflexive critical questions to everyday citizens, such as those about an imminent climate dystopia. Others opt for social innovation or rural development. Through all such practices alternative and potential futures are designed and started today (Kaiying et al. 2019).

Alternative futures rely on a pluriverse. Similar to Escobar, they propose moving away from the ontology of separation, designing towards a vision of positive societal transition. As the discussion of grand societal challenges and wicked problems grows in design (Cech and Farnham 2021), Escobar wonders:

“Can design be extricated from its embeddedness in modernist unsustainable and defuturing practices (...)? Moreover, could design become part of the tool kit for transitions toward the pluriverse?” (page 15, Escobar 2018)

He draws parallels with practitioners of speculative design who develop alternative futures such as Dunne and Raby (2013) or Pelle Ehn (*Making Futures*, Ehn et al. 2014), whose work I have touched upon in this study. Escobar proposes engaging in autonomous design, which is basically the same as intrinsic design (Bødker et al. 2016), the process in which communities design themselves. As demonstrated in this study, alternative futures are successfully designed by grassroots critical makers and local communities.

6.1.2.4 Issues with Volunteer-Based Projects

Many grassroots projects are volunteer based. As I will outline in the next subchapter, next to *nongkrong*, *gotong royong* (mutual aid) is also a key approach in Indonesian grassroots. Some projects combine paid and unpaid roles, as seen in the Community Network Project that received a multi-year “learning grant”. Indigenous people mainly work on the project for themselves, within the *gotong royong* system, for cultural reasons. However, the *nongkrong* period—a common grassroots practice—became a problem in the case of the hackathon. Care and efficiency (Rossitto et al. 2021) clashed at the beginning of the project, which is common in community-led initiatives. Although some might feel that such deliberations slow down processes, the efficiency and speed preferred by traditional innovation could be seen as an issue. Based on findings from Case Study I, time

(two years) was needed to learn, understand, and build trust for the cooperation. Here, respect and care were improved the projects, as opposed to simply deploying elimination design (Escobar 2018) or blueprint development (Smith et al. 2014). In everyday Indonesian grassroots practice, the temporal dimension is experienced differently because time is in abundance². The lengthy co-design period only became an issue because the hackathon project was taking place during the COVID-19 pandemic. NGOs were increasingly anxious to run the hackathon and develop solutions that could immediately save lives. As demonstrated in the artefact ecology of the hackathon, a much larger issue became the volunteer-based nature of the project; towards the end, partners started dropping out due to other commitments. Ultimately, this meant that some of the planned activities that were designed to support the sustainability of the projects and broader societal impact could not take place.

Time is thus needed to create trust, which leads to the grassroots being able to mobilize communities and the broader society, while creating trusted spaces for the exchange of critical views. Even today, this is a dangerous activity that can lead to the incarceration of critical activists (Janti 2022).

6.2 Co-Production of Critical Knowledge

The second research question concerned itself with exploring criticality in Indonesian grassroots communities. To briefly recap, the goal of the second research question has been to explore the critical knowledge co-produced by grassroots, to better understand the critical, inquisitive, and participatory process that might enhance reflexivity. My hypothesis was that (incremental, vernacular, critical) knowledge produced by grassroots has been traditionally neglected by academia and research. It might be seen as inconvenient, due to its criticality (Cox and Fominaya 2009). The chapter thus aims to explore knowledge production and sharing in grassroots communities. Three interrelated questions were posed: 1) What are the particularities of critical knowledge development and sharing? 2) Is the knowledge produced used to make wider society more reflexive (building on Buechler 2000 and Smith et al. 2017)? 3) How is the knowledge is translated to real-world projects?

² I would like to briefly note here that the Javanese experience time differently. For example, when I asked the founder of the Art-Science Collective, what deadline would be reasonable for the participants to fill out the survey, she responded with “*Haha! Javanese time. A week, but prepare yourself for two weeks*”. Interesting, cultural and linguistic understandings of time can be found (Rahmawanto and Suhardijanto 2020).

6.2.1 Analysis

6.2.1.1 Enhancing Reflexivity in Society

Before summarizing concrete examples of criticality, it is necessary to briefly explain how the practice of *gotong royong* (mutual aid) contributes to the Indonesian grassroots landscape and their critical practices. As explained in the last section, trust is necessary for cooperation, especially in the current political climate. If safe spaces for discussion are created through *nongkrong*, then safe spaces for collaboration are also created in the *gotong royong*. Most of the communities in this study mentioned some version of this practice. For example, learning from the mistakes of shuttered makerspaces, the East-Javanese makerspace turned to the knowledge of others in the Javanese grassroots to co-design their practice. They learned how it could be useful for the locals, and how they could frame their activities so people understand them (e.g., “tool library” instead of “makerspace”).

Helping each other creates a feeling of belonging or togetherness. In the case of the hackathon, for example, the projects started with the Art-Science Collective’s founder’s call for *gotong royong*, and the word “together” was mentioned in discussions multiple times. It was established that people could trust each other when expressing critical thoughts (i.e., worries that the Indonesian government was not reacting fast enough to the pandemic). The rupture in trust was between the government and the citizens, and not among the citizens. Its outcome was a *collaborative* hackathon, and as explained in the outcomes of the participants’ survey, *gotong royong* was an efficient way to encourage people who didn’t know each other before to collaborate and develop critical maker projects. This and the other cases show how these practices of grassroots—rooted in local culture—which are centered around creating space for discussions, and those discussions and practices are extended towards broader audiences through learning. These are tangible practices that contribute to enhancing reflexivity in society (Smith et al. 2017).

6.2.1.2 Reflexivity and Criticality in Grassroots Making

Knowledge about society and social issues is often uncovered in the process of *nongkrong*, when people bring their ideas together. This is a special type of approach to co-learning, or learning from each other. People uncover societal issues as they go, as seen in the organization of the hackathon. Techno-social realities, like the lack of electricity or internet and the power of the “free basics” over other platforms, wasn’t initially part of the strategy. But as participants signaled that they encountered problems, tactics to circumvent them became a part

of the artefact ecology. Moreover, the organizers decided to change the future by asking the participants to design open-source solutions. Individual problems and injustices experienced by communities are part of the knowledge of grassroots that shape their post-colonial repertoires of action. Reflective-discursive processes are built into the design of these actions.

Reflexivity became visible in multiple cases investigated in the study. As the cases are at different levels of development, it is clear that in most cases, both reflexivity and criticality have been part of their practice from the beginning, cultivated by *nongkrong*. Criticality is also plural and diverse, as it highly depends on the collective and what its members criticize and how.

In terms of critical projects contributing to the design of the pluriverse, criticality is expressed differently, based on different reflexive approaches to design. Through a cross-case analysis, this study identifies the following five types of critical-reflexive approaches, which are closely linked to technology appropriation or development:

Developing and Sharing Critical Knowledge: the grassroots learn from others in participatory ways. Through such processes, they collect, develop, and most importantly, openly share novel, critical knowledge:

- The Library of Banned Books problematizes the question of who does cartography, and what they map. To move away from the “Western gaze”—a colonial lens of what is included—they reinvent how their environment is mapped. Thereby, they contribute to locally relevant spatial knowledge production and sharing. In addition, in their newest project, they contribute novel knowledge about urbanization and lack of gender equality; the goal is to enable women to access public space exactly as men do.
- In the hackathon, team members who applied with similar ideas were grouped and asked to develop critical artefacts together. All developed projects were openly documented, and supposed to be shared in a booklet with a Creative Commons license (although this goal was not realized).
- In the case of the community network, issues arose from the infrastructure (such as cyber-attacks) or from access to the internet (like being bullied or accessing harmful content). They were addressed through learning, instead of shutting down the system.

Post-Colonial Design and Computing: Criticality through post-colonial approaches manifests either by amplifying the voices of the marginalized, reflecting on the local particularities that colonial powers did not take into consideration, or by inspiring post-colonial computing projects:

- The hackathon was designed to utilize the creative energies of everyday Indonesians, who similarly to the organizers, waited impatiently at home for action to be taken by the government. Through the hackathon, they had a chance to reflect on local issues and express their insights through critically made artefacts.
- The development and selection criteria of the hackathon were also shaped reflexively and in a post-colonial manner. The organizers' critical thinking skills contributed to a shared understanding that only those solutions that work in the local context stand a chance to become sustainable.
- The Library of Banned Books expresses post-colonial critique by de-colonizing the map of the city and also developing novel ideas about how cartography could be done if Indonesians rid themselves of colonial practice.

Oppositive Critique: What this study calls oppositive critique is built on or expresses opposition. It often takes the form of critiques of the practices of decision-makers. Examples from the cases include:

- In the case of the hackathon, the starting point was critique of the government, as issues in the COVID-19 response were being uncovered and people in the immediate environment of the organizers became victims of the slow response.
- Less deliberate or expressive critiques were directed towards how hackathons are organized in Western settings. They were run in a techno-fetishist way that only engaged people for a weekend or to serve corporations, and not to continue with their projects. The hackathon was set up in the exact opposite way.
- In the case of the DIYbio Collective, some projects were direct responses to immediate issues.
- To establish the basis of the Community Network Project, the Network Collective built rapport over years with the Indigenous community to avoid the pitfalls of a disembedded blueprint developmentalism.

Blurring Traditional Boundaries: Traditional expert/non-expert boundaries are constantly blurred by grassroots, whether by involving everyday people in technology development, or enabling Indigenous groups to design their own infrastructure ecosystem in a self-directed way:

- In the practices of the DIYbio Collective, a starting point for projects is immediate issues of concern for locals, which represent larger societal problems. The solution offered is collaborative experimentation, through which science is brought to broader audiences. Their events are either low-cost or free, yet financial scarcity both on the sides of the collective and the citizens leads to novel solutions.
- The Art-Science Collective is known for its open-door policy, allowing curious children from the neighborhood to spontaneously join their activities as they hack. They also have adult education programs, run together with a university that lacks access to workshops tools.
- In the case of the Indigenous community, access to the internet opened new avenues for self-empowerment. With the newly available information, farmers could take collective action and experiment with new crops to sell. At the same time, their needs and eco-religious beliefs were built into the non-exploitative business model that was mainly developed for the maintenance of the infrastructure itself.

Covert Goal as Critique: Covert critique is necessary when grassroots engage with vulnerable populations or topics that are taboo in the country. Addressing societal issues thus ends up becoming a long-term plan instead of a quick intervention and it is only communicated with trusted people:

- In the case of the anarcha-feminist activist group, workshops were organized in which women could learn how to make reusable sanitary pads. Whenever possible, these workshops were used to discuss women's rights issues, reproductive rights, and other taboo topics concerning women.
- As outlined in Case Study II, the long-term covert goal of the Community Network Project is for the Indigenous community to receive Indigenous rights.

6.2.1.3 Sustainability as Real-World Critical Design From Hopeful Interventions to Sustainable Self-directed Development

As I will outline in the discussion, when designing data collection for the study, I did not expect to find so much critical thinking and strategizing going into the projects, particularly about how development could be done sustainably. In the case of the hackathon, two of the co-organizers focused on deciding which social

entrepreneurial skills needed to be shared with the potential winners, and what socio-technical prerequisites the winning artefacts would have to fit. This was done to ensure that the projects could become successful—not in the Western sense, but in the sense that the blueprints of hopeful interventions would spread to other makers, and peoples' lives could be saved.

Some of those hopeful interventions were envisioned as similar to the prototypes of the DIYbio Collective. The water filtration system has been iterated and co-developed with local communities for years. This socially innovative critical maker project might still get new features, but it is now ready to be added to a line of the city's tap water system (which is currently not potable), in cooperation with the local government.

The water filtration system is developing into an artefact that could become part of a sustainable self-directed development ecosystem. As seen in the case of the community network, multiple socially innovative and appropriate technologies were already designed and maintained by the Indigenous community, some of which were prerequisites for the community network to become feasible. In parallel, a sustainable source of income also became necessary for the community, enabled by access to the internet. The complete intrinsically designed ecosystem relies on its carefully considered parts (Fig. 6.2).

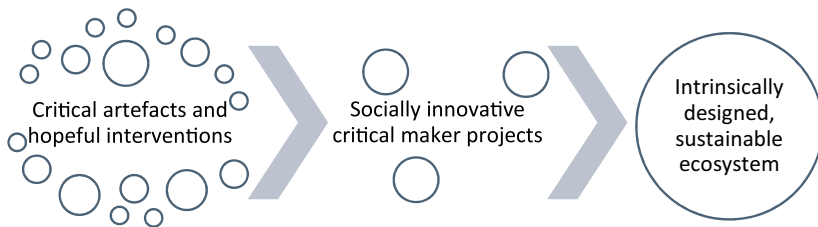


Fig. 6.2 The Fate of Ideas and Development of Critical Artefacts Over Time

Sustainability Strategies as Post-Colonial and Post-Development Practice

An important finding from the cases is that all involve local people—not as participants, but as autonomous, self-directed, intrinsic designers of their own futures. The collectives recognized that “imported” solutions (see the case of the Fab Lab)

and blueprint development (see the case of the Indigenous community) does not work sustainably because both are disembedded from the local context. Thus, both collectives in the two main cases learned that they had to support the development of sustainability strategies, even if it took longer than simply copying a solution that worked elsewhere. This approach is inherently critical post-colonial, and is reflected in the selection criteria of the hackathon. It asks critical makers to:

- Develop solutions in a multidisciplinary and collaborative manner (social dimension)
- Reflect on the local particularities (social dimension)
- Design technologies that are appropriate, affordable, and simple to produce (technological dimension)
- Ensure technologies are easy to reproduce based on an open-source blueprint, using locally available materials (technological and environmental dimension)
- Imagine alternative futures for a post-pandemic ecology (speculative)

In this understanding of the grassroots communities, critical artefacts and hopeful interventions that have these attributes can be adopted by wider audiences and communities and become parts of sustainable, good development ecosystems. In most of the cases, the solutions proposed are both socially just (addressing women's rights, Indigenous rights) and environmentally sustainable (reusable pads, circular ecology, powered by renewable sources). By inspiring the participants to think about “delayed sustainability” and that the solutions they develop could be complex, instead of simply techno-solutionist, the grassroots contribute to just transitions and fairer alternative futures.

Hackathon participants did not have the opportunity within the frameworks of the hackathon to test their hopeful interventions with the communities whom these were developed for. Still, the community network provides a case of how sustainability strategies could be co-developed with the community itself. It has already been established that blueprint developmentalism does not work. Instead, self-directed, intrinsic designing of sustainability strategies can be facilitated for good development. This entails four dimensions of financial, social, environmental, and technical sustainability (Fig. 6.3).

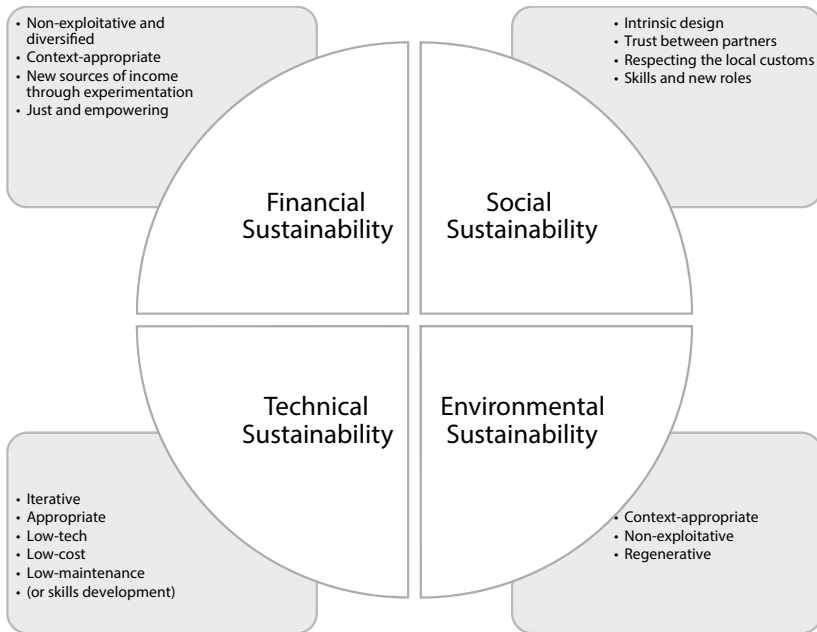


Fig. 6.3 Sustainability Dimensions of an Intrinsic Design Project

Finally, I would like to mention one more important strategy. As seen with the Community Network Project, and many other projects, project scale is small. This is a post-developmental and critical design approach of incremental innovation. The *kampung*- and village-scale is feasible and sustainable for the grassroots. However, as described in the case, findings can be adapted to other contexts, if the community is willing to learn how the other context works.

6.2.2 Discussion

This second research question concerns the exploration of the co-production of critical knowledge in grassroots communities. It was based on the scarcity of research of knowledge production practices in the “peripheries”, and aimed to

focus on how critical thinking (the first part of the term *critical making*) is exercised. Ultimately, from these questions I derive findings about socio-technical critique, reflexive-discursive processes of grassroots communities, and how the criticality feeds into real-world critical design projects.

Smith et al. defined three types of critical knowledge production methods (Smith et al. 2013), all of which have been demonstrated in this study. This study also uncovered how knowledge production is done in grassroots communities: by taking time and establishing trust, and therefore, creating a safe space for critical observations, thoughts, and ideas. Although in emergency situations collectives might rely on previously established trust, time is crucial for long-term and sustainable project development. Second, Indonesian grassroots, building on traditional practices, exercise non-transactional collaboration, which also creates trust and allows every member of the community to be useful. However, this volunteer-based approach might also cause problems if members can no longer lead the project together due to other obligations. Finally, knowledge production and critical maker projects in general are hierarchy free; everyone can bring ideas to the table, and roles are defined based on skills, forming an ad-hoc community of participants, who all iteratively work towards shared goals.

6.2.2.1 Non-Transactional Collaboration: Carrying Something Together

As this study explored the critical knowledge production practices of grassroots, it also aimed to shed light on why it is developed and how it gets embedded into the practices of grassroots critical makers. What became clear is that most of the practice is community centered. After trust is built, grassroots communities work together to participatorily and intrinsically develop critical artefacts that might lead to good development projects. In fact, the founding of many collectives (or grassroots communities) was inspired by the longstanding tradition and a key element of Indonesian culture: *gotong royong*. The term itself has been traced back to “several people carrying something together” (page 546, Bowen 1986). As Bowen also points out, in a village economy it is unquestioned that villagers engage in agriculture and other work without pay. Similarly, in the case of the Indigenous community, the villagers’ time is allocated to the rice, and the rest to the roles they play in the local society. The “minister of ICT” does not receive monetary compensation for running the TV and radio station, or the community network. Thus, the financial model of the community network was mainly developed for its sustainability. Funding goes towards maintaining the infrastructure and buying equipment from the “outside world”.

At first, one might think that the idea of mutual aid (or volunteerism) emerges from resource scarcity. However, it is clear from these cases that it is a cultural norm. As Escobar and Alvarez (1992) theorized, resource limitations in the Global South are not a limitation, as implied in the West (Smith et al. 2017). Rather, different forms of collective action can be influenced by economic, political, and cultural factors. Based on these cases, this study hypothesizes that the idea of mutual aid and volunteerism is not a workaround for resource scarcity, but a cultural phenomenon that feeds into the pluriverse of Indonesian grassroots activities.

How a diverse economy can move beyond mainstream-capitalist models is also demonstrated in the “iceberg model of the economy”. The iceberg symbolizes the visible part of the capitalist economy, and below the waterline lie all other practices people engage in, such as volunteering, unpaid work, cooperatives, and bartering (Gibson-Graham 2014). The practice of *gotong royong*, as seen in the case of the hackathon and the Community Network Project, engages people with the skills they are best at. It empowers them to work together towards a shared goal for the good of the community, without the need for a monetary transaction, reinforcing the iceberg model outlined above.

In conclusion, such non-transactional collaboration is not developed out of scarcity, but has historical roots. In practice, it generates trust, which creates safe spaces for critical thoughts to be shared. Trust also allows for intrinsically codesigned projects to be iterated while new knowledge is produced about the local context, leading to a higher chance of sustainability. Finally, the involvement of everyone blurs expert and non-expert boundaries to enable equal participation and learning. Hence, reflexivity, critical thinking, problem solving, and many other skills are enhanced through the production and sharing of critical knowledge in the community. This of course does not mean that grassroots innovators should remain underfunded: as we saw in the case of the Community Network Project, multi-year learning grants can help establish more sustainable and long-lasting projects by allowing participants to experiment. Instead, the learning here is that there exists an invisible layer of economy that is built on the sharing of non-monetary resources and has further positive social effects.

6.2.2.2 Criticality in Grassroots Critical Making

In this critical knowledge production and community-based critical thinking process, the grassroots uncover different social issues and injustices. This is an important difference between the Western understanding of critical thinking, which—in critical technical practice (Agre 1997) and critical making (Ratto and Hockema 2009)—is understood as a solitary activity. In the cases described in

this study, critical thoughts are more often developed and shared in the community. In “Pedagogy of the Oppressed”, Freire presents an iterative model of participatory communication theory to help understand how social movements organize themselves. It starts with forming a collective identity (in this study: *nongkrong*), facilitating collective action (in this study: *gotong royong*), and ends with reflexive dialogue (Freire 1970). In this study it “ends” with critical artefacts and reflexive projects that allow for grassroots innovators to come into reflexive dialogues with broader society.

In this cross-case analysis, I presented directions for Freirean reflexive dialogue in different communities, or different project stages. The first one—developing and sharing critical knowledge—is typical for many maker communities; amongst makers, openness is a prerequisite (Saari et al. 2021). Second, post-colonial computing and how technology is developed to inspire and promote alternative futures (Irani et al. 2010; Kaiying, et al. 2019), aligns with post-colonial design for the pluriverse (Escobar 2018). Third, there is what I call oppositional critique, which goes against a bad political practice, social injustice, or failure. Similar considerations have been made in the research of infrastructuring, where grassroots innovation is defined as a practice that stands in contrast to existing power and economic configurations (Vlachokyriakos et al. 2018). Fourth, multiple cases presented how traditional boundaries between experts and non-experts are blurred, and multidisciplinary partnerships are formed as everyday people cooperate with hackers, makers, scientists, and each other. Similar processes have been observed and celebrated in open science because they challenge institutionalized pathways of development (Smith et al. 2017). Finally, some projects have similar goals to those of activists. However, in the political climate, the collectives decided to keep those goals covert, as to not harm the people they are trying to help. Thus, the projects are framed as social innovation, which is more palatable for decision makers. That critical knowledge produced in such projects might be disregarded by decision makers because it is deemed too “uncomfortable” is not new (Cox and Fominaya 2009). The workaround of presenting the project as social innovation or rural development however is an interesting finding about the originality of grassroots.

All these different approaches to reflexive dialoguing are value-driven and highly political. The grassroots communities use critical thinking and critical discussions as a starting point for material practice, thereby establishing the *critical* in *critical making*.

It is an important finding of this study that the reflexive dialogue happens as grassroots communities think critically together while they design their critical artefacts. In the cases described in this study, critical thinking was always

combined with a material activity. Either they are combined in the planning or design phase, or in iterative processes that lead to the artefact's sustainability. As Hess pointed out, many societal issues are neglected by conventional scientific institutions, and are often addressed by activists, who develop alternative forms of technological change (Hess 2009a). Indeed, uncovering issues like the lack of potable water, everyday people left behind during the pandemic, or a lack of internet in rural areas is just the first step in grassroots criticality. Problems are alleviated by developing alternative technological solutions, such as water filtration systems.

6.2.2.3 Sustainability as Post-Colonial Strategy

The grassroots are political actors who act through technology design by opening alternative pathways of socio-technical development. They raise political questions about needs in their immediate environment—and through that, in wider society. Their design and infrastructuring work is built on lengthy discussions about injustices. Trusted spaces of critical thinking are a starting point for action, and subsequently, agreed-upon values shape the artefact ecology of critical making projects. As exemplified by the hackathon participant's description of the ideation process, it is lengthy, iterative, and reflexive, shedding light on critical societal injustices. These discussions are starting points for critical maker projects—hopeful interventions designed for alternative, just futures (Kaiying et al. 2019). Those hopeful interventions, as this study has demonstrated, are largely technosolutionist in themselves, thus, require further steps to become sustainable developments.

By questioning current socio-technical configurations, the grassroots propose appropriate directions of technological change. They question problematic structures like the “free basics” program that creates walled gardens around users (Ratto and Hockema 2009; Sen et al. 2017), or the failures of blueprint developmentalism in rural internet access (Purbo 2017). Indeed, such discourses about technology controversies serve as informal technology assessments (Rip 1986), and could inform technology development. However, instead of waiting for external help, the grassroots develop just technological answers themselves. They include the promotion of open-source solutions, positioning of appropriate technologies to be maintained by local communities, or low-cost alternatives to mainstream maker tools. Thereby, they also challenge Western assumptions about who is allowed to design—and is capable of designing—technology (Hess 2009a).

Their critique itself is often post-colonial. One example is questioning the technosolutionist approach of Fab Labs, built on the assumption that universal design can be deployed anywhere and people will make it work, regardless of local particularities and contexts. The solutions presented in this study were answers to failed attempts to deploy universalist solutions and participants have taken the next step in designing its opposite: *good development* (Pieterse 1998), which is reflexive, locally embedded, and context relevant. Of course, to understand the impact of any such practice, the most important question is whether those solutions are sustainably appropriated. This is an important question for HCI and participatory research too, as the sustainable appropriation of technologies (e.g., through values, communing, and caring) has been explored in the Global North (Teli et al. 2021), but less so in the Global South. The most detailed example was the case of the Indigenous community's network. It is clear that, similar to grassroots approaches in the Global North, values play a leading role in communing and care. However, there are also crucial differences between the North and the South. The fundamentals of Indonesian sustainable rural development were defined in the Ciamis model (Poerbo et al. 1995), outlined in the chapter on Developing Sustainable Strategies. These included social learning; information and communication to increase rationality; starting small; decentralized decision-making; catalytic partnerships; and limited PAR.

This study has uncovered recommendations of grassroots innovators, building on these key prerequisites by adding technological aspects, which were not key considerations in the original model:

- The technology must be co-designed in a way that reflects the local context. This means that historical, religious, political, or geographical contexts need to be taken in to account. The community is not a blank page, and it may reflect potential grievances, distrusts and other negative effects by previous disrespectful collaborators.
- Technology, if possible, should be low tech, low cost, and low maintenance. If these are not possible, then local skills need to be developed to deal with the discrepancy.
- Financial, environmental, social, technological sustainability models need to be co-developed with the community. Each model can be developed iteratively to ensure that they are not exploitative, unsustainable, or in any other way harmful.
- Any negative effects of the technologies on the local community need to be anticipated when possible, and immediately dealt with, as required by the community.

6.3 Making As Intrinsic Design

The third research question is concerned with making and understanding the intrinsic design processes of grassroots maker communities. My hypothesis was that by examining how intrinsic projects are designed and translated into real-world artefacts, the autonomously designed technology development process might illuminate how the critical thinking process is translated into material practice.

Building on the first two research questions, three interrelated questions were posed. 1) What does making look like in Indonesia, compared to Western countries? 2) What modes of making can be found and how do these address local needs? 3) Is there a relationship between values and what is “made” in grassroots critical making?

6.3.1 Analysis

6.3.1.1 Hacking and Making

Based on preliminary research and interactions with practitioners at the C&T workshop, I knew that it would be difficult to find people who identified as makers, despite the large number of makerspaces around the world. Although some people know about making, as demonstrated in the chapter Makerspaces in Indonesia, it is not a broadly known term, and Fab Labs and other Western makerspaces are rare. Hacking was used more broadly, as seen in the case of the Art-Science Collective, who earlier called themselves a hackerspace.

However, some Indonesian hardware hackers, who denounce the term maker, defined their own terms for their practice. The Indonesian word *ngoprek* means to tinker, and ultimately, that’s what they do. Again, not all collectives identify with this word, but it is interesting in light of *gambiarra* and *jugaad*. Tinkering encompasses activities like stealing mom’s wok to build a *wajan bolic* (Siagian 2016), but could also mean frugal innovation, like using an old motorbike to separate the rice from its shell (G.H.I. 2020c). Moreover, it also is closely linked with *nongkrong* (the time to tinker with ideas), meaning that the co-creative process of knowledge or artefact production is valued over the ownership of a specific outcome.

Finally, there is one more reason why people don’t identify with “making”. This co-productive DIY approach to creating things has seemingly *always been* a part of everyday Indonesian life. Things are partly made out of scarcity and are not always safe; an architect who works with the Art-Science Collective once

complained to me about the quick fix of a rooftop in the neighborhood. But sometimes making is a way to generate income, as in the case of seamsters who sit with a sewing machine next to the roadside, in case someone needs to fix a piece of clothing.

6.3.1.2 Modes of Indonesian Critical Making

Making can take many shapes and forms in Indonesia. It became clear in this study that very few of those shapes and forms would be understood as making in Western countries. Indonesian making is not mostly about Fab Labs built on Western blueprints, generating income by selling kits, or spaces where hardware startups are incubated. It is true that explicitly researching *critical* making resulted in the identification of grassroots projects with a more political undertone. Still, Western makerspaces do not make a lot of sense in Indonesia. As mentioned by the founder of the Network Collective, many makerspaces were started and failed because their business model was to sell kits. The Art-Science Collective's Fab Lab failed for similar reasons; their framing did not fit the local context. Makerspaces closest to the Western concept had to redesign their practice, starting with their name. In Indonesian, a "tool library" is clearer than a "makerspace". Their offerings were also altered, instead of incubating the next "unicorn", material-maker skills are supplemented with skills to navigate the informal economy, and people could sell what they make on online marketplaces.

The question remains: what *is* making in Indonesia? My answer is that it always was and remains part of the culture. Non-ideological and ideological making could be divided, as the founder of the Network Collective pointed out. Some Indonesian making is closer to frugal innovation or *jugaad*, and is done out of necessity. Such hacking inspired by resource scarcity creates new inventions and workarounds in the form of affordable DIY solutions. However, the focus of the study is the politically driven version of making. Based on the cross-case analysis, the following types of critical making were identified:

Vernacular technologies building on ancient knowledge. This type of making might be most widespread, but also the most vulnerable. Traditional houses woven from bamboo or clay are falling victim to the government's attempt to modernize the country. Recently, a new village was built on an island near Java using traditional stilt houses. It was used as tourist housing to inspire the locals to endorse them again. Similarly, the Belonging Collective's local handmade clay brick production might have to close because modern bricks are cheaper. Vernacular technologies are erased by some, while others attempt to preserve and

improve them³. In the case of the Indigenous community, experimentations with chili and wind vanes will continue, but might be extended with the IoT to record data.

Appropriated technology ecosystems. Such practices mainly became visible in the artefact ecologies of the hackathon, in which different platforms were hastily duct taped together to fulfill the basic requirements for the hackathon. A more sustainable and strategic appropriation is the artefact ecology of the community network, where a micro-hydro plant, solar panels, a router designed for emerging markets, and Chinese off-the-shelf products are hacked together for the local context. In both cases, new software or hardware could have been made, but there was simply no need for it. Making here can be understood as the appropriation of existing technologies in context-relevant ways.

Hardware hacking and making. These practices are closest to the Western understanding. They rapidly prototype by using microcontrollers, 3D printers, laser cutters, and robots. Most of the outcomes of the hackathon were such critically made artefacts, many of which used the aforementioned tools. These devices are also often used by the Art-Science to generate artwork or educate people, and by the DIYbio collective to further develop their water filtration system. Both also engage in making with bacteria instead of electronics. They are involved with re-nitrogenizing volcanic soil, fermenting safe alcohol, and biohacking. The collective that engages in producing Traditional Crafts and Robots also has a shared makerspace in their house that resembles Western workshops, where they create robots, simple electronic instruments, and light installations.

These types of making are sometimes combined or mixed, such as in the DIYbio collective, where an open-source, low-cost microscope or a self-developed microcontroller was developed in response to the high price of the originals. A low-cost alternative could be seen as representative of an appropriated system but also as hardware hacking for open science.

It is also important to note that some making is for fun. Robots that were built to participate in a hebocon and electronic instruments that generate music for parties are also a form of art that expresses a thought or emotion without exhibiting it. These are important creative outlets that generate new ideas, bring in new

³ According to one of my interviewees, some ancient knowledge is repackaged and sold to Indonesians in a neo-colonial model. When she told her grandmother about a new, sustainable approach to farming that was shared by a Westerner, her grandmother replied with “yes, that’s exactly what we did 70 years ago”. Unfortunately, much of such vernacular knowledge has been erased by colonialism and efforts of modernization.

members, and keep people excited about making. Moreover, such practices represent one of the important promises of maker culture; as the price of electronics and rapid prototyping sinks, an increasing number of people can afford to freely experiment and play.

6.3.1.3 Making as Intrinsic Design

Critical making in Indonesia is not done in commercial makerspaces. It is done in the houses of the collectives, where they keep the shared tools, and sometimes directly in the field. These free spaces allow them to develop their critical maker projects in autonomous, self-directed ways—as intrinsic design. Inspiration might come from different places. Some grassroots innovators might be inspired by external people coming to their international events, or from residencies, or through their own research. Others host lengthy hacker-researcher gatherings where open-source blueprints and projects for ideas are shared.

However, many ideas are intrinsically developed. Of course, they build on knowledge and experiences of the makers, and the co-design and ideation process is about sharing such existing critical observations, insights, and ideas. A major part of grassroots *making* is the design of the projects that have been explored in this study through ecologies of artefacts. Below I will compare the two intrinsic design cases. This comparative analysis aims to show how technology selection is negotiated based on histories, values, and shared strategies and tactics, and how these dictate the reflexive design process.

First, I would like to compare the differences between the two projects. Both collectives started around the same time, around the dawn of the post-authoritarian era. Their first activities were centered around practices they called “new media art”. Yet, they are headquartered in rather different cities. The Network Collective is based in a creative city, where the technical university is based, and design is part of the culture. The Art-Science Collective is based in the cultural capital, home to the fine arts university. As my interviewee put it, they approach social innovation from an artistic point of view. While this is true for many aesthetic technology artefacts the collective produces, in this case, their collaborative hardware hacking practices were highlighted. The Network Collective, however, has been developing into a network builder and facilitator since its conception.

The circumstances of the two intrinsic design projects are quite different too. The Community Network Project was built over a longer period, during a window of opportunity for co-creation with a trusted Indigenous community. They also received a learning grant to experiment with the community network, making this project somewhat relaxed and giving time to people to iterate. The hackathon was

however designed during a state of emergency at the beginning of the pandemic. In this case, it helped that the Art-Science Collective didn't settle on one direction in their practice; instead, they experimented with new things constantly, had a multidisciplinary community that could react quickly, and had the networks to invite two other grassroots communities to support the project.

The maturity of the projects and the circumstances had a major impact on the artefact ecology of the projects. The community network, being more mature, is significantly more strategy-heavy. The strategies have been collaboratively agreed upon, so that any happenstances that might occur are easy to navigate. In the case of the hackathon, some strategies were agreed upon, informed by reflexive problem framings. However, in the hasty organization, happenstances were answered more frequently with tactics. This means that while the Community Network Project had time to select technologies that are appropriate for their purposes, in the case of the hackathon, instead of open source and privacy protecting platforms, as originally envisioned, problematic "free basics" were duct taped together in an artefact ecology, leaving the participants' data open to being collected large tech companies.

The final major difference is that the hackathon projects were abruptly ended without delivering a final strategy—the fate of many volunteer-driven projects. From a critical techno-social perspective, this is an issue. Even though the hackathon participants' data away was sacrificed, the idea was that the openly documented and open-source artefacts would contribute to a utopian and alternative future where privacy is not a myth in Indonesia, but reality.

In terms of the similarities, what stands out is that the people who intrinsically design are from diverse disciplines. This multidisciplinary approach contributed to both projects in major ways, because they brought with them different knowledges, experiences, and values. Both had criticality as their starting points; one was against dis-embedded blueprint developmentalism, the other was critical of the government's approach to the pandemic. The hackathon organizers, without knowing, also expressed criticality towards Western hackathons, and felt like they needed to shape the program so that the issue-driven solutions would have a chance to become sustainable.

Sustainability was also a keyword in both projects. While there is a whole framework of sustainability built into the Community Network Project, in the case of the hackathon, this work remained unfinished. From the post-hackathon survey, however, it is clear that some participants aim to continue with their projects. There is yet a chance of those projects becoming sustainable without support from the organizers.

Sustainability was achieved in both cases by the organizers only playing the roles of facilitators. Thereby, they were able to empower everyday people to become critical makers and maintain their own self-directed projects, becoming “experts” within their own fields. Both also recognized the importance of open-source documentation to reach other audiences who could benefit from developing the solutions. While the hackathon organizers could not follow up on this, the Community Network Project has been sharing the knowledge produced openly in “rural ICT camps” and is now conducting a survey to set up a similar project elsewhere, at the invitation of another Indigenous community.

Both practices are critical and post-colonial. Through these actions, both grassroots communities envisioned alternative futures. One decided to inspire people to develop critical artefacts of local relevance. At the current stage, these are merely hopeful interventions, but carry the possibility to become projects similar to social innovations like the micro-hydro plant or the *wajan bolic* that have been crucial for the artefact ecology of the Community Network Project. Therefore, they too could become prerequisites of sustainable development.

6.3.2 Discussion

6.3.2.1 Making, but Rejecting the Maker Label

At the beginning of this study, I offered a provisional definition of making, and argued that is a technology-based extension of DIY culture. It is thus a material activity that uses analogue and digital tools, intersects with hacker culture, and is built on sharing and co-creation (Doyle 2013). Here I also outlined various problems with the so-called “maker movement” and “maker culture”, which is embedded in Western societies and has been criticized for its technosolutionist approach (Lindtner et al. 2016).

While this study has demonstrated that maker approaches similar to the Western definition exist in Indonesia, preliminary research has shown that in the Global South, the practice is situated and pluralistic (Sipos and Wenzelmann 2021). It can be concluded that Indonesian critical making projects share some attributes. Most are intrinsic design projects, which are value-driven and respond to societal issues. Indonesian making is an assemblage of practices that develop or appropriate technologies, and comes from a long history of DIY and self-made solutions. As is the custom in previous generations, they are based on “collectivism, open sharing, and tinkering” (page 1574, Kaiying et al. 2019). In a society where everyone makes, promises or practices of Western maker culture are difficult to separate from the everyday activities embedded in the culture. This is why

the Western term making is explicitly rejected by many. Those who attempted to endorse it, either failed or had to develop terms like “tool library” that were contextually and culturally relevant. As demonstrated in this study, the Western type of making is a hobby for the upper-middle classes. But selling pre-made kits and generating start-up companies rarely works in the Indonesian context. Neither does a technosolutionist approach to development, whether based on the promises of making or initiated by the government. In this sense, Indonesian making is closer to the historical German “open workshops”, many of which had political and countercultural roots (Sipos and Franzl 2020).

Hardware hacking and making is a relevant practice among collectives, as demonstrated throughout the study. It is, however, closer to Hertz’s vision of a critical making that brings criticality back into the apolitical practices of post-2010 maker culture (Hertz 2015a). The difference is that Indonesian grassroots have always *been* the counterculture, and therefore, most of their practices are already critical. Indonesian grassroots also do everything to be inclusive. They educate people for free, have open-door policies for women and children, develop low-cost alternatives for experimentation, and extend their offerings to rural areas. At the same time, there are many parallel problems to deal with, not only political and societal, but also environmental. As one of the founders of the DIYbio Collective said, problems of everyday people need to be prioritized based on how immediate these seem:

“[During] the earthquake last year, [...] liquefaction, [...] where do you go? [...] The standard procedure for earthquakes is look for fields, so no buildings, and then stay there. But you can’t because the land is going to swallow you [...], so where do we go? Tsunami is on the beach and on the hill liquefaction, and earthquake is [happening] ... [there is] no place to go. And then when we talk about the water quality of river, who cares? [People say] I won’t die because of that” (T.B. 2020).

Although their practice is inclusive and very open towards citizens, as opposed to Western making, it does not serve urban middle-class people. No wonder that Indonesian critical makers do not see themselves in the idealized picture of the maker Hertz describes (Hertz 2015a).

While there are apolitical maker activities, mainly due to resource scarcity, the political interventions explored in this study conducted by Indonesian grassroots respond to exploitation, control, and power (page 11, Lindtner and Lin 2017). The collectives are born out of post-authoritarian settings, experiment with new media and hacking, produce and collect zines, collaboratively develop artefacts in critical-reflexive-discursive settings, and develop post-colonial design and computing. These are all highly political activities. As discussed before, the grassroots

are political actors, emerging in a failed State (Pushkin 2020) in which they have had the freedom to experiment and invent their world. While such local particularities might be endorsed celebratorily to be co-opted by the West (ibid.; also see Kaiying et al.'s (2019) critique regarding such practices), this study showed that being a critical maker is inherently being a political actor, and that this role also comes with the need to carefully navigate political systems, re-build trust that had been broken, and diplomatic skills.

6.3.2.2 Social and Material in Co-production

In making, the invention of the world and of alternative futures starts with the co-productive powers of the social and the material (MacKenzie and Wajcman 1985; Bijker and Law 1992). How technology is produced highly depends on the technological style of its producer (Dienel 1995). In the Indonesian grassroots, that style is highly embedded in the community of makers, and the community of users of the technology. Indonesian making, and therefore, Indonesian approaches to sociotechnical change, are exceptionally participatory. Bottom-up technological solutions are collaboratively generated, often mobilizing wider publics, thereby generating alternative activities that address social issues (pp 3–4, Smith et al. 2017). Indonesian making is situated, as it considers the local situation, and also the interests and values of the communities that develop them.

The small scale of approaches (on the *kampung* or village level) allows for appropriate technologies and artefact ecosystems to be developed, which resonate with Schumacher's call for intermediate or appropriate technological solutions (Schumacher 1973). The concept and practice of making enables collectives and everyday people to collaboratively develop material solutions, and then quickly and cheaply iterate them. Indeed, as the study showed, context-relevant solutions can be brainstormed and their first prototypes can be developed within a few days' time. Therefore, critical making goes even further than Schumacher's vision. Technology in this sense is no longer merely a "choice", but co-developed with or by the community that will use it. Hence, the practice now allows for ancient knowledge and vernacular technologies to be embedded in it, be personalized to the local community who maintains it, and made sustainable. The shortcomings of how appropriate technologies were originally deployed—including universalist, generic solutions, lack of participatory practice, skills, time and humility (Chambers 1997) and the dependency on funds (Abrol 2005)—have been addressed in the solutions presented in this study. These shortcomings have all been addressed by grassroots maker concepts.

Moreover, Indonesian grassroots do not only develop appropriate technology ecosystems, but also sustainably appropriate technologies (Teli et al. 2021). In this

case, the solution is not self-made, but it is duct taped, hacked and bent to fit the local context. In the Community Network Project, the producer of any technology does not have to be a professional in the Western sense. Eglash et al. highlighted that Indigenous people's knowledge might be used and appropriated without compensation (Eglash et al. 2004). They might be seen as experts in their field, but not remunerated as experts would normally be. It is such exploitative methods that the Network Collective objected to by facilitating the self-directed embedding of Indigenous knowledge and customs into the design of the appropriated technology ecosystem for the Indigenous community.

6.3.2.3 Proposing Change: Beyond Grievances and Addressing Wicked Problems

It became clear from the artefact ecology analysis that values play a major role in how the communities shape their strategies. Moreover, the strategies for the development of projects signal that grassroots communities work together beyond their shared grievances (Smith et al. 2017). That is, they identify societal problems and share critical knowledge to solve societal issues. Making is used to support the material side of the critical practice by helping people develop hands-on solutions. This approach might seem technosolutionist at first, and it is clear that making or critical maker artefacts in themselves cannot solve major societal problems. The outcomes of the hackathon will remain prototypes of hopeful interventions. Yet, this is when grassroots critical maker projects can become a part of a solution for wicked problems. Ratto originally envisioned critical making as a practice to illustrate and discuss wicked problems (Rittel and Webber (1973) in Ratto and Hockema (2009). Since then, the term “wicked problems” has been increasingly used in community-oriented research (Cech and Farnham 2021). Moreover, to describe the extreme complexity of the climate crisis, the term “super wicked problems” has been coined (Levin et al. 2012). Indeed, the world's problems are becoming increasingly complex and increasingly interrelated. One wicked problem, like mining, can become a part of another wicked problem, like the climate crisis, increasing the problems of people who were already disadvantaged, like Indigenous people (de Castro Leal et al. 2021).

The partiality of solutions presents an ongoing dilemma of grassroots makers. Do they continue developing workarounds, and potentially contribute to wicked problems? Or can they truly generate change that is large enough to make a difference and break a “wicked cycle”? The hopeful interventions developed in the hackathon might help create a utopian ecology of artefacts like the village of the Indigenous community, in the case of the community network. Grassroots critical maker communities might benefit from the trust they build among each other and

local communities. They might exchange exactly the right piece of knowledge that leads to collaborative and sustainable intrinsic design projects, and in the long term, achieve good development. Reflexive, post-colonial collectives might be best equipped to address today's wicked problems, as they have demonstrated the ability in this study to identify them and already engage in values-based design.

What might hinder the process is a different wickedness. First, how do you address wicked problems after identifying them? Once a systemic issue is identified, how can critical knowledge be translated into action? This study presented some cases where knowledge is translated into action through critically made artefacts. However, the word *system* is key here: Smith et al. highlight that grassroots need to work within the systems to change those *same* systems that have deliberately created major structural disadvantages (page 67, Smith et al. 2014). Repairing the dominant system's harms allows it to perpetuate (Gregoire 2016).

Agre, when he coined the term "critical technical practice", seriously considered a revolution that would clear the ground of artificial intelligence. He imagined completely starting over by ridding himself of all the problems that were already embedded in the system (Agre 1997). This might not be possible or desirable here, but a transformation of wicked systems might be possible based on the alternatives grassroots critical makers offer—taking the time to build trust, adding a reflexive plurality to thinking about technological futures, and making social change, ultimately acting beyond shared grievances.

Conclusion

7

7.1 Towards a Grassroots Critical Making

The goal of this study was to extend the term critical making to grassroots communities by reflexively exploring whether critical making is a practice that Indonesian grassroots communities engage in. By mapping grassroots critical maker communities, conducting criterion-based sampling, and investigating their intrinsic design practices, the study has contributed to theories on grassroots innovation and Human-Computer Interaction (HCI), especially intrinsic design and post-colonial computing.

The research questions posed in this study were developed by breaking down the three attributes to be studied—*grassroots*, *critical*, and *making*—which were later woven back together in interrelated research sub-questions. This exercise allowed for a comprehensive study of how these attributes influence and build upon each other in practice.

Findings indicate that my hypothesis was correct; there indeed exists a grassroots critical making. However, it is plural, situated, and contextual, and individual communities should not be grouped as *movements*. My analysis showed that the term is not recognized by the practitioners, and even though “critical making” resonates with most communities, instead of grassroots, they use term collective (*kolektif*). Making is seen as a Western concept that many people refuse. Instead, local vocabulary that can be translated as hack (*retas*) and tinkering (*ngoprek*) is more widespread and accepted.

Beyond these initial insights, the study showed that Indonesian grassroots communities work collectively, cooperatively, and without hierarchies. Instead, their approaches involve local communities of everyday people as experts of

their own realities, and learn from and together with them. *Nongkrong* (hanging out and ideating) and *gotong royong* (mutual aid) are cultural traits that feed into reflexive exchange and critical knowledge development and sharing. Humility and time create trust and long-term collaborations, leading to the community-based development of critical artefacts and sustainability strategies that represent approaches to post-colonialism and post-development. In this making process, novel technologies *enhance* vernacular technologies and ancient knowledge, instead of erasing them. Grassroots making practices are highly critical of top-down approaches to development, and instead demonstrate that grassroots critical making can help local communities autonomously develop sustainable solutions. This study has showed that techno-solutionist approaches and artefacts will not solve the world's problems, but making can provide reflexive communities tools of intrinsic design. Supplemented with a broader understanding of wicked problems in society, they can contribute to self-directed good development practices.

The study has also shown that the projects of grassroots critical maker communities are not perfect, or have the universal solution to all that goes wrong. This would be idealistic and naïve. Volunteer-based projects struggle and the reflexive-iterative processes take time. What this study offered was an overview of how post-colonial hopeful interventions might be created, and how they could build sustainable artefact ecologies.

The study also added to the critical voices of other researchers who have already started challenging techno-solutionist, techno-fetishist promises of maker culture, and to those who have expressed critical views on dominant Western discourses about design and innovation. Dividing the world into North and South, center and periphery, and people into experts and non-experts only maintains the status quo. This study moved beyond such approaches and instead, showed the pluriverse of Indonesian grassroots critical makers in their own right.

7.2 Contributions

Given that this research builds on multiple disciplines, I would like to summarize the most important contributions it made.

In terms of methodological gaps, a new methodology needed to be developed for the analysis. The novelty of this study was that it aimed to combine grassroots innovation and critical making. Previously, for the analysis of critical making, participatory design (Ratto et al. 2014) had been proposed, while a

multidisciplinary analytical framework had been designed for analyzing grassroots innovation movements (GIM, Smith et al. 2017). To extend the term, the study combined three approaches: a criterion-based analysis based on the three predefined attributes, the grassroots innovation movements (GIM) analytical framework proposed by Smith et al. (2017), and Bødker et al.'s (2016) approach to community-led intrinsic design projects. Therefore, I was able to identify and analyze grassroots critical makers both through the lens of grassroots innovation and HCI. This study provides a starting point to better understand grassroots critical making. By combining GIM and critical making, this study contributes to a design research approach that can be refined and adopted by other researchers working with such communities.

In terms of human-computer interaction (HCI), there had been another methodological gap, which also included a population gap. Despite the growing research in HCI on participatory design, community-led and intrinsic design is still negligible. The research that has been conducted on autonomous or intrinsic design has mainly focused on Western populations. Research on a few Southeast Asian critical makers has focused on the political roles supported by post-colonial computing, but not analyzed their intrinsic design processes. In addition to furthering the scientific knowledge base on post-colonial computing, the study also contributed to an understanding of post-development from the perspective of the grassroots.

7.3 Future Research

My field research was restricted to Javanese projects due to the pandemic, therefore the study compared the analyzed practices with Western practices, based on the author's experience. However, in the future, field research in other parts of Indonesia and in other non-Western countries could contribute further insights. This study is only a first step towards engagement with grassroots practitioners, and there is clearly growing interest in academia. Researchers have proposed mapping of the pluriverse of grassroots innovations and practices (Szaniecki et al. 2020). But as mentioned before, reflexive and non-participatory research that explores the intrinsic design methods of bottom-up communities remains scarce.

This study raised several new research questions and hypotheses. Here I would like to summarize some conceptual questions that might be interesting in the future.

Comparative work. During my field research, the biggest surprise was the sheer number of projects I found. In addition to the ones described here, there

were another 30 projects I did not get the chance to visit. Some attribute this to the post-authoritarian failed State (Pushkin 2020), but I suspect that there is more behind it. Comparative research could produce findings on whether the political context influences grassroots critical making. In a paper, I have presented preliminary research on how history influenced the DIY practices of people in Germany, indicating that making is not only practiced in times of scarcity, but can also be politically motivated (Sipos and Franzl 2020).

It was also an interesting finding that making can be already embedded in some cultures, and critical makers refuse the maker label. Kaiying et al. (2019) already started exploring such questions by interviewing the mothers of Indonesian critical makers. In future research, it could be explored if making, as a part of the culture, when triggered by turbulent social contexts, leads to more grassroots critical making projects, or if other reasons might be behind the phenomenon.

Longitudinal research. Another area of future research is the fate of ideas and prototypes. As briefly indicated, Indonesian grassroots continuously develop new projects, and further develop old ones. This study only presented a snapshot of the Indonesian grassroots critical maker pluriverse from the period of the field research. During my field research, the water filtration system was to be adopted by the local government, and the Community Network Project was prepared for bringing the concept of community internet from West-Javanese mountains to an Eastern-Indonesian islands-based Indigenous community (thus, an entirely different social and geographical challenge). What effects the pandemic had on the projects and what new projects were developed in the meantime would also be interesting questions about grassroots innovation.

The study also raised the question how prototypes of critically made artefacts might eventually grow into sustainable development projects, and whether such developments might address—or even solve—wicked problems. The next layer of such research would look at adoption of such solutions by different communities in different contexts, and what such intrinsic processes teach us.

Reflexive reciprocity. Giving back is a direct response to exploitative colonial scholarship (Gupta and Kelly 2014). When I asked my interviewees how I could give back to them for their time, generosity, and labor, the answer I most often received was to send them a book of my research. Unfortunately, knowledge is still mainly produced in English and not in local languages, and books are often about a community rather than produced by them. It was also discussed in this study that while Indonesian grassroots communities might benefit from international visibility, due to the political-activistic opinions expressed in this study, some might be in danger. Hence, for this study, I decided to use pseudonyms. Following in the footsteps of Hertz, in a post-colonial manner, it would be

interesting to invite Indonesian critical makers to share their projects in their own words and let them decide how they are represented. A self-produced zine format would be a creative outlet that could be openly shared with others in Indonesian and in English.

7.4 Final Thoughts

“When the machines started breaking, we were happy” (R.D. 2020)

This quote embodies many things I wanted to reflect on in this study. The failures of universalist techno-solutionist approaches, as my other interviewee put, are disembedded from the local reality. It shows how a grassroots community received funding from the West to try a Western approach to generate something good, but they realized that they failed at it and were relieved when they did not have to engage with it anymore.

The blueprint of a Western Fab Lab is merely one symbol of many failed attempts at techno-solutionist blueprint development through ideological colonialism. My finding is simple: that no approach, solution, or innovation should be directly placed into any local context without considering how it makes sense there. As this study showed, even Java’s contexts are vastly different. How could “elimination design” or any universalist solution work here?

This study has used the diverse cases to contribute to theoretical propositions, not to produce generalized insights about communities. The value of extending the term critical making to the grassroots is extensive for research, to broaden fields focused on Western, industrial, and top-down methods of innovation.

As I showed, I’m not the first researcher who thinks that it is time to look beyond large-scale, industrial science and technology. Nevertheless, the first time I presented my research proposal four years ago, I received the snide remark that *these makers are nice, but they don’t invent the atomic bomb*. Examining this remark from a different perspective it is probably for the best that they don’t invent such technologies. From a societal perspective—as this study showed—it is even more relevant that instead grassroots innovators propose alternative, fairer, and more just futures through critical technical practice.

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